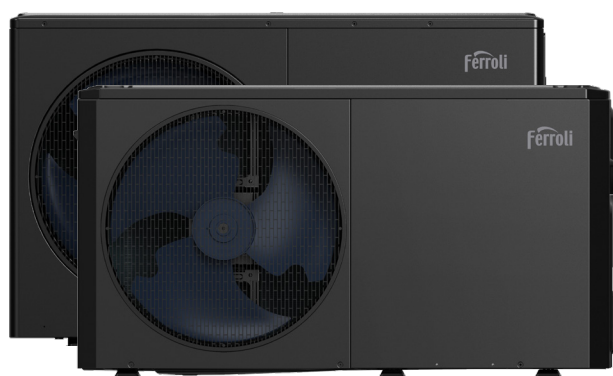


## OMNIA LIFE M MOD. 4-16T

POMPE DI CALORE REVERSIBILI PER INSTALLAZIONE ESTERNA CON COMPRESSORE DC INVERTER  
REVERSIBLE HEAT PUMP FOR OUTDOOR INSTALLATION WITH DC INVERTER COMPRESSOR



Cod. 3540002050 - Rev. 01 - 05/2025

IT

BOLLETTINO TECNICO

EN

TECHNICAL BULLETIN



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# 1. CARATTERISTICHE GENERALI

## 1.1 Descrizione del sistema

### Descrizione dell'unità

Questa serie di pompe di calore aria-acqua soddisfa le esigenze di climatizzazione invernale ed estiva di impianti residenziali e commerciali di piccola e media potenza.

Tutte le unità sono idonee per installazione esterna e potendo produrre acqua sino a 75°C possono essere impiegate in impianti radianti, a ventilconvettori, a radiatori e per la produzione indiretta di acqua calda sanitaria (ACS) tramite bollitore esterno (non fornito). Le unità si caratterizzano per l'utilizzo di un compressore DC inverter che permette di modularne la potenza erogata dal 30 al 120% della potenza nominale e sono complete di un kit idronico composto da tutti i componenti essenziali per un'installazione rapida e sicura. Le unità si caratterizzano per l'elevata efficienza energetica e per livelli sonori contenuti che ne consentono l'impiego come generatore unico a servizio dell'impianto o integrato con altre fonti energetiche quali resistenze elettriche di appoggio o caldaia.

Tutte le unità sono fornite con sonda temperatura acqua bollitore ACS (fornita di serie, montaggio a cura dell'installatore) e con sonda temperatura aria esterna (già installata sull'unità), per realizzare la regolazione climatica in riscaldamento e in raffreddamento. Tutte le unità sono accuratamente costruite e singolarmente collaudate in fabbrica. L'installazione richiede solamente i collegamenti elettrici ed idraulici.

**CIRCUITO FRIGORIFERO:** contenuto in un vano riparato dal flusso dell'aria per facilitare le operazioni di manutenzione, è dotato di compressore con motore DC INVERTER di tipo Twin rotary per garantire il maggior bilanciamento dinamico e ridurre le vibrazioni. Viene posizionato su supporti antivibranti in gomma e avvolto da un doppio strato di materiale fonoassorbente per ridurre il rumore. Il compressore inoltre è equipaggiato di resistenza olio carter. Il circuito si completa di scambiatore a piastre in acciaio inox saldobrasate completo di resistenza elettrica antigelo, valvola di espansione elettronica biflusso, valvola di inversione ciclo, ventilatori assiali con motore brushless DC completi di griglie di protezione antinfortunistiche, batteria alettata con trattamento anticorrosione costituita da tubi in rame e alette in alluminio idrofilico. Il circuito è controllato tramite sonde di temperatura e trasduttori di pressione e protetto tramite pressostati di alta pressione. Tutte le unità sono equipaggiate con controllo a velocità variabile dei ventilatori che ne consente il funzionamento con basse temperature esterne in raffreddamento e alte temperature esterne in riscaldamento.

**CIRCUITO IDRAULICO:** contenuto in un vano riparato dal flusso dell'aria per facilitare le operazioni di manutenzione, è dotato di circolatore elettronico a basso consumo con motore DC brushless a velocità variabile controllata da scheda elettronica, flussostato acqua, sfiato aria automatico, manometro acqua, vaso di espansione, valvola di sicurezza, filtro acqua a Y (fornito di serie, montaggio a cura dell'installatore). Lo scambiatore a piastre e tutte le tubazioni del circuito idraulico sono isolate termicamente per evitare la formazione di condensa e ridurre le dispersioni termiche.

Per le unità in versione con Riscaldatore elettrico impianto (IBH, (disponibile solo su alcuni modelli), nel circuito idraulico è installato a valle dello scambiatore a piastre un riscaldatore elettrico che può funzionare in integrazione (ad esempio con basse temperature esterne) o in sostituzione (ad esempio in caso di guasto) della potenza termica della pompa di calore.

### Accessori

- **DEGASATORE RAPIDO.** Si tratta di un dispositivo da installare sulla tubazione di mandata acqua impianto e ha lo scopo di garantire l'evacuazione rapida del refrigerante R290 in caso di rottura dello scambiatore a piastre. Questo consente di ottemperare ai requisiti dell'allegato GG punto 6 della norma IEC 60335-2-40
- **SERBATOIO INERZIALE ACCUMULO ACQUA.** E' costituito da un serbatoio da 60 litri in lamiera verniciata, termicamente isolato e inserito all'interno di una scatola in lamiera verniciata da posizionare sotto l'unità.
- **SONDA TEMPERATURA ACQUA.** La sonda di temperatura può essere collegata per svolgere le funzioni T1 / Tbt1 / Tbt2 / T5 / Tw2 / Tsolar (per maggiori dettagli fare riferimento al manuale di installazione e uso dell'unità).
- **ANTIVIBRANTI IN GOMMA.**

## 2. DATI ERP

| Modello             |                                         |                   | 4    | 6    | 8    | 10   | 12   | 14   | 16   | 12T  | 14T  | 16T  | UM     |
|---------------------|-----------------------------------------|-------------------|------|------|------|------|------|------|------|------|------|------|--------|
| Modo riscaldamento  | Bassa temperatura (acqua prodotta 35°C) | Efficienza        | 200  | 193  | 204  | 200  | 184  | 182  | 181  | 184  | 182  | 181  | ηs (%) |
|                     |                                         | Classe energetica | A+++ |      |      |      |      |      |      |      |      |      | class  |
|                     |                                         | SCOP              | 5,07 | 4,89 | 5,19 | 5,07 | 4,67 | 4,63 | 4,59 | 4,67 | 4,63 | 4,59 | W/W    |
|                     | Media temperatura (acqua prodotta 55°C) | Efficienza        | 149  | 150  | 150  | 150  | 142  | 141  | 140  | 142  | 141  | 140  | ηs (%) |
|                     |                                         | Classe energetica | A++  |      |      |      |      |      |      |      |      |      | class  |
|                     |                                         | SCOP              | 3,79 | 3,82 | 3,82 | 3,82 | 3,62 | 3,61 | 3,57 | 3,62 | 3,61 | 3,57 | W/W    |
| Modo raffreddamento | Acqua prodotta 7°C                      | Efficienza        | 206  | 210  | 231  | 219  | 204  | 204  | 202  | 204  | 204  | 202  | ηs (%) |
|                     |                                         | SEER              | 5,23 | 5,32 | 5,86 | 5,55 | 5,19 | 5,18 | 5,12 | 5,19 | 5,18 | 5,12 | W/W    |
|                     | Acqua prodotta 18°C                     | Efficienza        | 251  | 263  | 323  | 323  | 254  | 267  | 263  | 254  | 267  | 263  | ηs (%) |
|                     |                                         | SEER              | 6,36 | 6,65 | 8,14 | 8,16 | 6,42 | 6,75 | 6,65 | 6,42 | 6,75 | 6,65 | W/W    |

### NOTA:

I valori di efficienza, SCOP, SEER indicati in tabella si riferiscono al clima medio per unità prive di eventuali opzioni o accessori.

Classe di efficienza calcolata secondo regolamento europeo 811/2013.

### 3. DATI TECNICI

| Dati prestazioni |                                     |            | 4          | 6          | 8          | 10         | 12          | 14          | 16        | 12T         | 14T         | 16T       | UM        |
|------------------|-------------------------------------|------------|------------|------------|------------|------------|-------------|-------------|-----------|-------------|-------------|-----------|-----------|
| A7W35            | <b>Potenza termica</b>              | <b>nom</b> | <b>4,5</b> | <b>6,2</b> | <b>8,4</b> | <b>10</b>  | <b>12</b>   | <b>14</b>   | <b>15</b> | <b>12</b>   | <b>14</b>   | <b>15</b> | <b>kW</b> |
|                  | Potenza assorbita                   | nom        | 0,87       | 1,27       | 1,68       | 2,13       | 2,50        | 3,11        | 3,41      | 2,50        | 3,11        | 3,41      | kW        |
|                  | COP                                 |            | 5,15       | 4,90       | 5,00       | 4,70       | 4,80        | 4,50        | 4,40      | 4,80        | 4,50        | 4,40      | W/W       |
|                  | Portata acqua                       |            | 774        | 1066       | 1445       | 1720       | 2064        | 2408        | 2580      | 2064        | 2408        | 2580      | l/h       |
|                  | Pressione statica utile senza IBH * |            | 89         | 88         | 79         | 70         | 64          | 54          | 49        | 64          | 54          | 49        | kPa       |
|                  | Pressione statica utile con IBH *   |            | 84         | 82         | 76         | 67         | -           | -           | -         | 53          | 38          | 32        | kPa       |
| A7W45            | <b>Potenza termica</b>              | <b>nom</b> | <b>4,5</b> | <b>6,4</b> | <b>8,2</b> | <b>10</b>  | <b>12</b>   | <b>14</b>   | <b>15</b> | <b>12</b>   | <b>14</b>   | <b>15</b> | <b>kW</b> |
|                  | Potenza assorbita                   | nom        | 1,11       | 1,68       | 2,13       | 2,74       | 3,24        | 4,00        | 4,48      | 3,24        | 4,00        | 4,48      | kW        |
|                  | COP                                 |            | 4,05       | 3,80       | 3,85       | 3,65       | 3,70        | 3,50        | 3,35      | 3,70        | 3,50        | 3,35      | W/W       |
|                  | Portata acqua                       |            | 774        | 1101       | 1410       | 1720       | 2064        | 2408        | 2580      | 2064        | 2408        | 2580      | l/h       |
|                  | Pressione statica utile senza IBH * |            | 89         | 87         | 80         | 70         | 64          | 54          | 49        | 64          | 54          | 49        | kPa       |
|                  | Pressione statica utile con IBH *   |            | 84         | 82         | 77         | 67         | -           | -           | -         | 53          | 38          | 32        | kPa       |
| A7W55            | <b>Potenza termica</b>              | <b>nom</b> | <b>4,6</b> | <b>6,2</b> | <b>7,8</b> | <b>9,5</b> | <b>12</b>   | <b>14</b>   | <b>15</b> | <b>12</b>   | <b>14</b>   | <b>15</b> | <b>kW</b> |
|                  | Potenza assorbita                   | nom        | 1,44       | 2,00       | 2,44       | 3,12       | 3,87        | 4,67        | 5,26      | 3,87        | 4,67        | 5,26      | kW        |
|                  | COP                                 |            | 3,20       | 3,10       | 3,20       | 3,05       | 3,10        | 3,00        | 2,85      | 3,10        | 3,00        | 2,85      | W/W       |
|                  | Portata acqua                       |            | 495        | 667        | 839        | 1021       | 1290        | 1505        | 1613      | 1290        | 1505        | 1613      | l/h       |
|                  | Pressione statica utile senza IBH * |            | 89         | 89         | 86         | 86         | 86          | 81          | 78        | 86          | 81          | 78        | kPa       |
|                  | Pressione statica utile con IBH *   |            | 86         | 85         | 83         | 83         | -           | -           | -         | 82          | 76          | 72        | kPa       |
| A35W18           | <b>Potenza frigorifera</b>          | <b>nom</b> | <b>4,5</b> | <b>6,5</b> | <b>8,3</b> | <b>10</b>  | <b>12</b>   | <b>14</b>   | <b>16</b> | <b>12</b>   | <b>14</b>   | <b>16</b> | <b>kW</b> |
|                  | Potenza assorbita                   | nom        | 0,82       | 1,28       | 1,61       | 2,11       | 2,67        | 3,33        | 4,10      | 2,67        | 3,33        | 4,10      | kW        |
|                  | EER                                 |            | 5,50       | 5,10       | 5,15       | 4,75       | 4,50        | 4,20        | 3,90      | 4,50        | 4,20        | 3,90      | W/W       |
|                  | Portata acqua                       |            | 774        | 1118       | 1428       | 1720       | 2064        | 2408        | 2752      | 2064        | 2408        | 2752      | l/h       |
|                  | Pressione statica utile senza IBH * |            | 89         | 87         | 80         | 70         | 64          | 54          | 44        | 64          | 54          | 44        | kPa       |
|                  | Pressione statica utile con IBH *   |            | 84         | 82         | 77         | 67         | -           | -           | -         | 53          | 38          | 24        | kPa       |
| A35W7            | <b>Potenza frigorifera</b>          | <b>nom</b> | <b>4,7</b> | <b>6,8</b> | <b>7,5</b> | <b>8,9</b> | <b>11,5</b> | <b>12,7</b> | <b>14</b> | <b>11,5</b> | <b>12,7</b> | <b>14</b> | <b>kW</b> |
|                  | Potenza assorbita                   | nom        | 1,29       | 2,19       | 2,17       | 2,74       | 3,77        | 4,38        | 5,09      | 3,77        | 4,38        | 5,09      | kW        |
|                  | EER                                 |            | 3,65       | 3,10       | 3,45       | 3,25       | 3,05        | 2,90        | 2,75      | 3,05        | 2,90        | 2,75      | W/W       |
|                  | Portata acqua                       |            | 808        | 1170       | 1290       | 1531       | 1978        | 2184        | 2408      | 1978        | 2184        | 2408      | l/h       |
|                  | Pressione statica utile senza IBH * |            | 89         | 87         | 83         | 77         | 66          | 60          | 54        | 66          | 60          | 54        | kPa       |
|                  | Pressione statica utile con IBH *   |            | 84         | 81         | 80         | 74         | -           | -           | -         | 56          | 47          | 38        | kPa       |

I valori si riferiscono ad unità prive di eventuali opzioni o accessori.

Dati dichiarati secondo EN 14511:

**EER** (Energy Efficiency Ratio) = rapporto potenza frigorifera su potenza assorbita

**COP** (Coefficient Of Performance) = rapporto potenza termica su potenza assorbita

**A7W35** = sorgente : aria in 7°C b.s. 6°C b.u. / impianto : acqua in 30°C out 35°C

**A7W45** = sorgente : aria in 7°C b.s. 6°C b.u. / impianto : acqua in 40°C out 45°C

**A7W55** = sorgente : aria in 7°C b.s. 6°C b.u. / impianto : acqua in 47°C out 55°C

**A35W18** = sorgente : aria in 35°C b.s. / impianto : acqua in 23°C out 18°C

**A35W7** = sorgente : aria in 35°C b.s. / impianto : acqua in 12°C out 7°C

**IBH \*** = Riscaldatore elettrico impianto

| Dati tecnici generali                                                       |          |           | 4                         | 6    | 8             | 10   | 12             | 14   | 16 | 12T                      | 14T | 16T   | UM              |
|-----------------------------------------------------------------------------|----------|-----------|---------------------------|------|---------------|------|----------------|------|----|--------------------------|-----|-------|-----------------|
| Alimentazione elettrica                                                     |          |           | 220-240 V ~ / 50 Hz       |      |               |      |                |      |    | 380-415 V~ / 3+N / 50 Hz |     |       | -               |
| Tipo di compressore                                                         |          |           | Twin Rotary DC            |      |               |      |                |      |    |                          |     |       | -               |
| N° di compressori / N° circuiti frigoriferi                                 |          |           | 1 / 1                     |      |               |      |                |      |    |                          |     |       | n°              |
| Tipo scambiatore lato impianto                                              |          |           | piastre inox saldobrasate |      |               |      |                |      |    |                          |     |       | -               |
| Tipo scambiatore lato sorgente                                              |          |           | batteria alettata         |      |               |      |                |      |    |                          |     |       | -               |
| Tipo di ventilatori                                                         |          |           | assiale DC                |      |               |      |                |      |    |                          |     |       | -               |
| N° di ventilatori                                                           |          |           | 1                         |      |               |      |                |      |    |                          |     |       | n°              |
| Volume vaso di espansione                                                   |          |           | 8                         |      |               |      |                |      |    |                          |     |       | l               |
| Taratura valvola di sicurezza acqua                                         |          |           | 3                         |      |               |      |                |      |    |                          |     |       | bar             |
| Riscaldatore elettrico impianto<br>(disponibile solo per versione con IBH*) |          |           | 3,25                      |      |               | -    |                |      | 9  |                          |     | kW    |                 |
| Attacchi idraulici                                                          |          |           | 1                         |      | 1 1/4         |      |                |      |    |                          |     | “     |                 |
| Contenuto minino acqua impianto                                             |          |           | 40                        |      |               |      |                |      |    |                          |     |       | l               |
| Sup. minima serpentino eventuale bollitore ACS<br>(minima / raccomandata)   | acciaio  | 1,4 / 2,5 | 1,75 / 4,0                |      |               |      |                |      |    |                          |     | m²    |                 |
|                                                                             | smaltato | 1,7 / 3,0 | 2,5 / 5,6                 |      |               |      |                |      |    |                          |     | m²    |                 |
| Tipo di refrigerante                                                        |          |           | R290                      |      |               |      |                |      |    |                          |     |       | tipo            |
| GWP                                                                         |          |           | 3                         |      |               |      |                |      |    |                          |     |       | kg-CO2 eq.      |
| Carica refrigerante                                                         |          |           | 0,7<br>0,0021             |      | 1,1<br>0,0033 |      | 1,25<br>0,0375 |      |    |                          |     |       | kg<br>t-CO2 eq. |
| Tipo controllo                                                              |          |           | a filo remoto             |      |               |      |                |      |    |                          |     |       | -               |
| SWL - Livello di potenza sonora<br>in riscaldamento *                       | A7W35    | 56        | 58                        | 60   | 61            | 65   | 65             | 69   | 65 | 65                       | 69  | dB(A) |                 |
|                                                                             | Max      | 58        | 60                        | 62   | 63            | 67   | 68             | 70   | 67 | 68                       | 70  | dB(A) |                 |
|                                                                             | Sil. 1   | 54        | 56                        | 58   | 59            | 62   | 63             | 64   | 62 | 63                       | 64  | dB(A) |                 |
|                                                                             | Sil. 2   | 51        | 53                        | 55   | 56            | 58   | 59             | 60   | 58 | 59                       | 60  | dB(A) |                 |
| SWL - Livello di potenza<br>sonora in raffreddamento *                      | A35W18   | 56        | 58                        | 60   | 61            | 65   | 66             | 69   | 65 | 66                       | 69  | dB(A) |                 |
|                                                                             | Max      | 58        | 60                        | 62   | 63            | 66   | 67             | 70   | 66 | 67                       | 70  | dB(A) |                 |
|                                                                             | Sil. 1   | 54        | 56                        | 57   | 58            | 62   | 62             | 64   | 62 | 62                       | 64  | dB(A) |                 |
|                                                                             | Sil. 2   | 51        | 53                        | 56   | 55            | 58   | 59             | 60   | 58 | 59                       | 60  | dB(A) |                 |
| Corrente massima assorbita                                                  |          |           | 12                        | 13,5 | 16            | 17,5 | 25             | 26,5 | 28 | 8,5                      | 9   | 9,5   | A               |

\* : SWL = Livelli di potenza sonora, riferiti a 1x10-12 W con unità funzionante in condizioni:

**A7W35** = sorgente : aria in 7°C b.s. 6°C b.u. / impianto : acqua in 30°C out 35°C.

**A35W18** = sorgente : aria in 35°C b.s. / impianto : acqua in 23°C out 18°C

**Max** = alle condizioni massimali in modo riscaldamento / raffreddamento

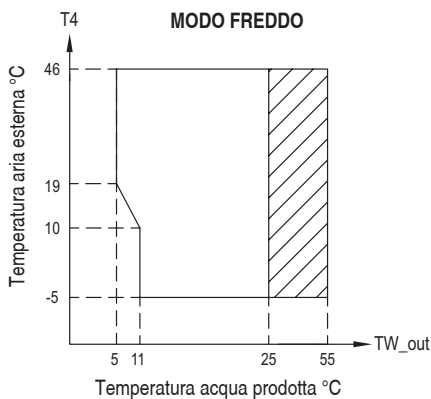
**Sil. 1** = se attivo livello silenzioso 1 in modo riscaldamento / raffreddamento

**Sil. 2** = se attivo livello silenzioso 2 in modo riscaldamento / raffreddamento

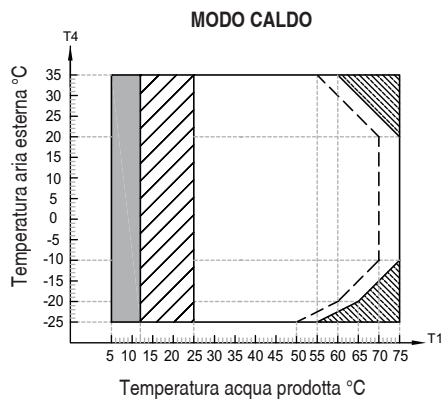
Il livello di potenza sonora Totale in dB(A) è misurato in accordo alla normativa ISO 9614.

**IBH \*** = Riscaldatore elettrico impianto

## 4. LIMITI OPERATIVI



▨ Campo di funzionamento con pompa di calore con possibile limitazione e protezione.

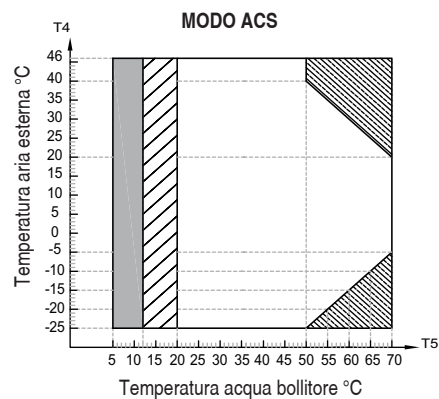


▨ Se l'impostazione IBH (resistenza elettrica) / AHS (caldaia) è abilitata, si attiva solo IBH / AHS; Se l'impostazione IBH / AHS non è abilitata, si attiva solo la pompa di calore con possibili limitazioni e protezioni.

▨ Campo di funzionamento con pompa di calore con possibile limitazione e protezione.

▨ La pompa di calore rimane spenta e si accende solo IBH (resistenza elettrica) / AHS (caldaia)

— Linea massima temperatura acqua in ingresso per funzionamento pompa di calore.



▨ Se l'impostazione TBH (resistenza elettrica bollitore) / IBH (resistenza elettrica) / AHS (caldaia) è abilitata, si attiva solo TBH / IBH / AHS.

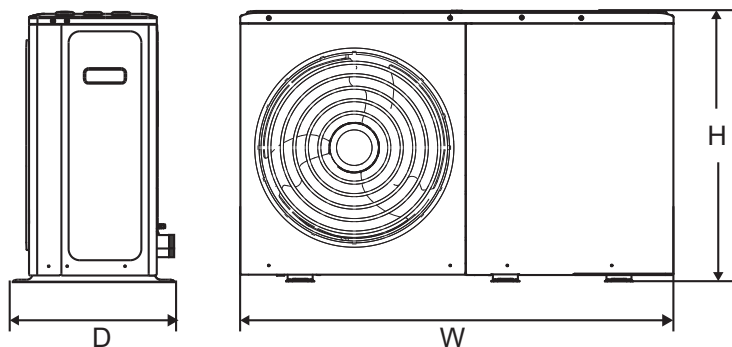
Se l'impostazione TBH / IBH / AHS non è abilitata, si attiva solo la pompa di calore con possibili limitazioni e protezioni.

▨ Campo di funzionamento con pompa di calore con possibile limitazione e protezione.

▨ La pompa di calore rimane spenta e si accende solo TBH (resistenza elettrica bollitore) / IBH (resistenza elettrica) / AHS (caldaia)

**NOTA MODO ACS:** la temperatura acqua massima indicata nel grafico, si basa su un corretto dimensionamento della superficie del serpentino dell'eventuale bollitore ACS.

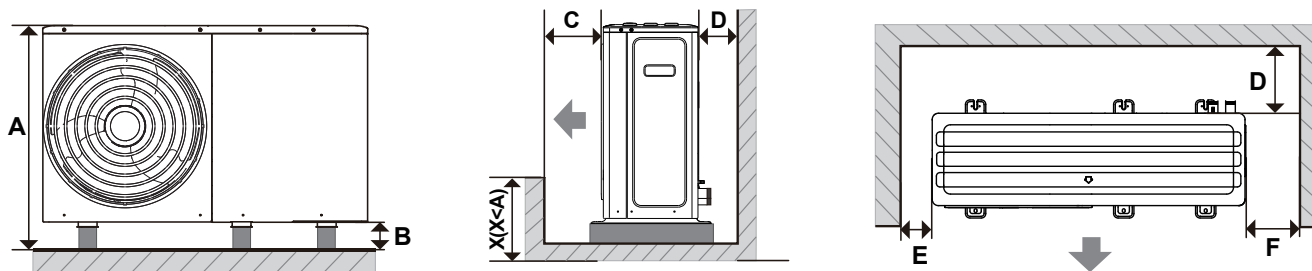
## 5. DATI DIMENSIONALI



| Modelli                                | 4                | 6 | 8                 | 10 | 12        | 14 | 16        | 12T | 14T | 16T | UM |
|----------------------------------------|------------------|---|-------------------|----|-----------|----|-----------|-----|-----|-----|----|
| Unità (WxHxD)                          | 1299 x 717 x 426 |   | 1385 x 865 x 523  |    |           |    |           |     |     |     | mm |
| Imballo (WxHxD)                        | 1375 x 885 x 475 |   | 1465 x 1035 x 560 |    |           |    |           |     |     |     | mm |
| Peso netto \ lordo versione senza IBH* | 90 / 110         |   | 117 / 139         |    | 135 / 157 |    | 137 / 159 |     |     |     | kg |
| Peso netto \ lordo versione con IBH*   | 95 / 115         |   | 122 / 144         |    | -         |    | 142 / 164 |     |     |     | kg |

IBH \* = Riscaldatore elettrico impianto

### 5.1 Spazio minimo di installazione



| Modelli | 4                      | 6 | 8 | 10 | 12 | 14                     | 16 | 12T | 14T | 16T | UM |
|---------|------------------------|---|---|----|----|------------------------|----|-----|-----|-----|----|
| A       | Altezza dell'unità + B |   |   |    |    | Altezza dell'unità + B |    |     |     |     | mm |
| B       | ≥ 100*                 |   |   |    |    | ≥ 100*                 |    |     |     |     | mm |
| C       | ≥ 1000                 |   |   |    |    | ≥ 1500                 |    |     |     |     | mm |
| D       | ≥ 300                  |   |   |    |    | ≥ 300                  |    |     |     |     | mm |
| E       | ≥ 500                  |   |   |    |    | ≥ 500                  |    |     |     |     | mm |
| F       | ≥ 500                  |   |   |    |    | ≥ 500                  |    |     |     |     | mm |

\* In caso di freddo, tenere conto della presenza di neve al suolo.

## 6. SISTEMA DI CONTROLLO

L'interfaccia utente è costituita da un controllore remoto cablato (2 fili, lunghezza max 50 m dall'unità) con menu in 18 lingue (IT italiano, EN inglese, FR francese, ES spagnolo, DE tedesco, NL olandese, PL polacco, RO rumeno, PT portoghese, RU russo, TR turco, GR Greco, SE Svedese, SI Sloveno, CS Ceco, SK Slovacco, HU Ungherese, HR Croato) che consente la gestione di:

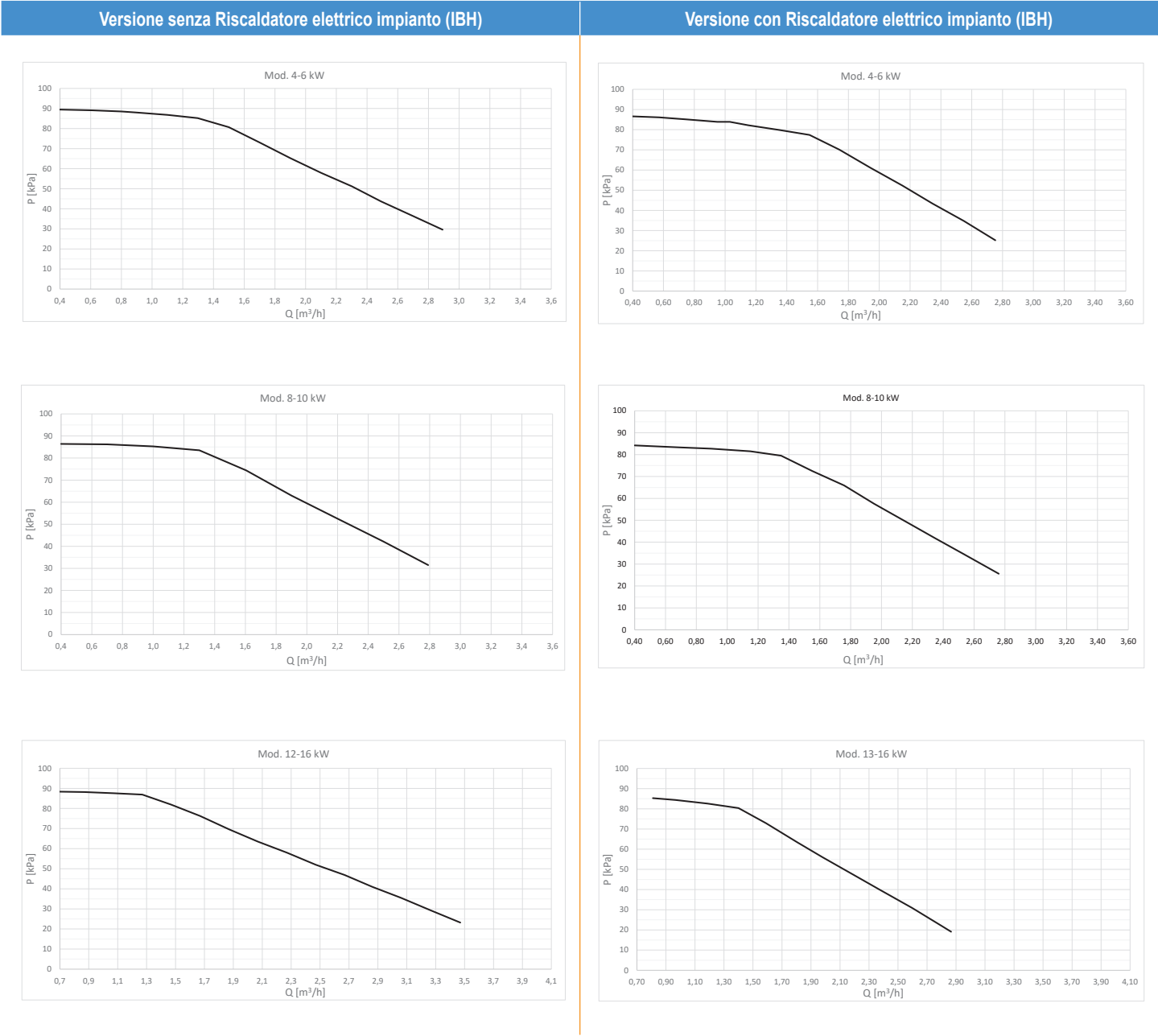
- **IMPIANTO RISCALDAMENTO E RAFFREDDAMENTO** dove la pompa di calore è l'unica fonte energetica. L'unità si attiva in modo caldo o freddo, funziona modulando la frequenza del compressore per mantenere la temperatura acqua prodotta al valore di setpoint impostato tramite controllore. Tramite parametrizzazione è possibile utilizzare il controllore remoto (ad es. per impianti monozona) come termostato ambiente.
- **PRODUZIONE ACQUA CALDA SANITARIA (ACS).** L'unità si attiva in modo caldo per mantenere la temperatura di un bollitore ACS (non fornito) al valore di setpoint impostato. Sono necessarie una valvola a 3 vie deviatrice (non fornita) e una sonda di temperatura (sonda T5, L=10m, fornita a corredo) da inserire in un pozzetto del bollitore ACS.
- **FONTI ENERGETICHE ADDIZIONALI** (caldaia o riscaldatore elettrico). In funzione dei parametri impostati, tali fonti possono essere attivati in Integrazione o Sostituzione della pompa di calore quando il sistema serve l'impianto in riscaldamento o per la produzione ACS. La scheda attiverà inoltre le fonti energetiche aggiuntive in caso la pompa di calore non funzioni. E' necessaria una sonda di temperatura (disponibile come accessorio) da inserire in un pozzetto della tubazione comune o dell'accumulo per la misurazione della temperatura dell'acqua del sistema.
- **RISCALDATORE ELETTRICO BOLLITORE ACS.** In modo sanitario può gestire un riscaldatore elettrico inserito nel bollitore ACS come integrazione alla pompa di calore, funzione antilegionella, o come fonte energetica di riserva per la produzione ACS in caso la pompa di calore non funzioni.
- **FAST ACS.** Funzione attivabile manualmente che consente di dare priorità al sanitario attivando tutte le fonti energetiche (pompa di calore, resistenze elettriche, caldaia) disponibili per il riscaldamento ACS per portare nel tempo minore possibile il bollitore ACS al setpoint impostato.
- **FUNZIONE ANTILEGIONELLA.** È possibile impostare da controllore cicli settimanali antilegionella. Per poter eseguire correttamente tali cicli la pompa di calore deve essere integrata con riscaldatore elettrico boiler ACS o caldaia.
- **MODO SILENZIATO.** Se attivo comporta una riduzione della massima frequenza del compressore e della velocità del ventilatore per ridurre il rumore emesso e la potenza assorbita dall'unità. Sono disponibili 2 livelli di silenziamento. Tramite programmazione oraria, è possibile definire per 2 fasce orarie giornaliere in cui il modo silenziato sarà attivo (ad es. di notte). Sono disponibili 2 livelli di silenziamento. Il livello selezionato verrà poi applicato ad entrambe le fasce orarie impostate.
- **ON/OFF tramite un contatto esterno.** L'unità può essere attivata e disattivata (ad es. termostato di zona / interruttore remoto) tramite un contatto esterno: in questo caso l'unità funzionerà nel modo impostato tramite tastiera controllore.
- **CALDO/FREDDO tramite contatti esterni.** L'unità può essere attivata e disattivata in modo freddo e modo caldo tramite 2 contatti esterni (ad es. termostato di zona che gestisce la richiesta di caldo e freddo / interruttore remoto).
- **ECO.** Possibilità di definire in modo caldo le fasce orarie e relativo setpoint per il modo ECO.
- **PROGRAMMAZIONE ORARIA SETTIMANALE.** Consente la programmazione oraria differenziata per ciascun giorno della settimana definendo per ogni fascia il modo (FREDDO/CALDO/ACS) e il setpoint di lavoro.
- **PROTEZIONE ANTIGELO.** Garantita sino a temperatura aria esterna di -25°C grazie alla gestione della scheda elettronica dell'unità che consente di riscaldare l'acqua usando il riscaldatore elettrico antigelo (di serie sullo scambiatore a piastre), la pompa di calore stessa funzionante in modo caldo, e il booster elettrico (se installato).
- **GESTIONE CASCATA DI PIÙ UNITÀ** - controllo in cascata di fino a 6 unità (1 master, 5 slave, anche di diversa potenza), mediante un unico controller collegato all'unità master. Possibilità di dedicare l'unità master alla funzione di produzione ACS (acqua calda sanitaria). In caso di guasto di una unità slave, le altre unità continuano a funzionare normalmente.
- **GESTIONE SINO A 2 ZONE (1 MISCELATA E 1 DIRETTA).** L'unità è in grado di gestire le pompe (non fornite) di entrambe le zone e, per la sola zona miscelata, anche la valvola miscelatrice (non fornita) e la sonda temperatura mandata acqua (disponibile come accessorio).
- **SOLARE TERMICO.** Gestione del solare termico a circolazione forzata (circolatore non fornito, sonda disponibile come accessorio) garantita dalle logiche di funzionamento della pompa di calore.
- **INPUT FOTOVOLTAICO E INPUT SMART GRID.** L'unità è dotata di 2 ingressi digitali per la gestione di un input da impianto fotovoltaico e da rete elettrica. Logica di funzionamento:
  - SG=OFF; EVU=OFF > Funzionamento normale
  - SG=ON; EVU=OFF > L'elettricità ha un costo elevato, PDC ed eventuali riscaldatori elettrici vengono spenti.
  - SG=OFF; EVU=ON > Energia elettrica disponibile a basso prezzo, ACS prioritario e riscaldatori elettrici ausiliari ON.
  - SG=ON; EVU=ON > L'unità attiva la modalità ACS con setpoint ACS = 70°C e (se disponibile) si accenderà il riscaldatore elettrico del serbatoio ACS. L'unità funziona in modalità raffreddamento / riscaldamento come la normale logica.
- **ENERGY MONITORING.** E' possibile monitorare i consumi, la potenza resa e la quota di rinnovabile prodotta dall'utilizzo della pompa di calore, visibili sia dall'interfaccia utente che dall'APP.
- **LIMITAZIONE DI CORRENTE DA PARAMETRO.**
- **CONTROLLO REMOTO DELL'UNITÀ VIA APP** (disponibile per IOS e Android).
- **DIAGNOSTICA ERRORI** dettagliata con storico allarmi.
- **VISUALIZZAZIONE DI TUTTI I PARAMETRI OPERATIVI.**



7. PREVALENZA UTILE

Si intende come prevalenza utile quella all'uscita dell'unità decurtata di tutte le perdite di carico interne all'unità.  
Il campo di funzionamento è delimitato dal valore minimo riportato nella tabella successiva.

| Modelli                  | 4   | 6 | 8 | 10 | 12  | 14 | 16 | 12T | 14T | 16T | UM |      |
|--------------------------|-----|---|---|----|-----|----|----|-----|-----|-----|----|------|
| Portata minima richiesta | 0,4 |   |   |    | 0,7 |    |    |     |     |     |    | m³/h |



NOTA  
P: Prevalenza utile  
Q: Portata acqua

8. TABELLE PRESTAZIONALI - MODO RISCALDAMENTO

8.1 Prestazioni riscaldamento Mod. 04

| Tw_out<br>°C | DB/WB<br>°C | Maximum |      |      | 100%(Normal) |             |             | 75%  |       |      | 50%  |       |      | 25%  |       |      | Minimum |       |      |
|--------------|-------------|---------|------|------|--------------|-------------|-------------|------|-------|------|------|-------|------|------|-------|------|---------|-------|------|
|              |             | HC      | COP  | PI   | HC           | COP         | PI          | HC   | COP   | PI   | HC   | COP   | PI   | HC   | COP   | PI   | HC      | COP   | PI   |
| 25           | -25         | 3.24    | 2.17 | 1.50 | 3.24         | 2.17        | 1.50        | 2.42 | 2.43  | 1.00 | 1.60 | 2.48  | 0.65 | 1.34 | 2.50  | 0.54 | 1.34    | 2.50  | 0.54 |
|              | -20         | 3.92    | 2.50 | 1.57 | 3.92         | 2.50        | 1.57        | 2.95 | 2.77  | 1.06 | 1.97 | 2.80  | 0.70 | 1.63 | 2.82  | 0.58 | 1.63    | 2.82  | 0.58 |
|              | -15         | 4.63    | 2.85 | 1.63 | 4.45         | 2.89        | 1.54        | 3.28 | 2.97  | 1.11 | 2.20 | 3.12  | 0.70 | 1.92 | 3.15  | 0.61 | 1.92    | 3.15  | 0.61 |
|              | -10         | 5.54    | 3.26 | 1.70 | 4.43         | 3.43        | 1.29        | 3.32 | 3.60  | 0.92 | 2.35 | 3.62  | 0.65 | 2.35 | 3.62  | 0.65 | 2.35    | 3.62  | 0.65 |
|              | -7          | 5.78    | 3.56 | 1.62 | 4.41         | 3.75        | 1.18        | 3.28 | 3.90  | 0.84 | 2.55 | 3.94  | 0.65 | 2.55 | 3.94  | 0.65 | 2.55    | 3.94  | 0.65 |
|              | -5          | 5.95    | 3.74 | 1.59 | 4.49         | 4.10        | 1.09        | 3.39 | 4.21  | 0.81 | 2.73 | 4.28  | 0.64 | 2.73 | 4.28  | 0.64 | 2.73    | 4.28  | 0.64 |
|              | -2          | 6.07    | 4.06 | 1.49 | 4.38         | 4.50        | 0.97        | 3.31 | 4.61  | 0.72 | 2.43 | 4.64  | 0.52 | 2.43 | 4.64  | 0.52 | 2.43    | 4.64  | 0.52 |
|              | 0           | 6.16    | 4.32 | 1.43 | 4.42         | 4.78        | 0.92        | 3.37 | 4.90  | 0.69 | 2.59 | 4.97  | 0.52 | 2.59 | 4.97  | 0.52 | 2.59    | 4.97  | 0.52 |
|              | 2           | 6.29    | 4.58 | 1.37 | 4.40         | 5.06        | 0.87        | 3.33 | 5.18  | 0.64 | 2.74 | 5.27  | 0.52 | 2.74 | 5.27  | 0.52 | 2.74    | 5.27  | 0.52 |
|              | 5           | 6.80    | 5.31 | 1.28 | 4.38         | 5.86        | 0.75        | 3.28 | 6.02  | 0.54 | 2.79 | 6.08  | 0.46 | 2.79 | 6.08  | 0.46 | 2.79    | 6.08  | 0.46 |
|              | 7           | 7.18    | 5.74 | 1.25 | 4.55         | 6.08        | 0.75        | 3.32 | 6.32  | 0.53 | 3.01 | 6.38  | 0.47 | 3.01 | 6.38  | 0.47 | 3.01    | 6.38  | 0.47 |
|              | 10          | 7.04    | 6.25 | 1.13 | 4.46         | 6.89        | 0.65        | 3.34 | 7.17  | 0.47 | 3.34 | 7.17  | 0.47 | 3.34 | 7.17  | 0.47 | 3.34    | 7.17  | 0.47 |
|              | 12          | 7.26    | 6.79 | 1.07 | 4.43         | 7.57        | 0.59        | 3.53 | 7.77  | 0.45 | 3.53 | 7.77  | 0.45 | 3.53 | 7.77  | 0.45 | 3.53    | 7.77  | 0.45 |
|              | 15          | 6.90    | 7.25 | 0.95 | 4.42         | 7.88        | 0.56        | 3.62 | 8.14  | 0.44 | 3.62 | 8.14  | 0.44 | 3.62 | 8.14  | 0.44 | 3.62    | 8.14  | 0.44 |
|              | 20          | 6.48    | 8.11 | 0.80 | 4.52         | 8.79        | 0.51        | 3.40 | 9.23  | 0.37 | 3.40 | 9.23  | 0.37 | 3.40 | 9.23  | 0.37 | 3.40    | 9.23  | 0.37 |
|              | 25          | 6.42    | 9.45 | 0.68 | 4.44         | 10.23       | 0.43        | 3.75 | 10.57 | 0.35 | 3.75 | 10.57 | 0.35 | 3.75 | 10.57 | 0.35 | 3.75    | 10.57 | 0.35 |
|              | 30          | /       | /    | /    | /            | /           | /           | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 35          | /       | /    | /    | /            | /           | /           | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 30           | -25         | 3.22    | 1.99 | 1.62 | 3.22         | 1.99        | 1.62        | 2.39 | 2.04  | 1.17 | 1.60 | 2.10  | 0.76 | 1.29 | 2.12  | 0.61 | 1.29    | 2.12  | 0.61 |
|              | -20         | 3.88    | 2.29 | 1.70 | 3.88         | 2.29        | 1.70        | 2.91 | 2.37  | 1.23 | 1.94 | 2.45  | 0.79 | 1.58 | 2.48  | 0.64 | 1.58    | 2.48  | 0.64 |
|              | -15         | 4.56    | 2.65 | 1.72 | 4.56         | 2.65        | 1.72        | 3.30 | 2.69  | 1.23 | 2.18 | 2.78  | 0.78 | 1.85 | 2.85  | 0.65 | 1.85    | 2.85  | 0.65 |
|              | -10         | 5.45    | 3.02 | 1.80 | 4.42         | 3.17        | 1.39        | 3.32 | 3.23  | 1.03 | 2.20 | 3.28  | 0.67 | 2.20 | 3.28  | 0.67 | 2.20    | 3.28  | 0.67 |
|              | -7          | 5.67    | 3.29 | 1.72 | 4.42         | 3.44        | 1.29        | 3.30 | 3.52  | 0.94 | 2.52 | 3.63  | 0.69 | 2.52 | 3.63  | 0.69 | 2.52    | 3.63  | 0.69 |
|              | -5          | 5.83    | 3.45 | 1.69 | 4.36         | 3.69        | 1.18        | 3.26 | 3.75  | 0.87 | 2.63 | 3.79  | 0.69 | 2.63 | 3.79  | 0.69 | 2.63    | 3.79  | 0.69 |
|              | -2          | 5.95    | 3.74 | 1.59 | 4.40         | 4.02        | 1.09        | 3.33 | 4.08  | 0.82 | 2.39 | 4.18  | 0.57 | 2.39 | 4.18  | 0.57 | 2.39    | 4.18  | 0.57 |
|              | 0           | 6.04    | 3.97 | 1.52 | 4.42         | 4.19        | 1.05        | 3.31 | 4.30  | 0.77 | 2.49 | 4.38  | 0.57 | 2.49 | 4.38  | 0.57 | 2.49    | 4.38  | 0.57 |
|              | 2           | 6.17    | 4.19 | 1.47 | 4.42         | 4.51        | 0.98        | 3.36 | 4.58  | 0.73 | 2.64 | 4.65  | 0.57 | 2.64 | 4.65  | 0.57 | 2.64    | 4.65  | 0.57 |
|              | 5           | 6.66    | 4.84 | 1.38 | 4.36         | 5.15        | 0.85        | 3.26 | 5.25  | 0.62 | 2.79 | 5.37  | 0.52 | 2.79 | 5.37  | 0.52 | 2.79    | 5.37  | 0.52 |
|              | 7           | 7.03    | 5.20 | 1.35 | 4.54         | 5.63        | 0.81        | 3.36 | 5.79  | 0.58 | 3.04 | 5.85  | 0.52 | 3.04 | 5.85  | 0.52 | 3.04    | 5.85  | 0.52 |
|              | 10          | 7.18    | 5.81 | 1.24 | 4.44         | 6.34        | 0.70        | 3.33 | 6.44  | 0.52 | 3.20 | 6.46  | 0.50 | 3.20 | 6.46  | 0.50 | 3.20    | 6.46  | 0.50 |
|              | 12          | 7.42    | 6.10 | 1.21 | 4.42         | 6.58        | 0.67        | 3.44 | 6.78  | 0.51 | 3.44 | 6.78  | 0.51 | 3.44 | 6.78  | 0.51 | 3.44    | 6.78  | 0.51 |
|              | 15          | 6.74    | 6.37 | 1.06 | 4.53         | 6.80        | 0.67        | 3.50 | 7.02  | 0.50 | 3.50 | 7.02  | 0.50 | 3.50 | 7.02  | 0.50 | 3.50    | 7.02  | 0.50 |
|              | 20          | 6.35    | 7.10 | 0.89 | 4.53         | 7.55        | 0.60        | 3.29 | 7.90  | 0.42 | 3.29 | 7.90  | 0.42 | 3.29 | 7.90  | 0.42 | 3.29    | 7.90  | 0.42 |
|              | 25          | 6.31    | 8.18 | 0.77 | 4.44         | 8.78        | 0.51        | 3.62 | 9.07  | 0.40 | 3.62 | 9.07  | 0.40 | 3.62 | 9.07  | 0.40 | 3.62    | 9.07  | 0.40 |
|              | 30          | 6.63    | 9.33 | 0.71 | 4.44         | 9.61        | 0.46        | 3.98 | 9.77  | 0.41 | 3.98 | 9.77  | 0.41 | 3.98 | 9.77  | 0.41 | 3.98    | 9.77  | 0.41 |
|              | 35          | /       | /    | /    | /            | /           | /           | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 35           | -25         | 3.19    | 1.86 | 1.71 | 3.19         | 1.86        | 1.71        | 2.40 | 1.92  | 1.26 | 1.58 | 2.01  | 0.79 | 1.24 | 2.05  | 0.61 | 1.24    | 2.05  | 0.61 |
|              | -20         | 3.82    | 2.14 | 1.79 | 3.82         | 2.14        | 1.79        | 2.87 | 2.18  | 1.31 | 1.92 | 2.28  | 0.84 | 1.52 | 2.32  | 0.65 | 1.52    | 2.32  | 0.65 |
|              | -15         | 4.47    | 2.46 | 1.82 | 4.42         | 2.40        | 1.84        | 3.31 | 2.49  | 1.33 | 2.20 | 2.64  | 0.83 | 1.79 | 2.66  | 0.67 | 1.79    | 2.66  | 0.67 |
|              | -10         | 5.35    | 2.81 | 1.90 | 4.42         | 2.86        | 1.55        | 3.27 | 2.97  | 1.10 | 2.18 | 3.08  | 0.71 | 2.18 | 3.08  | 0.71 | 2.18    | 3.08  | 0.71 |
|              | -7          | 5.56    | 3.04 | 1.83 | 4.50         | 3.10        | 1.45        | 3.32 | 3.12  | 1.06 | 2.36 | 3.21  | 0.73 | 2.36 | 3.21  | 0.73 | 2.36    | 3.21  | 0.73 |
|              | -5          | 5.70    | 3.18 | 1.80 | 4.37         | 3.32        | 1.32        | 3.30 | 3.37  | 0.98 | 2.53 | 3.43  | 0.74 | 2.53 | 3.43  | 0.74 | 2.53    | 3.43  | 0.74 |
|              | -2          | 5.82    | 3.44 | 1.69 | 4.27         | 3.58        | 1.19        | 3.18 | 3.67  | 0.87 | 2.31 | 3.74  | 0.62 | 2.31 | 3.74  | 0.62 | 2.31    | 3.74  | 0.62 |
|              | 0           | 5.90    | 3.64 | 1.62 | 4.38         | 3.78        | 1.16        | 3.28 | 3.88  | 0.85 | 2.39 | 3.95  | 0.61 | 2.39 | 3.95  | 0.61 | 2.39    | 3.95  | 0.61 |
|              | 2           | 6.05    | 3.84 | 1.57 | 4.40         | 4.10        | 1.07        | 3.34 | 4.14  | 0.81 | 2.53 | 4.19  | 0.60 | 2.53 | 4.19  | 0.60 | 2.53    | 4.19  | 0.60 |
|              | 5           | 6.50    | 4.40 | 1.48 | 4.33         | 4.61        | 0.94        | 3.24 | 4.70  | 0.69 | 2.68 | 4.75  | 0.56 | 2.68 | 4.75  | 0.56 | 2.68    | 4.75  | 0.56 |
|              | 7*          | 6.86    | 4.72 | 1.46 | <b>4.50</b>  | <b>5.15</b> | <b>0.87</b> | 3.34 | 5.25  | 0.64 | 2.93 | 5.32  | 0.55 | 2.93 | 5.32  | 0.55 | 2.93    | 5.32  | 0.55 |
|              | 10          | 7.00    | 5.21 | 1.34 | 4.43         | 5.58        | 0.79        | 3.37 | 5.67  | 0.59 | 3.24 | 5.78  | 0.56 | 3.24 | 5.78  | 0.56 | 3.24    | 5.78  | 0.56 |
|              | 12          | 7.23    | 5.59 | 1.29 | 4.49         | 6.01        | 0.75        | 3.48 | 6.14  | 0.57 | 3.45 | 6.22  | 0.56 | 3.45 | 6.22  | 0.56 | 3.45    | 6.22  | 0.56 |
|              | 15          | 6.86    | 5.87 | 1.17 | 4.49         | 6.20        | 0.72        | 3.55 | 6.36  | 0.56 | 3.55 | 6.36  | 0.56 | 3.55 | 6.36  | 0.56 | 3.55    | 6.36  | 0.56 |
|              | 20          | 6.20    | 6.23 | 0.99 | 4.52         | 6.52        | 0.69        | 3.17 | 6.66  | 0.48 | 3.17 | 6.66  | 0.48 | 3.17 | 6.66  | 0.48 | 3.17    | 6.66  | 0.48 |
|              | 25          | 6.31    | 7.10 | 0.89 | 4.46         | 7.48        | 0.60        | 3.50 | 7.69  | 0.46 | 3.50 | 7.69  | 0.46 | 3.50 | 7.69  | 0.46 | 3.50    | 7.69  | 0.46 |
|              | 30          | 6.51    | 8.05 | 0.81 | 4.45         | 8.58        | 0.52        | 3.84 | 8.75  | 0.44 | 3.84 | 8.75  | 0.44 | 3.84 | 8.75  | 0.44 | 3.84    | 8.75  | 0.44 |
|              | 35          | 6.76    | 9.10 | 0.74 | 4.40         | 10.04       | 0.44        | 4.20 | 10.05 | 0.42 | 4.20 | 10.05 | 0.42 | 4.20 | 10.05 | 0.42 | 4.20    | 10.05 | 0.42 |
| 40           | -25         | 3.15    | 1.79 | 1.76 | 3.15         | 1.79        | 1.76        | 2.36 | 1.84  | 1.29 | 1.56 | 1.94  | 0.81 | 1.26 | 1.95  | 0.65 | 1.26    | 1.95  | 0.65 |
|              | -20         | 3.77    | 2.04 | 1.85 | 3.77         | 2.04        | 1.85        | 2.83 | 2.08  | 1.36 | 1.89 | 2.19  | 0.87 | 1.54 | 2.22  | 0.69 | 1.54    | 2.22  | 0.69 |
|              | -15         | 4.38    | 2.29 | 1.92 | 4.38         | 2.29        | 1.92        | 3.27 | 2.32  | 1.41 | 2.17 | 2.43  | 0.89 | 1.82 | 2.46  | 0.74 | 1.82    | 2.46  | 0.74 |
|              | -10         | 5.22    | 2.63 | 1.99 | 4.50         | 2.65        | 1.70        | 3.40 | 2.69  | 1.26 | 2.26 | 2.73  | 0.83 | 2.16 | 2.75  | 0.79 | 2.16    | 2.75  | 0.79 |
|              | -7          | 5.44    | 2.81 | 1.94 | 4.50         | 2.89        | 1.56        | 3.34 | 2.97  | 1.13 | 2.39 | 3.07  | 0.78 | 2.39 | 3.07  | 0.78 | 2.39    | 3.07  | 0.78 |
|              | -5          | 5.59    | 2.93 | 1.91 | 4.45         | 3.02        | 1.47        | 3.31 | 3.06  | 1.08 | 2.56 | 3.20  | 0.80 | 2.56 | 3.20  | 0.80 | 2.56    | 3.20  | 0.80 |
|              | -2          | 5.70    | 3.17 | 1.80 | 4.29         | 3.28        | 1.31        | 3.22 | 3.34  | 0.97 | 2.57 | 3.36  | 0.76 | 2.57 | 3.36  | 0.76 | 2.57    | 3     |      |

| Tw_out<br>°C | DB/WB<br>°C | Maximum |      |      | 100%(Normal) |      |      | 75%  |      |      | 50%  |      |      | 25%  |      |      | Minimum |      |      |
|--------------|-------------|---------|------|------|--------------|------|------|------|------|------|------|------|------|------|------|------|---------|------|------|
|              |             | HC      | COP  | PI   | HC           | COP  | PI   | HC   | COP  | PI   | HC   | COP  | PI   | HC   | COP  | PI   | HC      | COP  | PI   |
| 50           | -25         | 3.03    | 1.58 | 1.92 | 3.03         | 1.58 | 1.92 | 2.27 | 1.61 | 1.41 | 1.52 | 1.66 | 0.92 | 1.52 | 1.66 | 0.92 | 1.52    | 1.66 | 0.92 |
|              | -20         | 3.61    | 1.78 | 2.03 | 3.61         | 1.78 | 2.03 | 2.69 | 1.81 | 1.48 | 1.81 | 1.88 | 0.96 | 1.81 | 1.88 | 0.96 | 1.81    | 1.88 | 0.96 |
|              | -15         | 4.16    | 1.96 | 2.12 | 4.16         | 1.96 | 2.12 | 3.12 | 1.99 | 1.57 | 2.12 | 2.06 | 1.03 | 2.12 | 2.06 | 1.03 | 2.12    | 2.06 | 1.03 |
|              | -10         | 4.96    | 2.22 | 2.24 | 4.61         | 2.28 | 2.02 | 3.45 | 2.32 | 1.49 | 2.52 | 2.37 | 1.06 | 2.52 | 2.37 | 1.06 | 2.52    | 2.37 | 1.06 |
|              | -7          | 5.17    | 2.38 | 2.17 | 4.64         | 2.42 | 1.92 | 3.41 | 2.50 | 1.37 | 2.77 | 2.55 | 1.09 | 2.77 | 2.55 | 1.09 | 2.77    | 2.55 | 1.09 |
|              | -5          | 5.31    | 2.47 | 2.15 | 4.51         | 2.57 | 1.75 | 3.41 | 2.60 | 1.31 | 2.99 | 2.63 | 1.13 | 2.99 | 2.63 | 1.13 | 2.99    | 2.63 | 1.13 |
|              | -2          | 5.41    | 2.66 | 2.04 | 4.43         | 2.71 | 1.64 | 3.33 | 2.74 | 1.22 | 3.04 | 2.79 | 1.09 | 3.04 | 2.79 | 1.09 | 3.04    | 2.79 | 1.09 |
|              | 0           | 5.50    | 2.80 | 1.97 | 4.54         | 2.90 | 1.56 | 3.43 | 2.87 | 1.19 | 3.24 | 2.94 | 1.10 | 3.24 | 2.94 | 1.10 | 3.24    | 2.94 | 1.10 |
|              | 2           | 5.63    | 2.96 | 1.90 | 4.50         | 3.00 | 1.50 | 3.40 | 3.03 | 1.12 | 3.40 | 3.03 | 1.12 | 3.40 | 3.03 | 1.12 | 3.40    | 3.03 | 1.12 |
|              | 5           | 6.02    | 3.20 | 1.88 | 4.44         | 3.34 | 1.33 | 3.90 | 3.34 | 1.17 | 3.90 | 3.34 | 1.17 | 3.90 | 3.34 | 1.17 | 3.90    | 3.34 | 1.17 |
|              | 7           | 6.35    | 3.51 | 1.81 | 4.57         | 3.54 | 1.29 | 4.26 | 3.58 | 1.19 | 4.26 | 3.58 | 1.19 | 4.26 | 3.58 | 1.19 | 4.26    | 3.58 | 1.19 |
|              | 10          | 6.46    | 3.79 | 1.70 | 4.54         | 3.90 | 1.16 | 4.30 | 3.92 | 1.10 | 4.30 | 3.92 | 1.10 | 4.30 | 3.92 | 1.10 | 4.30    | 3.92 | 1.10 |
|              | 12          | 6.69    | 4.03 | 1.66 | 4.63         | 4.12 | 1.13 | 4.60 | 4.16 | 1.11 | 4.60 | 4.16 | 1.11 | 4.60 | 4.16 | 1.11 | 4.60    | 4.16 | 1.11 |
|              | 15          | 6.31    | 4.17 | 1.51 | 4.77         | 4.25 | 1.12 | 4.73 | 4.29 | 1.10 | 4.73 | 4.29 | 1.10 | 4.73 | 4.29 | 1.10 | 4.73    | 4.29 | 1.10 |
|              | 20          | 5.95    | 4.51 | 1.32 | 4.51         | 4.55 | 0.99 | 4.13 | 4.59 | 0.90 | 4.13 | 4.59 | 0.90 | 4.13 | 4.59 | 0.90 | 4.13    | 4.59 | 0.90 |
|              | 25          | 5.95    | 4.98 | 1.19 | 4.61         | 5.06 | 0.91 | 4.56 | 5.08 | 0.90 | 4.56 | 5.08 | 0.90 | 4.56 | 5.08 | 0.90 | 4.56    | 5.08 | 0.90 |
|              | 30          | 6.26    | 5.46 | 1.15 | 5.08         | 5.70 | 0.89 | 5.03 | 5.71 | 0.88 | 5.03 | 5.71 | 0.88 | 5.03 | 5.71 | 0.88 | 5.03    | 5.71 | 0.88 |
|              | 35          | 6.53    | 6.05 | 1.08 | 5.58         | 6.16 | 0.91 | 5.54 | 6.22 | 0.89 | 5.54 | 6.22 | 0.89 | 5.54 | 6.22 | 0.89 | 5.54    | 6.22 | 0.89 |
|              | -25         | 2.87    | 1.44 | 2.00 | 2.87         | 1.44 | 2.00 | 2.15 | 1.46 | 1.47 | 1.55 | 1.50 | 1.03 | 1.55 | 1.50 | 1.03 | 1.55    | 1.50 | 1.03 |
|              | -20         | 3.33    | 1.59 | 2.10 | 3.33         | 1.59 | 2.10 | 2.56 | 1.65 | 1.55 | 1.87 | 1.69 | 1.11 | 1.87 | 1.69 | 1.11 | 1.87    | 1.69 | 1.11 |
| -15          | 3.93        | 1.77    | 2.22 | 3.93 | 1.77         | 2.22 | 2.93 | 1.81 | 1.62 | 2.19 | 1.85 | 1.18 | 2.19 | 1.85 | 1.18 | 2.19 | 1.85    | 1.18 |      |
| -10          | 4.66        | 1.98    | 2.36 | 4.60 | 1.99         | 2.32 | 3.44 | 2.00 | 1.72 | 2.59 | 2.04 | 1.27 | 2.59 | 2.04 | 1.27 | 2.59 | 2.04    | 1.27 |      |
| -7           | 4.85        | 2.14    | 2.27 | 4.70 | 2.14         | 2.20 | 3.41 | 2.17 | 1.57 | 2.86 | 2.20 | 1.30 | 2.86 | 2.20 | 1.30 | 2.86 | 2.20    | 1.30 |      |
| -5           | 5.00        | 2.22    | 2.25 | 4.55 | 2.25         | 2.02 | 3.42 | 2.28 | 1.50 | 3.07 | 2.30 | 1.33 | 3.07 | 2.30 | 1.33 | 3.07 | 2.30    | 1.33 |      |
| -2           | 5.11        | 2.37    | 2.16 | 4.46 | 2.50         | 1.78 | 3.35 | 2.51 | 1.33 | 3.25 | 2.57 | 1.27 | 3.25 | 2.57 | 1.27 | 3.25 | 2.57    | 1.27 |      |
| 0            | 5.37        | 2.58    | 2.08 | 4.61 | 2.61         | 1.76 | 3.49 | 2.65 | 1.32 | 3.49 | 2.65 | 1.32 | 3.49 | 2.65 | 1.32 | 3.49 | 2.65    | 1.32 |      |
| 2            | 5.50        | 2.70    | 2.04 | 4.60 | 2.70         | 1.70 | 3.67 | 2.73 | 1.34 | 3.67 | 2.73 | 1.34 | 3.67 | 2.73 | 1.34 | 3.67 | 2.73    | 1.34 |      |
| 5            | 5.88        | 2.98    | 1.97 | 4.48 | 3.03         | 1.48 | 4.17 | 3.05 | 1.37 | 4.17 | 3.05 | 1.37 | 4.17 | 3.05 | 1.37 | 4.17 | 3.05    | 1.37 |      |
| 7            | 6.20        | 3.10    | 2.00 | 4.60 | 3.20         | 1.44 | 4.55 | 3.23 | 1.41 | 4.55 | 3.23 | 1.41 | 4.55 | 3.23 | 1.41 | 4.55 | 3.23    | 1.41 |      |
| 10           | 6.32        | 3.38    | 1.87 | 4.59 | 3.45         | 1.33 | 4.26 | 3.53 | 1.21 | 4.26 | 3.53 | 1.21 | 4.26 | 3.53 | 1.21 | 4.26 | 3.53    | 1.21 |      |
| 12           | 6.50        | 3.55    | 1.83 | 4.69 | 3.69         | 1.27 | 4.55 | 3.74 | 1.22 | 4.55 | 3.74 | 1.22 | 4.55 | 3.74 | 1.22 | 4.55 | 3.74    | 1.22 |      |
| 15           | 6.15        | 3.77    | 1.63 | 4.83 | 3.80         | 1.27 | 4.68 | 3.85 | 1.22 | 4.68 | 3.85 | 1.22 | 4.68 | 3.85 | 1.22 | 4.68 | 3.85    | 1.22 |      |
| 20           | 5.78        | 4.04    | 1.43 | 4.60 | 4.10         | 1.12 | 4.11 | 4.07 | 1.01 | 4.11 | 4.07 | 1.01 | 4.11 | 4.07 | 1.01 | 4.11 | 4.07    | 1.01 |      |
| 25           | 5.76        | 4.42    | 1.30 | 4.71 | 4.51         | 1.04 | 4.56 | 4.55 | 1.00 | 4.56 | 4.55 | 1.00 | 4.56 | 4.55 | 1.00 | 4.56 | 4.55    | 1.00 |      |
| 30           | 6.07        | 4.85    | 1.25 | 5.19 | 4.90         | 1.06 | 5.03 | 4.91 | 1.02 | 5.03 | 4.91 | 1.02 | 5.03 | 4.91 | 1.02 | 5.03 | 4.91    | 1.02 |      |
| 35           | 6.32        | 5.00    | 1.26 | 5.73 | 5.06         | 1.13 | 5.54 | 5.08 | 1.09 | 5.54 | 5.08 | 1.09 | 5.54 | 5.08 | 1.09 | 5.54 | 5.08    | 1.09 |      |
| 55           | -25         | /       | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
|              | -20         | 3.29    | 1.49 | 2.21 | 3.29         | 1.49 | 2.21 | 2.47 | 1.52 | 1.63 | 1.94 | 1.53 | 1.27 | 1.94 | 1.53 | 1.27 | 1.94    | 1.53 | 1.27 |
|              | -15         | 3.78    | 1.62 | 2.34 | 3.78         | 1.62 | 2.34 | 2.83 | 1.66 | 1.70 | 2.27 | 1.69 | 1.35 | 2.27 | 1.69 | 1.35 | 2.27    | 1.69 | 1.35 |
|              | -10         | 4.47    | 1.80 | 2.49 | 4.47         | 1.80 | 2.49 | 3.43 | 1.84 | 1.86 | 2.67 | 1.86 | 1.43 | 2.67 | 1.86 | 1.43 | 2.67    | 1.86 | 1.43 |
|              | -7          | 4.68    | 1.93 | 2.43 | 4.61         | 1.93 | 2.39 | 3.45 | 1.96 | 1.76 | 2.94 | 1.99 | 1.48 | 2.94 | 1.99 | 1.48 | 2.94    | 1.99 | 1.48 |
|              | -5          | 4.82    | 2.00 | 2.40 | 4.58         | 2.03 | 2.26 | 3.47 | 2.10 | 1.65 | 3.16 | 2.13 | 1.48 | 3.16 | 2.13 | 1.48 | 3.16    | 2.13 | 1.48 |
|              | -2          | 4.93    | 2.16 | 2.29 | 4.44         | 2.28 | 1.95 | 3.34 | 2.32 | 1.44 | 3.10 | 2.36 | 1.31 | 3.10 | 2.36 | 1.31 | 3.10    | 2.36 | 1.31 |
|              | 0           | 5.20    | 2.34 | 2.22 | 4.41         | 2.35 | 1.88 | 3.31 | 2.42 | 1.37 | 3.31 | 2.42 | 1.37 | 3.31 | 2.42 | 1.37 | 3.31    | 2.42 | 1.37 |
|              | 2           | 5.32    | 2.44 | 2.18 | 4.37         | 2.51 | 1.74 | 3.50 | 2.55 | 1.37 | 3.50 | 2.55 | 1.37 | 3.50 | 2.55 | 1.37 | 3.50    | 2.55 | 1.37 |
|              | 5           | 5.67    | 2.73 | 2.08 | 4.56         | 2.80 | 1.63 | 3.97 | 2.84 | 1.40 | 3.97 | 2.84 | 1.40 | 3.97 | 2.84 | 1.40 | 3.97    | 2.84 | 1.40 |
|              | 7           | 5.97    | 2.89 | 2.07 | 4.58         | 2.94 | 1.56 | 4.34 | 2.98 | 1.46 | 4.34 | 2.98 | 1.46 | 4.34 | 2.98 | 1.46 | 4.34    | 2.98 | 1.46 |
|              | 10          | 6.06    | 3.02 | 2.01 | 4.57         | 3.10 | 1.47 | 4.22 | 3.15 | 1.34 | 4.22 | 3.15 | 1.34 | 4.22 | 3.15 | 1.34 | 4.22    | 3.15 | 1.34 |
|              | 12          | 6.26    | 3.09 | 2.03 | 4.66         | 3.24 | 1.44 | 4.51 | 3.27 | 1.38 | 4.51 | 3.27 | 1.38 | 4.51 | 3.27 | 1.38 | 4.51    | 3.27 | 1.38 |
|              | 15          | 5.91    | 3.35 | 1.77 | 4.79         | 3.39 | 1.41 | 4.65 | 3.43 | 1.35 | 4.65 | 3.43 | 1.35 | 4.65 | 3.43 | 1.35 | 4.65    | 3.43 | 1.35 |
|              | 20          | 5.52    | 3.56 | 1.55 | 4.62         | 3.66 | 1.26 | 3.91 | 3.70 | 1.06 | 3.91 | 3.70 | 1.06 | 3.91 | 3.70 | 1.06 | 3.91    | 3.70 | 1.06 |
|              | 25          | 5.48    | 3.88 | 1.41 | 4.60         | 3.92 | 1.17 | 4.32 | 3.97 | 1.09 | 4.32 | 3.97 | 1.09 | 4.32 | 3.97 | 1.09 | 4.32    | 3.97 | 1.09 |
|              | 30          | 5.78    | 4.17 | 1.39 | 4.95         | 4.25 | 1.17 | 4.78 | 4.28 | 1.11 | 4.78 | 4.28 | 1.11 | 4.78 | 4.28 | 1.11 | 4.78    | 4.28 | 1.11 |
|              | 35          | /       | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
|              | -25         | /       | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
|              | -20         | 3.29    | 1.49 | 2.21 | 3.29         | 1.49 | 2.21 | 2.47 | 1.52 | 1.63 | 1.94 | 1.53 | 1.27 | 1.94 | 1.53 | 1.27 | 1.94    | 1.53 | 1.27 |
| -15          | 3.78        | 1.62    | 2.34 | 3.78 | 1.62         | 2.34 | 2.83 | 1.66 | 1.70 | 2.27 | 1.69 | 1.35 | 2.27 | 1.69 | 1.35 | 2.27 | 1.69    | 1.35 |      |
| -10          | 4.47        | 1.80    | 2.49 | 4.47 | 1.80         | 2.49 | 3.43 | 1.84 | 1.86 | 2.67 | 1.86 | 1.43 | 2.67 | 1.86 |      |      |         |      |      |

## 8.2 Prestazioni riscaldamento Mod. 06

| Tw_out<br>°C | DBWB<br>°C | Maximum |      |      | 100%(Normal) |      |      | 75%  |       |      | 50%  |       |      | 25%  |       |      | Minimum |       |      |
|--------------|------------|---------|------|------|--------------|------|------|------|-------|------|------|-------|------|------|-------|------|---------|-------|------|
|              |            | HC      | COP  | PI   | HC           | COP  | PI   | HC   | COP   | PI   | HC   | COP   | PI   | HC   | COP   | PI   | HC      | COP   | PI   |
| 25           | -25        | 3.50    | 2.14 | 1.64 | 3.50         | 2.14 | 1.64 | 2.64 | 2.40  | 1.10 | 1.75 | 2.47  | 0.71 | 1.34 | 2.50  | 0.54 | 1.34    | 2.50  | 0.54 |
|              | -20        | 4.35    | 2.41 | 1.80 | 4.35         | 2.41 | 1.80 | 3.26 | 2.72  | 1.20 | 2.18 | 2.80  | 0.78 | 1.63 | 2.82  | 0.58 | 1.63    | 2.82  | 0.58 |
|              | -15        | 5.18    | 2.77 | 1.87 | 5.18         | 2.77 | 1.87 | 3.89 | 2.91  | 1.34 | 2.63 | 2.98  | 0.88 | 1.92 | 3.15  | 0.61 | 1.92    | 3.15  | 0.61 |
|              | -10        | 6.08    | 3.16 | 1.92 | 5.83         | 3.21 | 1.81 | 4.34 | 3.48  | 1.25 | 2.90 | 3.61  | 0.80 | 2.35 | 3.62  | 0.65 | 2.35    | 3.62  | 0.65 |
|              | -7         | 6.36    | 3.41 | 1.86 | 5.79         | 3.56 | 1.63 | 4.32 | 3.78  | 1.14 | 2.90 | 3.92  | 0.74 | 2.55 | 3.94  | 0.65 | 2.55    | 3.94  | 0.65 |
|              | -5         | 6.55    | 3.59 | 1.83 | 5.66         | 3.85 | 1.47 | 4.25 | 4.14  | 1.03 | 2.83 | 4.27  | 0.66 | 2.73 | 4.28  | 0.64 | 2.73    | 4.28  | 0.64 |
|              | -2         | 6.84    | 3.87 | 1.77 | 5.47         | 4.28 | 1.28 | 4.18 | 4.52  | 0.93 | 2.83 | 4.62  | 0.61 | 2.43 | 4.64  | 0.52 | 2.43    | 4.64  | 0.52 |
|              | 0          | 6.99    | 4.10 | 1.70 | 5.66         | 4.49 | 1.26 | 4.29 | 4.79  | 0.90 | 2.78 | 4.91  | 0.57 | 2.59 | 4.97  | 0.52 | 2.59    | 4.97  | 0.52 |
|              | 2          | 7.09    | 4.34 | 1.64 | 5.65         | 4.78 | 1.18 | 4.22 | 5.09  | 0.83 | 2.85 | 5.20  | 0.55 | 2.74 | 5.27  | 0.52 | 2.74    | 5.27  | 0.52 |
|              | 5          | 7.72    | 5.06 | 1.53 | 5.85         | 5.65 | 1.03 | 4.41 | 5.85  | 0.75 | 2.90 | 6.07  | 0.48 | 2.79 | 6.08  | 0.46 | 2.79    | 6.08  | 0.46 |
|              | 7          | 8.00    | 5.50 | 1.45 | 6.24         | 5.92 | 1.05 | 4.68 | 6.07  | 0.77 | 3.14 | 6.35  | 0.49 | 3.01 | 6.38  | 0.47 | 3.01    | 6.38  | 0.47 |
|              | 10         | 7.95    | 5.89 | 1.35 | 6.24         | 6.55 | 0.95 | 4.62 | 6.87  | 0.67 | 3.34 | 7.17  | 0.47 | 3.34 | 7.17  | 0.47 | 3.34    | 7.17  | 0.47 |
|              | 12         | 8.16    | 6.40 | 1.27 | 6.16         | 7.14 | 0.86 | 4.57 | 7.53  | 0.61 | 3.53 | 7.77  | 0.45 | 3.53 | 7.77  | 0.45 | 3.53    | 7.77  | 0.45 |
|              | 15         | 7.92    | 6.79 | 1.17 | 6.27         | 7.39 | 0.85 | 4.65 | 7.81  | 0.60 | 3.62 | 8.14  | 0.44 | 3.62 | 8.14  | 0.44 | 3.62    | 8.14  | 0.44 |
|              | 20         | 7.56    | 7.89 | 0.96 | 6.25         | 8.21 | 0.76 | 4.64 | 8.76  | 0.53 | 3.40 | 9.23  | 0.37 | 3.40 | 9.23  | 0.37 | 3.40    | 9.23  | 0.37 |
|              | 25         | 7.12    | 9.21 | 0.77 | 6.30         | 9.54 | 0.66 | 4.71 | 10.02 | 0.47 | 3.75 | 10.57 | 0.35 | 3.75 | 10.57 | 0.35 | 3.75    | 10.57 | 0.35 |
|              | 30         | /       | /    | /    | /            | /    | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 35         | /       | /    | /    | /            | /    | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 30           | -25        | 3.46    | 1.89 | 1.83 | 3.46         | 1.89 | 1.83 | 2.59 | 2.02  | 1.28 | 1.73 | 2.06  | 0.84 | 1.29 | 2.12  | 0.61 | 1.29    | 2.12  | 0.61 |
|              | -20        | 4.30    | 2.25 | 1.91 | 4.30         | 2.25 | 1.91 | 3.23 | 2.36  | 1.37 | 2.15 | 2.40  | 0.90 | 1.58 | 2.48  | 0.64 | 1.58    | 2.48  | 0.64 |
|              | -15        | 5.11    | 2.58 | 1.98 | 5.11         | 2.58 | 1.98 | 3.83 | 2.66  | 1.44 | 2.56 | 2.71  | 0.94 | 1.85 | 2.85  | 0.65 | 1.85    | 2.85  | 0.65 |
|              | -10        | 6.00    | 2.96 | 2.03 | 5.77         | 2.98 | 1.94 | 4.34 | 3.22  | 1.35 | 2.91 | 3.27  | 0.89 | 2.20 | 3.28  | 0.67 | 2.20    | 3.28  | 0.67 |
|              | -7         | 6.27    | 3.19 | 1.97 | 5.76         | 3.29 | 1.75 | 4.33 | 3.49  | 1.24 | 2.92 | 3.56  | 0.82 | 2.52 | 3.63  | 0.69 | 2.52    | 3.63  | 0.69 |
|              | -5         | 6.45    | 3.34 | 1.93 | 5.68         | 3.51 | 1.62 | 4.26 | 3.72  | 1.15 | 2.87 | 3.76  | 0.76 | 2.63 | 3.79  | 0.69 | 2.63    | 3.79  | 0.69 |
|              | -2         | 6.73    | 3.59 | 1.87 | 5.48         | 3.90 | 1.41 | 4.13 | 4.03  | 1.02 | 2.80 | 4.13  | 0.68 | 2.39 | 4.18  | 0.57 | 2.39    | 4.18  | 0.57 |
|              | 0          | 6.87    | 3.79 | 1.81 | 5.66         | 4.08 | 1.39 | 4.29 | 4.21  | 1.02 | 2.85 | 4.34  | 0.66 | 2.49 | 4.38  | 0.57 | 2.49    | 4.38  | 0.57 |
|              | 2          | 6.96    | 4.00 | 1.74 | 5.65         | 4.31 | 1.31 | 4.28 | 4.53  | 0.95 | 2.89 | 4.62  | 0.63 | 2.64 | 4.65  | 0.57 | 2.64    | 4.65  | 0.57 |
|              | 5          | 7.58    | 4.64 | 1.63 | 5.87         | 4.99 | 1.18 | 4.36 | 5.15  | 0.85 | 2.91 | 5.28  | 0.55 | 2.79 | 5.37  | 0.52 | 2.79    | 5.37  | 0.52 |
|              | 7          | 7.85    | 5.04 | 1.56 | 6.18         | 5.42 | 1.14 | 4.66 | 5.61  | 0.83 | 3.04 | 5.85  | 0.52 | 3.04 | 5.85  | 0.52 | 3.04    | 5.85  | 0.52 |
|              | 10         | 8.16    | 5.59 | 1.46 | 6.22         | 6.04 | 1.03 | 4.59 | 6.21  | 0.74 | 3.20 | 6.46  | 0.50 | 3.20 | 6.46  | 0.50 | 3.20    | 6.46  | 0.50 |
|              | 12         | 8.35    | 6.00 | 1.39 | 6.23         | 6.23 | 1.00 | 4.68 | 6.54  | 0.71 | 3.44 | 6.78  | 0.51 | 3.44 | 6.78  | 0.51 | 3.44    | 6.78  | 0.51 |
|              | 15         | 7.74    | 6.05 | 1.28 | 6.25         | 6.45 | 0.97 | 4.64 | 6.78  | 0.68 | 3.50 | 7.02  | 0.50 | 3.50 | 7.02  | 0.50 | 3.50    | 7.02  | 0.50 |
|              | 20         | 7.38    | 6.90 | 1.07 | 6.23         | 7.14 | 0.87 | 4.64 | 7.54  | 0.62 | 3.29 | 7.90  | 0.42 | 3.29 | 7.90  | 0.42 | 3.29    | 7.90  | 0.42 |
|              | 25         | 7.04    | 7.95 | 0.89 | 6.31         | 8.18 | 0.77 | 4.65 | 8.71  | 0.53 | 3.62 | 9.07  | 0.40 | 3.62 | 9.07  | 0.40 | 3.62    | 9.07  | 0.40 |
|              | 30         | 7.07    | 9.19 | 0.77 | 6.17         | 9.24 | 0.67 | 4.65 | 9.56  | 0.49 | 3.98 | 9.77  | 0.41 | 3.98 | 9.77  | 0.41 | 3.98    | 9.77  | 0.41 |
|              | 35         | /       | /    | /    | /            | /    | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 35           | -25        | 3.40    | 1.76 | 1.94 | 3.40         | 1.76 | 1.94 | 2.56 | 1.91  | 1.34 | 1.74 | 1.94  | 0.89 | 1.24 | 2.05  | 0.61 | 1.24    | 2.05  | 0.61 |
|              | -20        | 4.24    | 2.10 | 2.01 | 4.24         | 2.10 | 2.01 | 3.18 | 2.15  | 1.48 | 2.12 | 2.21  | 0.96 | 1.52 | 2.32  | 0.65 | 1.52    | 2.32  | 0.65 |
|              | -15        | 5.05    | 2.43 | 2.08 | 5.05         | 2.43 | 2.08 | 3.79 | 2.46  | 1.54 | 2.53 | 2.51  | 1.01 | 1.79 | 2.66  | 0.67 | 1.79    | 2.66  | 0.67 |
|              | -10        | 5.91    | 2.77 | 2.13 | 5.81         | 2.78 | 2.09 | 4.34 | 2.94  | 1.48 | 2.93 | 2.98  | 0.98 | 2.18 | 3.08  | 0.71 | 2.18    | 3.08  | 0.71 |
|              | -7         | 6.18    | 2.93 | 2.11 | 5.90         | 2.95 | 2.00 | 4.34 | 3.11  | 1.39 | 2.89 | 3.14  | 0.92 | 2.36 | 3.21  | 0.73 | 2.36    | 3.21  | 0.73 |
|              | -5         | 6.36    | 3.12 | 2.04 | 5.70         | 3.20 | 1.78 | 4.28 | 3.36  | 1.27 | 2.89 | 3.41  | 0.85 | 2.53 | 3.43  | 0.74 | 2.53    | 3.43  | 0.74 |
|              | -2         | 6.61    | 3.32 | 1.99 | 5.51         | 3.55 | 1.55 | 4.16 | 3.62  | 1.15 | 2.76 | 3.69  | 0.75 | 2.31 | 3.74  | 0.62 | 2.31    | 3.74  | 0.62 |
|              | 0          | 6.75    | 3.49 | 1.93 | 5.64         | 3.73 | 1.51 | 4.28 | 3.84  | 1.11 | 2.84 | 3.92  | 0.73 | 2.39 | 3.95  | 0.61 | 2.39    | 3.95  | 0.61 |
|              | 2          | 6.83    | 3.67 | 1.86 | 5.60         | 3.90 | 1.44 | 4.17 | 4.12  | 1.01 | 2.76 | 4.18  | 0.66 | 2.53 | 4.19  | 0.60 | 2.53    | 4.19  | 0.60 |
|              | 5          | 7.42    | 4.26 | 1.74 | 5.92         | 4.48 | 1.32 | 4.37 | 4.60  | 0.95 | 2.89 | 4.72  | 0.61 | 2.68 | 4.75  | 0.56 | 2.68    | 4.75  | 0.56 |
|              | 7*         | 7.70    | 4.61 | 1.67 | 6.20         | 4.90 | 1.27 | 4.64 | 5.12  | 0.91 | 3.15 | 5.26  | 0.60 | 2.93 | 5.32  | 0.55 | 2.93    | 5.32  | 0.55 |
|              | 10         | 8.00    | 5.08 | 1.57 | 6.19         | 5.36 | 1.15 | 4.69 | 5.56  | 0.84 | 3.24 | 5.78  | 0.56 | 3.24 | 5.78  | 0.56 | 3.24    | 5.78  | 0.56 |
|              | 12         | 8.17    | 5.43 | 1.50 | 6.24         | 5.77 | 1.08 | 4.63 | 5.97  | 0.78 | 3.45 | 6.22  | 0.56 | 3.45 | 6.22  | 0.56 | 3.45    | 6.22  | 0.56 |
|              | 15         | 7.92    | 5.66 | 1.40 | 6.25         | 5.96 | 1.05 | 4.62 | 6.18  | 0.75 | 3.55 | 6.36  | 0.56 | 3.55 | 6.36  | 0.56 | 3.55    | 6.36  | 0.56 |
|              | 20         | 7.53    | 6.21 | 1.21 | 5.97         | 6.30 | 0.95 | 4.62 | 6.51  | 0.71 | 3.17 | 6.66  | 0.48 | 3.17 | 6.66  | 0.48 | 3.17    | 6.66  | 0.48 |
|              | 25         | 6.96    | 6.92 | 1.01 | 6.31         | 7.10 | 0.89 | 4.59 | 7.43  | 0.62 | 3.50 | 7.69  | 0.46 | 3.50 | 7.69  | 0.46 | 3.50    | 7.69  | 0.46 |
|              | 30         | 6.94    | 7.95 | 0.87 | 6.17         | 8.10 | 0.76 | 4.59 | 8.55  | 0.54 | 3.98 | 8.74  | 0.46 | 3.84 | 8.75  | 0.44 | 3.84    | 8.75  | 0.44 |
|              | 35         | 7.12    | 9.07 | 0.79 | 6.14         | 9.51 | 0.65 | 4.73 | 9.88  | 0.48 | 4.20 | 10.05 | 0.42 | 4.20 | 10.05 | 0.42 | 4.20    | 10.05 | 0.42 |
| -25          | 3.34       | 1.65    | 2.03 | 3.34 | 1.65         | 2.03 | 2.51 | 1.82 | 1.38  | 1.73 | 1.86 | 0.93  | 1.26 | 1.95 | 0.65  | 1.26 | 1.95    | 0.65  |      |
| 40           | -20        | 4.17    | 1.98 | 2.11 | 4.17         | 1.98 | 2.11 | 3.20 | 2.06  | 1.55 | 2.08 | 2.09  | 1.00 | 1.54 | 2.22  | 0.69 | 1.54    | 2.22  | 0.69 |
|              | -15        | 4.96    | 2.21 | 2.24 | 4.96         | 2.21 | 2.24 | 3.72 | 2.31  | 1.61 | 2.48 | 2.33  | 1.06 | 1.82 | 2.46  | 0.74 | 1.82    | 2.46  | 0.74 |
|              | -10        | 5.83    | 2.53 | 2.30 | 5.60         | 2.61 | 2.15 | 4.21 | 2.67  | 1.5  |      |       |      |      |       |      |         |       |      |

| Tw_out<br>°C | DBWB<br>°C | Maximum |      |      | 100%(Normal) |      |      | 75%  |      |      | 50%  |      |      | 25%  |      |      | Minimum |      |      |
|--------------|------------|---------|------|------|--------------|------|------|------|------|------|------|------|------|------|------|------|---------|------|------|
|              |            | HC      | COP  | PI   | HC           | COP  | PI   | HC   | COP  | PI   | HC   | COP  | PI   | HC   | COP  | PI   | HC      | COP  | PI   |
| 55           | -25        | 3.00    | 1.40 | 2.14 | 3.00         | 1.40 | 2.14 | 2.25 | 1.44 | 1.56 | 1.55 | 1.50 | 1.03 | 1.55 | 1.50 | 1.03 | 1.55    | 1.50 | 1.03 |
|              | -20        | 3.66    | 1.57 | 2.34 | 3.66         | 1.57 | 2.34 | 2.77 | 1.63 | 1.70 | 1.87 | 1.69 | 1.11 | 1.87 | 1.69 | 1.11 | 1.87    | 1.69 | 1.11 |
|              | -15        | 4.46    | 1.71 | 2.61 | 4.46         | 1.71 | 2.61 | 3.34 | 1.79 | 1.87 | 2.19 | 1.85 | 1.18 | 2.19 | 1.85 | 1.18 | 2.19    | 1.85 | 1.18 |
|              | -10        | 5.23    | 1.91 | 2.74 | 5.12         | 1.92 | 2.67 | 3.81 | 1.99 | 1.92 | 2.59 | 2.04 | 1.27 | 2.59 | 2.04 | 1.27 | 2.59    | 2.04 | 1.27 |
|              | -7         | 5.47    | 2.13 | 2.57 | 5.20         | 2.15 | 2.42 | 3.74 | 2.16 | 1.73 | 2.86 | 2.20 | 1.30 | 2.86 | 2.20 | 1.30 | 2.86    | 2.20 | 1.30 |
|              | -5         | 5.59    | 2.15 | 2.60 | 5.17         | 2.20 | 2.35 | 3.88 | 2.27 | 1.71 | 3.07 | 2.30 | 1.33 | 3.07 | 2.30 | 1.33 | 3.07    | 2.30 | 1.33 |
|              | -2         | 5.81    | 2.30 | 2.53 | 5.28         | 2.35 | 2.24 | 4.01 | 2.51 | 1.60 | 3.25 | 2.57 | 1.27 | 3.25 | 2.57 | 1.27 | 3.25    | 2.57 | 1.27 |
|              | 0          | 6.16    | 2.50 | 2.46 | 5.34         | 2.60 | 2.06 | 4.04 | 2.63 | 1.53 | 3.49 | 2.65 | 1.32 | 3.49 | 2.65 | 1.32 | 3.49    | 2.65 | 1.32 |
|              | 2          | 6.22    | 2.60 | 2.39 | 5.80         | 2.65 | 2.19 | 4.41 | 2.72 | 1.62 | 3.67 | 2.73 | 1.34 | 3.67 | 2.73 | 1.34 | 3.67    | 2.73 | 1.34 |
|              | 5          | 6.72    | 2.88 | 2.33 | 5.99         | 2.95 | 2.03 | 4.42 | 3.04 | 1.46 | 4.17 | 3.05 | 1.37 | 4.17 | 3.05 | 1.37 | 4.17    | 3.05 | 1.37 |
|              | 7          | 6.97    | 3.00 | 2.32 | 6.20         | 3.10 | 2.00 | 4.72 | 3.19 | 1.48 | 4.55 | 3.23 | 1.41 | 4.55 | 3.23 | 1.41 | 4.55    | 3.23 | 1.41 |
|              | 10         | 7.24    | 3.32 | 2.18 | 6.22         | 3.40 | 1.83 | 4.59 | 3.48 | 1.32 | 4.26 | 3.53 | 1.21 | 4.26 | 3.53 | 1.21 | 4.26    | 3.53 | 1.21 |
|              | 12         | 7.40    | 3.52 | 2.10 | 6.20         | 3.58 | 1.73 | 4.69 | 3.69 | 1.27 | 4.55 | 3.74 | 1.22 | 4.55 | 3.74 | 1.22 | 4.55    | 3.74 | 1.22 |
|              | 15         | 7.14    | 3.67 | 1.95 | 6.25         | 3.72 | 1.68 | 4.83 | 3.80 | 1.27 | 4.68 | 3.85 | 1.22 | 4.68 | 3.85 | 1.22 | 4.68    | 3.85 | 1.22 |
|              | 20         | 6.77    | 3.95 | 1.71 | 6.24         | 3.98 | 1.57 | 4.60 | 4.06 | 1.13 | 4.11 | 4.07 | 1.01 | 4.11 | 4.07 | 1.01 | 4.11    | 4.07 | 1.01 |
|              | 25         | 6.39    | 4.32 | 1.48 | 6.26         | 4.35 | 1.44 | 4.71 | 4.51 | 1.04 | 4.56 | 4.55 | 1.00 | 4.56 | 4.55 | 1.00 | 4.56    | 4.55 | 1.00 |
|              | 30         | 6.50    | 4.72 | 1.38 | 6.22         | 4.75 | 1.31 | 5.03 | 4.91 | 1.02 | 5.03 | 4.91 | 1.02 | 5.03 | 4.91 | 1.02 | 5.03    | 4.91 | 1.02 |
|              | 35         | 6.69    | 4.95 | 1.35 | 6.18         | 4.98 | 1.24 | 5.54 | 5.08 | 1.09 | 5.54 | 5.08 | 1.09 | 5.54 | 5.08 | 1.09 | 5.54    | 5.08 | 1.09 |
|              | 60         | -25     | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
| -20          |            | 3.60    | 1.41 | 2.55 | 3.60         | 1.41 | 2.55 | 2.70 | 1.50 | 1.81 | 1.94 | 1.53 | 1.27 | 1.94 | 1.53 | 1.27 | 1.94    | 1.53 | 1.27 |
| -15          |            | 4.31    | 1.60 | 2.70 | 4.31         | 1.60 | 2.70 | 3.23 | 1.65 | 1.97 | 2.27 | 1.69 | 1.35 | 2.27 | 1.69 | 1.35 | 2.27    | 1.69 | 1.35 |
| -10          |            | 5.05    | 1.78 | 2.83 | 5.05         | 1.78 | 2.83 | 3.83 | 1.83 | 2.10 | 2.67 | 1.86 | 1.43 | 2.67 | 1.86 | 1.43 | 2.67    | 1.86 | 1.43 |
| -7           |            | 5.27    | 1.84 | 2.87 | 5.10         | 1.86 | 2.74 | 3.79 | 1.94 | 1.95 | 2.94 | 1.99 | 1.48 | 2.94 | 1.99 | 1.48 | 2.94    | 1.99 | 1.48 |
| -5           |            | 5.39    | 1.90 | 2.84 | 5.15         | 1.91 | 2.70 | 3.93 | 2.05 | 1.92 | 3.16 | 2.13 | 1.48 | 3.16 | 2.13 | 1.48 | 3.16    | 2.13 | 1.48 |
| -2           |            | 5.61    | 2.01 | 2.79 | 5.29         | 2.04 | 2.60 | 3.99 | 2.29 | 1.74 | 3.10 | 2.36 | 1.31 | 3.10 | 2.36 | 1.31 | 3.10    | 2.36 | 1.31 |
| 0            |            | 5.92    | 2.17 | 2.73 | 5.66         | 2.22 | 2.55 | 4.29 | 2.36 | 1.82 | 3.31 | 2.42 | 1.37 | 3.31 | 2.42 | 1.37 | 3.31    | 2.42 | 1.37 |
| 2            |            | 6.01    | 2.26 | 2.66 | 5.44         | 2.34 | 2.32 | 4.10 | 2.52 | 1.63 | 3.50 | 2.55 | 1.37 | 3.50 | 2.55 | 1.37 | 3.50    | 2.55 | 1.37 |
| 5            |            | 6.49    | 2.59 | 2.51 | 5.96         | 2.65 | 2.25 | 4.41 | 2.81 | 1.57 | 3.97 | 2.84 | 1.40 | 3.97 | 2.84 | 1.40 | 3.97    | 2.84 | 1.40 |
| 7            |            | 6.75    | 2.80 | 2.41 | 6.20         | 2.82 | 2.20 | 4.58 | 2.94 | 1.56 | 4.34 | 2.98 | 1.46 | 4.34 | 2.98 | 1.46 | 4.34    | 2.98 | 1.46 |
| 10           |            | 7.00    | 2.95 | 2.37 | 6.21         | 2.98 | 2.08 | 4.69 | 3.09 | 1.52 | 4.22 | 3.15 | 1.34 | 4.22 | 3.15 | 1.34 | 4.22    | 3.15 | 1.34 |
| 12           |            | 7.13    | 3.03 | 2.35 | 6.18         | 3.10 | 1.99 | 4.66 | 3.24 | 1.44 | 4.51 | 3.27 | 1.38 | 4.51 | 3.27 | 1.38 | 4.51    | 3.27 | 1.38 |
| 15           |            | 6.88    | 3.10 | 2.22 | 6.34         | 3.25 | 1.95 | 4.79 | 3.39 | 1.41 | 4.65 | 3.43 | 1.35 | 4.65 | 3.43 | 1.35 | 4.65    | 3.43 | 1.35 |
| 20           |            | 6.49    | 3.26 | 1.99 | 6.27         | 3.46 | 1.81 | 4.62 | 3.66 | 1.26 | 3.91 | 3.70 | 1.06 | 3.91 | 3.70 | 1.06 | 3.91    | 3.70 | 1.06 |
| 25           |            | 6.09    | 3.76 | 1.62 | 6.09         | 3.76 | 1.62 | 4.60 | 3.92 | 1.17 | 4.32 | 3.97 | 1.09 | 4.32 | 3.97 | 1.09 | 4.32    | 3.97 | 1.09 |
| 30           |            | 6.18    | 4.13 | 1.50 | 6.18         | 4.13 | 1.50 | 4.78 | 4.28 | 1.11 | 4.78 | 4.28 | 1.11 | 4.78 | 4.28 | 1.11 | 4.78    | 4.28 | 1.11 |
| 35           |            | /       | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
| 65           |            | -25     | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
|              | -20        | 3.44    | 1.29 | 2.66 | 3.44         | 1.29 | 2.66 | 2.58 | 1.37 | 1.89 | 1.86 | 1.39 | 1.34 | 1.86 | 1.39 | 1.34 | 1.86    | 1.39 | 1.34 |
|              | -15        | 4.12    | 1.38 | 2.99 | 4.12         | 1.38 | 2.99 | 3.09 | 1.50 | 2.06 | 2.16 | 1.51 | 1.43 | 2.16 | 1.51 | 1.43 | 2.16    | 1.51 | 1.43 |
|              | -10        | 4.83    | 1.56 | 3.10 | 4.83         | 1.56 | 3.10 | 3.61 | 1.65 | 2.18 | 2.55 | 1.69 | 1.51 | 2.55 | 1.69 | 1.51 | 2.55    | 1.69 | 1.51 |
|              | -7         | 5.05    | 1.69 | 2.99 | 5.05         | 1.69 | 2.99 | 3.81 | 1.73 | 2.20 | 2.79 | 1.79 | 1.56 | 2.79 | 1.79 | 1.56 | 2.79    | 1.79 | 1.56 |
|              | -5         | 4.93    | 1.76 | 2.80 | 4.93         | 1.76 | 2.80 | 3.77 | 1.84 | 2.05 | 3.00 | 1.88 | 1.59 | 3.00 | 1.88 | 1.59 | 3.00    | 1.88 | 1.59 |
|              | -2         | 5.31    | 1.89 | 2.81 | 5.31         | 1.89 | 2.81 | 4.02 | 2.12 | 1.89 | 2.91 | 2.15 | 1.36 | 2.91 | 2.15 | 1.36 | 2.91    | 2.15 | 1.36 |
|              | 0          | 5.68    | 1.96 | 2.90 | 5.35         | 1.99 | 2.68 | 4.05 | 2.17 | 1.87 | 3.11 | 2.20 | 1.41 | 3.11 | 2.20 | 1.41 | 3.11    | 2.20 | 1.41 |
|              | 2          | 5.78    | 2.04 | 2.84 | 5.41         | 2.07 | 2.61 | 4.13 | 2.20 | 1.88 | 3.29 | 2.22 | 1.48 | 3.29 | 2.22 | 1.48 | 3.29    | 2.22 | 1.48 |
|              | 5          | 6.23    | 2.30 | 2.71 | 5.96         | 2.32 | 2.57 | 4.44 | 2.43 | 1.83 | 3.74 | 2.46 | 1.52 | 3.74 | 2.46 | 1.52 | 3.74    | 2.46 | 1.52 |
|              | 7          | 6.46    | 2.45 | 2.64 | 6.19         | 2.47 | 2.51 | 4.63 | 2.57 | 1.80 | 4.10 | 2.59 | 1.58 | 4.10 | 2.59 | 1.58 | 4.10    | 2.59 | 1.58 |
|              | 10         | 6.68    | 2.60 | 2.57 | 6.22         | 2.63 | 2.37 | 4.58 | 2.75 | 1.67 | 3.94 | 2.80 | 1.41 | 3.94 | 2.80 | 1.41 | 3.94    | 2.80 | 1.41 |
|              | 12         | 6.82    | 2.72 | 2.51 | 6.20         | 2.75 | 2.25 | 4.58 | 2.85 | 1.60 | 4.21 | 2.89 | 1.46 | 4.21 | 2.89 | 1.46 | 4.21    | 2.89 | 1.46 |
|              | 15         | 6.57    | 2.80 | 2.35 | 6.25         | 2.82 | 2.22 | 4.72 | 2.90 | 1.63 | 4.33 | 2.97 | 1.46 | 4.33 | 2.97 | 1.46 | 4.33    | 2.97 | 1.46 |
|              | 20         | 6.18    | 3.05 | 2.03 | 6.18         | 3.05 | 2.03 | 4.64 | 3.15 | 1.47 | 3.66 | 3.17 | 1.16 | 3.66 | 3.17 | 1.16 | 3.66    | 3.17 | 1.16 |
|              | 25         | 5.74    | 3.27 | 1.76 | 5.74         | 3.27 | 1.76 | 4.55 | 3.38 | 1.34 | 4.05 | 3.41 | 1.19 | 4.05 | 3.41 | 1.19 | 4.05    | 3.41 | 1.19 |
|              | 30         | 5.78    | 3.55 | 1.63 | 5.78         | 3.55 | 1.63 | 4.66 | 3.70 | 1.26 | 4.47 | 3.75 | 1.19 | 4.47 | 3.75 | 1.19 | 4.47    | 3.75 | 1.19 |
|              | 35         | /       | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
|              | 70         | -25     | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
| -20          |            | /       | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
| -15          |            | 3.96    | 1.30 | 3.04 | 3.96         | 1.30 | 3.04 | 2.97 | 1.38 | 2.16 | 2.05 | 1.48 | 1.39 | 2.05 | 1.48 | 1.39 | 2.05    | 1.48 | 1.39 |
| -10          |            | 4.18    | 1.39 | 3.00 | 4.18         | 1.39 | 3.00 | 3.13 | 1.48 | 2.12 | 2.54 | 1.57 | 1.62 | 2.54 | 1.57 | 1.62 | 2.54    | 1.57 | 1.62 |
| -7           |            | 4.48    | 1.48 | 3.03 | 4.48         | 1.48 | 3.03 | 3.44 | 1.59 | 2.16 | 2.83 | 1.63 | 1.74 | 2.83 | 1.63 | 1.74 | 2.83    | 1.63 | 1.74 |
| -5           |            | 4.50    | 1.54 | 2.91 | 4.50         | 1.54 | 2.91 | 3.62 | 1.65 | 2.19 | 2.77 | 1.79 | 1.55 | 2.77 | 1.79 | 1.55 | 2.77    | 1.79 | 1.55 |
| -2           |            | 5.13    | 1.70 | 3.01 | 5.13         | 1.70 | 3.01 | 4.01 | 1.93 | 2.08 | 2.69 | 2.13 | 1.26 | 2.69 | 2.13 | 1.26 | 2.69    | 2.13 | 1.26 |
| 0            |            | 5.35    | 1.76 | 3.04 | 5.35         | 1.76 | 3.04 | 4.05 | 1.98 | 2.04 | 2.88 | 2.16 |      |      |      |      |         |      |      |

8.3 Prestazioni riscaldamento Mod. 08

| Tw_out<br>°C | DBWB<br>°C | Maximum |       |      | 100%(Normal) |       |      | 75%  |       |      | 50%  |       |      | 25%  |       |      | Minimum |       |      |
|--------------|------------|---------|-------|------|--------------|-------|------|------|-------|------|------|-------|------|------|-------|------|---------|-------|------|
|              |            | HC      | COP   | PI   | HC           | COP   | PI   | HC   | COP   | PI   | HC   | COP   | PI   | HC   | COP   | PI   | HC      | COP   | PI   |
| 25           | -25        | 5.15    | 2.16  | 2.39 | 5.15         | 2.16  | 2.39 | 3.86 | 2.49  | 1.55 | 2.57 | 2.79  | 0.92 | 2.19 | 2.85  | 0.77 | 2.19    | 2.85  | 0.77 |
|              | -20        | 6.00    | 2.46  | 2.44 | 6.00         | 2.46  | 2.44 | 4.63 | 2.70  | 1.71 | 3.09 | 3.01  | 1.02 | 2.61 | 3.10  | 0.84 | 2.61    | 3.10  | 0.84 |
|              | -15        | 7.43    | 2.86  | 2.60 | 7.13         | 2.88  | 2.47 | 5.15 | 3.10  | 1.66 | 3.43 | 3.50  | 0.98 | 3.05 | 3.56  | 0.85 | 3.05    | 3.56  | 0.85 |
|              | -10        | 8.61    | 3.07  | 2.80 | 7.01         | 3.65  | 1.92 | 5.38 | 3.83  | 1.41 | 3.65 | 4.23  | 0.86 | 3.36 | 4.25  | 0.79 | 3.36    | 4.25  | 0.79 |
|              | -7         | 8.93    | 3.41  | 2.62 | 6.95         | 3.81  | 1.83 | 5.29 | 4.03  | 1.31 | 3.80 | 4.42  | 0.86 | 3.80 | 4.42  | 0.86 | 3.80    | 4.42  | 0.86 |
|              | -5         | 9.07    | 3.61  | 2.51 | 7.03         | 4.02  | 1.75 | 5.27 | 4.35  | 1.21 | 4.07 | 4.62  | 0.88 | 4.07 | 4.62  | 0.88 | 4.07    | 4.62  | 0.88 |
|              | -2         | 9.34    | 3.97  | 2.35 | 7.10         | 4.42  | 1.60 | 5.40 | 4.82  | 1.12 | 3.64 | 5.25  | 0.69 | 3.64 | 5.25  | 0.69 | 3.64    | 5.25  | 0.69 |
|              | 0          | 9.29    | 4.16  | 2.24 | 7.08         | 4.69  | 1.51 | 5.39 | 4.98  | 1.08 | 3.82 | 5.31  | 0.72 | 3.82 | 5.31  | 0.72 | 3.82    | 5.31  | 0.72 |
|              | 2          | 9.42    | 4.46  | 2.11 | 7.11         | 5.02  | 1.42 | 5.41 | 5.32  | 1.02 | 4.03 | 5.71  | 0.71 | 4.03 | 5.71  | 0.71 | 4.03    | 5.71  | 0.71 |
|              | 5          | 10.18   | 5.39  | 1.89 | 7.66         | 5.89  | 1.30 | 5.77 | 6.27  | 0.92 | 4.23 | 6.52  | 0.65 | 4.23 | 6.52  | 0.65 | 4.23    | 6.52  | 0.65 |
|              | 7          | 10.76   | 5.84  | 1.84 | 8.28         | 6.53  | 1.27 | 6.25 | 6.95  | 0.90 | 4.62 | 7.15  | 0.65 | 4.62 | 7.15  | 0.65 | 4.62    | 7.15  | 0.65 |
|              | 10         | 11.16   | 6.79  | 1.64 | 8.36         | 7.50  | 1.12 | 6.25 | 7.84  | 0.80 | 5.08 | 8.11  | 0.63 | 5.08 | 8.11  | 0.63 | 5.08    | 8.11  | 0.63 |
|              | 12         | 11.40   | 7.62  | 1.50 | 8.28         | 8.41  | 0.99 | 6.31 | 8.85  | 0.71 | 5.42 | 9.01  | 0.60 | 5.42 | 9.01  | 0.60 | 5.42    | 9.01  | 0.60 |
|              | 15         | 11.17   | 8.07  | 1.38 | 8.23         | 8.83  | 0.93 | 6.12 | 9.38  | 0.65 | 5.50 | 9.46  | 0.58 | 5.50 | 9.46  | 0.58 | 5.50    | 9.46  | 0.58 |
|              | 20         | 10.82   | 8.80  | 1.23 | 8.10         | 9.64  | 0.84 | 6.18 | 10.02 | 0.62 | 5.17 | 10.24 | 0.50 | 5.17 | 10.24 | 0.50 | 5.17    | 10.24 | 0.50 |
|              | 25         | 10.65   | 9.75  | 1.09 | 8.14         | 10.30 | 0.79 | 6.16 | 10.68 | 0.58 | 5.70 | 10.83 | 0.53 | 5.70 | 10.83 | 0.53 | 5.70    | 10.83 | 0.53 |
|              | 30         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 35         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 30           | -25        | 5.00    | 2.04  | 2.45 | 5.00         | 2.04  | 2.45 | 3.75 | 2.33  | 1.61 | 2.52 | 2.58  | 0.98 | 2.14 | 2.64  | 0.81 | 2.14    | 2.64  | 0.81 |
|              | -20        | 5.95    | 2.34  | 2.54 | 5.95         | 2.34  | 2.54 | 4.55 | 2.63  | 1.73 | 3.03 | 2.89  | 1.05 | 2.56 | 3.01  | 0.85 | 2.56    | 3.01  | 0.85 |
|              | -15        | 7.36    | 2.56  | 2.87 | 7.18         | 2.58  | 2.78 | 5.15 | 2.92  | 1.76 | 3.43 | 3.20  | 1.07 | 3.00 | 3.30  | 0.91 | 3.00    | 3.30  | 0.91 |
|              | -10        | 8.53    | 2.88  | 2.96 | 7.08         | 3.11  | 2.28 | 5.39 | 3.43  | 1.57 | 3.26 | 3.80  | 0.86 | 3.26 | 3.80  | 0.86 | 3.26    | 3.80  | 0.86 |
|              | -7         | 8.78    | 3.10  | 2.84 | 6.93         | 3.43  | 2.02 | 5.31 | 3.70  | 1.44 | 3.70 | 4.04  | 0.92 | 3.70 | 4.04  | 0.92 | 3.70    | 4.04  | 0.92 |
|              | -5         | 8.99    | 3.30  | 2.72 | 6.97         | 3.70  | 1.89 | 5.32 | 3.99  | 1.33 | 3.98 | 4.35  | 0.92 | 3.98 | 4.35  | 0.92 | 3.98    | 4.35  | 0.92 |
|              | -2         | 9.26    | 3.54  | 2.62 | 7.10         | 4.10  | 1.73 | 5.35 | 4.42  | 1.21 | 3.56 | 4.70  | 0.76 | 3.56 | 4.70  | 0.76 | 3.56    | 4.70  | 0.76 |
|              | 0          | 9.18    | 3.74  | 2.46 | 7.10         | 4.26  | 1.67 | 5.31 | 4.59  | 1.16 | 3.73 | 4.92  | 0.76 | 3.73 | 4.92  | 0.76 | 3.73    | 4.92  | 0.76 |
|              | 2          | 9.30    | 4.00  | 2.32 | 7.10         | 4.49  | 1.58 | 5.35 | 4.86  | 1.10 | 3.94 | 5.18  | 0.76 | 3.94 | 5.18  | 0.76 | 3.94    | 5.18  | 0.76 |
|              | 5          | 10.03   | 4.58  | 2.19 | 7.70         | 5.02  | 1.53 | 5.82 | 5.41  | 1.08 | 4.13 | 5.87  | 0.70 | 4.13 | 5.87  | 0.70 | 4.13    | 5.87  | 0.70 |
|              | 7          | 10.60   | 5.01  | 2.11 | 8.34         | 5.42  | 1.54 | 6.31 | 5.88  | 1.07 | 4.52 | 6.12  | 0.74 | 4.52 | 6.12  | 0.74 | 4.52    | 6.12  | 0.74 |
|              | 10         | 11.02   | 5.64  | 1.95 | 8.41         | 6.21  | 1.35 | 6.30 | 6.60  | 0.96 | 4.98 | 6.79  | 0.73 | 4.98 | 6.79  | 0.73 | 4.98    | 6.79  | 0.73 |
|              | 12         | 11.27   | 5.97  | 1.89 | 8.32         | 6.50  | 1.28 | 6.24 | 6.82  | 0.92 | 5.32 | 7.12  | 0.75 | 5.32 | 7.12  | 0.75 | 5.32    | 7.12  | 0.75 |
|              | 15         | 11.03   | 6.47  | 1.70 | 8.28         | 7.06  | 1.17 | 6.22 | 7.41  | 0.84 | 5.41 | 7.59  | 0.71 | 5.41 | 7.59  | 0.71 | 5.41    | 7.59  | 0.71 |
|              | 20         | 10.71   | 7.48  | 1.43 | 8.24         | 7.89  | 1.04 | 6.13 | 8.25  | 0.74 | 5.06 | 8.30  | 0.61 | 5.06 | 8.30  | 0.61 | 5.06    | 8.30  | 0.61 |
|              | 25         | 10.51   | 8.70  | 1.21 | 8.05         | 9.06  | 0.89 | 6.10 | 9.77  | 0.62 | 5.61 | 9.75  | 0.58 | 5.61 | 9.75  | 0.58 | 5.61    | 9.75  | 0.58 |
|              | 30         | 10.50   | 9.80  | 1.07 | 8.21         | 10.35 | 0.79 | 6.21 | 10.78 | 0.58 | 6.16 | 11.02 | 0.56 | 6.16 | 11.02 | 0.56 | 6.16    | 11.02 | 0.56 |
|              | 35         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 35           | -25        | 4.71    | 1.98  | 2.38 | 4.71         | 1.98  | 2.38 | 3.67 | 2.05  | 1.79 | 2.41 | 2.25  | 1.07 | 2.09 | 2.35  | 0.89 | 2.09    | 2.35  | 0.89 |
|              | -20        | 5.91    | 2.22  | 2.67 | 5.91         | 2.22  | 2.67 | 4.46 | 2.34  | 1.90 | 2.93 | 2.62  | 1.12 | 2.50 | 2.64  | 0.95 | 2.50    | 2.64  | 0.95 |
|              | -15        | 7.25    | 2.40  | 3.02 | 7.25         | 2.40  | 3.02 | 5.40 | 2.56  | 2.11 | 3.60 | 2.87  | 1.25 | 2.91 | 2.96  | 0.98 | 2.91    | 2.96  | 0.98 |
|              | -10        | 8.02    | 2.75  | 2.91 | 7.22         | 2.86  | 2.53 | 5.44 | 3.04  | 1.79 | 3.16 | 3.28  | 0.96 | 3.16 | 3.28  | 0.96 | 3.16    | 3.28  | 0.96 |
|              | -7         | 8.32    | 2.85  | 2.92 | 7.00         | 3.00  | 2.33 | 5.35 | 3.43  | 1.56 | 3.60 | 3.52  | 1.02 | 3.60 | 3.52  | 1.02 | 3.60    | 3.52  | 1.02 |
|              | -5         | 8.57    | 3.11  | 2.76 | 7.03         | 3.36  | 2.09 | 5.25 | 3.62  | 1.45 | 3.88 | 3.72  | 1.04 | 3.88 | 3.72  | 1.04 | 3.88    | 3.72  | 1.04 |
|              | -2         | 8.88    | 3.37  | 2.64 | 7.10         | 3.70  | 1.92 | 5.40 | 3.97  | 1.36 | 3.47 | 4.16  | 0.84 | 3.47 | 4.16  | 0.84 | 3.47    | 4.16  | 0.84 |
|              | 0          | 8.77    | 3.45  | 2.54 | 7.12         | 3.80  | 1.87 | 5.36 | 4.08  | 1.31 | 3.66 | 4.37  | 0.84 | 3.66 | 4.37  | 0.84 | 3.66    | 4.37  | 0.84 |
|              | 2          | 8.90    | 3.60  | 2.47 | 7.10         | 3.85  | 1.84 | 5.38 | 4.32  | 1.25 | 3.84 | 4.48  | 0.86 | 3.84 | 4.48  | 0.86 | 3.84    | 4.48  | 0.86 |
|              | 5          | 9.63    | 4.17  | 2.31 | 7.72         | 4.55  | 1.70 | 5.90 | 4.85  | 1.22 | 4.01 | 5.16  | 0.78 | 4.01 | 5.16  | 0.78 | 4.01    | 5.16  | 0.78 |
|              | 7*         | 10.17   | 4.59  | 2.22 | 8.40         | 5.00  | 1.68 | 6.37 | 5.39  | 1.18 | 4.40 | 5.64  | 0.78 | 4.40 | 5.64  | 0.78 | 4.40    | 5.64  | 0.78 |
|              | 10         | 10.63   | 5.10  | 2.09 | 8.45         | 5.60  | 1.51 | 6.36 | 5.94  | 1.07 | 4.86 | 6.43  | 0.76 | 4.86 | 6.43  | 0.76 | 4.86    | 6.43  | 0.76 |
|              | 12         | 10.92   | 5.50  | 1.99 | 8.34         | 6.14  | 1.36 | 6.31 | 6.51  | 0.97 | 5.19 | 6.61  | 0.79 | 5.19 | 6.61  | 0.79 | 5.19    | 6.61  | 0.79 |
|              | 15         | 10.69   | 5.83  | 1.83 | 8.31         | 6.35  | 1.31 | 6.30 | 7.00  | 0.94 | 5.31 | 6.90  | 0.77 | 5.31 | 6.90  | 0.77 | 5.31    | 6.90  | 0.77 |
|              | 20         | 10.41   | 6.58  | 1.58 | 8.26         | 7.01  | 1.18 | 6.23 | 7.36  | 0.85 | 4.92 | 7.57  | 0.65 | 4.92 | 7.57  | 0.65 | 4.92    | 7.57  | 0.65 |
|              | 25         | 10.25   | 7.65  | 1.34 | 8.12         | 8.17  | 0.99 | 6.22 | 8.48  | 0.73 | 5.46 | 8.64  | 0.63 | 5.46 | 8.64  | 0.63 | 5.46    | 8.64  | 0.63 |
|              | 30         | 10.43   | 8.94  | 1.17 | 8.10         | 9.55  | 0.85 | 6.10 | 9.94  | 0.61 | 6.00 | 10.06 | 0.60 | 6.00 | 10.06 | 0.60 | 6.00    | 10.06 | 0.60 |
|              | 35         | 10.30   | 10.10 | 1.02 | 8.19         | 10.58 | 0.77 | 6.60 | 11.13 | 0.59 | 6.60 | 11.13 | 0.59 | 6.60 | 11.13 | 0.59 | 6.60    | 11.13 | 0.59 |
| 40           | -25        | 4.72    | 1.67  | 2.82 | 4.31         | 1.70  | 2.54 | 3.23 | 1.83  | 1.77 | 2.14 | 2.19  | 0.98 | 2.14 | 2.19  | 0.98 | 2.14    | 2.19  | 0.98 |
|              | -20        | 6.33    | 2.12  | 2.98 | 5.82         | 2.14  | 2.71 | 4.37 | 2.26  | 1.93 | 2.52 | 2.47  | 1.02 | 2.52 | 2.47  | 1.02 | 2.52    | 2.47  | 1.02 |
|              | -15        | 7.22    | 2.25  | 3.21 | 7.22         | 2.25  | 3.21 | 5.44 | 2.35  | 2.31 | 2.98 | 2.86  | 1.04 | 2.98 | 2.86  | 1.04 | 2.98    | 2.86  | 1.04 |
|              | -10        | 7.92    | 2.59  | 3.06 | 7.7          |       |      |      |       |      |      |       |      |      |       |      |         |       |      |

| Tw_out<br>°C | DBWB<br>°C | Maximum |      |      | 100%(Normal) |      |      | 75%  |      |      | 50%  |      |      | 25%  |      |      | Minimum |      |      |
|--------------|------------|---------|------|------|--------------|------|------|------|------|------|------|------|------|------|------|------|---------|------|------|
|              |            | HC      | COP  | PI   | HC           | COP  | PI   | HC   | COP  | PI   | HC   | COP  | PI   | HC   | COP  | PI   | HC      | COP  | PI   |
| 55           | -25        | 4.34    | 1.37 | 3.16 | 4.34         | 1.37 | 3.16 | 3.25 | 1.56 | 2.09 | 2.59 | 1.67 | 1.55 | 2.59 | 1.67 | 1.55 | 2.59    | 1.67 | 1.55 |
|              | -20        | 6.08    | 1.63 | 3.72 | 5.50         | 1.68 | 3.29 | 4.13 | 1.78 | 2.32 | 2.77 | 2.07 | 1.34 | 2.77 | 2.07 | 1.34 | 2.77    | 2.07 | 1.34 |
|              | -15        | 6.84    | 1.78 | 3.85 | 6.27         | 1.80 | 3.49 | 4.77 | 1.92 | 2.49 | 3.29 | 2.12 | 1.55 | 3.29 | 2.12 | 1.55 | 3.29    | 2.12 | 1.55 |
|              | -10        | 7.56    | 1.98 | 3.82 | 7.23         | 2.05 | 3.53 | 5.46 | 2.22 | 2.46 | 4.08 | 2.38 | 1.71 | 4.08 | 2.38 | 1.71 | 4.08    | 2.38 | 1.71 |
|              | -7         | 7.89    | 2.06 | 3.83 | 6.90         | 2.15 | 3.21 | 5.14 | 2.39 | 2.15 | 4.61 | 2.43 | 1.90 | 4.61 | 2.43 | 1.90 | 4.61    | 2.43 | 1.90 |
|              | -5         | 8.18    | 2.16 | 3.78 | 7.10         | 2.27 | 3.12 | 5.56 | 2.52 | 2.21 | 4.92 | 2.60 | 1.89 | 4.92 | 2.60 | 1.89 | 4.92    | 2.60 | 1.89 |
|              | -2         | 8.56    | 2.20 | 3.89 | 7.32         | 2.36 | 3.11 | 5.70 | 2.60 | 2.19 | 5.00 | 2.73 | 1.83 | 5.00 | 2.73 | 1.83 | 5.00    | 2.73 | 1.83 |
|              | 0          | 8.17    | 2.44 | 3.35 | 7.32         | 2.53 | 2.90 | 5.73 | 2.68 | 2.14 | 5.03 | 2.78 | 1.81 | 5.03 | 2.78 | 1.81 | 5.03    | 2.78 | 1.81 |
|              | 2          | 8.25    | 2.51 | 3.28 | 7.80         | 2.55 | 3.06 | 5.74 | 2.86 | 2.01 | 5.33 | 2.93 | 1.82 | 5.33 | 2.93 | 1.82 | 5.33    | 2.93 | 1.82 |
|              | 5          | 8.87    | 2.91 | 3.05 | 7.87         | 2.97 | 2.65 | 5.96 | 3.12 | 1.91 | 5.96 | 3.12 | 1.91 | 5.96 | 3.12 | 1.91 | 5.96    | 3.12 | 1.91 |
|              | 7          | 9.27    | 3.13 | 2.97 | 7.80         | 3.20 | 2.44 | 6.52 | 3.39 | 1.92 | 6.52 | 3.42 | 1.91 | 6.52 | 3.42 | 1.91 | 6.52    | 3.42 | 1.91 |
|              | 10         | 9.69    | 3.39 | 2.86 | 8.00         | 3.53 | 2.26 | 6.11 | 3.74 | 1.63 | 6.11 | 3.70 | 1.65 | 6.11 | 3.70 | 1.65 | 6.11    | 3.70 | 1.65 |
|              | 12         | 9.88    | 3.55 | 2.78 | 8.05         | 3.75 | 2.15 | 6.55 | 4.01 | 1.63 | 6.55 | 3.97 | 1.65 | 6.55 | 3.97 | 1.65 | 6.55    | 3.97 | 1.65 |
|              | 15         | 9.67    | 3.68 | 2.63 | 7.85         | 3.91 | 2.01 | 6.71 | 4.10 | 1.64 | 6.71 | 4.06 | 1.65 | 6.71 | 4.06 | 1.65 | 6.71    | 4.06 | 1.65 |
|              | 20         | 9.31    | 4.13 | 2.25 | 7.78         | 4.25 | 1.83 | 6.04 | 4.44 | 1.36 | 6.04 | 4.44 | 1.36 | 6.04 | 4.44 | 1.36 | 6.04    | 4.44 | 1.36 |
|              | 25         | 9.14    | 4.62 | 1.98 | 7.83         | 4.77 | 1.64 | 6.66 | 4.94 | 1.35 | 6.66 | 4.94 | 1.35 | 6.66 | 4.94 | 1.35 | 6.66    | 4.94 | 1.35 |
|              | 30         | 9.30    | 5.19 | 1.79 | 7.86         | 5.36 | 1.47 | 7.35 | 5.47 | 1.34 | 7.35 | 5.47 | 1.34 | 7.35 | 5.47 | 1.34 | 7.35    | 5.47 | 1.34 |
|              | 35         | 9.13    | 5.55 | 1.65 | 8.10         | 5.78 | 1.40 | 8.10 | 5.84 | 1.39 | 8.10 | 5.84 | 1.39 | 8.10 | 5.84 | 1.39 | 8.10    | 5.84 | 1.39 |
| 60           | -25        | /       | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
|              | -20        | 6.00    | 1.52 | 3.96 | 5.48         | 1.58 | 3.47 | 4.11 | 1.81 | 2.27 | 2.92 | 1.99 | 1.47 | 2.92 | 1.99 | 1.47 | 2.92    | 1.99 | 1.47 |
|              | -15        | 6.21    | 1.71 | 3.63 | 6.21         | 1.71 | 3.63 | 4.65 | 1.93 | 2.42 | 3.45 | 2.04 | 1.69 | 3.45 | 2.04 | 1.69 | 3.45    | 2.04 | 1.69 |
|              | -10        | 7.52    | 1.96 | 3.84 | 6.96         | 1.99 | 3.49 | 5.22 | 2.25 | 2.32 | 4.27 | 2.46 | 1.73 | 4.27 | 2.46 | 1.73 | 4.27    | 2.46 | 1.73 |
|              | -7         | 7.86    | 2.01 | 3.91 | 6.98         | 2.08 | 3.36 | 5.30 | 2.35 | 2.26 | 4.80 | 2.50 | 1.92 | 4.80 | 2.50 | 1.92 | 4.80    | 2.50 | 1.92 |
|              | -5         | 8.13    | 2.13 | 3.81 | 7.17         | 2.22 | 3.24 | 5.45 | 2.41 | 2.26 | 5.14 | 2.57 | 2.00 | 5.14 | 2.57 | 2.00 | 5.14    | 2.57 | 2.00 |
|              | -2         | 8.38    | 2.19 | 3.83 | 7.29         | 2.30 | 3.17 | 5.58 | 2.45 | 2.28 | 4.83 | 2.60 | 1.86 | 4.83 | 2.60 | 1.86 | 4.83    | 2.60 | 1.86 |
|              | 0          | 7.97    | 2.32 | 3.44 | 7.44         | 2.36 | 3.15 | 5.75 | 2.61 | 2.20 | 4.82 | 2.73 | 1.77 | 4.82 | 2.73 | 1.77 | 4.82    | 2.73 | 1.77 |
|              | 2          | 8.05    | 2.40 | 3.36 | 7.66         | 2.43 | 3.15 | 5.82 | 2.71 | 2.15 | 5.10 | 2.87 | 1.78 | 5.10 | 2.87 | 1.78 | 5.10    | 2.87 | 1.78 |
|              | 5          | 8.63    | 2.73 | 3.16 | 7.94         | 2.79 | 2.85 | 5.99 | 3.05 | 1.96 | 5.99 | 3.05 | 1.96 | 5.99 | 3.05 | 1.96 | 5.99    | 3.05 | 1.96 |
|              | 7          | 8.99    | 2.92 | 3.08 | 8.31         | 2.98 | 2.79 | 6.24 | 3.29 | 1.89 | 6.24 | 3.29 | 1.89 | 6.24 | 3.29 | 1.89 | 6.24    | 3.29 | 1.89 |
|              | 10         | 9.37    | 3.13 | 3.00 | 8.04         | 3.30 | 2.43 | 6.10 | 3.61 | 1.69 | 6.10 | 3.61 | 1.69 | 6.10 | 3.61 | 1.69 | 6.10    | 3.61 | 1.69 |
|              | 12         | 9.59    | 3.37 | 2.85 | 8.07         | 3.56 | 2.27 | 6.51 | 3.82 | 1.70 | 6.51 | 3.82 | 1.70 | 6.51 | 3.82 | 1.70 | 6.51    | 3.82 | 1.70 |
|              | 15         | 9.36    | 3.51 | 2.67 | 7.89         | 3.65 | 2.16 | 6.68 | 3.98 | 1.68 | 6.68 | 3.98 | 1.68 | 6.68 | 3.98 | 1.68 | 6.68    | 3.98 | 1.68 |
|              | 20         | 9.01    | 3.79 | 2.37 | 8.01         | 3.97 | 2.02 | 5.99 | 4.36 | 1.37 | 5.99 | 4.36 | 1.37 | 5.99 | 4.36 | 1.37 | 5.99    | 4.36 | 1.37 |
|              | 25         | 8.79    | 4.21 | 2.09 | 7.90         | 4.38 | 1.80 | 6.35 | 4.63 | 1.37 | 6.35 | 4.63 | 1.37 | 6.35 | 4.63 | 1.37 | 6.35    | 4.63 | 1.37 |
|              | 30         | 8.91    | 4.63 | 1.92 | 7.88         | 4.83 | 1.63 | 7.01 | 4.99 | 1.41 | 7.01 | 4.99 | 1.41 | 7.01 | 4.99 | 1.41 | 7.01    | 4.99 | 1.41 |
|              | 35         | /       | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
| 65           | -25        | /       | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
|              | -20        | 5.43    | 1.48 | 3.67 | 5.43         | 1.48 | 3.67 | 4.07 | 1.53 | 2.66 | 2.82 | 1.62 | 1.74 | 2.82 | 1.62 | 1.74 | 2.82    | 1.62 | 1.74 |
|              | -15        | 6.15    | 1.60 | 3.84 | 6.15         | 1.60 | 3.84 | 4.62 | 1.66 | 2.78 | 3.45 | 1.87 | 1.84 | 3.45 | 1.87 | 1.84 | 3.45    | 1.87 | 1.84 |
|              | -10        | 7.49    | 1.83 | 4.08 | 6.96         | 1.87 | 3.73 | 5.22 | 1.89 | 2.76 | 4.10 | 2.12 | 1.93 | 4.10 | 2.12 | 1.93 | 4.10    | 2.12 | 1.93 |
|              | -7         | 7.80    | 1.96 | 3.97 | 7.21         | 2.02 | 3.57 | 5.47 | 2.11 | 2.59 | 4.63 | 2.17 | 2.13 | 4.63 | 2.17 | 2.13 | 4.63    | 2.17 | 2.13 |
|              | -5         | 8.09    | 2.02 | 4.00 | 7.21         | 2.06 | 3.50 | 5.40 | 2.16 | 2.50 | 4.98 | 2.29 | 2.18 | 4.98 | 2.29 | 2.18 | 4.98    | 2.29 | 2.18 |
|              | -2         | 8.27    | 2.08 | 3.97 | 7.33         | 2.12 | 3.46 | 5.63 | 2.20 | 2.56 | 4.64 | 2.41 | 1.93 | 4.64 | 2.41 | 1.93 | 4.64    | 2.41 | 1.93 |
|              | 0          | 7.80    | 2.19 | 3.56 | 7.45         | 2.19 | 3.40 | 5.68 | 2.23 | 2.55 | 4.58 | 2.51 | 1.82 | 4.58 | 2.51 | 1.82 | 4.58    | 2.51 | 1.82 |
|              | 2          | 7.84    | 2.26 | 3.47 | 7.48         | 2.27 | 3.30 | 5.75 | 2.35 | 2.45 | 4.86 | 2.48 | 1.96 | 4.86 | 2.48 | 1.96 | 4.86    | 2.48 | 1.96 |
|              | 5          | 8.37    | 2.41 | 3.47 | 8.02         | 2.44 | 3.28 | 6.05 | 2.56 | 2.36 | 5.44 | 2.71 | 2.01 | 5.44 | 2.71 | 2.01 | 5.44    | 2.71 | 2.01 |
|              | 7          | 8.71    | 2.57 | 3.39 | 8.11         | 2.67 | 3.03 | 6.17 | 2.79 | 2.21 | 5.95 | 2.89 | 2.06 | 5.95 | 2.89 | 2.06 | 5.95    | 2.89 | 2.06 |
|              | 10         | 9.05    | 2.78 | 3.26 | 8.06         | 2.90 | 2.78 | 6.11 | 3.02 | 2.02 | 5.75 | 3.14 | 1.83 | 5.75 | 3.14 | 1.83 | 5.75    | 3.14 | 1.83 |
|              | 12         | 9.28    | 2.94 | 3.16 | 8.07         | 3.08 | 2.62 | 6.16 | 3.23 | 1.91 | 6.16 | 3.20 | 1.92 | 6.16 | 3.20 | 1.92 | 6.16    | 3.20 | 1.92 |
|              | 15         | 9.02    | 3.04 | 2.97 | 7.87         | 3.17 | 2.48 | 6.31 | 3.30 | 1.91 | 6.31 | 3.30 | 1.91 | 6.31 | 3.30 | 1.91 | 6.31    | 3.30 | 1.91 |
|              | 20         | 8.61    | 3.29 | 2.62 | 7.96         | 3.35 | 2.38 | 5.99 | 3.55 | 1.69 | 5.39 | 3.67 | 1.47 | 5.39 | 3.67 | 1.47 | 5.39    | 3.67 | 1.47 |
|              | 25         | 8.38    | 3.66 | 2.29 | 8.03         | 3.72 | 2.16 | 5.99 | 3.96 | 1.51 | 5.99 | 3.96 | 1.51 | 5.99 | 3.96 | 1.51 | 5.99    | 3.96 | 1.51 |
|              | 30         | 8.40    | 3.95 | 2.13 | 7.86         | 3.99 | 1.97 | 6.61 | 4.01 | 1.65 | 6.61 | 4.01 | 1.65 | 6.61 | 4.01 | 1.65 | 6.61    | 4.01 | 1.65 |
|              | 35         | /       | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
| 70           | -25        | /       | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
|              | -20        | /       | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
|              | -15        | 6.16    | 1.53 | 4.03 | 6.16         | 1.53 | 4.03 | 4.62 | 1.66 | 2.78 | 3.19 | 1.93 | 1.66 | 3.19 | 1.93 | 1.66 | 3.19    | 1.93 | 1.66 |
|              | -10        | 7.46    | 1.76 | 4.24 | 6.96         | 1.80 | 3.87 | 5.28 | 1.92 | 2.75 | 3.99 | 2.12 | 1.88 | 3.99 | 2.12 | 1.88 | 3.99    | 2.12 | 1.88 |
|              | -7         | 7.79    | 1.86 | 4.20 | 6.98         | 1.87 | 3.73 | 5.36 | 1.99 | 2.69 | 4.55 | 2.13 | 2.13 | 4.55 | 2.13 | 2.13 | 4.55    | 2.13 | 2.13 |
|              | -5         | 8.04    | 1.94 | 4.14 | 7.10         | 1.98 | 3.58 | 5.46 | 2.10 | 2.59 | 4.87 | 2.25 | 2.17 | 4.87 | 2.25 | 2.17 | 4.87    | 2.25 | 2.17 |
|              | -2         | 8.25    | 1.98 | 4.16 | 7.39         | 1.99 | 3.71 | 5.61 | 2.13 | 2.63 | 4.46 | 2.32 | 1.92 | 4.46 | 2.32 | 1.92 | 4.46    | 2.32 | 1.92 |
|              | 0          | 7.71    | 2.04 | 3.78 | 7.46         | 2.05 | 3.63 | 5.69 | 2.19 | 2.60 | 4.50 |      |      |      |      |      |         |      |      |

## 8.4 Prestazioni riscaldamento Mod. 10

| Tw_out<br>°C | DB/WB<br>°C | Maximum |      |      | 100%(Normal) |        |      | 75%  |       |      | 50%  |       |      | 25%  |       |      | Minimum |       |      |
|--------------|-------------|---------|------|------|--------------|--------|------|------|-------|------|------|-------|------|------|-------|------|---------|-------|------|
|              |             | HC      | COP  | PI   | HC           | COP    | PI   | HC   | COP   | PI   | HC   | COP   | PI   | HC   | COP   | PI   | HC      | COP   | PI   |
| 25           | -25         | 5.55    | 2.14 | 2.59 | 5.55         | 2.14   | 2.59 | 4.16 | 2.48  | 1.83 | 2.75 | 2.77  | 1.17 | 2.19 | 2.85  | 0.77 | 2.19    | 2.85  | 0.77 |
|              | -20         | 6.71    | 2.34 | 2.87 | 6.71         | 2.34   | 2.87 | 5.03 | 2.67  | 1.88 | 3.34 | 3.00  | 1.23 | 2.61 | 3.10  | 0.84 | 2.61    | 3.10  | 0.84 |
|              | -15         | 7.66    | 2.70 | 2.83 | 7.66         | 2.70   | 2.83 | 5.74 | 3.01  | 1.93 | 3.83 | 3.45  | 1.24 | 3.05 | 3.56  | 0.85 | 3.05    | 3.56  | 0.85 |
|              | -10         | 8.76    | 3.05 | 2.87 | 8.05         | 3.34   | 2.41 | 6.06 | 3.74  | 1.60 | 4.04 | 4.17  | 1.03 | 3.36 | 4.25  | 0.79 | 3.36    | 4.25  | 0.79 |
|              | -7          | 9.12    | 3.26 | 2.80 | 8.03         | 3.60   | 2.23 | 6.11 | 3.95  | 1.56 | 4.12 | 4.40  | 0.97 | 3.80 | 4.42  | 0.86 | 3.80    | 4.42  | 0.86 |
|              | -5          | 9.34    | 3.45 | 2.71 | 8.03         | 3.81   | 2.11 | 6.09 | 4.22  | 1.44 | 4.12 | 4.61  | 0.89 | 4.07 | 4.62  | 0.88 | 4.07    | 4.62  | 0.88 |
|              | -2          | 9.70    | 3.90 | 2.49 | 8.09         | 4.28   | 1.89 | 6.11 | 4.71  | 1.30 | 4.10 | 5.20  | 0.79 | 3.64 | 5.25  | 0.69 | 3.64    | 5.25  | 0.69 |
|              | 0           | 9.68    | 4.08 | 2.37 | 8.20         | 4.44   | 1.84 | 6.15 | 4.85  | 1.27 | 4.13 | 5.28  | 0.78 | 3.82 | 5.31  | 0.72 | 3.82    | 5.31  | 0.72 |
|              | 2           | 9.84    | 4.37 | 2.25 | 8.21         | 4.78   | 1.72 | 6.21 | 5.22  | 1.19 | 4.17 | 5.70  | 0.73 | 4.03 | 5.71  | 0.71 | 4.03    | 5.71  | 0.71 |
|              | 5           | 10.73   | 5.28 | 2.03 | 8.80         | 5.69   | 1.55 | 6.65 | 6.13  | 1.08 | 4.50 | 6.51  | 0.69 | 4.23 | 6.52  | 0.65 | 4.23    | 6.52  | 0.65 |
|              | 7           | 11.46   | 5.70 | 2.01 | 9.97         | 6.26   | 1.59 | 7.48 | 6.70  | 1.12 | 5.07 | 7.06  | 0.72 | 4.62 | 7.15  | 0.65 | 4.62    | 7.15  | 0.65 |
|              | 10          | 11.86   | 6.65 | 1.78 | 9.67         | 7.22   | 1.34 | 7.41 | 7.66  | 0.97 | 5.08 | 8.11  | 0.63 | 5.08 | 8.11  | 0.63 | 5.08    | 8.11  | 0.63 |
|              | 12          | 12.16   | 7.55 | 1.61 | 9.70         | 8.21   | 1.18 | 7.44 | 8.59  | 0.87 | 5.42 | 9.01  | 0.60 | 5.42 | 9.01  | 0.60 | 5.42    | 9.01  | 0.60 |
|              | 15          | 11.93   | 7.92 | 1.51 | 9.54         | 8.55   | 1.12 | 7.15 | 9.11  | 0.78 | 5.50 | 9.46  | 0.58 | 5.50 | 9.46  | 0.58 | 5.50    | 9.46  | 0.58 |
|              | 20          | 11.73   | 8.77 | 1.34 | 9.62         | 9.33   | 1.03 | 7.21 | 9.75  | 0.74 | 5.17 | 10.24 | 0.50 | 5.17 | 10.24 | 0.50 | 5.17    | 10.24 | 0.50 |
|              | 25          | 11.87   | 9.45 | 1.26 | 9.67         | 9.98   | 0.97 | 7.19 | 10.51 | 0.68 | 5.70 | 10.83 | 0.48 | 5.70 | 10.83 | 0.53 | 5.70    | 10.83 | 0.53 |
|              | 30          | /       | /    | /    | /            | /      | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 35          | /       | /    | /    | /            | /      | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 30           | -25         | 5.46    | 2.01 | 2.71 | 5.46         | 2.01   | 2.71 | 4.09 | 2.30  | 1.78 | 2.69 | 2.58  | 1.04 | 2.14 | 2.64  | 0.81 | 2.14    | 2.64  | 0.81 |
|              | -20         | 6.62    | 2.31 | 2.86 | 6.62         | 2.31   | 2.86 | 4.97 | 2.59  | 1.92 | 3.26 | 2.87  | 1.13 | 2.56 | 3.01  | 0.85 | 2.56    | 3.01  | 0.85 |
|              | -15         | 7.57    | 2.54 | 2.98 | 7.57         | 2.54   | 2.98 | 5.67 | 2.85  | 1.99 | 3.78 | 3.18  | 1.19 | 3.00 | 3.30  | 0.91 | 3.00    | 3.30  | 0.91 |
|              | -10         | 8.65    | 2.86 | 3.03 | 8.05         | 2.94   | 2.73 | 6.06 | 3.28  | 1.85 | 4.04 | 3.65  | 1.11 | 3.26 | 3.80  | 0.86 | 3.26    | 3.80  | 0.86 |
|              | -7          | 9.00    | 3.06 | 2.94 | 8.03         | 3.20   | 2.51 | 6.11 | 3.60  | 1.70 | 4.12 | 3.93  | 1.05 | 3.70 | 4.04  | 0.92 | 3.70    | 4.04  | 0.92 |
|              | -5          | 9.24    | 3.23 | 2.86 | 8.03         | 3.45   | 2.33 | 6.09 | 3.84  | 1.59 | 4.12 | 4.20  | 0.98 | 3.98 | 4.35  | 0.92 | 3.98    | 4.35  | 0.92 |
|              | -2          | 9.60    | 3.48 | 2.76 | 8.04         | 3.86   | 2.08 | 6.14 | 4.22  | 1.45 | 4.16 | 4.57  | 0.91 | 3.56 | 4.70  | 0.76 | 3.56    | 4.70  | 0.76 |
|              | 0           | 9.56    | 3.68 | 2.60 | 8.20         | 4.00   | 2.05 | 6.20 | 4.34  | 1.43 | 4.20 | 4.73  | 0.89 | 3.73 | 4.92  | 0.76 | 3.73    | 4.92  | 0.76 |
|              | 2           | 9.70    | 3.91 | 2.48 | 8.21         | 4.27   | 1.92 | 6.26 | 4.63  | 1.35 | 4.25 | 5.02  | 0.85 | 3.94 | 5.18  | 0.76 | 3.94    | 5.18  | 0.76 |
|              | 5           | 10.56   | 4.49 | 2.35 | 8.77         | 4.82   | 1.82 | 6.62 | 5.27  | 1.25 | 4.53 | 5.69  | 0.80 | 4.13 | 5.87  | 0.70 | 4.13    | 5.87  | 0.70 |
|              | 7           | 11.24   | 4.82 | 2.33 | 9.92         | 5.16   | 1.92 | 7.46 | 5.68  | 1.31 | 4.96 | 5.97  | 0.83 | 4.52 | 6.12  | 0.74 | 4.52    | 6.12  | 0.74 |
|              | 10          | 11.65   | 5.50 | 2.12 | 9.66         | 5.94   | 1.63 | 7.38 | 6.36  | 1.16 | 4.98 | 6.79  | 0.73 | 4.98 | 6.79  | 0.73 | 4.98    | 6.79  | 0.73 |
|              | 12          | 11.95   | 5.85 | 2.04 | 9.71         | 6.28   | 1.55 | 7.45 | 6.55  | 1.14 | 5.32 | 7.12  | 0.75 | 5.32 | 7.12  | 0.75 | 5.32    | 7.12  | 0.75 |
|              | 15          | 11.73   | 6.35 | 1.85 | 9.53         | 6.78   | 1.41 | 7.17 | 7.22  | 0.99 | 5.41 | 7.59  | 0.71 | 5.41 | 7.59  | 0.71 | 5.41    | 7.59  | 0.71 |
|              | 20          | 11.56   | 7.33 | 1.58 | 9.62         | 7.77   | 1.24 | 7.24 | 8.01  | 0.90 | 5.06 | 8.30  | 0.61 | 5.06 | 8.30  | 0.61 | 5.06    | 8.30  | 0.61 |
|              | 25          | 11.71   | 8.45 | 1.39 | 9.68         | 8.80   | 1.10 | 7.25 | 9.50  | 0.76 | 5.61 | 9.75  | 0.58 | 5.61 | 9.75  | 0.58 | 5.61    | 9.75  | 0.58 |
|              | 30          | 11.47   | 9.49 | 1.21 | 9.57         | 10.06  | 0.95 | 7.35 | 10.48 | 0.70 | 6.16 | 11.02 | 0.56 | 6.16 | 11.02 | 0.56 | 6.16    | 11.02 | 0.56 |
|              | 35          | /       | /    | /    | /            | /      | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 35           | -25         | 5.37    | 1.92 | 2.80 | 5.37         | 1.92   | 2.80 | 3.96 | 2.01  | 1.97 | 2.62 | 2.20  | 1.19 | 2.09 | 2.35  | 0.89 | 2.09    | 2.35  | 0.89 |
|              | -20         | 6.54    | 2.19 | 2.99 | 6.54         | 2.19   | 2.99 | 4.83 | 2.27  | 2.13 | 3.22 | 2.57  | 1.25 | 2.50 | 2.64  | 0.95 | 2.50    | 2.64  | 0.95 |
|              | -15         | 7.45    | 2.39 | 3.12 | 7.45         | 2.39   | 3.12 | 5.78 | 2.50  | 2.31 | 3.88 | 2.72  | 1.43 | 2.91 | 2.96  | 0.98 | 2.91    | 2.96  | 0.98 |
|              | -10         | 8.54    | 2.69 | 3.18 | 7.60         | 2.82   | 2.70 | 5.74 | 3.01  | 1.91 | 3.92 | 3.12  | 1.26 | 3.16 | 3.28  | 0.96 | 3.16    | 3.28  | 0.96 |
|              | -7          | 8.89    | 2.78 | 3.20 | 8.00         | 2.85   | 2.81 | 6.00 | 3.22  | 1.86 | 4.00 | 3.45  | 1.16 | 3.60 | 3.52  | 1.02 | 3.60    | 3.52  | 1.02 |
|              | -5          | 9.12    | 3.04 | 3.00 | 8.00         | 3.22   | 2.49 | 6.32 | 3.49  | 1.81 | 4.24 | 3.71  | 1.14 | 3.88 | 3.72  | 1.04 | 3.88    | 3.72  | 1.04 |
|              | -2          | 9.49    | 3.29 | 2.88 | 8.02         | 3.55   | 2.26 | 6.40 | 3.83  | 1.67 | 4.26 | 4.05  | 1.05 | 3.47 | 4.16  | 0.84 | 3.47    | 4.16  | 0.84 |
|              | 0           | 9.42    | 3.37 | 2.80 | 8.06         | 3.65   | 2.21 | 6.55 | 3.90  | 1.68 | 4.36 | 4.18  | 1.04 | 3.66 | 4.37  | 0.84 | 3.66    | 4.37  | 0.84 |
|              | 2           | 9.57    | 3.55 | 2.69 | 8.20         | 3.65   | 2.25 | 6.10 | 4.03  | 1.52 | 4.18 | 4.44  | 0.94 | 3.84 | 4.48  | 0.86 | 3.84    | 4.48  | 0.86 |
|              | 5           | 10.40   | 4.00 | 2.60 | 8.81         | 4.35   | 2.02 | 6.59 | 4.70  | 1.40 | 4.53 | 5.12  | 0.88 | 4.01 | 5.16  | 0.78 | 4.01    | 5.16  | 0.78 |
|              | 7*          | 11.05   | 4.40 | 2.51 | 10.00        | 4.70   | 2.13 | 7.50 | 5.12  | 1.47 | 4.97 | 5.52  | 0.90 | 4.40 | 5.64  | 0.78 | 4.40    | 5.64  | 0.78 |
|              | 10          | 11.44   | 4.93 | 2.32 | 9.63         | 5.32   | 1.81 | 7.37 | 5.71  | 1.29 | 5.02 | 6.40  | 0.78 | 4.86 | 6.43  | 0.76 | 4.86    | 6.43  | 0.76 |
|              | 12          | 11.73   | 5.43 | 2.16 | 9.70         | 5.81   | 1.67 | 7.43 | 6.26  | 1.19 | 5.19 | 6.61  | 0.79 | 5.19 | 6.61  | 0.79 | 5.19    | 6.61  | 0.79 |
|              | 15          | 11.53   | 5.66 | 2.04 | 9.82         | 6.03   | 1.63 | 7.16 | 6.44  | 1.11 | 5.31 | 6.90  | 0.77 | 5.31 | 6.90  | 0.77 | 5.31    | 6.90  | 0.77 |
|              | 20          | 11.35   | 6.40 | 1.78 | 9.60         | 6.67   | 1.44 | 7.25 | 7.11  | 1.02 | 4.92 | 7.57  | 0.65 | 4.92 | 7.57  | 0.65 | 4.92    | 7.57  | 0.65 |
|              | 25          | 11.48   | 7.46 | 1.54 | 9.66         | 7.84   | 1.23 | 7.28 | 8.19  | 0.89 | 5.46 | 8.64  | 0.63 | 5.46 | 8.64  | 0.63 | 5.46    | 8.64  | 0.63 |
|              | 30          | 11.25   | 8.83 | 1.27 | 9.66         | 9.25   | 1.04 | 7.21 | 9.69  | 0.74 | 6.00 | 10.06 | 0.60 | 6.00 | 10.06 | 0.60 | 6.00    | 10.06 | 0.60 |
|              | 35          | 10.81   | 9.95 | 1.09 | 9.67         | 10.92  | 0.89 | 7.41 | 10.95 | 0.68 | 6.60 | 11.13 | 0.59 | 6.60 | 11.13 | 0.59 | 6.60    | 11.13 | 0.59 |
| 40           | -25         | 5.29    | 1.84 | 3.23 | 5.29         | 1.84   | 3.23 | 3.88 | 1.72  | 2.25 | 2.56 | 2.08  | 1.23 | 2.14 | 2.19  | 0.98 | 2.14    | 2.19  | 0.98 |
|              | -20         | 6.46    | 2.10 | 3.08 | 6.46         | 2.10   | 3.08 | 4.89 | 2.21  | 2.21 | 3.19 | 2.40  | 1.33 | 2.52 | 2.47  | 1.02 | 2.52    | 2.47  | 1.02 |
|              | -15         | 7.35    | 2.25 | 3.28 | 7.35         | 2.25   | 3.28 | 5.65 | 2.33  | 2.42 | 3.77 | 2.72  | 1.39 | 2.98 | 2.86  | 1.04 | 2.98    | 2.86  | 1.04 |
|              | -10         | 8.43    | 2.51 | 3.35 | 7.86         | 2.62</ |      |      |       |      |      |       |      |      |       |      |         |       |      |

| Tw_out<br>°C | DBWB<br>°C | Maximum |      |      | 100%(Normal) |      |      | 75%  |      |      | 50%  |      |      | 25%  |      |      | Minimum |      |      |
|--------------|------------|---------|------|------|--------------|------|------|------|------|------|------|------|------|------|------|------|---------|------|------|
|              |            | HC      | COP  | PI   | HC           | COP  | PI   | HC   | COP  | PI   | HC   | COP  | PI   | HC   | COP  | PI   | HC      | COP  | PI   |
| 55           | -25        | 5.00    | 1.36 | 3.67 | 5.00         | 1.36 | 3.67 | 3.78 | 1.53 | 2.47 | 2.59 | 1.67 | 1.55 | 2.59 | 1.67 | 1.55 | 2.59    | 1.67 | 1.55 |
|              | -20        | 6.21    | 1.55 | 3.99 | 6.21         | 1.55 | 3.99 | 4.66 | 1.74 | 2.68 | 3.10 | 1.87 | 1.66 | 2.77 | 2.07 | 1.34 | 2.77    | 2.07 | 1.34 |
|              | -15        | 6.98    | 1.69 | 4.13 | 6.98         | 1.69 | 4.13 | 5.33 | 1.89 | 2.82 | 3.58 | 1.98 | 1.81 | 3.29 | 2.12 | 1.55 | 3.29    | 2.12 | 1.55 |
|              | -10        | 8.14    | 1.89 | 4.31 | 7.34         | 2.02 | 3.64 | 5.48 | 2.20 | 2.49 | 4.08 | 2.38 | 1.71 | 4.08 | 2.38 | 1.71 | 4.08    | 2.38 | 1.71 |
|              | -7         | 8.61    | 2.01 | 4.28 | 7.40         | 2.10 | 3.52 | 5.54 | 2.34 | 2.37 | 4.61 | 2.43 | 1.90 | 4.61 | 2.43 | 1.90 | 4.61    | 2.43 | 1.90 |
|              | -5         | 8.96    | 2.14 | 4.20 | 7.67         | 2.18 | 3.52 | 5.84 | 2.50 | 2.33 | 4.92 | 2.60 | 1.89 | 4.92 | 2.60 | 1.89 | 4.92    | 2.60 | 1.89 |
|              | -2         | 9.22    | 2.19 | 4.22 | 8.01         | 2.23 | 3.59 | 6.07 | 2.58 | 2.35 | 5.00 | 2.73 | 1.83 | 5.00 | 2.73 | 1.83 | 5.00    | 2.73 | 1.83 |
|              | 0          | 8.85    | 2.40 | 3.68 | 8.01         | 2.46 | 3.26 | 6.10 | 2.67 | 2.28 | 5.03 | 2.78 | 1.81 | 5.03 | 2.78 | 1.81 | 5.03    | 2.78 | 1.81 |
|              | 2          | 8.96    | 2.46 | 3.65 | 8.40         | 2.50 | 3.36 | 6.32 | 2.81 | 2.25 | 5.33 | 2.93 | 1.82 | 5.33 | 2.93 | 1.82 | 5.33    | 2.93 | 1.82 |
|              | 5          | 9.68    | 2.86 | 3.39 | 8.62         | 2.92 | 2.95 | 6.53 | 3.12 | 2.09 | 5.96 | 3.12 | 1.91 | 5.96 | 3.12 | 1.91 | 5.96    | 3.12 | 1.91 |
|              | 7          | 10.17   | 3.00 | 3.39 | 9.50         | 3.05 | 3.11 | 7.13 | 3.33 | 2.14 | 6.52 | 3.42 | 1.91 | 6.52 | 3.42 | 1.91 | 6.52    | 3.42 | 1.91 |
|              | 10         | 10.37   | 3.15 | 3.29 | 9.17         | 3.35 | 2.74 | 7.02 | 3.60 | 1.95 | 6.11 | 3.70 | 1.65 | 6.11 | 3.70 | 1.65 | 6.11    | 3.70 | 1.65 |
|              | 12         | 10.72   | 3.30 | 3.25 | 9.24         | 3.50 | 2.64 | 6.93 | 3.95 | 1.75 | 6.55 | 3.97 | 1.65 | 6.55 | 3.97 | 1.65 | 6.55    | 3.97 | 1.65 |
|              | 15         | 10.48   | 3.53 | 2.97 | 9.24         | 3.79 | 2.44 | 6.92 | 4.04 | 1.71 | 6.71 | 4.06 | 1.65 | 6.71 | 4.06 | 1.65 | 6.71    | 4.06 | 1.65 |
|              | 20         | 10.25   | 3.85 | 2.66 | 9.07         | 4.10 | 2.21 | 6.81 | 4.37 | 1.56 | 6.04 | 4.44 | 1.36 | 6.04 | 4.44 | 1.36 | 6.04    | 4.44 | 1.36 |
|              | 25         | 10.35   | 4.20 | 2.46 | 9.08         | 4.60 | 1.97 | 6.81 | 4.92 | 1.38 | 6.66 | 4.94 | 1.35 | 6.66 | 4.94 | 1.35 | 6.66    | 4.94 | 1.35 |
|              | 30         | 10.04   | 4.87 | 2.06 | 9.09         | 5.16 | 1.76 | 7.35 | 5.47 | 1.34 | 7.35 | 5.47 | 1.34 | 7.35 | 5.47 | 1.34 | 7.35    | 5.47 | 1.34 |
|              | 35         | 9.59    | 5.45 | 1.76 | 9.13         | 5.49 | 1.66 | 8.10 | 5.84 | 1.39 | 8.10 | 5.84 | 1.39 | 8.10 | 5.84 | 1.39 | 8.10    | 5.84 | 1.39 |
|              | 60         | -25     | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
| -20          |            | 6.12    | 1.50 | 4.08 | 5.97         | 1.57 | 3.80 | 4.57 | 1.77 | 2.58 | 3.04 | 1.97 | 1.54 | 2.92 | 1.99 | 1.47 | 2.92    | 1.99 | 1.47 |
| -15          |            | 6.94    | 1.61 | 4.31 | 6.94         | 1.61 | 4.31 | 5.33 | 1.81 | 2.95 | 3.57 | 2.02 | 1.77 | 3.45 | 2.04 | 1.69 | 3.45    | 2.04 | 1.69 |
| -10          |            | 8.12    | 1.89 | 4.30 | 7.10         | 1.97 | 3.61 | 5.42 | 2.20 | 2.46 | 4.27 | 2.46 | 1.73 | 4.27 | 2.46 | 1.73 | 4.27    | 2.46 | 1.73 |
| -7           |            | 8.58    | 1.95 | 4.40 | 7.12         | 2.03 | 3.51 | 5.31 | 2.35 | 2.26 | 4.80 | 2.50 | 1.92 | 4.80 | 2.50 | 1.92 | 4.80    | 2.50 | 1.92 |
| -5           |            | 8.90    | 2.04 | 4.35 | 7.46         | 2.16 | 3.46 | 5.68 | 2.38 | 2.39 | 5.14 | 2.57 | 2.00 | 5.14 | 2.57 | 2.00 | 5.14    | 2.57 | 2.00 |
| -2           |            | 9.18    | 2.06 | 4.44 | 7.46         | 2.22 | 3.36 | 5.68 | 2.42 | 2.35 | 4.83 | 2.60 | 1.86 | 4.83 | 2.60 | 1.86 | 4.83    | 2.60 | 1.86 |
| 0            |            | 8.73    | 2.27 | 3.85 | 8.01         | 2.32 | 3.45 | 6.10 | 2.53 | 2.42 | 4.82 | 2.73 | 1.77 | 4.82 | 2.73 | 1.77 | 4.82    | 2.73 | 1.77 |
| 2            |            | 8.82    | 2.33 | 3.78 | 8.35         | 2.36 | 3.54 | 6.37 | 2.60 | 2.45 | 5.10 | 2.87 | 1.78 | 5.10 | 2.87 | 1.78 | 5.10    | 2.87 | 1.78 |
| 5            |            | 9.46    | 2.56 | 3.69 | 8.65         | 2.65 | 3.27 | 6.52 | 2.90 | 2.24 | 5.99 | 3.05 | 1.96 | 5.99 | 3.05 | 1.96 | 5.99    | 3.05 | 1.96 |
| 7            |            | 9.90    | 2.75 | 3.60 | 9.27         | 2.82 | 3.28 | 6.99 | 3.13 | 2.23 | 6.24 | 3.29 | 1.89 | 6.24 | 3.29 | 1.89 | 6.24    | 3.29 | 1.89 |
| 10           |            | 10.18   | 2.94 | 3.46 | 9.28         | 3.01 | 3.08 | 7.01 | 3.45 | 2.03 | 6.10 | 3.61 | 1.69 | 6.10 | 3.61 | 1.69 | 6.10    | 3.61 | 1.69 |
| 12           |            | 10.41   | 3.10 | 3.36 | 9.23         | 3.29 | 2.81 | 6.90 | 3.78 | 1.83 | 6.51 | 3.82 | 1.70 | 6.51 | 3.82 | 1.70 | 6.51    | 3.82 | 1.70 |
| 15           |            | 10.17   | 3.21 | 3.16 | 9.22         | 3.40 | 2.71 | 6.68 | 3.98 | 1.68 | 6.68 | 3.98 | 1.68 | 6.68 | 3.98 | 1.68 | 6.68    | 3.98 | 1.68 |
| 20           |            | 9.93    | 3.45 | 2.87 | 9.31         | 3.67 | 2.54 | 6.85 | 4.18 | 1.64 | 5.99 | 4.36 | 1.37 | 5.99 | 4.36 | 1.37 | 5.99    | 4.36 | 1.37 |
| 25           |            | 9.99    | 3.76 | 2.65 | 9.10         | 4.04 | 2.25 | 6.89 | 4.58 | 1.50 | 6.35 | 4.63 | 1.37 | 6.35 | 4.63 | 1.37 | 6.35    | 4.63 | 1.37 |
| 30           |            | 9.64    | 4.36 | 2.21 | 9.26         | 4.49 | 2.06 | 7.01 | 4.99 | 1.41 | 7.01 | 4.99 | 1.41 | 7.01 | 4.99 | 1.41 | 7.01    | 4.99 | 1.41 |
| 35           |            | /       | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
| 65           |            | -25     | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
|              | -20        | 6.10    | 1.47 | 4.16 | 6.10         | 1.47 | 4.16 | 4.62 | 1.50 | 3.08 | 3.18 | 1.60 | 1.99 | 2.82 | 1.62 | 1.74 | 2.82    | 1.62 | 1.74 |
|              | -15        | 6.81    | 1.57 | 4.34 | 6.81         | 1.57 | 4.34 | 5.10 | 1.63 | 3.13 | 3.45 | 1.87 | 1.84 | 3.45 | 1.87 | 1.84 | 3.45    | 1.87 | 1.84 |
|              | -10        | 8.01    | 1.78 | 4.49 | 7.40         | 1.84 | 4.03 | 5.60 | 1.87 | 2.99 | 4.10 | 2.12 | 1.93 | 4.10 | 2.12 | 1.93 | 4.10    | 2.12 | 1.93 |
|              | -7         | 8.56    | 1.87 | 4.58 | 7.39         | 1.99 | 3.71 | 5.62 | 2.09 | 2.69 | 4.63 | 2.17 | 2.13 | 4.63 | 2.17 | 2.13 | 4.63    | 2.17 | 2.13 |
|              | -5         | 8.82    | 1.94 | 4.55 | 7.60         | 2.05 | 3.72 | 5.78 | 2.12 | 2.73 | 4.98 | 2.29 | 2.18 | 4.98 | 2.29 | 2.18 | 4.98    | 2.29 | 2.18 |
|              | -2         | 9.14    | 2.01 | 4.55 | 7.87         | 2.10 | 3.75 | 6.00 | 2.16 | 2.77 | 4.64 | 2.41 | 1.93 | 4.64 | 2.41 | 1.93 | 4.64    | 2.41 | 1.93 |
|              | 0          | 8.58    | 2.09 | 4.10 | 8.15         | 2.18 | 3.74 | 6.21 | 2.20 | 2.82 | 4.58 | 2.51 | 1.82 | 4.58 | 2.51 | 1.82 | 4.58    | 2.51 | 1.82 |
|              | 2          | 8.64    | 2.19 | 3.94 | 8.28         | 2.24 | 3.69 | 6.32 | 2.30 | 2.74 | 4.86 | 2.48 | 1.96 | 4.86 | 2.48 | 1.96 | 4.86    | 2.48 | 1.96 |
|              | 5          | 9.21    | 2.37 | 3.89 | 8.66         | 2.41 | 3.60 | 6.53 | 2.53 | 2.58 | 5.44 | 2.71 | 2.01 | 5.44 | 2.71 | 2.01 | 5.44    | 2.71 | 2.01 |
|              | 7          | 9.61    | 2.50 | 3.85 | 9.26         | 2.55 | 3.64 | 6.98 | 2.69 | 2.59 | 5.95 | 2.89 | 2.06 | 5.95 | 2.89 | 2.06 | 5.95    | 2.89 | 2.06 |
|              | 10         | 9.84    | 2.72 | 3.62 | 9.20         | 2.77 | 3.32 | 6.98 | 2.91 | 2.40 | 5.75 | 3.14 | 1.83 | 5.75 | 3.14 | 1.83 | 5.75    | 3.14 | 1.83 |
|              | 12         | 10.05   | 2.88 | 3.49 | 9.28         | 2.94 | 3.16 | 7.00 | 3.08 | 2.27 | 6.16 | 3.20 | 1.92 | 6.16 | 3.20 | 1.92 | 6.16    | 3.20 | 1.92 |
|              | 15         | 9.81    | 2.97 | 3.31 | 9.30         | 3.03 | 3.07 | 6.84 | 3.27 | 2.09 | 6.31 | 3.30 | 1.91 | 6.31 | 3.30 | 1.91 | 6.31    | 3.30 | 1.91 |
|              | 20         | 9.56    | 3.21 | 2.98 | 9.30         | 3.26 | 2.86 | 6.91 | 3.45 | 2.00 | 5.39 | 3.67 | 1.47 | 5.39 | 3.67 | 1.47 | 5.39    | 3.67 | 1.47 |
|              | 25         | 9.60    | 3.50 | 2.74 | 9.22         | 3.54 | 2.61 | 7.00 | 3.74 | 1.87 | 5.99 | 3.96 | 1.51 | 5.99 | 3.96 | 1.51 | 5.99    | 3.96 | 1.51 |
|              | 30         | 9.18    | 3.90 | 2.36 | 9.18         | 3.90 | 2.36 | 6.61 | 4.01 | 1.65 | 6.61 | 4.01 | 1.65 | 6.61 | 4.01 | 1.65 | 6.61    | 4.01 | 1.65 |
|              | 35         | /       | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
|              | 70         | -25     | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
| -20          |            | /       | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
| -15          |            | 6.86    | 1.45 | 4.72 | 6.86         | 1.45 | 4.72 | 5.32 | 1.61 | 3.30 | 3.19 | 1.93 | 1.66 | 3.19 | 1.93 | 1.66 | 3.19    | 1.93 | 1.66 |
| -10          |            | 7.86    | 1.59 | 4.94 | 7.11         | 1.75 | 4.06 | 5.43 | 1.91 | 2.84 | 3.99 | 2.12 | 1.88 | 3.99 | 2.12 | 1.88 | 3.99    | 2.12 | 1.88 |
| -7           |            | 8.21    | 1.69 | 4.85 | 7.01         | 1.87 | 3.76 | 5.36 | 1.99 | 2.69 | 4.55 | 2.13 | 2.13 | 4.55 | 2.13 | 2.13 | 4.55    | 2.13 | 2.13 |
| -5           |            | 8.43    | 1.78 | 4.73 | 7.20         | 1.96 | 3.67 | 5.46 | 2.10 | 2.59 | 4.87 | 2.25 | 2.17 | 4.87 | 2.25 | 2.17 | 4.87    | 2.25 | 2.17 |
| -2           |            | 8.78    | 1.82 | 4.83 | 7.39         | 1.99 | 3.71 | 5.61 | 2.13 | 2.63 | 4.46 | 2.32 | 1.92 | 4.46 | 2.32 | 1.92 | 4.46    | 2.32 | 1.92 |
| 0            |            | 8.37    | 1.91 | 4.38 | 7.58         |      |      |      |      |      |      |      |      |      |      |      |         |      |      |

## 8.5 Prestazioni riscaldamento Mod. 12

| Tw_out<br>°C | DB/WB<br>°C | Maximum |      |      | 100%(Normal) |       |      | 75%  |       |      | 50%  |       |      | 25%  |       |      | Minimum |       |      |
|--------------|-------------|---------|------|------|--------------|-------|------|------|-------|------|------|-------|------|------|-------|------|---------|-------|------|
|              |             | HC      | COP  | PI   | HC           | COP   | PI   | HC   | COP   | PI   | HC   | COP   | PI   | HC   | COP   | PI   | HC      | COP   | PI   |
| 25           | -25         | 7.62    | 2.27 | 3.36 | 7.62         | 2.27  | 3.36 | 7.62 | 2.27  | 3.36 | 5.06 | 2.42  | 2.09 | 2.98 | 2.56  | 1.17 | 2.98    | 2.56  | 1.17 |
|              | -20         | 9.15    | 2.60 | 3.52 | 9.15         | 2.60  | 3.52 | 7.54 | 2.74  | 2.75 | 5.00 | 2.95  | 1.70 | 3.77 | 3.11  | 1.21 | 3.77    | 3.11  | 1.21 |
|              | -15         | 10.59   | 2.90 | 3.66 | 10.06        | 2.99  | 3.37 | 7.52 | 3.28  | 2.29 | 5.02 | 3.44  | 1.46 | 4.37 | 3.77  | 1.16 | 4.37    | 3.77  | 1.16 |
|              | -10         | 11.28   | 3.03 | 3.72 | 9.93         | 3.19  | 3.11 | 7.57 | 3.54  | 2.14 | 5.18 | 3.96  | 1.31 | 4.69 | 3.99  | 1.18 | 4.69    | 3.99  | 1.18 |
|              | -7          | 11.50   | 3.26 | 3.53 | 9.92         | 3.61  | 2.75 | 7.57 | 4.11  | 1.84 | 5.41 | 4.44  | 1.22 | 5.41 | 4.44  | 1.22 | 5.41    | 4.44  | 1.22 |
|              | -5          | 11.55   | 3.43 | 3.36 | 9.62         | 3.75  | 2.57 | 7.28 | 4.11  | 1.77 | 5.60 | 4.55  | 1.23 | 5.60 | 4.55  | 1.23 | 5.60    | 4.55  | 1.23 |
|              | -2          | 12.14   | 3.73 | 3.25 | 9.52         | 4.22  | 2.26 | 7.16 | 4.59  | 1.56 | 4.86 | 5.22  | 0.93 | 4.86 | 5.22  | 0.93 | 4.86    | 5.22  | 0.93 |
|              | 0           | 12.40   | 3.96 | 3.13 | 9.36         | 4.63  | 2.02 | 7.00 | 5.01  | 1.40 | 5.02 | 5.42  | 0.93 | 5.02 | 5.42  | 0.93 | 5.02    | 5.42  | 0.93 |
|              | 2           | 12.73   | 4.20 | 3.03 | 9.12         | 5.00  | 1.83 | 6.90 | 5.35  | 1.29 | 5.38 | 5.80  | 0.93 | 5.38 | 5.80  | 0.93 | 5.38    | 5.80  | 0.93 |
|              | 5           | 14.25   | 4.89 | 2.91 | 10.79        | 5.64  | 1.91 | 8.19 | 6.44  | 1.27 | 5.85 | 7.00  | 0.84 | 5.85 | 7.00  | 0.84 | 5.85    | 7.00  | 0.84 |
|              | 7           | 15.69   | 5.47 | 2.87 | 12.09        | 6.29  | 1.92 | 8.94 | 6.86  | 1.30 | 6.24 | 7.44  | 0.84 | 6.24 | 7.44  | 0.84 | 6.24    | 7.44  | 0.84 |
|              | 10          | 17.24   | 6.38 | 2.70 | 12.08        | 7.65  | 1.58 | 9.13 | 8.47  | 1.08 | 7.25 | 8.95  | 0.81 | 7.25 | 8.95  | 0.81 | 7.25    | 8.95  | 0.81 |
|              | 12          | 17.67   | 7.04 | 2.51 | 11.95        | 8.48  | 1.41 | 9.16 | 9.37  | 0.98 | 7.71 | 9.79  | 0.79 | 7.71 | 9.79  | 0.79 | 7.71    | 9.79  | 0.79 |
|              | 15          | 16.62   | 7.45 | 2.23 | 12.08        | 8.76  | 1.38 | 9.03 | 9.77  | 0.92 | 7.78 | 10.26 | 0.76 | 7.78 | 10.26 | 0.76 | 7.78    | 10.26 | 0.76 |
|              | 20          | 14.83   | 8.21 | 1.81 | 12.03        | 9.39  | 1.28 | 9.09 | 10.71 | 0.85 | 6.62 | 11.24 | 0.59 | 6.62 | 11.24 | 0.59 | 6.62    | 11.24 | 0.59 |
|              | 25          | 14.66   | 9.04 | 1.62 | 11.92        | 10.23 | 1.17 | 9.14 | 11.50 | 0.79 | 7.40 | 11.79 | 0.63 | 7.40 | 11.79 | 0.63 | 7.40    | 11.79 | 0.63 |
|              | 30          | /       | /    | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 35          | /       | /    | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 30           | -25         | 7.20    | 2.03 | 3.55 | 7.20         | 2.03  | 3.55 | 7.20 | 2.03  | 3.55 | 4.88 | 2.12  | 2.31 | 2.78 | 2.25  | 1.24 | 2.78    | 2.25  | 1.24 |
|              | -20         | 8.71    | 2.34 | 3.72 | 8.71         | 2.34  | 3.72 | 7.53 | 2.44  | 3.09 | 4.99 | 2.55  | 1.96 | 3.56 | 2.76  | 1.29 | 3.56    | 2.76  | 1.29 |
|              | -15         | 10.38   | 2.68 | 3.87 | 9.87         | 2.74  | 3.60 | 7.39 | 2.92  | 2.53 | 5.05 | 3.15  | 1.61 | 4.19 | 3.36  | 1.25 | 4.19    | 3.36  | 1.25 |
|              | -10         | 10.65   | 2.72 | 3.91 | 9.86         | 2.82  | 3.50 | 7.41 | 3.10  | 2.39 | 4.98 | 3.44  | 1.45 | 4.58 | 3.58  | 1.28 | 4.58    | 3.58  | 1.28 |
|              | -7          | 11.30   | 3.01 | 3.75 | 9.92         | 3.17  | 3.13 | 7.62 | 3.59  | 2.13 | 5.39 | 4.03  | 1.34 | 5.39 | 4.03  | 1.34 | 5.39    | 4.03  | 1.34 |
|              | -5          | 11.11   | 3.10 | 3.59 | 9.54         | 3.29  | 2.90 | 7.17 | 3.67  | 1.95 | 5.38 | 4.10  | 1.31 | 5.38 | 4.10  | 1.31 | 5.38    | 4.10  | 1.31 |
|              | -2          | 11.67   | 3.35 | 3.48 | 9.37         | 3.70  | 2.53 | 7.16 | 3.98  | 1.80 | 4.84 | 4.30  | 1.13 | 4.66 | 4.56  | 1.02 | 4.66    | 4.56  | 1.02 |
|              | 0           | 11.93   | 3.56 | 3.35 | 9.23         | 4.05  | 2.28 | 6.94 | 4.31  | 1.61 | 4.74 | 4.63  | 1.02 | 4.74 | 4.63  | 1.02 | 4.74    | 4.63  | 1.02 |
|              | 2           | 12.25   | 3.76 | 3.26 | 9.00         | 4.35  | 2.07 | 6.93 | 4.55  | 1.52 | 5.16 | 5.02  | 1.03 | 5.16 | 5.02  | 1.03 | 5.16    | 5.02  | 1.03 |
|              | 5           | 13.72   | 4.36 | 3.15 | 10.84        | 5.13  | 2.11 | 8.16 | 5.45  | 1.50 | 5.54 | 5.92  | 0.93 | 5.54 | 5.92  | 0.93 | 5.54    | 5.92  | 0.93 |
|              | 7           | 15.02   | 4.86 | 3.09 | 12.02        | 5.43  | 2.21 | 9.10 | 5.80  | 1.57 | 5.98 | 6.33  | 0.94 | 5.98 | 6.33  | 0.94 | 5.98    | 6.33  | 0.94 |
|              | 10          | 16.43   | 5.61 | 2.93 | 11.91        | 6.48  | 1.84 | 9.12 | 7.03  | 1.30 | 6.91 | 7.52  | 0.92 | 6.91 | 7.52  | 0.92 | 6.91    | 7.52  | 0.92 |
|              | 12          | 16.81   | 6.12 | 2.75 | 11.84        | 7.12  | 1.66 | 8.96 | 7.79  | 1.15 | 7.32 | 8.10  | 0.90 | 7.32 | 8.10  | 0.90 | 7.32    | 8.10  | 0.90 |
|              | 15          | 15.87   | 6.43 | 2.47 | 11.82        | 7.45  | 1.59 | 9.07 | 8.44  | 1.07 | 7.59 | 8.73  | 0.87 | 7.59 | 8.73  | 0.87 | 7.59    | 8.73  | 0.87 |
|              | 20          | 14.25   | 7.18 | 1.98 | 12.00        | 8.01  | 1.50 | 8.96 | 9.19  | 0.97 | 6.37 | 9.36  | 0.68 | 6.37 | 9.36  | 0.68 | 6.37    | 9.36  | 0.68 |
|              | 25          | 14.13   | 7.83 | 1.80 | 11.96        | 8.50  | 1.41 | 9.04 | 9.58  | 0.94 | 7.11 | 9.62  | 0.74 | 7.11 | 9.62  | 0.74 | 7.11    | 9.62  | 0.74 |
|              | 30          | 14.12   | 9.33 | 1.51 | 11.93        | 9.98  | 1.20 | 8.97 | 11.20 | 0.80 | 7.78 | 11.27 | 0.69 | 7.78 | 11.27 | 0.69 | 7.78    | 11.27 | 0.69 |
|              | 35          | /       | /    | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 35           | -25         | 7.00    | 1.88 | 3.73 | 7.00         | 1.88  | 3.73 | 7.00 | 1.88  | 3.73 | 4.85 | 1.92  | 2.52 | 2.68 | 2.02  | 1.33 | 2.68    | 2.02  | 1.33 |
|              | -20         | 8.53    | 2.18 | 3.92 | 8.53         | 2.18  | 3.92 | 7.53 | 2.25  | 3.35 | 5.07 | 2.32  | 2.18 | 3.47 | 2.50  | 1.39 | 3.47    | 2.50  | 1.39 |
|              | -15         | 10.21   | 2.50 | 4.08 | 10.07        | 2.51  | 4.01 | 7.55 | 2.66  | 2.83 | 5.13 | 2.79  | 1.84 | 4.16 | 2.97  | 1.40 | 4.16    | 2.97  | 1.40 |
|              | -10         | 10.44   | 2.52 | 4.14 | 9.84         | 2.59  | 3.80 | 7.43 | 2.81  | 2.64 | 5.08 | 3.20  | 1.58 | 4.50 | 3.28  | 1.37 | 4.50    | 3.28  | 1.37 |
|              | -7          | 11.10   | 2.79 | 3.98 | 10.00        | 2.80  | 3.57 | 7.56 | 3.19  | 2.37 | 5.25 | 3.58  | 1.47 | 5.25 | 3.58  | 1.47 | 5.25    | 3.58  | 1.47 |
|              | -5          | 10.91   | 2.86 | 3.81 | 9.66         | 2.98  | 3.24 | 7.25 | 3.25  | 2.23 | 5.08 | 3.62  | 1.40 | 5.08 | 3.62  | 1.40 | 5.08    | 3.62  | 1.40 |
|              | -2          | 11.46   | 3.09 | 3.71 | 9.42         | 3.34  | 2.82 | 7.24 | 3.58  | 2.03 | 4.85 | 4.14  | 1.17 | 4.57 | 4.12  | 1.11 | 4.57    | 4.12  | 1.11 |
|              | 0           | 11.72   | 3.28 | 3.58 | 9.28         | 3.61  | 2.57 | 7.05 | 3.85  | 1.83 | 4.73 | 4.33  | 1.09 | 4.73 | 4.33  | 1.09 | 4.73    | 4.33  | 1.09 |
|              | 2           | 12.04   | 3.45 | 3.49 | 9.10         | 3.80  | 2.39 | 7.05 | 4.04  | 1.75 | 5.07 | 4.50  | 1.13 | 5.07 | 4.50  | 1.13 | 5.07    | 4.50  | 1.13 |
|              | 5           | 13.47   | 3.99 | 3.38 | 10.77        | 4.59  | 2.34 | 8.13 | 4.83  | 1.68 | 5.36 | 5.18  | 1.03 | 5.36 | 5.18  | 1.03 | 5.36    | 5.18  | 1.03 |
|              | 7*          | 14.72   | 4.42 | 3.33 | 12.00        | 4.80  | 2.50 | 9.06 | 5.11  | 1.77 | 5.87 | 5.61  | 1.05 | 5.87 | 5.61  | 1.05 | 5.87    | 5.61  | 1.05 |
|              | 10          | 15.57   | 4.92 | 3.17 | 12.00        | 5.45  | 2.20 | 9.13 | 5.80  | 1.57 | 6.42 | 6.22  | 1.03 | 6.42 | 6.22  | 1.03 | 6.42    | 6.22  | 1.03 |
|              | 12          | 15.92   | 5.33 | 2.99 | 12.12        | 5.91  | 2.05 | 9.04 | 6.35  | 1.42 | 6.82 | 6.71  | 1.02 | 6.82 | 6.71  | 1.02 | 6.82    | 6.71  | 1.02 |
|              | 15          | 15.03   | 5.61 | 2.68 | 12.06        | 6.34  | 1.90 | 8.99 | 6.90  | 1.30 | 7.09 | 7.20  | 0.99 | 7.09 | 7.20  | 0.99 | 7.09    | 7.20  | 0.99 |
|              | 20          | 13.54   | 6.47 | 2.09 | 11.96        | 6.95  | 1.72 | 9.12 | 7.72  | 1.18 | 5.97 | 7.99  | 0.75 | 5.97 | 7.99  | 0.75 | 5.97    | 7.99  | 0.75 |
|              | 25          | 13.43   | 7.08 | 1.90 | 11.84        | 7.50  | 1.58 | 9.06 | 8.23  | 1.10 | 6.68 | 8.47  | 0.79 | 6.68 | 8.47  | 0.79 | 6.68    | 8.47  | 0.79 |
|              | 30          | 13.45   | 8.04 | 1.67 | 12.13        | 8.44  | 1.44 | 9.02 | 8.93  | 1.01 | 7.30 | 9.00  | 0.81 | 7.30 | 9.00  | 0.81 | 7.30    | 9.00  | 0.81 |
|              | 35          | 13.22   | 9.58 | 1.38 | 12.02        | 9.89  | 1.22 | 9.23 | 10.37 | 0.89 | 7.97 | 10.45 | 0.76 | 7.97 | 10.45 | 0.76 | 7.97    | 10.45 | 0.76 |
| 40           | -25         | 6.82    | 1.75 | 3.90 | 6.82         | 1.75  | 3.90 | 6.82 | 1.75  | 3.90 | 5.13 | 1.80  | 2.84 | 2.74 | 1.89  | 1.45 | 2.74    | 1.89  | 1.45 |
|              | -20         | 8.34    | 2.03 | 4.11 | 8.34         | 2.03  | 4.11 | 7.54 | 2.08  | 3.62 | 5.08 | 2.18  | 2.33 | 3.57 | 2.31  | 1.55 | 3.57    | 2.31  | 1.55 |
|              | -15         | 10.03   | 2.34 | 4.29 | 10.03        | 2.34  | 4.29 | 7.59 | 2.45  | 3.10 | 5.20 | 2.50  | 2.08 | 3.99 | 2.59  | 1.54 | 3.99    | 2.59  | 1.54 |

| Tw_out<br>°C | DBWB<br>°C | Maximum |      |      | 100%(Normal) |      |      | 75%   |      |      | 50%   |      |      | 25%   |      |      | Minimum |      |      |
|--------------|------------|---------|------|------|--------------|------|------|-------|------|------|-------|------|------|-------|------|------|---------|------|------|
|              |            | HC      | COP  | PI   | HC           | COP  | PI   | HC    | COP  | PI   | HC    | COP  | PI   | HC    | COP  | PI   | HC      | COP  | PI   |
| 55           | -25        | 6.35    | 1.43 | 4.44 | 6.35         | 1.43 | 4.44 | 6.35  | 1.43 | 4.44 | 5.14  | 1.49 | 3.46 | 3.71  | 1.53 | 2.43 | 3.71    | 1.53 | 2.43 |
|              | -20        | 7.93    | 1.69 | 4.70 | 7.93         | 1.69 | 4.70 | 7.81  | 1.69 | 4.63 | 5.11  | 1.74 | 2.93 | 4.73  | 1.85 | 2.55 | 4.73    | 1.85 | 2.55 |
|              | -15        | 9.62    | 1.94 | 4.96 | 9.62         | 1.94 | 4.96 | 7.92  | 1.99 | 3.99 | 5.39  | 2.06 | 2.62 | 5.39  | 2.06 | 2.62 | 5.39    | 2.06 | 2.62 |
|              | -10        | 10.06   | 1.98 | 5.09 | 10.06        | 1.98 | 5.09 | 7.95  | 2.05 | 3.88 | 6.01  | 2.28 | 2.64 | 6.01  | 2.28 | 2.64 | 6.01    | 2.28 | 2.64 |
|              | -7         | 10.40   | 2.15 | 4.84 | 10.40        | 2.15 | 4.84 | 8.01  | 2.22 | 3.61 | 6.53  | 2.32 | 2.82 | 6.53  | 2.32 | 2.82 | 6.53    | 2.32 | 2.82 |
|              | -5         | 10.24   | 2.16 | 4.75 | 10.24        | 2.16 | 4.75 | 7.99  | 2.25 | 3.55 | 6.80  | 2.40 | 2.83 | 6.80  | 2.40 | 2.83 | 6.80    | 2.40 | 2.83 |
|              | -2         | 10.69   | 2.29 | 4.67 | 10.69        | 2.29 | 4.67 | 8.36  | 2.39 | 3.50 | 6.91  | 2.58 | 2.68 | 6.91  | 2.58 | 2.68 | 6.91    | 2.58 | 2.68 |
|              | 0          | 10.91   | 2.40 | 4.54 | 10.91        | 2.40 | 4.54 | 8.59  | 2.51 | 3.42 | 7.28  | 2.69 | 2.71 | 7.28  | 2.69 | 2.71 | 7.28    | 2.69 | 2.71 |
|              | 2          | 11.30   | 2.55 | 4.43 | 11.30        | 2.55 | 4.43 | 8.51  | 2.67 | 3.19 | 7.64  | 2.80 | 2.73 | 7.64  | 2.80 | 2.73 | 7.64    | 2.80 | 2.73 |
|              | 5          | 12.49   | 2.84 | 4.40 | 11.86        | 2.99 | 3.96 | 9.12  | 3.21 | 2.84 | 9.12  | 3.21 | 2.84 | 9.12  | 3.21 | 2.84 | 9.12    | 3.21 | 2.84 |
|              | 7          | 13.61   | 3.11 | 4.37 | 12.00        | 3.10 | 3.87 | 9.75  | 3.39 | 2.88 | 9.75  | 3.39 | 2.88 | 9.75  | 3.39 | 2.88 | 9.75    | 3.39 | 2.88 |
|              | 10         | 14.25   | 3.37 | 4.23 | 12.08        | 3.52 | 3.43 | 9.05  | 3.70 | 2.44 | 9.05  | 3.70 | 2.44 | 9.05  | 3.70 | 2.44 | 9.05    | 3.70 | 2.44 |
|              | 12         | 14.54   | 3.59 | 4.05 | 11.96        | 3.75 | 3.19 | 9.59  | 3.90 | 2.46 | 9.59  | 3.90 | 2.46 | 9.59  | 3.90 | 2.46 | 9.59    | 3.90 | 2.46 |
|              | 15         | 13.74   | 3.71 | 3.71 | 11.92        | 3.90 | 3.06 | 10.36 | 4.13 | 2.51 | 10.06 | 4.14 | 2.43 | 10.06 | 4.14 | 2.43 | 10.06   | 4.14 | 2.43 |
|              | 20         | 12.80   | 4.05 | 3.16 | 11.87        | 4.20 | 2.83 | 9.13  | 4.43 | 2.06 | 7.98  | 4.46 | 1.79 | 7.98  | 4.46 | 1.79 | 7.98    | 4.46 | 1.79 |
|              | 25         | 12.67   | 4.39 | 2.88 | 11.99        | 4.50 | 2.66 | 9.15  | 4.74 | 1.93 | 9.12  | 4.77 | 1.91 | 9.12  | 4.77 | 1.91 | 9.12    | 4.77 | 1.91 |
|              | 30         | 12.64   | 5.02 | 2.52 | 11.88        | 5.13 | 2.32 | 9.92  | 5.19 | 1.91 | 9.92  | 5.19 | 1.91 | 9.92  | 5.19 | 1.91 | 9.92    | 5.19 | 1.91 |
|              | 35         | 12.42   | 5.57 | 2.23 | 11.85        | 5.67 | 2.09 | 10.78 | 5.69 | 1.89 | 10.78 | 5.69 | 1.89 | 10.78 | 5.69 | 1.89 | 10.78   | 5.69 | 1.89 |
| 60           | -25        | /       | /    | /    | /            | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /    | /       | /    | /    |
|              | -20        | 7.69    | 1.56 | 4.93 | 7.69         | 1.56 | 4.93 | 7.69  | 1.56 | 4.93 | 5.14  | 1.66 | 3.10 | 4.92  | 1.68 | 2.92 | 4.92    | 1.68 | 2.92 |
|              | -15        | 9.38    | 1.75 | 5.35 | 9.38         | 1.75 | 5.35 | 7.90  | 1.83 | 4.32 | 5.70  | 1.88 | 3.03 | 5.70  | 1.88 | 3.03 | 5.70    | 1.88 | 3.03 |
|              | -10        | 10.01   | 1.87 | 5.36 | 10.01        | 1.87 | 5.36 | 8.04  | 1.94 | 4.14 | 6.39  | 2.12 | 3.01 | 6.39  | 2.12 | 3.01 | 6.39    | 2.12 | 3.01 |
|              | -7         | 10.35   | 1.99 | 5.20 | 10.35        | 1.99 | 5.20 | 7.78  | 2.09 | 3.72 | 7.54  | 2.17 | 3.47 | 7.54  | 2.17 | 3.47 | 7.54    | 2.17 | 3.47 |
|              | -5         | 10.15   | 2.02 | 5.03 | 10.15        | 2.02 | 5.03 | 7.96  | 2.12 | 3.75 | 7.08  | 2.25 | 3.15 | 7.08  | 2.25 | 3.15 | 7.08    | 2.25 | 3.15 |
|              | -2         | 10.44   | 2.11 | 4.96 | 10.44        | 2.11 | 4.96 | 8.18  | 2.15 | 3.80 | 7.12  | 2.31 | 3.08 | 7.12  | 2.31 | 3.08 | 7.12    | 2.31 | 3.08 |
|              | 0          | 10.59   | 2.19 | 4.84 | 10.59        | 2.19 | 4.84 | 8.41  | 2.27 | 3.71 | 7.52  | 2.41 | 3.12 | 7.52  | 2.41 | 3.12 | 7.52    | 2.41 | 3.12 |
|              | 2          | 10.87   | 2.28 | 4.77 | 10.87        | 2.28 | 4.77 | 8.46  | 2.39 | 3.54 | 7.89  | 2.51 | 3.15 | 7.89  | 2.51 | 3.15 | 7.89    | 2.51 | 3.15 |
|              | 5          | 12.75   | 2.65 | 4.81 | 11.76        | 2.68 | 4.38 | 8.91  | 2.87 | 3.10 | 8.91  | 2.87 | 3.10 | 8.91  | 2.87 | 3.10 | 8.91    | 2.87 | 3.10 |
|              | 7          | 13.13   | 2.84 | 4.62 | 12.06        | 2.88 | 4.18 | 9.55  | 3.04 | 3.14 | 9.55  | 3.04 | 3.14 | 9.55  | 3.04 | 3.14 | 9.55    | 3.04 | 3.14 |
|              | 10         | 13.75   | 3.03 | 4.54 | 12.01        | 3.12 | 3.85 | 9.12  | 3.29 | 2.77 | 9.12  | 3.29 | 2.77 | 9.12  | 3.29 | 2.77 | 9.12    | 3.29 | 2.77 |
|              | 12         | 14.03   | 3.22 | 4.36 | 12.18        | 3.32 | 3.67 | 9.66  | 3.45 | 2.80 | 9.66  | 3.45 | 2.80 | 9.66  | 3.45 | 2.80 | 9.66    | 3.45 | 2.80 |
|              | 15         | 13.28   | 3.33 | 3.99 | 12.03        | 3.46 | 3.48 | 10.19 | 3.68 | 2.77 | 10.19 | 3.68 | 2.77 | 10.19 | 3.68 | 2.77 | 10.19   | 3.68 | 2.77 |
|              | 20         | 12.49   | 3.64 | 3.43 | 11.98        | 3.71 | 3.23 | 9.09  | 3.88 | 2.34 | 8.16  | 3.96 | 2.06 | 8.16  | 3.96 | 2.06 | 8.16    | 3.96 | 2.06 |
|              | 25         | 12.36   | 3.95 | 3.13 | 11.91        | 4.00 | 2.98 | 9.40  | 4.25 | 2.21 | 9.40  | 4.25 | 2.21 | 9.40  | 4.25 | 2.21 | 9.40    | 4.25 | 2.21 |
|              | 30         | 12.30   | 4.45 | 2.77 | 11.82        | 4.54 | 2.60 | 10.20 | 4.59 | 2.22 | 10.20 | 4.59 | 2.22 | 10.20 | 4.59 | 2.22 | 10.20   | 4.59 | 2.22 |
|              | 35         | /       | /    | /    | /            | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /    | /       | /    | /    |
| 65           | -25        | /       | /    | /    | /            | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /    | /       | /    | /    |
|              | -20        | 7.42    | 1.43 | 5.19 | 7.42         | 1.43 | 5.19 | 7.42  | 1.43 | 5.19 | 5.15  | 1.45 | 3.54 | 4.66  | 1.51 | 3.08 | 4.66    | 1.51 | 3.08 |
|              | -15        | 8.99    | 1.60 | 5.61 | 8.99         | 1.60 | 5.61 | 7.85  | 1.68 | 4.68 | 5.64  | 1.79 | 3.15 | 5.64  | 1.79 | 3.15 | 5.64    | 1.79 | 3.15 |
|              | -10        | 9.76    | 1.78 | 5.48 | 9.76         | 1.78 | 5.48 | 8.07  | 1.82 | 4.43 | 6.46  | 2.01 | 3.22 | 6.46  | 2.01 | 3.22 | 6.46    | 2.01 | 3.22 |
|              | -7         | 10.27   | 1.83 | 5.61 | 10.27        | 1.83 | 5.61 | 7.88  | 1.93 | 4.08 | 6.68  | 2.02 | 3.31 | 6.68  | 2.02 | 3.31 | 6.68    | 2.02 | 3.31 |
|              | -5         | 9.83    | 1.86 | 5.28 | 9.83         | 1.86 | 5.28 | 7.93  | 1.99 | 3.98 | 6.94  | 2.06 | 3.37 | 6.94  | 2.06 | 3.37 | 6.94    | 2.06 | 3.37 |
|              | -2         | 10.06   | 1.96 | 5.14 | 10.06        | 1.96 | 5.14 | 8.09  | 2.04 | 3.97 | 6.90  | 2.10 | 3.29 | 6.90  | 2.10 | 3.29 | 6.90    | 2.10 | 3.29 |
|              | 0          | 9.97    | 2.01 | 4.96 | 9.97         | 2.01 | 4.96 | 8.19  | 2.07 | 3.95 | 7.22  | 2.16 | 3.34 | 7.22  | 2.16 | 3.34 | 7.22    | 2.16 | 3.34 |
|              | 2          | 10.37   | 2.06 | 5.03 | 10.37        | 2.06 | 5.03 | 8.36  | 2.12 | 3.94 | 7.57  | 2.24 | 3.38 | 7.57  | 2.24 | 3.38 | 7.57    | 2.24 | 3.38 |
|              | 5          | 12.26   | 2.39 | 5.14 | 11.77        | 2.39 | 4.91 | 8.74  | 2.47 | 3.55 | 8.52  | 2.56 | 3.32 | 8.52  | 2.56 | 3.32 | 8.52    | 2.56 | 3.32 |
|              | 7          | 12.53   | 2.59 | 4.84 | 12.02        | 2.63 | 4.58 | 9.06  | 2.79 | 3.25 | 9.06  | 2.79 | 3.25 | 9.06  | 2.79 | 3.25 | 9.06    | 2.79 | 3.25 |
|              | 10         | 13.19   | 2.71 | 4.87 | 12.07        | 2.78 | 4.34 | 9.01  | 2.90 | 3.11 | 8.74  | 2.92 | 2.99 | 8.74  | 2.92 | 2.99 | 8.74    | 2.92 | 2.99 |
|              | 12         | 13.45   | 2.87 | 4.69 | 12.04        | 2.93 | 4.11 | 9.28  | 3.07 | 3.03 | 9.28  | 3.07 | 3.03 | 9.28  | 3.07 | 3.03 | 9.28    | 3.07 | 3.03 |
|              | 15         | 12.73   | 2.96 | 4.31 | 11.88        | 3.06 | 3.88 | 10.16 | 3.28 | 3.10 | 9.84  | 3.28 | 3.00 | 9.84  | 3.28 | 3.00 | 9.84    | 3.28 | 3.00 |
|              | 20         | 12.14   | 3.24 | 3.75 | 11.84        | 3.30 | 3.59 | 9.02  | 3.53 | 2.55 | 7.87  | 3.53 | 2.23 | 7.87  | 3.53 | 2.23 | 7.87    | 3.53 | 2.23 |
|              | 25         | 11.99   | 3.52 | 3.41 | 11.99        | 3.52 | 3.41 | 9.14  | 3.78 | 2.42 | 9.14  | 3.78 | 2.42 | 9.14  | 3.78 | 2.42 | 9.14    | 3.78 | 2.42 |
|              | 30         | 11.95   | 3.96 | 3.02 | 11.95        | 3.96 | 3.02 | 9.92  | 4.07 | 2.44 | 9.92  | 4.07 | 2.44 | 9.92  | 4.07 | 2.44 | 9.92    | 4.07 | 2.44 |
|              | 35         | /       | /    | /    | /            | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /    | /       | /    | /    |
| 70           | -25        | /       | /    | /    | /            | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /    | /       | /    | /    |
|              | -20        | /       | /    | /    | /            | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /    | /       | /    | /    |
|              | -15        | 8.45    | 1.57 | 5.40 | 8.45         | 1.57 | 5.40 | 7.90  | 1.61 | 4.92 | 5.47  | 1.65 | 3.32 | 5.47  | 1.65 | 3.32 | 5.47    | 1.65 | 3.32 |
|              | -10        | 9.40    | 1.72 | 5.47 | 9.40         | 1.72 | 5.47 | 8.38  | 1.75 | 4.80 | 6.50  | 1.92 | 3.40 | 6.50  | 1.92 | 3.40 | 6.50    | 1.92 | 3.40 |
|              | -7         | 9.86    | 1.80 | 5.47 | 9.86         | 1.80 | 5.47 | 8.00  | 1.85 | 4.33 | 7.04  | 1.96 | 3.59 | 6.90  | 1.96 | 3.52 | 6.90    | 1.96 | 3.52 |
|              | -5         | 9.52    | 1.82 | 5.24 | 9.52         | 1.82 | 5.24 | 8.04  | 1.87 | 4.30 | 7.13  | 1.97 | 3.61 | 7.13  | 1.97 | 3.61 | 7.13    | 1.97 | 3.61 |
|              | -2         | 9.71    | 1.85 | 5.24 | 9.71         | 1.85 | 5.24 | 8.31  | 1.89 | 4.40 | 6.98  | 2.00 | 3.49 |       |      |      |         |      |      |

## 8.6 Prestazioni riscaldamento Mod. 14

| Tw_out<br>°C | DBWB<br>°C | Maximum |      |      | 100%(Normal) |      |      | 75%   |       |      | 50%  |       |      | 25%  |       |      | Minimum |       |      |
|--------------|------------|---------|------|------|--------------|------|------|-------|-------|------|------|-------|------|------|-------|------|---------|-------|------|
|              |            | HC      | COP  | PI   | HC           | COP  | PI   | HC    | COP   | PI   | HC   | COP   | PI   | HC   | COP   | PI   | HC      | COP   | PI   |
| 25           | -25        | 8.42    | 2.18 | 3.87 | 8.42         | 2.18 | 3.87 | 8.42  | 2.18  | 3.87 | 5.74 | 2.40  | 2.40 | 2.98 | 2.56  | 1.17 | 2.98    | 2.56  | 1.17 |
|              | -20        | 10.04   | 2.48 | 4.04 | 10.04        | 2.48 | 4.04 | 8.57  | 2.69  | 3.19 | 5.61 | 2.89  | 1.94 | 3.77 | 3.11  | 1.21 | 3.77    | 3.11  | 1.21 |
|              | -15        | 11.52   | 2.77 | 4.16 | 11.52        | 2.77 | 4.16 | 8.55  | 3.19  | 2.68 | 5.60 | 3.41  | 1.64 | 4.37 | 3.77  | 1.16 | 4.37    | 3.77  | 1.16 |
|              | -10        | 12.24   | 2.90 | 4.22 | 11.59        | 2.95 | 3.93 | 8.58  | 3.38  | 2.54 | 5.83 | 3.69  | 1.58 | 4.69 | 3.99  | 1.18 | 4.69    | 3.99  | 1.18 |
|              | -7         | 12.49   | 3.06 | 4.09 | 11.40        | 3.29 | 3.47 | 8.55  | 3.93  | 2.17 | 5.73 | 4.38  | 1.31 | 5.41 | 4.44  | 1.22 | 5.41    | 4.44  | 1.22 |
|              | -5         | 12.71   | 3.27 | 3.89 | 11.35        | 3.46 | 3.28 | 8.54  | 3.94  | 2.17 | 5.60 | 4.55  | 1.23 | 5.60 | 4.55  | 1.23 | 5.60    | 4.55  | 1.23 |
|              | -2         | 13.13   | 3.58 | 3.67 | 11.12        | 3.90 | 2.85 | 8.49  | 4.42  | 1.92 | 5.53 | 4.92  | 1.13 | 4.86 | 5.22  | 0.93 | 4.86    | 5.22  | 0.93 |
|              | 0          | 13.32   | 3.80 | 3.50 | 10.82        | 4.28 | 2.53 | 8.37  | 4.76  | 1.76 | 5.40 | 5.36  | 1.01 | 5.02 | 5.42  | 0.93 | 5.02    | 5.42  | 0.93 |
|              | 2          | 13.53   | 4.03 | 3.36 | 10.81        | 4.64 | 2.33 | 8.06  | 5.11  | 1.58 | 5.54 | 5.72  | 0.97 | 5.38 | 5.80  | 0.93 | 5.38    | 5.80  | 0.93 |
|              | 5          | 15.22   | 4.68 | 3.25 | 12.76        | 5.26 | 2.42 | 9.53  | 6.13  | 1.55 | 6.35 | 6.94  | 0.91 | 5.85 | 7.00  | 0.84 | 5.85    | 7.00  | 0.84 |
|              | 7          | 16.96   | 5.23 | 3.24 | 13.97        | 5.88 | 2.38 | 10.49 | 6.55  | 1.60 | 6.99 | 7.42  | 0.94 | 6.24 | 7.44  | 0.84 | 6.24    | 7.44  | 0.84 |
|              | 10         | 18.26   | 6.13 | 2.98 | 14.12        | 7.28 | 1.94 | 10.55 | 8.00  | 1.32 | 7.25 | 8.95  | 0.81 | 7.25 | 8.95  | 0.81 | 7.25    | 8.95  | 0.81 |
|              | 12         | 18.97   | 6.64 | 2.86 | 14.04        | 8.00 | 1.75 | 10.45 | 8.90  | 1.17 | 7.71 | 9.79  | 0.79 | 7.71 | 9.79  | 0.79 | 7.71    | 9.79  | 0.79 |
|              | 15         | 17.72   | 7.06 | 2.51 | 13.93        | 8.15 | 1.71 | 10.58 | 9.26  | 1.14 | 7.78 | 10.26 | 0.76 | 7.78 | 10.26 | 0.76 | 7.78    | 10.26 | 0.76 |
|              | 20         | 15.66   | 8.00 | 1.96 | 13.97        | 8.66 | 1.61 | 10.45 | 10.28 | 1.02 | 6.95 | 11.20 | 0.62 | 6.62 | 11.24 | 0.59 | 6.62    | 11.24 | 0.59 |
|              | 25         | 15.59   | 8.79 | 1.77 | 13.92        | 9.59 | 1.45 | 10.55 | 11.10 | 0.95 | 7.40 | 11.79 | 0.63 | 7.40 | 11.79 | 0.63 | 7.40    | 11.79 | 0.63 |
|              | 30         | /       | /    | /    | /            | /    | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 35         | /       | /    | /    | /            | /    | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 30           | -25        | 7.97    | 1.97 | 4.05 | 7.97         | 1.97 | 4.05 | 7.97  | 1.97  | 4.05 | 5.53 | 2.10  | 2.63 | 2.78 | 2.25  | 1.24 | 2.78    | 2.25  | 1.24 |
|              | -20        | 9.57    | 2.25 | 4.25 | 9.57         | 2.25 | 4.25 | 8.37  | 2.38  | 3.52 | 5.59 | 2.54  | 2.20 | 3.56 | 2.76  | 1.29 | 3.56    | 2.76  | 1.29 |
|              | -15        | 10.92   | 2.51 | 4.34 | 10.92        | 2.51 | 4.34 | 8.48  | 2.84  | 2.99 | 5.63 | 3.00  | 1.88 | 4.19 | 3.36  | 1.25 | 4.19    | 3.36  | 1.25 |
|              | -10        | 11.57   | 2.62 | 4.42 | 11.57        | 2.62 | 4.42 | 8.59  | 2.96  | 2.90 | 5.84 | 3.24  | 1.80 | 4.58 | 3.58  | 1.28 | 4.58    | 3.58  | 1.28 |
|              | -7         | 12.30   | 2.84 | 4.33 | 11.60        | 3.05 | 3.80 | 8.47  | 3.45  | 2.46 | 5.80 | 3.84  | 1.51 | 5.39 | 4.03  | 1.34 | 5.39    | 4.03  | 1.34 |
|              | -5         | 12.24   | 2.96 | 4.14 | 11.11        | 3.16 | 3.52 | 8.39  | 3.52  | 2.38 | 5.60 | 3.85  | 1.45 | 5.38 | 4.10  | 1.31 | 5.38    | 4.10  | 1.31 |
|              | -2         | 12.65   | 3.23 | 3.92 | 10.90        | 3.46 | 3.15 | 8.33  | 3.87  | 2.15 | 5.57 | 4.18  | 1.33 | 4.66 | 4.56  | 1.02 | 4.66    | 4.56  | 1.02 |
|              | 0          | 12.84   | 3.43 | 3.74 | 10.87        | 3.72 | 2.93 | 8.26  | 4.15  | 1.99 | 5.47 | 4.52  | 1.21 | 4.74 | 4.63  | 1.02 | 4.74    | 4.63  | 1.02 |
|              | 2          | 13.04   | 3.62 | 3.60 | 10.62        | 4.04 | 2.63 | 8.05  | 4.47  | 1.80 | 5.46 | 4.82  | 1.13 | 5.16 | 5.02  | 1.03 | 5.16    | 5.02  | 1.03 |
|              | 5          | 14.66   | 4.20 | 3.50 | 12.70        | 4.55 | 2.79 | 9.56  | 5.25  | 1.82 | 6.42 | 5.75  | 1.12 | 5.54 | 5.92  | 0.93 | 5.54    | 5.92  | 0.93 |
|              | 7          | 16.25   | 4.66 | 3.48 | 14.10        | 5.01 | 2.82 | 10.58 | 5.58  | 1.89 | 7.07 | 6.16  | 1.15 | 5.98 | 6.33  | 0.94 | 5.98    | 6.33  | 0.94 |
|              | 10         | 17.24   | 5.39 | 3.20 | 13.93        | 6.19 | 2.25 | 10.46 | 6.73  | 1.55 | 6.91 | 7.52  | 0.92 | 6.91 | 7.52  | 0.92 | 6.91    | 7.52  | 0.92 |
|              | 12         | 18.09   | 5.84 | 3.10 | 14.05        | 6.76 | 2.08 | 10.43 | 7.41  | 1.41 | 7.32 | 8.10  | 0.90 | 7.32 | 8.10  | 0.90 | 7.32    | 8.10  | 0.90 |
|              | 15         | 16.91   | 6.15 | 2.75 | 13.94        | 6.98 | 2.00 | 10.44 | 7.77  | 1.34 | 7.59 | 8.73  | 0.87 | 7.59 | 8.73  | 0.87 | 7.59    | 8.73  | 0.87 |
|              | 20         | 15.03   | 6.94 | 2.17 | 13.82        | 7.53 | 1.83 | 10.51 | 8.56  | 1.23 | 6.92 | 9.26  | 0.75 | 6.37 | 9.36  | 0.68 | 6.37    | 9.36  | 0.68 |
|              | 25         | 15.04   | 7.58 | 1.98 | 14.13        | 7.93 | 1.78 | 10.39 | 9.18  | 1.13 | 7.11 | 9.62  | 0.74 | 7.11 | 9.62  | 0.74 | 7.11    | 9.62  | 0.74 |
|              | 30         | 14.61   | 9.22 | 1.58 | 14.12        | 9.53 | 1.48 | 10.47 | 10.81 | 0.97 | 7.78 | 11.27 | 0.69 | 7.78 | 11.27 | 0.69 | 7.78    | 11.27 | 0.69 |
|              | 35         | /       | /    | /    | /            | /    | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 35           | -25        | 7.78    | 1.85 | 4.22 | 7.78         | 1.85 | 4.22 | 7.78  | 1.85  | 4.22 | 5.58 | 1.92  | 2.91 | 2.68 | 2.02  | 1.33 | 2.68    | 2.02  | 1.33 |
|              | -20        | 9.38    | 2.11 | 4.45 | 9.38         | 2.11 | 4.45 | 8.42  | 2.19  | 3.84 | 5.61 | 2.31  | 2.43 | 3.47 | 2.50  | 1.39 | 3.47    | 2.50  | 1.39 |
|              | -15        | 10.75   | 2.35 | 4.57 | 10.75        | 2.35 | 4.57 | 8.63  | 2.61  | 3.31 | 5.71 | 2.75  | 2.07 | 4.16 | 2.97  | 1.40 | 4.16    | 2.97  | 1.40 |
|              | -10        | 11.41   | 2.54 | 4.49 | 11.1         | 2.56 | 4.34 | 8.69  | 2.69  | 3.23 | 6.03 | 3.14  | 1.92 | 4.50 | 3.28  | 1.37 | 4.50    | 3.28  | 1.37 |
|              | -7         | 11.75   | 2.73 | 4.30 | 11.5         | 2.76 | 4.17 | 8.60  | 3.03  | 2.84 | 5.76 | 3.50  | 1.64 | 5.25 | 3.58  | 1.47 | 5.25    | 3.58  | 1.47 |
|              | -5         | 12.12   | 2.87 | 4.23 | 11.33        | 2.99 | 3.79 | 8.43  | 3.12  | 2.70 | 5.68 | 3.58  | 1.59 | 5.08 | 3.62  | 1.40 | 5.08    | 3.62  | 1.40 |
|              | -2         | 12.45   | 2.99 | 4.16 | 10.92        | 3.15 | 3.47 | 8.39  | 3.50  | 2.40 | 5.69 | 3.73  | 1.53 | 4.57 | 4.12  | 1.11 | 4.57    | 4.12  | 1.11 |
|              | 0          | 12.63   | 3.17 | 3.99 | 11.14        | 3.35 | 3.33 | 8.34  | 3.72  | 2.24 | 5.46 | 4.01  | 1.36 | 4.73 | 4.33  | 1.09 | 4.73    | 4.33  | 1.09 |
|              | 2          | 12.84   | 3.35 | 3.83 | 10.80        | 3.50 | 3.09 | 8.23  | 3.98  | 2.07 | 5.45 | 4.24  | 1.28 | 5.07 | 4.50  | 1.13 | 5.07    | 4.50  | 1.13 |
|              | 5          | 14.43   | 3.85 | 3.75 | 12.60        | 4.11 | 3.07 | 9.51  | 4.69  | 2.03 | 6.42 | 5.04  | 1.28 | 5.36 | 5.18  | 1.03 | 5.36    | 5.18  | 1.03 |
|              | 7*         | 15.95   | 4.26 | 3.74 | 14.00        | 4.50 | 3.11 | 10.53 | 4.97  | 2.12 | 7.07 | 5.38  | 1.32 | 5.87 | 5.61  | 1.05 | 5.87    | 5.61  | 1.05 |
|              | 10         | 16.52   | 4.76 | 3.47 | 13.53        | 5.27 | 2.57 | 10.31 | 5.63  | 1.83 | 6.96 | 6.15  | 1.13 | 6.42 | 6.22  | 1.03 | 6.42    | 6.22  | 1.03 |
|              | 12         | 17.19   | 5.13 | 3.35 | 13.98        | 5.72 | 2.44 | 10.41 | 6.12  | 1.70 | 7.10 | 6.64  | 1.07 | 6.82 | 6.71  | 1.02 | 6.82    | 6.71  | 1.02 |
|              | 15         | 16.04   | 5.36 | 2.99 | 13.98        | 5.85 | 2.39 | 10.46 | 6.39  | 1.64 | 7.09 | 7.20  | 0.99 | 7.09 | 7.20  | 0.99 | 7.09    | 7.20  | 0.99 |
|              | 20         | 14.32   | 6.15 | 2.33 | 14.13        | 6.47 | 2.18 | 10.36 | 7.26  | 1.43 | 6.93 | 7.84  | 0.88 | 5.97 | 7.99  | 0.75 | 5.97    | 7.99  | 0.75 |
|              | 25         | 14.32   | 6.82 | 2.10 | 13.88        | 6.96 | 1.99 | 10.35 | 7.98  | 1.30 | 7.01 | 8.32  | 0.84 | 6.68 | 8.47  | 0.79 | 6.68    | 8.47  | 0.79 |
|              | 30         | 13.95   | 7.83 | 1.78 | 13.95        | 7.83 | 1.78 | 10.45 | 8.73  | 1.20 | 7.30 | 9.00  | 0.81 | 7.30 | 9.00  | 0.81 | 7.30    | 9.00  | 0.81 |
|              | 35         | 13.22   | 9.58 | 1.38 | 13.22        | 9.58 | 1.38 | 10.47 | 10.32 | 1.01 | 7.97 | 10.45 | 0.76 | 7.97 | 10.45 | 0.76 | 7.97    | 10.45 | 0.76 |
| 40           | -25        | 7.59    | 1.70 | 4.47 | 7.59         | 1.70 | 4.47 | 7.59  | 1.70  | 4.47 | 5.82 | 1.78  | 3.28 | 2.74 | 1.89  | 1.45 | 2.74    | 1.89  | 1.45 |
|              | -20        | 9.22    | 1.98 | 4.65 | 9.22         | 1.98 | 4.65 | 8.57  | 2.02  | 4.24 | 5.67 | 2.11  | 2.68 | 3.57 | 2.31  | 1.55 | 3.57    | 2.31  | 1.55 |
|              | -15        | 10.70   | 2.20 | 4.86 | 10.60        | 2.23 | 4.76 | 8.77  | 2.35  | 3.74 | 5.66 | 2.47  | 2.29 |      |       |      |         |       |      |

| Tw_out<br>°C | DBWB<br>°C | Maximum |      |      | 100%(Normal) |      |      | 75%   |      |      | 50%   |      |      | 25%   |      |      | Minimum |      |      |
|--------------|------------|---------|------|------|--------------|------|------|-------|------|------|-------|------|------|-------|------|------|---------|------|------|
|              |            | HC      | COP  | PI   | HC           | COP  | PI   | HC    | COP  | PI   | HC    | COP  | PI   | HC    | COP  | PI   | HC      | COP  | PI   |
| 55           | -25        | 7.14    | 1.41 | 5.08 | 7.14         | 1.41 | 5.08 | 7.14  | 1.41 | 5.08 | 5.87  | 1.47 | 4.00 | 3.71  | 1.53 | 2.43 | 3.71    | 1.53 | 2.43 |
|              | -20        | 8.78    | 1.66 | 5.28 | 8.78         | 1.66 | 5.28 | 8.56  | 1.66 | 5.14 | 5.89  | 1.80 | 3.26 | 4.73  | 1.85 | 2.55 | 4.73    | 1.85 | 2.55 |
|              | -15        | 10.58   | 1.88 | 5.62 | 10.58        | 1.88 | 5.62 | 8.51  | 1.94 | 4.38 | 5.89  | 2.03 | 2.91 | 5.39  | 2.06 | 2.62 | 5.39    | 2.06 | 2.62 |
|              | -10        | 10.79   | 1.93 | 5.58 | 10.79        | 1.93 | 5.58 | 8.72  | 2.01 | 4.34 | 6.01  | 2.28 | 2.64 | 6.01  | 2.28 | 2.64 | 6.01    | 2.28 | 2.64 |
|              | -7         | 11.30   | 2.10 | 5.38 | 11.30        | 2.10 | 5.38 | 8.66  | 2.18 | 3.97 | 6.53  | 2.32 | 2.82 | 6.53  | 2.32 | 2.82 | 6.53    | 2.32 | 2.82 |
|              | -5         | 11.12   | 2.14 | 5.21 | 11.12        | 2.14 | 5.21 | 8.57  | 2.22 | 3.87 | 6.80  | 2.40 | 2.83 | 6.80  | 2.40 | 2.83 | 6.80    | 2.40 | 2.83 |
|              | -2         | 11.64   | 2.22 | 5.24 | 11.64        | 2.22 | 5.24 | 8.86  | 2.37 | 3.74 | 6.91  | 2.58 | 2.68 | 6.91  | 2.58 | 2.68 | 6.91    | 2.58 | 2.68 |
|              | 0          | 11.83   | 2.35 | 5.04 | 11.83        | 2.35 | 5.04 | 8.98  | 2.50 | 3.60 | 7.28  | 2.69 | 2.71 | 7.28  | 2.69 | 2.71 | 7.28    | 2.69 | 2.71 |
|              | 2          | 12.00   | 2.50 | 4.80 | 12.00        | 2.50 | 4.80 | 9.06  | 2.62 | 3.46 | 7.64  | 2.80 | 2.73 | 7.64  | 2.80 | 2.73 | 7.64    | 2.80 | 2.73 |
|              | 5          | 13.46   | 2.79 | 4.83 | 13.33        | 2.80 | 4.77 | 9.72  | 3.08 | 3.15 | 9.12  | 3.21 | 2.84 | 9.12  | 3.21 | 2.84 | 9.12    | 3.21 | 2.84 |
|              | 7          | 14.81   | 3.04 | 4.87 | 14.00        | 3.00 | 4.67 | 10.46 | 3.23 | 3.24 | 9.75  | 3.39 | 2.88 | 9.75  | 3.39 | 2.88 | 9.75    | 3.39 | 2.88 |
|              | 10         | 15.19   | 3.31 | 4.59 | 14.05        | 3.38 | 4.16 | 10.63 | 3.56 | 2.98 | 9.05  | 3.70 | 2.44 | 9.05  | 3.70 | 2.44 | 9.05    | 3.70 | 2.44 |
|              | 12         | 15.75   | 3.50 | 4.50 | 13.93        | 3.64 | 3.83 | 10.50 | 3.81 | 2.75 | 9.59  | 3.90 | 2.46 | 9.59  | 3.90 | 2.46 | 9.59    | 3.90 | 2.46 |
|              | 15         | 14.75   | 3.63 | 4.07 | 14.16        | 3.68 | 3.85 | 10.55 | 4.05 | 2.61 | 10.06 | 4.14 | 2.43 | 10.06 | 4.14 | 2.43 | 10.06   | 4.14 | 2.43 |
|              | 20         | 13.59   | 3.97 | 3.43 | 13.59        | 4.00 | 3.40 | 10.52 | 4.31 | 2.44 | 7.98  | 4.46 | 1.79 | 7.98  | 4.46 | 1.79 | 7.98    | 4.46 | 1.79 |
|              | 25         | 13.54   | 4.27 | 3.17 | 13.54        | 4.27 | 3.17 | 10.61 | 4.69 | 2.26 | 9.12  | 4.77 | 1.91 | 9.12  | 4.77 | 1.91 | 9.12    | 4.77 | 1.91 |
|              | 30         | 13.13   | 4.91 | 2.67 | 13.13        | 4.91 | 2.67 | 10.76 | 5.13 | 2.10 | 9.92  | 5.19 | 1.91 | 9.92  | 5.19 | 1.91 | 9.92    | 5.19 | 1.91 |
|              | 35         | 12.42   | 5.57 | 2.23 | 12.42        | 5.57 | 2.23 | 10.78 | 5.69 | 1.89 | 10.78 | 5.69 | 1.89 | 10.78 | 5.69 | 1.89 | 10.78   | 5.69 | 1.89 |
| 60           | -25        | /       | /    | /    | /            | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /    | /       | /    | /    |
|              | -20        | 8.54    | 1.54 | 5.55 | 8.54         | 1.54 | 5.55 | 8.33  | 1.54 | 5.39 | 5.86  | 1.64 | 3.57 | 4.92  | 1.68 | 2.92 | 4.92    | 1.68 | 2.92 |
|              | -15        | 10.11   | 1.70 | 5.94 | 10.11        | 1.70 | 5.94 | 8.45  | 1.76 | 4.80 | 5.70  | 1.88 | 3.03 | 5.70  | 1.88 | 3.03 | 5.70    | 1.88 | 3.03 |
|              | -10        | 10.52   | 1.84 | 5.71 | 10.52        | 1.84 | 5.71 | 8.65  | 1.92 | 4.51 | 6.39  | 2.12 | 3.01 | 6.39  | 2.12 | 3.01 | 6.39    | 2.12 | 3.01 |
|              | -7         | 11.24   | 1.95 | 5.75 | 11.24        | 1.95 | 5.75 | 8.68  | 2.04 | 4.26 | 7.54  | 2.17 | 3.47 | 7.54  | 2.17 | 3.47 | 7.54    | 2.17 | 3.47 |
|              | -5         | 10.65   | 1.99 | 5.35 | 10.65        | 1.99 | 5.35 | 8.67  | 2.08 | 4.18 | 7.08  | 2.25 | 3.15 | 7.08  | 2.25 | 3.15 | 7.08    | 2.25 | 3.15 |
|              | -2         | 11.17   | 2.07 | 5.39 | 11.17        | 2.07 | 5.39 | 8.79  | 2.14 | 4.10 | 7.12  | 2.31 | 3.08 | 7.12  | 2.31 | 3.08 | 7.12    | 2.31 | 3.08 |
|              | 0          | 11.52   | 2.15 | 5.36 | 11.52        | 2.15 | 5.36 | 8.92  | 2.25 | 3.96 | 7.52  | 2.41 | 3.12 | 7.52  | 2.41 | 3.12 | 7.52    | 2.41 | 3.12 |
|              | 2          | 11.69   | 2.25 | 5.20 | 11.69        | 2.25 | 5.20 | 8.99  | 2.36 | 3.81 | 7.89  | 2.51 | 3.15 | 7.89  | 2.51 | 3.15 | 7.89    | 2.51 | 3.15 |
|              | 5          | 13.37   | 2.58 | 5.18 | 13.37        | 2.58 | 5.18 | 9.90  | 2.76 | 3.58 | 8.91  | 2.87 | 3.10 | 8.91  | 2.87 | 3.10 | 8.91    | 2.87 | 3.10 |
|              | 7          | 14.36   | 2.77 | 5.19 | 14.00        | 2.77 | 5.05 | 10.46 | 2.93 | 3.57 | 9.55  | 3.04 | 3.14 | 9.55  | 3.04 | 3.14 | 9.55    | 3.04 | 3.14 |
|              | 10         | 14.70   | 2.99 | 4.92 | 13.94        | 3.02 | 4.62 | 10.41 | 3.17 | 3.28 | 9.12  | 3.29 | 2.77 | 9.12  | 3.29 | 2.77 | 9.12    | 3.29 | 2.77 |
|              | 12         | 15.23   | 3.15 | 4.83 | 14.02        | 3.22 | 4.36 | 10.49 | 3.37 | 3.11 | 9.66  | 3.45 | 2.80 | 9.66  | 3.45 | 2.80 | 9.66    | 3.45 | 2.80 |
|              | 15         | 14.26   | 3.25 | 4.38 | 14.07        | 3.27 | 4.30 | 10.46 | 3.60 | 2.90 | 10.19 | 3.68 | 2.77 | 10.19 | 3.68 | 2.77 | 10.19   | 3.68 | 2.77 |
|              | 20         | 13.27   | 3.55 | 3.74 | 13.27        | 3.56 | 3.73 | 10.45 | 3.82 | 2.74 | 8.16  | 3.96 | 2.06 | 8.16  | 3.96 | 2.06 | 8.16    | 3.96 | 2.06 |
|              | 25         | 13.23   | 3.84 | 3.44 | 13.23        | 3.84 | 3.44 | 10.56 | 4.17 | 2.53 | 9.40  | 4.25 | 2.21 | 9.40  | 4.25 | 2.21 | 9.40    | 4.25 | 2.21 |
|              | 30         | 12.80   | 4.38 | 2.92 | 12.80        | 4.38 | 2.92 | 10.53 | 4.58 | 2.30 | 10.20 | 4.59 | 2.22 | 10.20 | 4.59 | 2.22 | 10.20   | 4.59 | 2.22 |
|              | 35         | /       | /    | /    | /            | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /    | /       | /    | /    |
| 65           | -25        | /       | /    | /    | /            | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /    | /       | /    | /    |
|              | -20        | 8.14    | 1.42 | 5.75 | 8.14         | 1.42 | 5.75 | 8.14  | 1.42 | 5.75 | 5.80  | 1.44 | 4.01 | 4.66  | 1.51 | 3.08 | 4.66    | 1.51 | 3.08 |
|              | -15        | 9.46    | 1.54 | 6.15 | 9.46         | 1.54 | 6.15 | 8.29  | 1.65 | 5.03 | 5.64  | 1.79 | 3.15 | 5.64  | 1.79 | 3.15 | 5.64    | 1.79 | 3.15 |
|              | -10        | 10.25   | 1.75 | 5.85 | 10.25        | 1.75 | 5.85 | 8.71  | 1.80 | 4.85 | 6.46  | 2.01 | 3.22 | 6.46  | 2.01 | 3.22 | 6.46    | 2.01 | 3.22 |
|              | -7         | 10.74   | 1.80 | 5.96 | 10.74        | 1.80 | 5.96 | 8.58  | 1.90 | 4.51 | 6.68  | 2.02 | 3.31 | 6.68  | 2.02 | 3.31 | 6.68    | 2.02 | 3.31 |
|              | -5         | 10.18   | 1.84 | 5.53 | 10.18        | 1.84 | 5.53 | 8.67  | 1.95 | 4.44 | 6.94  | 2.06 | 3.37 | 6.94  | 2.06 | 3.37 | 6.94    | 2.06 | 3.37 |
|              | -2         | 10.77   | 1.93 | 5.59 | 10.77        | 1.93 | 5.59 | 8.86  | 2.01 | 4.40 | 6.90  | 2.10 | 3.29 | 6.90  | 2.10 | 3.29 | 6.90    | 2.10 | 3.29 |
|              | 0          | 10.70   | 1.98 | 5.40 | 10.70        | 1.98 | 5.40 | 8.85  | 2.05 | 4.32 | 7.22  | 2.16 | 3.34 | 7.22  | 2.16 | 3.34 | 7.22    | 2.16 | 3.34 |
|              | 2          | 11.08   | 2.04 | 5.42 | 11.08        | 2.04 | 5.42 | 8.87  | 2.10 | 4.22 | 7.57  | 2.24 | 3.38 | 7.57  | 2.24 | 3.38 | 7.57    | 2.24 | 3.38 |
|              | 5          | 13.21   | 2.36 | 5.60 | 13.21        | 2.36 | 5.60 | 9.95  | 2.45 | 4.07 | 8.52  | 2.56 | 3.32 | 8.52  | 2.56 | 3.32 | 8.52    | 2.56 | 3.32 |
|              | 7          | 13.71   | 2.51 | 5.47 | 13.71        | 2.51 | 5.47 | 10.45 | 2.69 | 3.88 | 9.06  | 2.79 | 3.25 | 9.06  | 2.79 | 3.25 | 9.06    | 2.79 | 3.25 |
|              | 10         | 13.94   | 2.69 | 5.19 | 13.94        | 2.69 | 5.19 | 10.50 | 2.85 | 3.69 | 8.74  | 2.92 | 2.99 | 8.74  | 2.92 | 2.99 | 8.74    | 2.92 | 2.99 |
|              | 12         | 14.63   | 2.79 | 5.24 | 14.04        | 2.82 | 4.98 | 10.53 | 2.96 | 3.55 | 9.28  | 3.07 | 3.03 | 9.28  | 3.07 | 3.03 | 9.28    | 3.07 | 3.03 |
|              | 15         | 13.72   | 2.91 | 4.71 | 13.72        | 2.91 | 4.71 | 10.38 | 3.23 | 3.22 | 9.84  | 3.28 | 3.00 | 9.84  | 3.28 | 3.00 | 9.84    | 3.28 | 3.00 |
|              | 20         | 12.90   | 3.17 | 4.06 | 12.90        | 3.17 | 4.06 | 10.55 | 3.42 | 3.08 | 7.87  | 3.53 | 2.23 | 7.87  | 3.53 | 2.23 | 7.87    | 3.53 | 2.23 |
|              | 25         | 12.85   | 3.44 | 3.74 | 12.85        | 3.44 | 3.74 | 10.47 | 3.71 | 2.82 | 9.14  | 3.78 | 2.42 | 9.14  | 3.78 | 2.42 | 9.14    | 3.78 | 2.42 |
|              | 30         | 12.43   | 3.90 | 3.18 | 12.43        | 3.90 | 3.18 | 10.45 | 4.04 | 2.59 | 9.92  | 4.07 | 2.44 | 9.92  | 4.07 | 2.44 | 9.92    | 4.07 | 2.44 |
|              | 35         | /       | /    | /    | /            | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /    | /       | /    | /    |
| 70           | -25        | /       | /    | /    | /            | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /    | /       | /    | /    |
|              | -20        | /       | /    | /    | /            | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /    | /       | /    | /    |
|              | -15        | 8.93    | 1.54 | 5.79 | 8.93         | 1.54 | 5.79 | 8.22  | 1.56 | 5.27 | 5.47  | 1.65 | 3.32 | 5.47  | 1.65 | 3.32 | 5.47    | 1.65 | 3.32 |
|              | -10        | 9.87    | 1.68 | 5.87 | 9.87         | 1.68 | 5.87 | 8.72  | 1.74 | 5.02 | 6.50  | 1.92 | 3.40 | 6.50  | 1.92 | 3.40 | 6.50    | 1.92 | 3.40 |
|              | -7         | 10.07   | 1.75 | 5.74 | 10.07        | 1.75 | 5.74 | 8.60  | 1.82 | 4.73 | 7.04  | 1.96 | 3.59 | 6.90  | 1.96 | 3.52 | 6.90    | 1.96 | 3.52 |
|              | -5         | 9.75    | 1.77 | 5.52 | 9.75         | 1.77 | 5.52 | 8.68  | 1.84 | 4.71 | 7.13  | 1.97 | 3.61 | 7.13  | 1.97 | 3.61 | 7.13    | 1.97 | 3.61 |
|              | -2         | 10.33   | 1.80 | 5.74 | 10.33        | 1.80 | 5.74 | 8.83  | 1.87 |      |       |      |      |       |      |      |         |      |      |

## 8.7 Prestazioni riscaldamento Mod. 16

| Tw_out<br>°C | DBWB<br>°C | Maximum |      |      | 100%(Normal) |      |      | 75%   |       |      | 50%  |       |      | 25%  |       |      | Minimum |       |      |
|--------------|------------|---------|------|------|--------------|------|------|-------|-------|------|------|-------|------|------|-------|------|---------|-------|------|
|              |            | HC      | COP  | PI   | HC           | COP  | PI   | HC    | COP   | PI   | HC   | COP   | PI   | HC   | COP   | PI   | HC      | COP   | PI   |
| 25           | -25        | 9.50    | 1.99 | 4.77 | 9.50         | 1.99 | 4.77 | 9.50  | 1.99  | 4.77 | 6.47 | 2.26  | 2.87 | 2.98 | 2.56  | 1.17 | 2.98    | 2.56  | 1.17 |
|              | -20        | 11.26   | 2.37 | 4.75 | 11.26        | 2.37 | 4.75 | 9.73  | 2.54  | 3.83 | 6.53 | 2.72  | 2.40 | 3.77 | 3.11  | 1.21 | 3.77    | 3.11  | 1.21 |
|              | -15        | 12.59   | 2.63 | 4.78 | 12.59        | 2.63 | 4.78 | 9.45  | 3.07  | 3.08 | 6.18 | 3.26  | 1.90 | 4.37 | 3.77  | 1.16 | 4.37    | 3.77  | 1.16 |
|              | -10        | 13.33   | 2.76 | 4.84 | 12.68        | 2.84 | 4.47 | 9.54  | 3.24  | 2.95 | 6.32 | 3.54  | 1.78 | 4.69 | 3.99  | 1.18 | 4.69    | 3.99  | 1.18 |
|              | -7         | 13.66   | 2.87 | 4.75 | 12.66        | 3.06 | 4.14 | 9.31  | 3.74  | 2.49 | 6.35 | 4.10  | 1.55 | 5.41 | 4.44  | 1.22 | 5.41    | 4.44  | 1.22 |
|              | -5         | 14.12   | 3.11 | 4.53 | 12.76        | 3.30 | 3.87 | 9.48  | 3.75  | 2.53 | 6.34 | 4.23  | 1.50 | 5.60 | 4.55  | 1.23 | 5.60    | 4.55  | 1.23 |
|              | -2         | 14.53   | 3.42 | 4.25 | 12.62        | 3.68 | 3.43 | 9.63  | 4.20  | 2.29 | 6.36 | 4.71  | 1.35 | 4.86 | 5.22  | 0.93 | 4.86    | 5.22  | 0.93 |
|              | 0          | 14.83   | 3.64 | 4.08 | 12.56        | 3.96 | 3.17 | 9.45  | 4.59  | 2.06 | 6.27 | 5.10  | 1.23 | 5.02 | 5.42  | 0.93 | 5.02    | 5.42  | 0.93 |
|              | 2          | 15.03   | 3.84 | 3.91 | 12.65        | 4.24 | 2.98 | 9.46  | 4.93  | 1.92 | 6.45 | 5.41  | 1.19 | 5.38 | 5.80  | 0.93 | 5.38    | 5.80  | 0.93 |
|              | 5          | 16.61   | 4.48 | 3.71 | 14.00        | 4.99 | 2.80 | 10.59 | 5.95  | 1.78 | 7.06 | 6.67  | 1.06 | 5.85 | 7.00  | 0.84 | 5.85    | 7.00  | 0.84 |
|              | 7          | 18.67   | 5.02 | 3.72 | 14.95        | 5.68 | 2.63 | 11.19 | 6.44  | 1.74 | 7.50 | 7.28  | 1.03 | 6.24 | 7.44  | 0.84 | 6.24    | 7.44  | 0.84 |
|              | 10         | 20.15   | 5.81 | 3.47 | 15.07        | 7.12 | 2.12 | 11.31 | 7.86  | 1.44 | 7.58 | 8.85  | 0.86 | 7.25 | 8.95  | 0.81 | 7.25    | 8.95  | 0.81 |
|              | 12         | 20.80   | 6.32 | 3.29 | 15.03        | 7.78 | 1.93 | 11.26 | 8.70  | 1.29 | 7.71 | 9.79  | 0.79 | 7.71 | 9.79  | 0.79 | 7.71    | 9.79  | 0.79 |
|              | 15         | 19.18   | 6.60 | 2.91 | 15.08        | 8.04 | 1.88 | 11.23 | 9.16  | 1.23 | 7.78 | 10.26 | 0.76 | 7.78 | 10.26 | 0.76 | 7.78    | 10.26 | 0.76 |
|              | 20         | 16.84   | 7.51 | 2.24 | 15.04        | 8.19 | 1.84 | 11.32 | 9.88  | 1.15 | 7.46 | 11.00 | 0.68 | 6.62 | 11.24 | 0.59 | 6.62    | 11.24 | 0.59 |
|              | 25         | 17.20   | 8.27 | 2.08 | 14.89        | 9.12 | 1.63 | 11.38 | 10.81 | 1.05 | 7.48 | 11.73 | 0.64 | 7.40 | 11.79 | 0.63 | 7.40    | 11.79 | 0.63 |
|              | 30         | /       | /    | /    | /            | /    | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 35         | /       | /    | /    | /            | /    | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 30           | -25        | 9.03    | 1.82 | 4.96 | 9.03         | 1.82 | 4.96 | 9.03  | 1.82  | 4.96 | 6.06 | 1.99  | 3.05 | 2.78 | 2.25  | 1.24 | 2.78    | 2.25  | 1.24 |
|              | -20        | 10.78   | 2.07 | 5.21 | 10.78        | 2.07 | 5.21 | 9.38  | 2.17  | 4.32 | 6.19 | 2.40  | 2.58 | 3.56 | 2.76  | 1.29 | 3.56    | 2.76  | 1.29 |
|              | -15        | 12.13   | 2.41 | 5.03 | 12.13        | 2.41 | 5.03 | 9.32  | 2.72  | 3.43 | 6.13 | 2.85  | 2.15 | 4.19 | 3.36  | 1.25 | 4.19    | 3.36  | 1.25 |
|              | -10        | 12.81   | 2.50 | 5.12 | 12.71        | 2.52 | 5.05 | 9.40  | 2.86  | 3.29 | 6.24 | 3.10  | 2.01 | 4.58 | 3.58  | 1.28 | 4.58    | 3.58  | 1.28 |
|              | -7         | 13.43   | 2.67 | 5.02 | 12.64        | 2.80 | 4.51 | 9.31  | 3.33  | 2.80 | 6.37 | 3.72  | 1.71 | 5.39 | 4.03  | 1.34 | 5.39    | 4.03  | 1.34 |
|              | -5         | 13.60   | 2.82 | 4.82 | 12.65        | 2.93 | 4.32 | 9.42  | 3.30  | 2.85 | 6.20 | 3.66  | 1.70 | 5.38 | 4.10  | 1.31 | 5.38    | 4.10  | 1.31 |
|              | -2         | 14.02   | 3.10 | 4.53 | 12.40        | 3.29 | 3.77 | 9.47  | 3.68  | 2.57 | 6.47 | 3.99  | 1.62 | 4.66 | 4.56  | 1.02 | 4.66    | 4.56  | 1.02 |
|              | 0          | 14.31   | 3.29 | 4.35 | 12.54        | 3.51 | 3.58 | 9.45  | 3.98  | 2.37 | 6.18 | 4.35  | 1.42 | 4.74 | 4.63  | 1.02 | 4.74    | 4.63  | 1.02 |
|              | 2          | 14.50   | 3.47 | 4.18 | 12.64        | 3.72 | 3.40 | 9.50  | 4.26  | 2.23 | 6.21 | 4.61  | 1.35 | 5.16 | 5.02  | 1.03 | 5.16    | 5.02  | 1.03 |
|              | 5          | 16.01   | 4.03 | 3.97 | 14.18        | 4.33 | 3.28 | 10.37 | 5.16  | 2.01 | 7.04 | 5.64  | 1.25 | 5.54 | 5.92  | 0.93 | 5.54    | 5.92  | 0.93 |
|              | 7          | 17.89   | 4.49 | 3.98 | 15.23        | 4.86 | 3.13 | 11.25 | 5.50  | 2.04 | 7.52 | 6.08  | 1.24 | 5.98 | 6.33  | 0.94 | 5.98    | 6.33  | 0.94 |
|              | 10         | 19.22   | 5.15 | 3.73 | 15.19        | 5.93 | 2.56 | 11.17 | 6.62  | 1.69 | 7.43 | 7.46  | 1.00 | 6.91 | 7.52  | 0.92 | 6.91    | 7.52  | 0.92 |
|              | 12         | 19.84   | 5.59 | 3.55 | 15.03        | 6.65 | 2.26 | 11.20 | 7.29  | 1.54 | 7.72 | 8.01  | 0.96 | 7.32 | 8.10  | 0.90 | 7.32    | 8.10  | 0.90 |
|              | 15         | 18.32   | 5.80 | 3.16 | 15.03        | 6.80 | 2.21 | 11.43 | 7.56  | 1.51 | 7.59 | 8.73  | 0.87 | 7.59 | 8.73  | 0.87 | 7.59    | 8.73  | 0.87 |
|              | 20         | 16.17   | 6.55 | 2.47 | 15.00        | 7.26 | 2.07 | 11.25 | 8.36  | 1.35 | 7.50 | 9.22  | 0.81 | 6.37 | 9.36  | 0.68 | 6.37    | 9.36  | 0.68 |
|              | 25         | 16.57   | 7.17 | 2.31 | 15.04        | 7.58 | 1.98 | 11.19 | 8.99  | 1.24 | 7.46 | 9.53  | 0.78 | 7.11 | 9.62  | 0.74 | 7.11    | 9.62  | 0.74 |
|              | 30         | 15.63   | 8.79 | 1.78 | 14.89        | 9.34 | 1.59 | 11.34 | 10.62 | 1.07 | 7.86 | 11.26 | 0.70 | 7.78 | 11.27 | 0.69 | 7.78    | 11.27 | 0.69 |
|              | 35         | /       | /    | /    | /            | /    | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 35           | -25        | 8.83    | 1.74 | 5.08 | 8.83         | 1.74 | 5.08 | 8.83  | 1.76  | 5.02 | 6.21 | 1.85  | 3.36 | 2.68 | 2.02  | 1.33 | 2.68    | 2.02  | 1.33 |
|              | -20        | 10.56   | 1.96 | 5.39 | 10.56        | 1.96 | 5.39 | 9.30  | 2.07  | 4.50 | 6.25 | 2.20  | 2.85 | 3.47 | 2.50  | 1.39 | 3.47    | 2.50  | 1.39 |
|              | -15        | 11.96   | 2.27 | 5.28 | 11.96        | 2.23 | 5.37 | 9.25  | 2.50  | 3.70 | 6.35 | 2.62  | 2.42 | 4.16 | 2.97  | 1.40 | 4.16    | 2.97  | 1.40 |
|              | -10        | 12.47   | 2.43 | 5.13 | 12.33        | 2.46 | 5.02 | 9.39  | 2.56  | 3.66 | 6.32 | 2.88  | 2.19 | 4.50 | 3.28  | 1.37 | 4.50    | 3.28  | 1.37 |
|              | -7         | 13.06   | 2.71 | 4.82 | 12.43        | 2.74 | 4.54 | 9.41  | 2.90  | 3.24 | 6.44 | 3.35  | 1.92 | 5.25 | 3.58  | 1.47 | 5.25    | 3.58  | 1.47 |
|              | -5         | 13.25   | 2.79 | 4.75 | 12.55        | 2.91 | 4.31 | 9.44  | 3.01  | 3.14 | 6.27 | 3.28  | 1.91 | 5.08 | 3.62  | 1.40 | 5.08    | 3.62  | 1.40 |
|              | -2         | 13.61   | 2.90 | 4.70 | 12.67        | 3.06 | 4.14 | 9.51  | 3.33  | 2.86 | 6.33 | 3.60  | 1.76 | 4.57 | 4.12  | 1.11 | 4.57    | 4.12  | 1.11 |
|              | 0          | 14.05   | 3.03 | 4.64 | 12.80        | 3.17 | 4.03 | 9.64  | 3.56  | 2.71 | 6.30 | 3.87  | 1.63 | 4.73 | 4.33  | 1.09 | 4.73    | 4.33  | 1.09 |
|              | 2          | 14.27   | 3.20 | 4.46 | 12.80        | 3.20 | 4.00 | 9.72  | 3.80  | 2.55 | 6.35 | 4.09  | 1.55 | 5.07 | 4.50  | 1.13 | 5.07    | 4.50  | 1.13 |
|              | 5          | 15.77   | 3.72 | 4.24 | 14.20        | 3.92 | 3.63 | 10.49 | 4.57  | 2.30 | 7.02 | 4.96  | 1.42 | 5.36 | 5.18  | 1.03 | 5.36    | 5.18  | 1.03 |
|              | 7*         | 17.57   | 4.13 | 4.25 | 15.00        | 4.40 | 3.41 | 11.17 | 4.89  | 2.28 | 7.50 | 5.31  | 1.41 | 5.87 | 5.61  | 1.05 | 5.87    | 5.61  | 1.05 |
|              | 10         | 18.31   | 4.58 | 4.00 | 14.94        | 5.04 | 2.97 | 11.24 | 5.51  | 2.04 | 7.49 | 6.09  | 1.23 | 6.42 | 6.22  | 1.03 | 6.42    | 6.22  | 1.03 |
|              | 12         | 18.87   | 4.93 | 3.82 | 15.01        | 5.52 | 2.72 | 11.29 | 5.99  | 1.88 | 8.35 | 6.45  | 1.29 | 6.82 | 6.71  | 1.02 | 6.82    | 6.71  | 1.02 |
|              | 15         | 17.44   | 5.10 | 3.42 | 15.03        | 5.61 | 2.68 | 11.37 | 6.23  | 1.83 | 7.44 | 6.98  | 1.07 | 7.09 | 7.20  | 0.99 | 7.09    | 7.20  | 0.99 |
|              | 20         | 15.44   | 5.70 | 2.71 | 15.00        | 6.09 | 2.46 | 11.25 | 6.92  | 1.62 | 7.50 | 7.48  | 1.00 | 5.97 | 7.99  | 0.75 | 5.97    | 7.99  | 0.75 |
|              | 25         | 15.82   | 6.18 | 2.56 | 14.97        | 6.45 | 2.32 | 11.11 | 7.48  | 1.49 | 7.51 | 8.30  | 0.90 | 6.68 | 8.47  | 0.79 | 6.68    | 8.47  | 0.79 |
|              | 30         | 14.95   | 7.47 | 2.00 | 14.95        | 7.47 | 2.00 | 11.30 | 8.60  | 1.31 | 7.66 | 8.91  | 0.86 | 7.30 | 9.00  | 0.81 | 7.30    | 9.00  | 0.81 |
|              | 35         | 13.82   | 9.45 | 1.46 | 13.82        | 9.45 | 1.46 | 11.09 | 10.10 | 1.10 | 8.07 | 10.43 | 0.77 | 7.97 | 10.45 | 0.76 | 7.97    | 10.45 | 0.76 |
| 40           | -25        | 8.64    | 1.56 | 5.53 | 8.64         | 1.56 | 5.53 | 8.64  | 1.56  | 5.53 | 6.26 | 1.74  | 3.60 | 2.74 | 1.89  | 1.45 | 2.74    | 1.89  | 1.45 |
|              | -20        | 10.39   | 1.83 | 5.68 | 10.39        | 1.83 | 5.68 | 9.32  | 1.90  | 4.91 | 6.33 | 2.09  | 3.03 | 3.57 | 2.31  | 1.55 | 3.57    | 2.31  | 1.55 |
|              | -15        | 11.78   | 2.05 | 5.74 | 11.78        | 2.05 | 5.74 | 9.30  | 2.24  | 4.14 | 6.47 | 2.41  |      |      |       |      |         |       |      |

| Tw_out<br>°C | DBWB<br>°C | Maximum |      |      | 100%(Normal) |      |      | 75%   |      |      | 50%   |      |      | 25%   |      |      | Minimum |      |      |
|--------------|------------|---------|------|------|--------------|------|------|-------|------|------|-------|------|------|-------|------|------|---------|------|------|
|              |            | HC      | COP  | PI   | HC           | COP  | PI   | HC    | COP  | PI   | HC    | COP  | PI   | HC    | COP  | PI   | HC      | COP  | PI   |
| 55           | -25        | 8.18    | 1.30 | 6.30 | 8.18         | 1.30 | 6.30 | 8.18  | 1.30 | 6.30 | 6.36  | 1.44 | 4.42 | 3.71  | 1.53 | 2.43 | 3.71    | 1.53 | 2.43 |
|              | -20        | 9.82    | 1.56 | 6.28 | 9.82         | 1.56 | 6.28 | 9.50  | 1.58 | 6.00 | 6.32  | 1.76 | 3.58 | 4.73  | 1.85 | 2.55 | 4.73    | 1.85 | 2.55 |
|              | -15        | 11.19   | 1.79 | 6.26 | 11.19        | 1.79 | 6.26 | 9.60  | 1.89 | 5.07 | 6.29  | 2.02 | 3.11 | 5.39  | 2.06 | 2.62 | 5.39    | 2.06 | 2.62 |
|              | -10        | 11.30   | 1.90 | 5.95 | 11.30        | 1.90 | 5.95 | 9.43  | 2.02 | 4.67 | 6.20  | 2.24 | 2.77 | 6.01  | 2.28 | 2.64 | 6.01    | 2.28 | 2.64 |
|              | -7         | 12.40   | 2.05 | 6.05 | 12.40        | 2.05 | 6.05 | 9.46  | 2.16 | 4.38 | 7.20  | 2.26 | 3.18 | 6.53  | 2.32 | 2.82 | 6.53    | 2.32 | 2.82 |
|              | -5         | 11.85   | 2.09 | 5.68 | 11.85        | 2.09 | 5.68 | 9.33  | 2.18 | 4.28 | 6.80  | 2.40 | 2.83 | 6.80  | 2.40 | 2.83 | 6.80    | 2.40 | 2.83 |
|              | -2         | 12.28   | 2.18 | 5.64 | 12.28        | 2.18 | 5.64 | 9.53  | 2.33 | 4.08 | 6.91  | 2.58 | 2.68 | 6.91  | 2.58 | 2.68 | 6.91    | 2.58 | 2.68 |
|              | 0          | 12.91   | 2.26 | 5.70 | 12.91        | 2.26 | 5.70 | 9.81  | 2.45 | 4.01 | 7.28  | 2.69 | 2.71 | 7.28  | 2.69 | 2.71 | 7.28    | 2.69 | 2.71 |
|              | 2          | 13.10   | 2.45 | 5.35 | 13.10        | 2.45 | 5.35 | 9.92  | 2.57 | 3.85 | 7.64  | 2.80 | 2.73 | 7.64  | 2.80 | 2.73 | 7.64    | 2.80 | 2.73 |
|              | 5          | 14.74   | 2.66 | 5.54 | 14.27        | 2.75 | 5.19 | 10.82 | 3.05 | 3.55 | 9.12  | 3.21 | 2.84 | 9.12  | 3.21 | 2.84 | 9.12    | 3.21 | 2.84 |
|              | 7          | 16.00   | 2.81 | 5.70 | 15.00        | 2.85 | 5.26 | 11.22 | 3.17 | 3.54 | 9.75  | 3.39 | 2.88 | 9.75  | 3.39 | 2.88 | 9.75    | 3.39 | 2.88 |
|              | 10         | 16.89   | 3.14 | 5.38 | 14.99        | 3.34 | 4.49 | 11.19 | 3.52 | 3.18 | 9.05  | 3.70 | 2.44 | 9.05  | 3.70 | 2.44 | 9.05    | 3.70 | 2.44 |
|              | 12         | 17.36   | 3.38 | 5.13 | 15.00        | 3.56 | 4.22 | 11.20 | 3.78 | 2.97 | 9.59  | 3.90 | 2.46 | 9.59  | 3.90 | 2.46 | 9.59    | 3.90 | 2.46 |
|              | 15         | 16.14   | 3.53 | 4.58 | 14.95        | 3.61 | 4.14 | 11.29 | 3.85 | 2.93 | 10.06 | 4.14 | 2.43 | 10.06 | 4.14 | 2.43 | 10.06   | 4.14 | 2.43 |
|              | 20         | 14.72   | 3.79 | 3.88 | 14.72        | 3.79 | 3.88 | 11.05 | 4.09 | 2.70 | 7.98  | 4.46 | 1.79 | 7.98  | 4.46 | 1.79 | 7.98    | 4.46 | 1.79 |
|              | 25         | 15.00   | 4.14 | 3.63 | 15.00        | 4.14 | 3.63 | 11.31 | 4.62 | 2.45 | 9.12  | 4.77 | 1.91 | 9.12  | 4.77 | 1.91 | 9.12    | 4.77 | 1.91 |
|              | 30         | 14.09   | 4.75 | 2.97 | 14.09        | 4.75 | 2.97 | 11.10 | 5.06 | 2.19 | 9.92  | 5.19 | 1.91 | 9.92  | 5.19 | 1.91 | 9.92    | 5.19 | 1.91 |
|              | 35         | 12.99   | 5.50 | 2.36 | 12.99        | 5.50 | 2.36 | 11.27 | 5.60 | 2.01 | 10.78 | 5.63 | 1.91 | 10.78 | 5.69 | 1.89 | 10.78   | 5.69 | 1.89 |
| 60           | -25        | /       | /    | /    | /            | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /    | /       | /    | /    |
|              | -20        | 9.37    | 1.45 | 6.45 | 9.37         | 1.45 | 6.45 | 9.37  | 1.45 | 6.45 | 6.26  | 1.62 | 3.87 | 4.92  | 1.68 | 2.92 | 4.92    | 1.68 | 2.92 |
|              | -15        | 10.45   | 1.67 | 6.24 | 10.45        | 1.67 | 6.24 | 9.37  | 1.73 | 5.42 | 6.16  | 1.85 | 3.33 | 5.70  | 1.88 | 3.03 | 5.70    | 1.88 | 3.03 |
|              | -10        | 10.85   | 1.75 | 6.20 | 10.85        | 1.79 | 6.07 | 9.49  | 1.87 | 5.08 | 6.39  | 2.12 | 3.01 | 6.39  | 2.12 | 3.01 | 6.39    | 2.12 | 3.01 |
|              | -7         | 11.65   | 1.84 | 6.32 | 11.65        | 1.88 | 6.19 | 9.49  | 1.96 | 4.85 | 7.54  | 2.17 | 3.47 | 7.54  | 2.17 | 3.47 | 7.54    | 2.17 | 3.47 |
|              | -5         | 11.39   | 1.87 | 6.10 | 11.39        | 1.91 | 5.98 | 9.32  | 2.02 | 4.61 | 7.08  | 2.25 | 3.15 | 7.08  | 2.25 | 3.15 | 7.08    | 2.25 | 3.15 |
|              | -2         | 11.54   | 1.97 | 5.86 | 11.54        | 2.01 | 5.74 | 9.46  | 2.08 | 4.56 | 7.12  | 2.31 | 3.08 | 7.12  | 2.31 | 3.08 | 7.12    | 2.31 | 3.08 |
|              | 0          | 11.91   | 2.04 | 5.83 | 11.91        | 2.08 | 5.72 | 9.74  | 2.17 | 4.48 | 7.52  | 2.41 | 3.12 | 7.52  | 2.41 | 3.12 | 7.52    | 2.41 | 3.12 |
|              | 2          | 12.18   | 2.14 | 5.70 | 12.18        | 2.18 | 5.59 | 9.85  | 2.27 | 4.33 | 7.89  | 2.51 | 3.15 | 7.89  | 2.51 | 3.15 | 7.89    | 2.51 | 3.15 |
|              | 5          | 14.19   | 2.49 | 5.70 | 14.19        | 2.49 | 5.70 | 10.75 | 2.67 | 4.03 | 8.91  | 2.87 | 3.10 | 8.91  | 2.87 | 3.10 | 8.91    | 2.87 | 3.10 |
|              | 7          | 15.59   | 2.63 | 5.93 | 15.10        | 2.68 | 5.64 | 11.10 | 2.88 | 3.85 | 9.55  | 3.04 | 3.14 | 9.55  | 3.04 | 3.14 | 9.55    | 3.04 | 3.14 |
|              | 10         | 16.34   | 2.82 | 5.79 | 15.00        | 2.94 | 5.11 | 11.25 | 3.13 | 3.59 | 9.12  | 3.29 | 2.77 | 9.12  | 3.29 | 2.77 | 9.12    | 3.29 | 2.77 |
|              | 12         | 16.78   | 3.05 | 5.50 | 14.97        | 3.17 | 4.72 | 11.17 | 3.35 | 3.34 | 9.66  | 3.45 | 2.80 | 9.66  | 3.45 | 2.80 | 9.66    | 3.45 | 2.80 |
|              | 15         | 15.61   | 3.14 | 4.97 | 15.04        | 3.20 | 4.69 | 11.26 | 3.43 | 3.29 | 10.19 | 3.68 | 2.77 | 10.19 | 3.68 | 2.77 | 10.19   | 3.68 | 2.77 |
|              | 20         | 14.03   | 3.35 | 4.18 | 14.03        | 3.35 | 4.18 | 11.18 | 3.60 | 3.10 | 8.16  | 3.96 | 2.06 | 8.16  | 3.96 | 2.06 | 8.16    | 3.96 | 2.06 |
|              | 25         | 14.67   | 3.73 | 3.93 | 14.67        | 3.73 | 3.93 | 11.25 | 4.09 | 2.75 | 9.40  | 4.25 | 2.21 | 9.40  | 4.25 | 2.21 | 9.40    | 4.25 | 2.21 |
|              | 30         | 13.75   | 4.24 | 3.24 | 13.75        | 4.24 | 3.24 | 11.31 | 4.50 | 2.51 | 10.20 | 4.59 | 2.22 | 10.20 | 4.59 | 2.22 | 10.20   | 4.59 | 2.22 |
|              | 35         | /       | /    | /    | /            | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /    | /       | /    | /    |
| 65           | -25        | /       | /    | /    | /            | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /    | /       | /    | /    |
|              | -20        | 8.66    | 1.39 | 6.21 | 8.66         | 1.39 | 6.21 | 8.66  | 1.39 | 6.21 | 6.29  | 1.41 | 4.47 | 4.66  | 1.51 | 3.08 | 4.66    | 1.51 | 3.08 |
|              | -15        | 9.67    | 1.52 | 6.36 | 9.67         | 1.52 | 6.36 | 9.20  | 1.55 | 5.93 | 6.22  | 1.73 | 3.60 | 5.64  | 1.79 | 3.15 | 5.64    | 1.79 | 3.15 |
|              | -10        | 10.37   | 1.72 | 6.02 | 10.37        | 1.72 | 6.02 | 9.44  | 1.76 | 5.37 | 6.46  | 2.01 | 3.22 | 6.46  | 2.01 | 3.22 | 6.46    | 2.01 | 3.22 |
|              | -7         | 10.85   | 1.79 | 6.05 | 10.85        | 1.79 | 6.05 | 9.40  | 1.82 | 5.16 | 6.68  | 2.02 | 3.31 | 6.68  | 2.02 | 3.31 | 6.68    | 2.02 | 3.31 |
|              | -5         | 10.83   | 1.81 | 5.98 | 10.83        | 1.81 | 5.98 | 9.43  | 1.90 | 4.96 | 6.94  | 2.06 | 3.37 | 6.94  | 2.06 | 3.37 | 6.94    | 2.06 | 3.37 |
|              | -2         | 10.89   | 1.90 | 5.72 | 10.89        | 1.90 | 5.72 | 9.52  | 1.94 | 4.92 | 6.90  | 2.10 | 3.29 | 6.90  | 2.10 | 3.29 | 6.90    | 2.10 | 3.29 |
|              | 0          | 11.29   | 1.94 | 5.83 | 11.29        | 1.94 | 5.83 | 9.67  | 2.00 | 4.84 | 7.22  | 2.16 | 3.34 | 7.22  | 2.16 | 3.34 | 7.22    | 2.16 | 3.34 |
|              | 2          | 11.32   | 2.01 | 5.63 | 11.32        | 2.01 | 5.63 | 9.85  | 2.08 | 4.74 | 7.57  | 2.24 | 3.38 | 7.57  | 2.24 | 3.38 | 7.57    | 2.24 | 3.38 |
|              | 5          | 13.21   | 2.36 | 5.60 | 13.21        | 2.36 | 5.60 | 10.62 | 2.42 | 4.38 | 8.52  | 2.56 | 3.32 | 8.52  | 2.56 | 3.32 | 8.52    | 2.56 | 3.32 |
|              | 7          | 14.71   | 2.43 | 6.06 | 14.71        | 2.43 | 6.06 | 11.39 | 2.55 | 4.46 | 9.06  | 2.79 | 3.25 | 9.06  | 2.79 | 3.25 | 9.06    | 2.79 | 3.25 |
|              | 10         | 15.00   | 2.61 | 5.75 | 15.00        | 2.61 | 5.75 | 11.20 | 2.70 | 4.14 | 8.74  | 2.92 | 2.99 | 8.74  | 2.92 | 2.99 | 8.74    | 2.92 | 2.99 |
|              | 12         | 15.35   | 2.72 | 5.65 | 15.17        | 2.74 | 5.53 | 11.26 | 2.90 | 3.88 | 9.28  | 3.07 | 3.03 | 9.28  | 3.07 | 3.03 | 9.28    | 3.07 | 3.03 |
|              | 15         | 15.04   | 2.81 | 5.36 | 15.04        | 2.81 | 5.36 | 11.17 | 3.03 | 3.69 | 9.84  | 3.28 | 3.00 | 9.84  | 3.28 | 3.00 | 9.84    | 3.28 | 3.00 |
|              | 20         | 13.09   | 3.06 | 4.27 | 13.09        | 3.06 | 4.27 | 11.25 | 3.20 | 3.52 | 7.87  | 3.53 | 2.23 | 7.87  | 3.53 | 2.23 | 7.87    | 3.53 | 2.23 |
|              | 25         | 14.27   | 3.31 | 4.32 | 14.27        | 3.31 | 4.32 | 11.13 | 3.63 | 3.07 | 9.14  | 3.78 | 2.42 | 9.14  | 3.78 | 2.42 | 9.14    | 3.78 | 2.42 |
|              | 30         | 13.37   | 3.79 | 3.52 | 13.37        | 3.79 | 3.52 | 11.21 | 3.98 | 2.82 | 9.92  | 4.07 | 2.44 | 9.92  | 4.07 | 2.44 | 9.92    | 4.07 | 2.44 |
|              | 35         | /       | /    | /    | /            | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /    | /       | /    | /    |
| 70           | -25        | /       | /    | /    | /            | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /    | /       | /    | /    |
|              | -20        | /       | /    | /    | /            | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /    | /       | /    | /    |
|              | -15        | 9.17    | 1.53 | 5.99 | 9.17         | 1.53 | 5.99 | 9.17  | 1.53 | 5.99 | 6.35  | 1.61 | 3.93 | 5.47  | 1.65 | 3.32 | 5.47    | 1.65 | 3.32 |
|              | -10        | 9.99    | 1.67 | 5.98 | 9.99         | 1.67 | 5.98 | 9.96  | 1.69 | 5.88 | 6.50  | 1.92 | 3.40 | 6.50  | 1.92 | 3.40 | 6.50    | 1.92 | 3.40 |
|              | -7         | 10.61   | 1.73 | 6.14 | 10.61        | 1.73 | 6.14 | 9.57  | 1.78 | 5.38 | 7.04  | 1.96 | 3.59 | 6.90  | 1.96 | 3.52 | 6.90    | 1.96 | 3.52 |
|              | -5         | 10.38   | 1.75 | 5.92 | 10.38        | 1.75 | 5.92 | 9.59  | 1.79 | 5.36 | 7.13  | 1.97 | 3.61 | 7.13  | 1.97 | 3.61 | 7.13    | 1.97 | 3.61 |
|              | -2         | 10.45   | 1.79 | 5.84 | 10.45        | 1.79 | 5.84 | 9.64  | 1.84 |      |       |      |      |       |      |      |         |      |      |

9. TABELLE PRESTAZIONALI - MODO RAFFREDDAMENTO

9.1 Prestazioni raffreddamento Mod. 04

| Tw_out<br>°C | DBWB<br>°C | Maximum |       |      | 100%(Normal) |       |      | 75%   |       |      | 50%   |       |      | 25%   |       |      | Minimum |       |      |
|--------------|------------|---------|-------|------|--------------|-------|------|-------|-------|------|-------|-------|------|-------|-------|------|---------|-------|------|
|              |            | CC      | EER   | PI   | CC           | EER   | PI   | CC    | EER   | PI   | CC    | EER   | PI   | CC    | EER   | PI   | CC      | EER   | PI   |
| 5            | -5         | /       | /     | /    | /            | /     | /    | /     | /     | /    | /     | /     | /    | /     | /     | /    | /       | /     | /    |
|              | 0          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /     | /     | /    | /     | /     | /    | /       | /     | /    |
|              | 5          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /     | /     | /    | /     | /     | /    | /       | /     | /    |
|              | 10         | /       | /     | /    | /            | /     | /    | /     | /     | /    | /     | /     | /    | /     | /     | /    | /       | /     | /    |
|              | 15         | /       | /     | /    | /            | /     | /    | /     | /     | /    | /     | /     | /    | /     | /     | /    | /       | /     | /    |
|              | 19         | 4.48    | 5.70  | 0.79 | 4.48         | 5.70  | 0.79 | 3.49  | 6.09  | 0.57 | 2.53  | 6.25  | 0.40 | 2.53  | 6.25  | 0.40 | 2.53    | 6.25  | 0.40 |
|              | 20         | 4.53    | 5.50  | 0.82 | 4.53         | 5.50  | 0.82 | 3.53  | 5.90  | 0.60 | 2.60  | 6.15  | 0.42 | 2.50  | 6.20  | 0.40 | 2.50    | 6.20  | 0.40 |
|              | 25         | 4.96    | 4.59  | 1.08 | 4.69         | 4.68  | 1.00 | 3.55  | 5.02  | 0.71 | 2.45  | 5.29  | 0.46 | 2.38  | 5.32  | 0.45 | 2.38    | 5.32  | 0.45 |
|              | 30         | 5.23    | 3.92  | 1.33 | 4.68         | 4.09  | 1.15 | 3.49  | 4.38  | 0.80 | 2.31  | 4.52  | 0.51 | 2.24  | 4.55  | 0.49 | 2.24    | 4.55  | 0.49 |
|              | 35         | 5.31    | 3.38  | 1.57 | 4.68         | 3.52  | 1.33 | 3.54  | 3.70  | 0.96 | 2.35  | 3.85  | 0.61 | 2.14  | 3.88  | 0.55 | 2.14    | 3.88  | 0.55 |
|              | 40         | 4.99    | 2.96  | 1.69 | 4.70         | 3.02  | 1.56 | 3.50  | 3.26  | 1.07 | 2.35  | 3.38  | 0.70 | 1.99  | 3.45  | 0.58 | 1.99    | 3.45  | 0.58 |
|              | 43         | 4.80    | 2.71  | 1.77 | 4.66         | 2.75  | 1.69 | 3.52  | 2.98  | 1.18 | 2.34  | 3.23  | 0.72 | 1.92  | 3.31  | 0.58 | 1.92    | 3.31  | 0.58 |
| 46           | 4.49       | 2.55    | 1.76  | 4.49 | 2.55         | 1.76  | 3.37 | 2.68  | 1.26  | 2.24 | 2.85  | 0.79  | 1.88 | 2.93  | 0.64  | 1.88 | 2.93    | 0.64  |      |
| 7            | -5         | /       | /     | /    | /            | /     | /    | /     | /     | /    | /     | /     | /    | /     | /     | /    | /       | /     | /    |
|              | 0          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /     | /     | /    | /     | /     | /    | /       | /     | /    |
|              | 5          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /     | /     | /    | /     | /     | /    | /       | /     | /    |
|              | 10         | /       | /     | /    | /            | /     | /    | /     | /     | /    | /     | /     | /    | /     | /     | /    | /       | /     | /    |
|              | 15         | /       | /     | /    | /            | /     | /    | /     | /     | /    | /     | /     | /    | /     | /     | /    | /       | /     | /    |
|              | 19         | 4.76    | 6.06  | 0.78 | 4.68         | 6.12  | 0.77 | 3.53  | 6.60  | 0.53 | 2.70  | 6.72  | 0.40 | 2.70  | 6.72  | 0.40 | 2.70    | 6.72  | 0.40 |
|              | 20         | 4.81    | 5.84  | 0.82 | 4.74         | 5.90  | 0.80 | 3.49  | 6.38  | 0.55 | 2.75  | 6.59  | 0.42 | 2.67  | 6.60  | 0.41 | 2.67    | 6.60  | 0.41 |
|              | 25         | 5.28    | 4.85  | 1.09 | 4.72         | 5.06  | 0.93 | 3.51  | 5.39  | 0.65 | 2.62  | 5.72  | 0.46 | 2.54  | 5.73  | 0.44 | 2.54    | 5.73  | 0.44 |
|              | 30         | 5.57    | 4.13  | 1.35 | 4.73         | 4.40  | 1.07 | 3.55  | 4.68  | 0.76 | 2.48  | 4.86  | 0.51 | 2.41  | 4.87  | 0.49 | 2.41    | 4.87  | 0.49 |
|              | 35         | 5.66    | 3.54  | 1.60 | 4.70         | 3.65  | 1.29 | 3.53  | 3.99  | 0.89 | 2.33  | 4.35  | 0.54 | 2.29  | 4.38  | 0.52 | 2.29    | 4.38  | 0.52 |
|              | 40         | 5.33    | 3.10  | 1.72 | 4.71         | 3.24  | 1.45 | 3.50  | 3.49  | 1.00 | 2.36  | 3.65  | 0.65 | 2.15  | 3.67  | 0.59 | 2.15    | 3.67  | 0.59 |
|              | 43         | 5.13    | 2.83  | 1.81 | 4.68         | 2.95  | 1.59 | 3.53  | 3.15  | 1.12 | 2.35  | 3.29  | 0.71 | 2.06  | 3.31  | 0.62 | 2.06    | 3.31  | 0.62 |
| 46           | 4.79       | 2.67    | 1.80  | 4.79 | 2.67         | 1.80  | 3.47 | 2.91  | 1.19  | 2.33 | 3.01  | 0.78  | 2.03 | 3.03  | 0.67  | 2.03 | 3.03    | 0.67  |      |
| 10           | -5         | /       | /     | /    | /            | /     | /    | /     | /     | /    | /     | /     | /    | /     | /     | /    | /       | /     | /    |
|              | 0          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /     | /     | /    | /     | /     | /    | /       | /     | /    |
|              | 5          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /     | /     | /    | /     | /     | /    | /       | /     | /    |
|              | 10         | /       | /     | /    | /            | /     | /    | /     | /     | /    | /     | /     | /    | /     | /     | /    | /       | /     | /    |
|              | 15         | /       | /     | /    | /            | /     | /    | /     | /     | /    | /     | /     | /    | /     | /     | /    | /       | /     | /    |
|              | 19         | 4.75    | 7.31  | 0.65 | 4.63         | 7.58  | 0.61 | 3.52  | 7.91  | 0.44 | 3.08  | 8.04  | 0.38 | 3.08  | 8.04  | 0.38 | 3.08    | 8.04  | 0.38 |
|              | 20         | 5.23    | 6.53  | 0.80 | 4.61         | 6.69  | 0.69 | 3.44  | 6.98  | 0.49 | 3.07  | 7.31  | 0.42 | 2.99  | 7.34  | 0.41 | 2.99    | 7.34  | 0.41 |
|              | 25         | 5.30    | 6.33  | 0.84 | 4.59         | 6.40  | 0.72 | 3.52  | 6.72  | 0.52 | 3.04  | 7.08  | 0.43 | 2.95  | 7.10  | 0.42 | 2.95    | 7.10  | 0.42 |
|              | 30         | 5.81    | 5.21  | 1.12 | 4.58         | 5.44  | 0.84 | 3.45  | 5.79  | 0.59 | 2.89  | 6.16  | 0.47 | 2.81  | 6.17  | 0.46 | 2.81    | 6.17  | 0.46 |
|              | 35         | 6.13    | 4.46  | 1.37 | 4.62         | 4.70  | 0.98 | 3.51  | 5.10  | 0.69 | 2.75  | 5.38  | 0.51 | 2.67  | 5.39  | 0.50 | 2.67    | 5.39  | 0.50 |
|              | 40         | 6.22    | 3.80  | 1.64 | 4.59         | 4.13  | 1.11 | 3.51  | 4.33  | 0.81 | 2.59  | 4.53  | 0.57 | 2.51  | 4.55  | 0.55 | 2.51    | 4.55  | 0.55 |
|              | 43         | 5.86    | 3.32  | 1.77 | 4.59         | 3.57  | 1.28 | 3.50  | 3.79  | 0.92 | 2.43  | 4.03  | 0.60 | 2.39  | 4.05  | 0.59 | 2.39    | 4.05  | 0.59 |
| 46           | 5.63       | 3.05    | 1.85  | 4.67 | 3.28         | 1.42  | 3.46 | 3.51  | 0.99  | 2.34 | 3.67  | 0.64  | 2.30 | 3.68  | 0.62  | 2.30 | 3.68    | 0.62  |      |
| 46           | 5.27       | 2.88    | 1.83  | 4.65 | 3.00         | 1.55  | 3.51 | 3.20  | 1.10  | 2.34 | 3.35  | 0.70  | 2.25 | 3.37  | 0.67  | 2.25 | 3.37    | 0.67  |      |
| 11           | -5         | /       | /     | /    | /            | /     | /    | /     | /     | /    | /     | /     | /    | /     | /     | /    | /       | /     | /    |
|              | 0          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /     | /     | /    | /     | /     | /    | /       | /     | /    |
|              | 5          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /     | /     | /    | /     | /     | /    | /       | /     | /    |
|              | 10         | 4.40    | 8.40  | 0.52 | 4.37         | 8.50  | 0.51 | 4.01  | 8.80  | 0.46 | 4.01  | 8.80  | 0.46 | 4.01  | 8.80  | 0.46 | 4.01    | 8.80  | 0.46 |
|              | 15         | 4.89    | 7.51  | 0.65 | 4.58         | 7.68  | 0.60 | 3.52  | 8.00  | 0.44 | 3.17  | 8.18  | 0.39 | 3.17  | 8.18  | 0.39 | 3.17    | 8.18  | 0.39 |
|              | 19         | 5.40    | 6.78  | 0.80 | 4.56         | 7.22  | 0.63 | 3.44  | 7.54  | 0.46 | 3.17  | 7.74  | 0.41 | 3.09  | 7.76  | 0.40 | 3.09    | 7.76  | 0.40 |
|              | 20         | 5.46    | 6.42  | 0.85 | 4.62         | 6.89  | 0.67 | 3.52  | 7.24  | 0.49 | 3.14  | 7.34  | 0.43 | 3.05  | 7.35  | 0.42 | 3.05    | 7.35  | 0.42 |
|              | 25         | 5.99    | 5.42  | 1.11 | 4.61         | 5.78  | 0.80 | 3.45  | 6.20  | 0.56 | 2.99  | 6.23  | 0.48 | 2.90  | 6.29  | 0.46 | 2.90    | 6.29  | 0.46 |
|              | 30         | 6.32    | 4.57  | 1.38 | 4.66         | 4.92  | 0.95 | 3.52  | 5.25  | 0.67 | 2.84  | 5.56  | 0.51 | 2.76  | 5.59  | 0.49 | 2.76    | 5.59  | 0.49 |
|              | 35         | 6.41    | 3.89  | 1.65 | 4.64         | 4.31  | 1.08 | 3.43  | 4.65  | 0.74 | 2.67  | 4.78  | 0.56 | 2.60  | 4.79  | 0.54 | 2.60    | 4.79  | 0.54 |
|              | 40         | 6.04    | 3.39  | 1.78 | 4.56         | 3.80  | 1.20 | 3.43  | 4.03  | 0.85 | 2.52  | 4.17  | 0.60 | 2.47  | 4.20  | 0.59 | 2.47    | 4.20  | 0.59 |
|              | 43         | 5.82    | 3.12  | 1.86 | 4.65         | 3.41  | 1.36 | 3.50  | 3.64  | 0.96 | 2.42  | 3.80  | 0.64 | 2.37  | 3.82  | 0.62 | 2.37    | 3.82  | 0.62 |
| 46           | 5.44       | 2.92    | 1.86  | 4.63 | 3.11         | 1.49  | 3.45 | 3.31  | 1.04  | 2.33 | 3.45  | 0.68  | 2.33 | 3.45  | 0.68  | 2.33 | 3.45    | 0.68  |      |
| -5           | 5.34       | 11.68   | 0.46  | 5.34 | 11.68        | 0.46  | 5.11 | 11.85 | 0.43  | 5.11 | 11.85 | 0.43  | 5.11 | 11.85 | 0.43  | 5.11 | 11.85   | 0.43  |      |
| 15           | 0          | 5.27    | 11.35 | 0.46 | 5.27         | 11.35 | 0.46 | 4.70  | 11.60 | 0.41 | 4.70  | 11.60 | 0.41 | 4.70  | 11.60 | 0.41 | 4.70    | 11.60 | 0.41 |
|              | 5          | 5.36    | 10.88 | 0.49 | 5.36         | 10.88 | 0.49 | 4.60  | 11.03 | 0.42 | 4.60  | 11.03 | 0.42 | 4.60  | 11.03 | 0.42 | 4.60    | 11.03 | 0.42 |
|              | 10         | 5.16    | 10.44 | 0.49 | 5.16         | 10.44 | 0.49 | 4.46  | 10.66 | 0.42 | 4.46  | 10.66 | 0.42 | 4.46  | 10.66 | 0.42 | 4.46    | 10.66 | 0.42 |
|              | 15         | 5.39    | 9.21  | 0.59 | 4.60         | 9.34  | 0.49 | 3.62  | 9.69  | 0.37 | 3.50  | 9.71  | 0.36 | 3.50  | 9.71  | 0.36 | 3.50    | 9.71  | 0.36 |
|              | 19         | 5.96    | 8.12  | 0.73 | 4.53         | 8.29  | 0.55 | 3.55  | 8.66  | 0.41 | 3.42  | 8.68  | 0.39 | 3.42  | 8.68  | 0.39 | 3.42    | 8.68  | 0.39 |
|              | 20         | 6.03    | 7.42  | 0.81 | 4.49         | 7.61  | 0.59 | 3.51  | 7.94  | 0.44 | 3.37  | 7.96  | 0.42 | 3.37  | 7.96  | 0.42 | 3.37    | 7.96  | 0.42 |
|              | 25         | 6.75    | 6.01  | 1.12 | 4.50         | 6.60  | 0.68 | 3.45  | 6.96  | 0.50 | 3.28  | 7.09  | 0.46 | 3.28  | 7.09  | 0.46 | 3.28    | 7.09  | 0.46 |
|              | 30         | 7.13    | 5.04  | 1.41 | 4.56         | 5.78  | 0.79 | 3.41  | 6.20  | 0.55 | 3.11  | 6.29  | 0.49 | 3.05  | 6.31  | 0.48 | 3.05    | 6.31  | 0.48 |
|              | 35         | 7.18    | 4.24  | 1.70 | 4.57         | 4.93  | 0.93 | 3.43  | 5.29  | 0.65 | 3.04  | 5.47  | 0.56 | 2.98  | 5.50  | 0.54 | 2.98    | 5.50  | 0.54 |
|              | 40         | 6.79    | 3.69  | 1.84 | 4.54         | 4.27  | 1.06 | 3.47  | 4.60  | 0.76 | 2.87  | 4.64  | 0.62 | 2.80  | 4.65  | 0.60 | 2.80    | 4.65  | 0.60 |
|              | 43         | 6.51    | 3.33  | 1.96 | 4.58         | 3.90  | 1.17 | 3.47  | 4.18  | 0.83 | 2.77  | 4.21  | 0.66 | 2.71  | 4.22  | 0.64 | 2.71    | 4.22  |      |

## 9.2 Prestazioni raffreddamento Mod. 06

| Tw_out<br>°C | DBWB<br>°C | Maximum |       |      | 100%(Normal) |       |      | 75%  |       |      | 50%  |       |      | 25%  |       |      | Minimum |       |      |
|--------------|------------|---------|-------|------|--------------|-------|------|------|-------|------|------|-------|------|------|-------|------|---------|-------|------|
|              |            | CC      | EER   | PI   | CC           | EER   | PI   | CC   | EER   | PI   | CC   | EER   | PI   | CC   | EER   | PI   | CC      | EER   | PI   |
| 5            | -5         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 0          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 15         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 19         | 5.40    | 5.30  | 1.02 | 5.40         | 5.30  | 1.02 | 4.05 | 5.71  | 0.71 | 2.70 | 6.22  | 0.43 | 2.53 | 6.25  | 0.40 | 2.53    | 6.25  | 0.40 |
|              | 20         | 5.54    | 5.12  | 1.08 | 5.54         | 5.12  | 1.08 | 4.15 | 5.53  | 0.75 | 2.77 | 6.10  | 0.45 | 2.50 | 6.20  | 0.40 | 2.50    | 6.20  | 0.40 |
|              | 25         | 6.10    | 4.22  | 1.45 | 6.10         | 4.22  | 1.45 | 4.57 | 4.69  | 0.98 | 3.05 | 5.15  | 0.59 | 2.38 | 5.32  | 0.45 | 2.38    | 5.32  | 0.45 |
|              | 30         | 6.75    | 3.48  | 1.94 | 6.75         | 3.48  | 1.94 | 5.09 | 3.83  | 1.33 | 3.40 | 4.39  | 0.77 | 2.24 | 4.55  | 0.49 | 2.24    | 4.55  | 0.49 |
|              | 35         | 6.77    | 2.94  | 2.30 | 6.77         | 2.94  | 2.30 | 5.04 | 3.30  | 1.53 | 3.40 | 3.68  | 0.92 | 2.14 | 3.88  | 0.55 | 2.14    | 3.88  | 0.55 |
|              | 40         | 5.44    | 2.80  | 1.95 | 5.44         | 2.80  | 1.95 | 4.08 | 3.06  | 1.33 | 2.72 | 3.32  | 0.82 | 1.99 | 3.45  | 0.58 | 1.99    | 3.45  | 0.58 |
| 7            | 43         | 4.86    | 2.70  | 1.80 | 4.86         | 2.70  | 1.80 | 3.64 | 2.90  | 1.25 | 2.43 | 3.20  | 0.76 | 1.92 | 3.31  | 0.58 | 1.92    | 3.31  | 0.58 |
|              | 46         | 4.49    | 2.55  | 1.76 | 4.49         | 2.55  | 1.76 | 3.37 | 2.68  | 1.26 | 2.24 | 2.85  | 0.79 | 1.88 | 2.93  | 0.64 | 1.88    | 2.93  | 0.64 |
|              | -5         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 0          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 15         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 19         | 5.76    | 5.68  | 1.01 | 5.76         | 5.68  | 1.01 | 4.32 | 6.13  | 0.70 | 2.88 | 6.65  | 0.43 | 2.70 | 6.72  | 0.40 | 2.70    | 6.72  | 0.40 |
|              | 20         | 5.90    | 5.47  | 1.08 | 5.90         | 5.47  | 1.08 | 4.43 | 5.91  | 0.75 | 2.95 | 6.41  | 0.46 | 2.67 | 6.60  | 0.41 | 2.67    | 6.60  | 0.41 |
|              | 25         | 6.50    | 4.52  | 1.44 | 6.50         | 4.52  | 1.44 | 4.88 | 4.93  | 0.99 | 3.25 | 5.42  | 0.60 | 2.54 | 5.73  | 0.44 | 2.54    | 5.73  | 0.44 |
|              | 30         | 7.11    | 3.69  | 1.93 | 6.75         | 3.77  | 1.79 | 5.09 | 4.14  | 1.23 | 3.37 | 4.69  | 0.72 | 2.41 | 4.87  | 0.49 | 2.41    | 4.87  | 0.49 |
| 10           | 35         | 7.14    | 3.01  | 2.38 | 6.80         | 3.10  | 2.19 | 5.05 | 3.55  | 1.42 | 3.38 | 4.01  | 0.84 | 2.29 | 4.38  | 0.52 | 2.29    | 4.38  | 0.52 |
|              | 40         | 5.82    | 2.94  | 1.98 | 5.82         | 2.94  | 1.98 | 4.36 | 3.26  | 1.34 | 2.91 | 3.61  | 0.81 | 2.15 | 3.67  | 0.59 | 2.15    | 3.67  | 0.59 |
|              | 43         | 5.19    | 2.82  | 1.84 | 5.19         | 2.82  | 1.84 | 3.89 | 3.07  | 1.27 | 2.59 | 3.27  | 0.79 | 2.06 | 3.31  | 0.62 | 2.06    | 3.31  | 0.62 |
|              | 46         | 4.79    | 2.67  | 1.80 | 4.79         | 2.67  | 1.80 | 3.59 | 2.89  | 1.24 | 2.33 | 3.01  | 0.78 | 2.03 | 3.03  | 0.67 | 2.03    | 3.03  | 0.67 |
|              | -5         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 0          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 15         | 5.72    | 7.05  | 0.81 | 5.72         | 7.05  | 0.81 | 4.33 | 7.60  | 0.57 | 3.08 | 8.04  | 0.38 | 3.08 | 8.04  | 0.38 | 3.08    | 8.04  | 0.38 |
|              | 19         | 6.31    | 6.17  | 1.02 | 6.31         | 6.17  | 1.02 | 4.73 | 6.65  | 0.71 | 3.20 | 7.30  | 0.44 | 2.99 | 7.34  | 0.41 | 2.99    | 7.34  | 0.41 |
|              | 20         | 6.47    | 5.94  | 1.09 | 6.47         | 5.94  | 1.09 | 4.89 | 6.37  | 0.77 | 3.26 | 7.05  | 0.46 | 2.95 | 7.10  | 0.42 | 2.95    | 7.10  | 0.42 |
| 11           | 25         | 7.16    | 4.80  | 1.49 | 6.64         | 4.90  | 1.36 | 4.96 | 5.42  | 0.92 | 3.40 | 5.99  | 0.57 | 2.81 | 6.17  | 0.46 | 2.81    | 6.17  | 0.46 |
|              | 30         | 7.80    | 3.96  | 1.97 | 6.75         | 4.17  | 1.62 | 5.04 | 4.67  | 1.08 | 3.36 | 5.23  | 0.64 | 2.67 | 5.39  | 0.50 | 2.67    | 5.39  | 0.50 |
|              | 35         | 7.80    | 3.27  | 2.39 | 6.65         | 3.58  | 1.86 | 5.05 | 3.97  | 1.27 | 3.37 | 4.35  | 0.77 | 2.51 | 4.55  | 0.55 | 2.51    | 4.55  | 0.55 |
|              | 40         | 6.40    | 3.17  | 2.02 | 6.40         | 3.17  | 2.02 | 4.80 | 3.48  | 1.38 | 3.20 | 3.81  | 0.84 | 2.39 | 4.05  | 0.59 | 2.39    | 4.05  | 0.59 |
|              | 43         | 5.70    | 3.01  | 1.89 | 5.70         | 3.01  | 1.89 | 4.26 | 3.32  | 1.28 | 2.81 | 3.55  | 0.79 | 2.30 | 3.68  | 0.62 | 2.30    | 3.68  | 0.62 |
|              | 46         | 5.27    | 2.88  | 1.83 | 5.27         | 2.88  | 1.83 | 3.96 | 3.15  | 1.25 | 2.64 | 3.22  | 0.82 | 2.25 | 3.37  | 0.67 | 2.25    | 3.37  | 0.67 |
|              | -5         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 0          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 15         | 4.94    | 8.30  | 0.59 | 4.94         | 8.30  | 0.59 | 4.01 | 8.80  | 0.46 | 4.01 | 8.80  | 0.46 | 4.01 | 8.80  | 0.46 | 4.01    | 8.80  | 0.46 |
| 15           | 19         | 5.89    | 7.19  | 0.82 | 5.89         | 7.19  | 0.82 | 4.42 | 7.73  | 0.57 | 3.17 | 8.18  | 0.39 | 3.17 | 8.18  | 0.39 | 3.17    | 8.18  | 0.39 |
|              | 20         | 6.50    | 6.30  | 1.03 | 6.50         | 6.60  | 0.98 | 4.87 | 7.14  | 0.68 | 3.30 | 7.71  | 0.43 | 3.09 | 7.76  | 0.40 | 3.09    | 7.76  | 0.40 |
|              | 25         | 6.68    | 5.93  | 1.13 | 6.68         | 6.23  | 1.07 | 5.05 | 6.73  | 0.75 | 3.38 | 7.27  | 0.46 | 3.05 | 7.35  | 0.42 | 3.05    | 7.35  | 0.42 |
|              | 30         | 7.39    | 4.87  | 1.52 | 6.65         | 5.26  | 1.26 | 5.02 | 5.65  | 0.89 | 3.31 | 6.23  | 0.53 | 2.90 | 6.29  | 0.46 | 2.90    | 6.29  | 0.46 |
|              | 35         | 7.98    | 4.01  | 1.99 | 6.64         | 4.31  | 1.54 | 5.01 | 4.87  | 1.03 | 3.37 | 5.44  | 0.62 | 2.76 | 5.59  | 0.49 | 2.76    | 5.59  | 0.49 |
|              | 40         | 8.04    | 3.30  | 2.44 | 6.67         | 3.71  | 1.80 | 5.06 | 4.15  | 1.22 | 3.39 | 4.70  | 0.72 | 2.60 | 4.79  | 0.54 | 2.60    | 4.79  | 0.54 |
|              | 43         | 6.61    | 3.21  | 2.06 | 6.61         | 3.21  | 2.06 | 4.95 | 3.60  | 1.37 | 3.30 | 4.05  | 0.82 | 2.47 | 4.20  | 0.59 | 2.47    | 4.20  | 0.59 |
|              | 46         | 5.88    | 3.05  | 1.93 | 5.88         | 3.05  | 1.93 | 4.41 | 3.45  | 1.28 | 2.94 | 3.70  | 0.79 | 2.37 | 3.82  | 0.62 | 2.37    | 3.82  | 0.62 |
|              | 40         | 5.44    | 2.92  | 1.86 | 5.44         | 2.92  | 1.86 | 4.08 | 3.27  | 1.25 | 2.72 | 3.41  | 0.80 | 2.33 | 3.45  | 0.68 | 2.33    | 3.45  | 0.68 |
|              | -5         | 5.89    | 11.26 | 0.52 | 5.89         | 11.26 | 0.52 | 5.11 | 11.85 | 0.43 | 5.11 | 11.85 | 0.43 | 5.11 | 11.85 | 0.43 | 5.11    | 11.85 | 0.43 |
|              | 0          | 5.78    | 11.00 | 0.53 | 5.78         | 11.00 | 0.53 | 5.00 | 11.56 | 0.43 | 4.70 | 11.60 | 0.41 | 4.70 | 11.60 | 0.41 | 4.70    | 11.60 | 0.41 |
| 18           | 5          | 5.54    | 10.52 | 0.53 | 5.54         | 10.52 | 0.53 | 4.80 | 11.00 | 0.44 | 4.60 | 11.03 | 0.42 | 4.60 | 11.03 | 0.42 | 4.60    | 11.03 | 0.42 |
|              | 10         | 5.40    | 10.18 | 0.53 | 5.40         | 10.18 | 0.53 | 4.46 | 10.66 | 0.42 | 4.46 | 10.66 | 0.42 | 4.46 | 10.66 | 0.42 | 4.46    | 10.66 | 0.42 |
|              | 15         | 6.45    | 8.91  | 0.72 | 6.45         | 8.91  | 0.72 | 4.84 | 9.21  | 0.53 | 3.50 | 9.71  | 0.36 | 3.50 | 9.71  | 0.36 | 3.50    | 9.71  | 0.36 |
|              | 19         | 7.13    | 7.66  | 0.93 | 6.55         | 7.73  | 0.85 | 4.89 | 8.17  | 0.60 | 3.42 | 8.68  | 0.39 | 3.42 | 8.68  | 0.39 | 3.42    | 8.68  | 0.39 |
|              | 20         | 7.32    | 7.00  | 1.05 | 6.49         | 7.28  | 0.89 | 4.96 | 7.52  | 0.66 | 3.37 | 7.96  | 0.42 | 3.37 | 7.96  | 0.42 | 3.37    | 7.96  | 0.42 |
|              | 25         | 8.30    | 5.47  | 1.52 | 6.50         | 6.03  | 1.08 | 4.86 | 6.57  | 0.74 | 3.28 | 7.09  | 0.46 | 3.28 | 7.09  | 0.46 | 3.28    | 7.09  | 0.46 |
|              | 30         | 8.99    | 4.51  | 1.99 | 6.57         | 5.12  | 1.28 | 4.91 | 5.74  | 0.85 | 3.27 | 6.28  | 0.52 | 3.05 | 6.31  | 0.48 | 3.05    | 6.31  | 0.48 |
|              | 35         | 8.99    | 3.70  | 2.43 | 6.52         | 4.42  | 1.48 | 4.92 | 4.91  | 1.00 | 3.31 | 5.40  | 0.61 | 2.98 | 5.50  | 0.54 | 2.98    | 5.50  | 0.54 |
|              | 40         | 7.42    | 3.44  | 2.16 | 6.58         | 3.70  | 1.78 | 4.95 | 4.16  | 1.19 | 3.33 | 4.61  | 0.72 | 2.80 | 4.65  | 0.60 | 2.80    | 4.65  | 0.60 |
|              | 43         | 6.60    | 3.30  | 2.00 | 6.51         | 3.33. |      |      |       |      |      |       |      |      |       |      |         |       |      |

### 9.3 Prestazioni raffreddamento Mod. 08

| Tw_out<br>°C | DBWB<br>°C | Maximum |       |      | 100%(Normal) |       |      | 75%  |       |      | 50%  |       |      | 25%  |       |      | Minimum |       |      |
|--------------|------------|---------|-------|------|--------------|-------|------|------|-------|------|------|-------|------|------|-------|------|---------|-------|------|
|              |            | CC      | EER   | PI   | CC           | EER   | PI   | CC   | EER   | PI   | CC   | EER   | PI   | CC   | EER   | PI   | CC      | EER   | PI   |
| 5            | -5         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 0          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 15         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 19         | 6.76    | 5.39  | 1.26 | 6.76         | 5.39  | 1.26 | 5.07 | 6.11  | 0.83 | 3.50 | 6.81  | 0.51 | 3.50 | 6.81  | 0.51 | 3.50    | 6.81  | 0.51 |
|              | 20         | 6.82    | 5.18  | 1.32 | 6.82         | 5.18  | 1.32 | 5.11 | 5.80  | 0.88 | 3.47 | 6.56  | 0.53 | 3.47 | 6.56  | 0.53 | 3.47    | 6.56  | 0.53 |
|              | 25         | 7.29    | 4.26  | 1.71 | 7.29         | 4.26  | 1.71 | 5.62 | 4.85  | 1.16 | 3.74 | 5.51  | 0.68 | 3.35 | 5.54  | 0.61 | 3.35    | 5.54  | 0.61 |
|              | 30         | 7.67    | 3.57  | 2.15 | 7.48         | 3.63  | 2.06 | 5.61 | 4.40  | 1.28 | 3.72 | 4.82  | 0.77 | 3.21 | 4.92  | 0.65 | 3.21    | 4.92  | 0.65 |
|              | 35         | 7.69    | 3.02  | 2.55 | 7.44         | 3.09  | 2.41 | 5.63 | 3.55  | 1.59 | 3.75 | 4.02  | 0.93 | 3.03 | 4.35  | 0.70 | 3.03    | 4.35  | 0.70 |
|              | 40         | 6.76    | 2.77  | 2.44 | 6.76         | 2.77  | 2.44 | 4.92 | 3.10  | 1.59 | 3.14 | 3.43  | 0.91 | 2.85 | 3.62  | 0.79 | 2.85    | 3.62  | 0.79 |
| 7            | 43         | 6.18    | 2.63  | 2.35 | 6.18         | 2.63  | 2.35 | 4.57 | 2.95  | 1.55 | 3.08 | 3.05  | 1.01 | 2.73 | 3.13  | 0.87 | 2.73    | 3.13  | 0.87 |
|              | 46         | 5.26    | 2.55  | 2.06 | 5.26         | 2.55  | 2.06 | 3.95 | 2.75  | 1.44 | 2.61 | 2.96  | 0.88 | 2.61 | 2.96  | 0.88 | 2.61    | 2.96  | 0.88 |
|              | -5         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 0          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 15         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 19         | 7.18    | 5.73  | 1.25 | 7.18         | 5.73  | 1.25 | 5.50 | 6.60  | 0.83 | 3.75 | 7.42  | 0.51 | 3.75 | 7.42  | 0.51 | 3.75    | 7.42  | 0.51 |
|              | 20         | 7.24    | 5.51  | 1.31 | 7.24         | 5.51  | 1.31 | 5.53 | 6.38  | 0.87 | 3.72 | 7.14  | 0.52 | 3.72 | 7.14  | 0.52 | 3.72    | 7.14  | 0.52 |
|              | 25         | 7.78    | 4.50  | 1.73 | 7.44         | 4.63  | 1.61 | 5.55 | 5.34  | 1.04 | 3.73 | 5.96  | 0.63 | 3.59 | 5.98  | 0.60 | 3.59    | 5.98  | 0.60 |
|              | 30         | 8.17    | 3.76  | 2.17 | 7.48         | 3.99  | 1.87 | 5.56 | 4.51  | 1.24 | 3.71 | 4.93  | 0.75 | 3.44 | 4.98  | 0.69 | 3.44    | 4.98  | 0.69 |
| 10           | 35         | 8.19    | 3.17  | 2.59 | 7.50         | 3.45  | 2.17 | 5.58 | 3.88  | 1.44 | 3.78 | 4.21  | 0.90 | 3.25 | 4.42  | 0.74 | 3.25    | 4.42  | 0.74 |
|              | 40         | 7.23    | 2.92  | 2.48 | 7.23         | 2.92  | 2.48 | 5.58 | 3.30  | 1.69 | 3.69 | 3.66  | 1.01 | 3.06 | 3.80  | 0.80 | 3.06    | 3.80  | 0.80 |
|              | 43         | 6.60    | 2.76  | 2.39 | 6.60         | 2.76  | 2.39 | 4.95 | 3.15  | 1.57 | 3.30 | 3.33  | 0.99 | 2.94 | 3.37  | 0.87 | 2.94    | 3.37  | 0.87 |
|              | 46         | 5.63    | 2.68  | 2.10 | 5.63         | 2.68  | 2.10 | 4.22 | 2.95  | 1.43 | 2.81 | 3.10  | 0.91 | 2.81 | 3.10  | 0.91 | 2.81    | 3.10  | 0.91 |
|              | -5         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 0          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 15         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 19         | 7.20    | 7.00  | 1.03 | 7.20         | 7.00  | 1.03 | 5.40 | 8.00  | 0.68 | 4.19 | 8.85  | 0.47 | 4.19 | 8.85  | 0.47 | 4.19    | 8.85  | 0.47 |
|              | 20         | 7.91    | 6.26  | 1.26 | 7.63         | 6.30  | 1.21 | 5.73 | 7.47  | 0.77 | 4.15 | 8.10  | 0.51 | 4.15 | 8.10  | 0.51 | 4.15    | 8.10  | 0.51 |
| 11           | 25         | 7.97    | 6.06  | 1.32 | 7.71         | 6.11  | 1.26 | 5.68 | 7.00  | 0.81 | 4.12 | 7.25  | 0.57 | 4.12 | 7.25  | 0.57 | 4.12    | 7.25  | 0.57 |
|              | 30         | 8.54    | 4.88  | 1.75 | 7.67         | 5.24  | 1.46 | 5.75 | 6.21  | 0.93 | 3.98 | 6.75  | 0.59 | 3.98 | 6.75  | 0.59 | 3.98    | 6.75  | 0.59 |
|              | 35         | 9.00    | 3.90  | 2.31 | 7.62         | 4.51  | 1.69 | 5.79 | 5.13  | 1.13 | 3.81 | 5.80  | 0.66 | 3.81 | 5.80  | 0.66 | 3.81    | 5.80  | 0.66 |
|              | 40         | 8.98    | 3.34  | 2.69 | 7.67         | 3.79  | 2.03 | 5.73 | 4.38  | 1.31 | 3.90 | 4.82  | 0.81 | 3.61 | 4.96  | 0.73 | 3.61    | 4.96  | 0.73 |
|              | 43         | 7.95    | 3.13  | 2.54 | 7.64         | 3.19  | 2.39 | 5.72 | 3.70  | 1.55 | 3.81 | 4.34  | 0.88 | 3.40 | 4.38  | 0.78 | 3.40    | 4.38  | 0.78 |
|              | 46         | 7.26    | 2.96  | 2.45 | 7.26         | 2.96  | 2.45 | 5.44 | 3.43  | 1.59 | 3.63 | 4.04  | 0.90 | 3.25 | 4.09  | 0.79 | 3.25    | 4.09  | 0.79 |
|              | 46         | 6.20    | 2.89  | 2.14 | 6.20         | 2.89  | 2.14 | 4.65 | 3.12  | 1.49 | 3.12 | 3.61  | 0.86 | 3.12 | 3.61  | 0.86 | 3.12    | 3.61  | 0.86 |
|              | -5         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 0          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 15         | 10      | 6.21  | 8.12 | 0.77         | 5.49  | 8.22 | 0.67 | 5.15  | 9.21 | 0.56 | 5.15  | 9.21 | 0.56 | 5.15  | 9.21 | 0.56    | 5.15  | 9.21 |
| 15           |            | 7.33    | 7.34  | 1.00 | 6.20         | 7.57  | 0.82 | 4.65 | 8.40  | 0.55 | 4.09 | 8.96  | 0.46 | 4.09 | 8.96  | 0.46 | 4.09    | 8.96  | 0.46 |
| 19           |            | 8.04    | 6.28  | 1.28 | 7.10         | 6.70  | 1.06 | 5.33 | 7.92  | 0.67 | 4.06 | 8.48  | 0.48 | 4.06 | 8.48  | 0.48 | 4.06    | 8.48  | 0.48 |
| 20           |            | 8.29    | 6.12  | 1.35 | 7.72         | 6.25  | 1.23 | 5.91 | 7.01  | 0.84 | 4.26 | 7.62  | 0.56 | 4.26 | 7.62  | 0.56 | 4.26    | 7.62  | 0.56 |
| 25           |            | 8.83    | 5.02  | 1.76 | 7.80         | 5.27  | 1.48 | 5.81 | 6.24  | 0.93 | 4.12 | 7.02  | 0.59 | 4.12 | 7.02  | 0.59 | 4.12    | 7.02  | 0.59 |
| 30           |            | 9.25    | 4.03  | 2.29 | 7.74         | 4.48  | 1.73 | 5.85 | 5.19  | 1.13 | 3.94 | 6.02  | 0.66 | 3.94 | 6.02  | 0.66 | 3.94    | 6.02  | 0.66 |
| 35           |            | 9.26    | 3.47  | 2.67 | 7.80         | 3.82  | 2.04 | 5.79 | 4.54  | 1.27 | 3.88 | 5.06  | 0.77 | 3.74 | 5.10  | 0.73 | 3.74    | 5.10  | 0.73 |
| 40           |            | 8.17    | 3.19  | 2.57 | 7.78         | 3.30  | 2.36 | 5.80 | 3.83  | 1.51 | 3.94 | 4.20  | 0.94 | 3.52 | 4.25  | 0.83 | 3.52    | 4.25  | 0.83 |
| 43           |            | 7.50    | 3.03  | 2.47 | 7.50         | 3.03  | 2.47 | 5.80 | 3.46  | 1.67 | 3.94 | 3.92  | 1.00 | 3.38 | 3.96  | 0.85 | 3.38    | 3.96  | 0.85 |
| 46           |            | 7.20    | 2.92  | 2.47 | 7.20         | 2.92  | 2.47 | 5.40 | 3.23  | 1.67 | 3.60 | 3.43  | 1.05 | 3.23 | 3.45  | 0.94 | 3.23    | 3.45  | 0.94 |
| 46           |            | 7.35    | 12.19 | 0.60 | 7.35         | 12.19 | 0.60 | 6.54 | 12.60 | 0.52 | 6.54 | 12.60 | 0.52 | 6.54 | 12.60 | 0.52 | 6.54    | 12.60 | 0.52 |
| 18           | 0          | 6.42    | 10.86 | 0.59 | 6.42         | 10.86 | 0.59 | 5.71 | 11.45 | 0.50 | 5.71 | 11.45 | 0.50 | 5.71 | 11.45 | 0.50 | 5.71    | 11.45 | 0.50 |
|              | 5          | 6.23    | 9.90  | 0.63 | 6.23         | 9.90  | 0.63 | 5.58 | 10.52 | 0.53 | 5.58 | 10.52 | 0.53 | 5.58 | 10.52 | 0.53 | 5.58    | 10.52 | 0.53 |
|              | 10         | 6.48    | 9.12  | 0.71 | 6.26         | 9.20  | 0.68 | 5.14 | 9.85  | 0.53 | 5.14 | 9.85  | 0.53 | 5.14 | 9.85  | 0.53 | 5.14    | 9.85  | 0.53 |
|              | 15         | 8.27    | 7.98  | 1.04 | 8.08         | 8.17  | 0.99 | 6.06 | 9.22  | 0.66 | 4.71 | 9.43  | 0.50 | 4.71 | 9.43  | 0.50 | 4.71    | 9.43  | 0.50 |
|              | 19         | 8.48    | 7.05  | 1.20 | 8.08         | 7.20  | 1.12 | 6.05 | 8.24  | 0.73 | 4.87 | 8.91  | 0.55 | 4.87 | 8.91  | 0.55 | 4.87    | 8.91  | 0.55 |
|              | 20         | 8.71    | 6.45  | 1.35 | 8.08         | 6.76  | 1.20 | 6.06 | 7.78  | 0.78 | 4.84 | 8.40  | 0.58 | 4.84 | 8.40  | 0.58 | 4.84    | 8.40  | 0.58 |
|              | 25         | 9.93    | 5.59  | 1.78 | 8.04         | 6.20  | 1.30 | 6.12 | 7.23  | 0.85 | 4.67 | 7.71  | 0.61 | 4.67 | 7.71  | 0.61 | 4.67    | 7.71  | 0.61 |
|              | 30         | 10.38   | 4.55  | 2.28 | 8.01         | 5.19  | 1.54 | 6.04 | 6.20  | 0.97 | 4.48 | 6.52  | 0.69 | 4.48 | 6.52  | 0.69 | 4.48    | 6.52  | 0.69 |
|              | 35         | 10.45   | 3.79  | 2.76 | 8.03         | 4.60  | 1.75 | 6.05 | 5.24  | 1.15 | 4.27 | 5.62  | 0.76 | 4.27 | 5.62  | 0.76 | 4.27    | 5.62  | 0.76 |
|              | 40         | 9.25    | 3.49  | 2.65 | 8.10         | 3.82  | 2.12 | 6.10 | 4.44  | 1.37 | 4.03 | 4.73  | 0.85 | 4.03 | 4.73  | 0.85 | 4.03    | 4.73  | 0.85 |
|              | 43         | 8.49    | 3.33  | 2.55 | 8.03         | 3.44  | 2    |      |       |      |      |       |      |      |       |      |         |       |      |

## 9.4 Prestazioni raffreddamento Mod. 10

| Tw_out<br>°C | DBWB<br>°C | Maximum |       |      | 100%(Normal) |       |      | 75%  |       |      | 50%  |       |      | 25%  |       |      | Minimum |       |      |
|--------------|------------|---------|-------|------|--------------|-------|------|------|-------|------|------|-------|------|------|-------|------|---------|-------|------|
|              |            | CC      | EER   | PI   | CC           | EER   | PI   | CC   | EER   | PI   | CC   | EER   | PI   | CC   | EER   | PI   | CC      | EER   | PI   |
| 5            | -5         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 0          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 15         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 19         | 7.06    | 5.21  | 1.36 | 7.06         | 5.21  | 1.36 | 5.30 | 6.05  | 0.88 | 3.50 | 6.81  | 0.51 | 3.50 | 6.81  | 0.51 | 3.50    | 6.81  | 0.51 |
|              | 20         | 7.23    | 4.97  | 1.45 | 7.23         | 4.97  | 1.45 | 5.41 | 5.75  | 0.94 | 3.61 | 6.53  | 0.55 | 3.47 | 6.56  | 0.53 | 3.47    | 6.56  | 0.53 |
|              | 25         | 7.81    | 4.09  | 1.91 | 7.81         | 4.09  | 1.91 | 5.86 | 4.78  | 1.22 | 3.88 | 5.48  | 0.71 | 3.35 | 5.54  | 0.61 | 3.35    | 5.54  | 0.61 |
|              | 30         | 8.20    | 3.41  | 2.40 | 8.20         | 3.41  | 2.40 | 6.15 | 4.32  | 1.42 | 4.06 | 4.76  | 0.85 | 3.21 | 4.92  | 0.65 | 3.21    | 4.92  | 0.65 |
|              | 35         | 8.23    | 2.88  | 2.86 | 8.23         | 2.88  | 2.86 | 6.14 | 3.42  | 1.79 | 4.06 | 3.98  | 1.02 | 3.03 | 4.35  | 0.70 | 3.03    | 4.35  | 0.70 |
| 7            | 40         | 6.76    | 2.77  | 2.44 | 6.76         | 2.77  | 2.44 | 5.02 | 3.06  | 1.64 | 3.32 | 3.42  | 0.97 | 2.85 | 3.62  | 0.79 | 2.85    | 3.62  | 0.79 |
|              | 43         | 6.18    | 2.63  | 2.35 | 6.18         | 2.63  | 2.35 | 4.57 | 2.95  | 1.55 | 3.08 | 3.05  | 1.01 | 2.73 | 3.13  | 0.87 | 2.73    | 3.13  | 0.87 |
|              | 46         | 5.26    | 2.55  | 2.06 | 5.26         | 2.55  | 2.06 | 3.95 | 2.75  | 1.44 | 2.61 | 2.96  | 0.88 | 2.61 | 2.96  | 0.88 | 2.61    | 2.96  | 0.88 |
|              | -5         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 0          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 15         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 19         | 7.54    | 5.55  | 1.36 | 7.54         | 5.55  | 1.36 | 5.70 | 6.52  | 0.87 | 3.75 | 7.42  | 0.51 | 3.75 | 7.42  | 0.51 | 3.75    | 7.42  | 0.51 |
|              | 20         | 7.72    | 5.29  | 1.46 | 7.72         | 5.29  | 1.46 | 5.65 | 6.32  | 0.89 | 3.86 | 7.07  | 0.55 | 3.72 | 7.14  | 0.52 | 3.72    | 7.14  | 0.52 |
| 10           | 25         | 8.33    | 4.32  | 1.93 | 8.33         | 4.32  | 1.93 | 6.25 | 5.20  | 1.20 | 4.15 | 5.90  | 0.70 | 3.59 | 5.98  | 0.60 | 3.59    | 5.98  | 0.60 |
|              | 30         | 8.73    | 3.58  | 2.44 | 8.73         | 3.58  | 2.44 | 6.55 | 4.33  | 1.51 | 4.37 | 4.90  | 0.89 | 3.44 | 4.98  | 0.69 | 3.44    | 4.98  | 0.69 |
|              | 35         | 9.23    | 3.06  | 3.01 | 8.90         | 3.25  | 2.74 | 6.57 | 3.62  | 1.81 | 4.38 | 4.15  | 1.06 | 3.25 | 4.42  | 0.74 | 3.25    | 4.42  | 0.74 |
|              | 40         | 7.23    | 2.92  | 2.48 | 7.23         | 2.92  | 2.48 | 5.58 | 3.30  | 1.69 | 3.69 | 3.66  | 1.01 | 3.06 | 3.80  | 0.80 | 3.06    | 3.80  | 0.80 |
|              | 43         | 6.60    | 2.76  | 2.39 | 6.60         | 2.76  | 2.39 | 4.95 | 3.15  | 1.57 | 3.30 | 3.33  | 0.99 | 2.94 | 3.37  | 0.87 | 2.94    | 3.37  | 0.87 |
|              | 46         | 5.63    | 2.68  | 2.10 | 5.63         | 2.68  | 2.10 | 4.22 | 2.95  | 1.43 | 2.81 | 3.10  | 0.91 | 2.81 | 3.10  | 0.91 | 2.81    | 3.10  | 0.91 |
|              | -5         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 0          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 11           | 15         | 7.48    | 6.95  | 1.08 | 7.48         | 6.95  | 1.08 | 5.55 | 7.96  | 0.70 | 4.19 | 8.85  | 0.47 | 4.19 | 8.85  | 0.47 | 4.19    | 8.85  | 0.47 |
|              | 19         | 8.30    | 6.11  | 1.36 | 8.30         | 6.11  | 1.36 | 6.16 | 7.28  | 0.85 | 4.15 | 8.10  | 0.51 | 4.15 | 8.10  | 0.51 | 4.15    | 8.10  | 0.51 |
|              | 20         | 8.49    | 5.83  | 1.46 | 8.49         | 5.83  | 1.46 | 6.37 | 6.82  | 0.93 | 4.28 | 7.21  | 0.59 | 4.12 | 7.25  | 0.57 | 4.12    | 7.25  | 0.57 |
|              | 25         | 9.15    | 4.68  | 1.96 | 9.15         | 4.68  | 1.96 | 6.76 | 5.90  | 1.15 | 4.59 | 6.61  | 0.69 | 3.98 | 6.75  | 0.59 | 3.98    | 6.75  | 0.59 |
|              | 30         | 9.58    | 3.84  | 2.49 | 9.21         | 3.89  | 2.37 | 6.88 | 4.84  | 1.42 | 4.56 | 5.53  | 0.82 | 3.81 | 5.80  | 0.66 | 3.81    | 5.80  | 0.66 |
|              | 35         | 9.60    | 3.22  | 2.98 | 9.20         | 3.30  | 2.79 | 6.87 | 4.06  | 1.69 | 4.57 | 4.63  | 0.99 | 3.61 | 4.96  | 0.73 | 3.61    | 4.96  | 0.73 |
|              | 40         | 7.95    | 3.13  | 2.54 | 7.95         | 3.13  | 2.54 | 5.94 | 3.64  | 1.63 | 3.95 | 4.30  | 0.92 | 3.40 | 4.38  | 0.78 | 3.40    | 4.38  | 0.78 |
|              | 43         | 7.26    | 2.96  | 2.45 | 7.26         | 2.96  | 2.45 | 5.55 | 3.42  | 1.62 | 3.67 | 4.00  | 0.92 | 3.25 | 4.09  | 0.79 | 3.25    | 4.09  | 0.79 |
|              | 46         | 6.20    | 2.89  | 2.14 | 6.20         | 2.89  | 2.14 | 4.65 | 3.12  | 1.49 | 3.12 | 3.61  | 0.86 | 3.12 | 3.61  | 0.86 | 3.12    | 3.61  | 0.86 |
|              | -5         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 15           | 0          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10         | 6.74    | 7.92  | 0.85 | 6.74         | 7.92  | 0.85 | 5.15 | 9.21  | 0.56 | 5.15 | 9.21  | 0.56 | 5.15 | 9.21  | 0.56 | 5.15    | 9.21  | 0.56 |
|              | 15         | 7.71    | 7.21  | 1.07 | 7.71         | 7.21  | 1.07 | 5.10 | 8.37  | 0.61 | 4.33 | 8.92  | 0.49 | 4.09 | 8.96  | 0.46 | 4.09    | 8.96  | 0.46 |
|              | 19         | 8.56    | 6.16  | 1.39 | 8.56         | 6.16  | 1.39 | 6.36 | 7.65  | 0.83 | 4.29 | 8.41  | 0.51 | 4.06 | 8.48  | 0.48 | 4.06    | 8.48  | 0.48 |
|              | 20         | 8.76    | 5.97  | 1.47 | 8.76         | 5.97  | 1.47 | 6.59 | 6.89  | 0.96 | 4.43 | 7.60  | 0.58 | 4.26 | 7.62  | 0.56 | 4.26    | 7.62  | 0.56 |
|              | 25         | 9.43    | 4.88  | 1.93 | 9.43         | 4.88  | 1.93 | 6.93 | 5.96  | 1.16 | 4.59 | 6.87  | 0.67 | 4.12 | 7.02  | 0.59 | 4.12    | 7.02  | 0.59 |
|              | 30         | 9.86    | 3.93  | 2.51 | 9.30         | 4.02  | 2.31 | 6.98 | 4.87  | 1.43 | 4.71 | 5.74  | 0.82 | 3.94 | 6.02  | 0.66 | 3.94    | 6.02  | 0.66 |
|              | 35         | 9.90    | 3.29  | 3.01 | 9.26         | 3.52  | 2.63 | 6.98 | 4.26  | 1.64 | 4.60 | 4.83  | 0.95 | 3.74 | 5.10  | 0.73 | 3.74    | 5.10  | 0.73 |
|              | 40         | 8.17    | 3.19  | 2.57 | 8.17         | 3.19  | 2.57 | 6.13 | 3.68  | 1.67 | 4.09 | 4.13  | 0.99 | 3.52 | 4.25  | 0.83 | 3.52    | 4.25  | 0.83 |
| 18           | 43         | 7.50    | 3.03  | 2.47 | 7.50         | 3.03  | 2.47 | 6.13 | 3.45  | 1.78 | 4.08 | 3.87  | 1.05 | 3.38 | 3.96  | 0.85 | 3.38    | 3.96  | 0.85 |
|              | 46         | 7.20    | 2.92  | 2.47 | 7.20         | 2.92  | 2.47 | 5.40 | 3.23  | 1.67 | 3.60 | 3.43  | 1.05 | 3.23 | 3.45  | 0.94 | 3.23    | 3.45  | 0.94 |
|              | -5         | 8.12    | 12.09 | 0.67 | 8.12         | 12.09 | 0.67 | 6.54 | 12.60 | 0.52 | 6.54 | 12.60 | 0.52 | 6.54 | 12.60 | 0.52 | 6.54    | 12.60 | 0.52 |
|              | 0          | 8.05    | 10.55 | 0.76 | 8.05         | 10.55 | 0.76 | 5.71 | 11.45 | 0.50 | 5.71 | 11.45 | 0.50 | 5.71 | 11.45 | 0.50 | 5.71    | 11.45 | 0.50 |
|              | 5          | 7.82    | 9.60  | 0.81 | 7.82         | 9.60  | 0.81 | 5.58 | 10.52 | 0.53 | 5.58 | 10.52 | 0.53 | 5.58 | 10.52 | 0.53 | 5.58    | 10.52 | 0.53 |
|              | 10         | 7.65    | 8.84  | 0.86 | 7.65         | 8.84  | 0.86 | 5.14 | 9.65  | 0.53 | 5.14 | 9.65  | 0.53 | 5.14 | 9.65  | 0.53 | 5.14    | 9.65  | 0.53 |
|              | 15         | 8.79    | 7.79  | 1.13 | 8.79         | 8.02  | 1.10 | 7.25 | 8.86  | 0.82 | 4.71 | 9.43  | 0.50 | 4.71 | 9.43  | 0.50 | 4.71    | 9.43  | 0.50 |
|              | 19         | 9.69    | 6.82  | 1.42 | 9.69         | 6.82  | 1.42 | 7.21 | 7.87  | 0.92 | 4.87 | 8.91  | 0.55 | 4.87 | 8.91  | 0.55 | 4.87    | 8.91  | 0.55 |
|              | 20         | 9.91    | 6.10  | 1.62 | 9.61         | 6.32  | 1.52 | 7.31 | 7.35  | 0.99 | 4.84 | 8.40  | 0.58 | 4.84 | 8.40  | 0.58 | 4.84    | 8.40  | 0.58 |
|              | 25         | 10.66   | 5.35  | 1.99 | 9.67         | 5.74  | 1.68 | 7.29 | 6.85  | 1.06 | 4.85 | 7.65  | 0.63 | 4.67 | 7.71  | 0.61 | 4.67    | 7.71  | 0.61 |
| 20           | 30         | 11.11   | 4.31  | 2.58 | 9.68         | 4.87  | 1.99 | 7.31 | 5.83  | 1.25 | 4.82 | 6.41  | 0.75 | 4.48 | 6.72  | 0.67 | 4.48    | 6.72  | 0.67 |
|              | 35         | 11.16   | 3.58  | 3.12 | 9.67         | 4.02  | 2.41 | 7.20 | 4.79  | 1.50 | 4.91 | 5.31  | 0.92 | 4.27 | 5.62  | 0.76 | 4.27    | 5.62  | 0.76 |
|              | 40         | 9.25    | 3.49  | 2.65 | 9.25         | 3.49  | 2.65 | 6.82 | 4.22  | 1.62 | 4.66 | 4.59  | 1.02 | 4.03 | 4.73  | 0.85 | 4.03    | 4.73  | 0.85 |
|              | 43         | 8.49    | 3.33  | 2.55 | 8.49         | 3.33  | 2.55 | 6.34 | 3.92  | 1.62 | 4.17 | 4.06  | 1.03 | 3.87 | 4.27  | 0.91 |         |       |      |

## 9.5 Prestazioni raffreddamento Mod. 12

| Tw_out<br>°C | DBWB<br>°C | Maximum |       |      | 100%(Normal) |       |      | 75%   |       |      | 50%  |       |      | 25%  |       |      | Minimum |       |      |
|--------------|------------|---------|-------|------|--------------|-------|------|-------|-------|------|------|-------|------|------|-------|------|---------|-------|------|
|              |            | CC      | EER   | PI   | CC           | EER   | PI   | CC    | EER   | PI   | CC   | EER   | PI   | CC   | EER   | PI   | CC      | EER   | PI   |
| 5            | -5         | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 0          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10         | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 15         | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 19         | 9.56    | 5.20  | 1.84 | 9.56         | 5.20  | 1.84 | 8.60  | 5.37  | 1.60 | 5.84 | 5.72  | 1.02 | 4.46 | 5.93  | 0.75 | 4.46    | 5.93  | 0.75 |
|              | 20         | 9.82    | 4.94  | 1.99 | 9.82         | 4.94  | 1.99 | 8.59  | 5.21  | 1.65 | 5.79 | 5.53  | 1.05 | 4.44 | 5.74  | 0.77 | 4.44    | 5.74  | 0.77 |
|              | 25         | 10.63   | 3.97  | 2.68 | 10.63        | 3.97  | 2.68 | 8.71  | 4.41  | 1.97 | 5.71 | 4.71  | 1.21 | 4.27 | 4.92  | 0.87 | 4.27    | 4.92  | 0.87 |
|              | 30         | 11.13   | 3.28  | 3.39 | 11.13        | 3.28  | 3.39 | 8.64  | 3.78  | 2.28 | 5.85 | 4.01  | 1.46 | 4.12 | 4.26  | 0.97 | 4.12    | 4.26  | 0.97 |
|              | 35         | 11.25   | 2.79  | 4.03 | 11.25        | 2.79  | 4.03 | 8.62  | 3.21  | 2.69 | 5.76 | 3.51  | 1.64 | 3.90 | 3.66  | 1.07 | 3.90    | 3.66  | 1.07 |
| 7            | 40         | 9.88    | 2.56  | 3.86 | 9.88         | 2.56  | 3.86 | 8.66  | 2.76  | 3.14 | 5.75 | 3.02  | 1.90 | 3.71 | 3.17  | 1.17 | 3.71    | 3.17  | 1.17 |
|              | 43         | 8.60    | 2.45  | 3.51 | 8.60         | 2.45  | 3.51 | 8.55  | 2.48  | 3.45 | 5.73 | 2.79  | 2.05 | 3.57 | 2.89  | 1.23 | 3.57    | 2.89  | 1.23 |
|              | 46         | 6.64    | 2.31  | 2.88 | 6.64         | 2.31  | 2.88 | 6.64  | 2.31  | 2.88 | 5.75 | 2.45  | 2.34 | 3.48 | 2.67  | 1.30 | 3.48    | 2.67  | 1.30 |
|              | -5         | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 0          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10         | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 15         | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 19         | 10.19   | 5.53  | 1.84 | 10.19        | 5.53  | 1.84 | 8.59  | 6.02  | 1.43 | 5.69 | 6.18  | 0.92 | 4.79 | 6.39  | 0.75 | 4.79    | 6.39  | 0.75 |
|              | 20         | 10.46   | 5.24  | 1.99 | 10.46        | 5.24  | 1.99 | 8.51  | 5.64  | 1.51 | 5.68 | 6.04  | 0.94 | 4.76 | 6.18  | 0.77 | 4.76    | 6.18  | 0.77 |
| 10           | 25         | 11.32   | 4.18  | 2.71 | 11.32        | 4.18  | 2.71 | 8.67  | 4.90  | 1.77 | 5.71 | 5.11  | 1.12 | 4.57 | 5.28  | 0.87 | 4.57    | 5.28  | 0.87 |
|              | 30         | 11.85   | 3.44  | 3.44 | 11.56        | 3.51  | 3.29 | 8.71  | 4.22  | 2.06 | 5.68 | 4.35  | 1.31 | 4.40 | 4.55  | 0.97 | 4.40    | 4.55  | 0.97 |
|              | 35         | 11.96   | 2.99  | 4.00 | 11.50        | 3.05  | 3.77 | 8.57  | 3.60  | 2.38 | 5.81 | 3.76  | 1.54 | 4.17 | 3.90  | 1.07 | 4.17    | 3.90  | 1.07 |
|              | 40         | 10.57   | 2.70  | 3.91 | 10.57        | 2.70  | 3.91 | 8.58  | 3.05  | 2.81 | 5.82 | 3.23  | 1.80 | 3.97 | 3.37  | 1.18 | 3.97    | 3.37  | 1.18 |
|              | 43         | 9.13    | 2.55  | 3.58 | 9.13         | 2.55  | 3.58 | 8.61  | 2.64  | 3.26 | 5.82 | 2.98  | 1.95 | 3.82 | 3.07  | 1.24 | 3.82    | 3.07  | 1.24 |
|              | 46         | 7.08    | 2.42  | 2.93 | 7.08         | 2.42  | 2.93 | 7.08  | 2.42  | 2.93 | 5.75 | 2.60  | 2.21 | 3.72 | 2.83  | 1.31 | 3.72    | 2.83  | 1.31 |
|              | -5         | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 0          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10         | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 11           | 15         | 10.09   | 7.20  | 1.40 | 10.09        | 7.20  | 1.40 | 8.78  | 7.75  | 1.13 | 5.85 | 8.34  | 0.70 | 5.41 | 8.37  | 0.65 | 5.41    | 8.37  | 0.65 |
|              | 19         | 11.19   | 6.08  | 1.84 | 11.19        | 6.08  | 1.84 | 8.80  | 6.85  | 1.28 | 5.70 | 7.27  | 0.78 | 5.33 | 7.37  | 0.72 | 5.33    | 7.37  | 0.72 |
|              | 20         | 11.49   | 5.74  | 2.00 | 11.49        | 5.74  | 2.00 | 8.72  | 6.62  | 1.32 | 5.85 | 7.01  | 0.84 | 5.29 | 7.10  | 0.74 | 5.29    | 7.10  | 0.74 |
|              | 25         | 12.40   | 4.51  | 2.75 | 11.69        | 4.73  | 2.47 | 8.73  | 5.56  | 1.57 | 5.89 | 5.83  | 1.01 | 5.05 | 5.89  | 0.86 | 5.05    | 5.89  | 0.86 |
|              | 30         | 12.97   | 3.70  | 3.51 | 11.71        | 4.03  | 2.91 | 8.81  | 4.68  | 1.88 | 5.85 | 4.86  | 1.20 | 4.85 | 5.04  | 0.96 | 4.85    | 5.04  | 0.96 |
|              | 35         | 13.05   | 3.10  | 4.21 | 11.63        | 3.43  | 3.39 | 8.71  | 4.08  | 2.13 | 5.81 | 4.20  | 1.38 | 4.60 | 4.29  | 1.07 | 4.60    | 4.29  | 1.07 |
|              | 40         | 11.55   | 2.80  | 4.12 | 10.57        | 3.02  | 3.50 | 8.80  | 3.45  | 2.55 | 5.88 | 3.57  | 1.65 | 4.38 | 3.69  | 1.19 | 4.38    | 3.69  | 1.19 |
|              | 43         | 10.04   | 2.74  | 3.66 | 10.04        | 2.74  | 3.66 | 8.69  | 2.99  | 2.91 | 5.90 | 3.28  | 1.80 | 4.22 | 3.35  | 1.26 | 4.22    | 3.35  | 1.26 |
|              | 46         | 7.80    | 2.61  | 2.99 | 7.80         | 2.61  | 2.99 | 7.80  | 2.61  | 2.99 | 5.82 | 2.99  | 1.95 | 4.12 | 3.10  | 1.33 | 4.12    | 3.10  | 1.33 |
|              | -5         | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 15           | 0          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10         | 8.97    | 9.13  | 0.98 | 8.97         | 9.13  | 0.98 | 8.76  | 9.40  | 0.93 | 7.23 | 9.99  | 0.72 | 7.23 | 9.99  | 0.72 | 7.23    | 9.99  | 0.72 |
|              | 15         | 10.40   | 7.44  | 1.40 | 10.40        | 7.44  | 1.40 | 8.83  | 8.15  | 1.08 | 6.05 | 8.72  | 0.69 | 5.60 | 8.76  | 0.64 | 5.60    | 8.76  | 0.64 |
|              | 19         | 11.54   | 6.27  | 1.84 | 11.54        | 6.27  | 1.84 | 8.85  | 7.17  | 1.24 | 5.77 | 7.60  | 0.76 | 5.48 | 7.63  | 0.72 | 5.48    | 7.63  | 0.72 |
|              | 20         | 11.84   | 5.91  | 2.00 | 11.64        | 6.00  | 1.94 | 8.77  | 6.92  | 1.27 | 5.81 | 7.37  | 0.79 | 5.47 | 7.41  | 0.74 | 5.47    | 7.41  | 0.74 |
|              | 25         | 12.78   | 4.63  | 2.76 | 11.67        | 4.97  | 2.35 | 8.80  | 5.78  | 1.52 | 5.78 | 6.13  | 0.94 | 5.25 | 6.17  | 0.85 | 5.25    | 6.17  | 0.85 |
|              | 30         | 13.36   | 3.78  | 3.53 | 11.72        | 4.22  | 2.78 | 8.69  | 5.01  | 1.73 | 5.71 | 5.16  | 1.10 | 5.00 | 5.21  | 0.96 | 5.00    | 5.21  | 0.96 |
|              | 35         | 13.44   | 3.17  | 4.24 | 11.68        | 3.59  | 3.25 | 8.81  | 4.25  | 2.07 | 5.64 | 4.41  | 1.28 | 4.79 | 4.47  | 1.07 | 4.79    | 4.47  | 1.07 |
|              | 40         | 11.89   | 2.93  | 4.05 | 11.76        | 2.96  | 3.97 | 8.74  | 3.52  | 2.48 | 5.78 | 3.79  | 1.52 | 4.61 | 3.86  | 1.20 | 4.61    | 3.86  | 1.20 |
| 18           | 43         | 10.34   | 2.80  | 3.69 | 10.34        | 2.80  | 3.69 | 8.81  | 3.10  | 2.84 | 5.74 | 3.42  | 1.68 | 4.45 | 3.51  | 1.27 | 4.45    | 3.51  | 1.27 |
|              | 46         | 8.05    | 2.68  | 3.01 | 8.05         | 2.68  | 3.01 | 8.05  | 2.68  | 3.01 | 5.84 | 3.08  | 1.90 | 4.26 | 3.19  | 1.33 | 4.26    | 3.19  | 1.33 |
|              | -5         | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 0          | 9.88    | 12.96 | 0.76 | 9.88         | 12.96 | 0.76 | 9.88  | 12.96 | 0.76 | 8.90 | 13.65 | 0.65 | 8.90 | 13.65 | 0.65 | 8.90    | 13.65 | 0.65 |
|              | 5          | 10.00   | 12.18 | 0.82 | 10.00        | 12.18 | 0.82 | 10.00 | 12.34 | 0.81 | 8.91 | 12.86 | 0.69 | 8.91 | 12.86 | 0.69 | 8.91    | 12.86 | 0.69 |
|              | 10         | 10.27   | 10.97 | 0.94 | 10.27        | 10.97 | 0.94 | 8.89  | 11.83 | 0.75 | 8.89 | 11.83 | 0.75 | 8.89 | 11.83 | 0.75 | 8.89    | 11.83 | 0.75 |
|              | 15         | 10.10   | 9.60  | 1.05 | 10.10        | 9.60  | 1.05 | 8.80  | 10.40 | 0.85 | 8.19 | 10.75 | 0.76 | 8.19 | 10.75 | 0.76 | 8.19    | 10.75 | 0.76 |
|              | 19         | 11.68   | 8.26  | 1.41 | 11.68        | 8.26  | 1.41 | 8.94  | 10.20 | 0.88 | 6.39 | 10.59 | 0.60 | 6.39 | 10.59 | 0.60 | 6.39    | 10.59 | 0.60 |
|              | 20         | 13.03   | 7.14  | 1.83 | 11.73        | 7.55  | 1.55 | 8.99  | 8.74  | 1.03 | 6.28 | 9.19  | 0.68 | 6.28 | 9.19  | 0.68 | 6.28    | 9.19  | 0.68 |
|              | 25         | 13.35   | 6.68  | 2.00 | 11.85        | 7.16  | 1.65 | 8.91  | 8.41  | 1.06 | 6.22 | 8.77  | 0.71 | 6.22 | 8.77  | 0.71 | 6.22    | 8.77  | 0.71 |
| 20           | 30         | 14.35   | 5.11  | 2.81 | 11.86        | 6.04  | 1.97 | 9.00  | 6.85  | 1.31 | 5.95 | 7.18  | 0.83 | 5.95 | 7.18  | 0.83 | 5.95    | 7.18  | 0.83 |
|              | 35         | 15.04   | 4.16  | 3.62 | 11.83        | 5.12  | 2.31 | 8.95  | 5.86  | 1.53 | 6.02 | 6.21  | 0.97 | 5.69 | 6.34  | 0.90 | 5.69    | 6.34  | 0.90 |
|              | 40         | 14.86   | 3.50  | 4.24 | 11.84        | 4.17  | 2.84 | 8.93  | 4.95  | 1.81 | 5.95 | 5.12  | 1.16 | 5.39 | 5.18  | 1.04 | 5.39    | 5.18  | 1.04 |
|              | 43         | 13.31   | 3.19  | 4.17 | 12.01        | 3.47  | 3.46 | 8.91  | 4.14  | 2.15 | 6.05 | 4.30  | 1.41 | 5.11 | 4.36  | 1.   |         |       |      |

## 9.6 Prestazioni raffreddamento Mod. 14

| Tw_out<br>°C | DBWB<br>°C | Maximum |       |      | 100%(Normal) |       |      | 75%   |       |      | 50%  |       |      | 25%  |       |      | Minimum |       |      |
|--------------|------------|---------|-------|------|--------------|-------|------|-------|-------|------|------|-------|------|------|-------|------|---------|-------|------|
|              |            | CC      | EER   | PI   | CC           | EER   | PI   | CC    | EER   | PI   | CC   | EER   | PI   | CC   | EER   | PI   | CC      | EER   | PI   |
| 5            | -5         | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 0          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10         | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 15         | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 19         | 10.25   | 4.96  | 2.06 | 10.25        | 4.96  | 2.06 | 9.55  | 5.19  | 1.84 | 6.43 | 5.59  | 1.15 | 4.46 | 5.93  | 0.75 | 4.46    | 5.93  | 0.75 |
|              | 20         | 10.32   | 4.77  | 2.16 | 10.32        | 4.77  | 2.16 | 9.64  | 4.99  | 1.93 | 6.37 | 5.41  | 1.18 | 4.44 | 5.74  | 0.77 | 4.44    | 5.74  | 0.77 |
|              | 25         | 11.21   | 3.79  | 2.96 | 11.21        | 3.79  | 2.96 | 9.52  | 4.25  | 2.24 | 6.44 | 4.60  | 1.40 | 4.27 | 4.92  | 0.87 | 4.27    | 4.92  | 0.87 |
|              | 30         | 11.92   | 3.11  | 3.83 | 11.92        | 3.11  | 3.83 | 9.56  | 3.63  | 2.63 | 6.36 | 3.97  | 1.60 | 4.12 | 4.26  | 0.97 | 4.12    | 4.26  | 0.97 |
|              | 35         | 11.98   | 2.65  | 4.52 | 11.98        | 2.65  | 4.52 | 9.54  | 3.11  | 3.07 | 6.40 | 3.45  | 1.86 | 3.90 | 3.66  | 1.07 | 3.90    | 3.66  | 1.07 |
| 7            | 0          | 9.88    | 2.56  | 3.86 | 9.88         | 2.56  | 3.86 | 9.54  | 2.63  | 3.62 | 6.34 | 2.98  | 2.13 | 3.71 | 3.17  | 1.17 | 3.71    | 3.17  | 1.17 |
|              | 40         | 8.60    | 2.45  | 3.51 | 8.60         | 2.45  | 3.51 | 8.60  | 2.45  | 3.51 | 6.30 | 2.72  | 2.31 | 3.57 | 2.89  | 1.23 | 3.57    | 2.89  | 1.23 |
|              | 46         | 6.64    | 2.31  | 2.88 | 6.64         | 2.31  | 2.88 | 6.64  | 2.31  | 2.88 | 6.35 | 2.37  | 2.68 | 3.48 | 2.67  | 1.30 | 3.48    | 2.67  | 1.30 |
|              | -5         | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 0          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10         | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 15         | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 19         | 10.92   | 5.26  | 2.07 | 10.92        | 5.26  | 2.07 | 9.62  | 5.74  | 1.68 | 6.44 | 6.11  | 1.05 | 4.79 | 6.39  | 0.75 | 4.79    | 6.39  | 0.75 |
|              | 20         | 11.00   | 5.06  | 2.17 | 11.00        | 5.06  | 2.17 | 9.54  | 5.57  | 1.71 | 6.38 | 5.78  | 1.10 | 4.76 | 6.18  | 0.77 | 4.76    | 6.18  | 0.77 |
| 10           | 25         | 11.93   | 3.99  | 2.99 | 11.93        | 3.99  | 2.99 | 9.63  | 4.65  | 2.07 | 6.31 | 5.00  | 1.26 | 4.57 | 5.28  | 0.87 | 4.57    | 5.28  | 0.87 |
|              | 30         | 12.67   | 3.25  | 3.89 | 12.67        | 3.25  | 3.89 | 9.54  | 3.99  | 2.39 | 6.43 | 4.26  | 1.51 | 4.40 | 4.55  | 0.97 | 4.40    | 4.55  | 0.97 |
|              | 35         | 12.70   | 2.90  | 4.38 | 12.70        | 2.90  | 4.38 | 9.56  | 3.39  | 2.82 | 6.32 | 3.71  | 1.71 | 4.17 | 3.90  | 1.07 | 4.17    | 3.90  | 1.07 |
|              | 40         | 10.57   | 2.70  | 3.91 | 10.57        | 2.70  | 3.91 | 9.51  | 2.90  | 3.28 | 6.30 | 3.19  | 1.98 | 3.97 | 3.37  | 1.18 | 3.97    | 3.37  | 1.18 |
|              | 43         | 9.13    | 2.55  | 3.58 | 9.13         | 2.55  | 3.58 | 9.13  | 2.55  | 3.58 | 6.28 | 2.88  | 2.18 | 3.82 | 3.07  | 1.24 | 3.82    | 3.07  | 1.24 |
|              | 46         | 7.08    | 2.42  | 2.93 | 7.08         | 2.42  | 2.93 | 7.08  | 2.42  | 2.93 | 6.35 | 2.50  | 2.54 | 3.72 | 2.83  | 1.31 | 3.72    | 2.83  | 1.31 |
|              | -5         | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 0          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10         | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 11           | 15         | 10.50   | 6.95  | 1.51 | 10.50        | 6.95  | 1.51 | 9.86  | 7.26  | 1.36 | 6.49 | 8.21  | 0.79 | 5.41 | 8.37  | 0.65 | 5.41    | 8.37  | 0.65 |
|              | 19         | 11.99   | 5.77  | 2.08 | 11.99        | 5.77  | 2.08 | 9.86  | 6.51  | 1.52 | 6.64 | 7.15  | 0.93 | 5.33 | 7.37  | 0.72 | 5.33    | 7.37  | 0.72 |
|              | 20         | 12.07   | 5.53  | 2.18 | 12.07        | 5.53  | 2.18 | 9.77  | 6.30  | 1.55 | 6.59 | 6.89  | 0.96 | 5.29 | 7.10  | 0.74 | 5.29    | 7.10  | 0.74 |
|              | 25         | 13.07   | 4.30  | 3.04 | 13.07        | 4.30  | 3.04 | 9.71  | 5.25  | 1.85 | 6.50 | 5.74  | 1.13 | 5.05 | 5.89  | 0.86 | 5.05    | 5.89  | 0.86 |
|              | 30         | 13.86   | 3.48  | 3.98 | 13.11        | 3.65  | 3.59 | 9.76  | 4.54  | 2.15 | 6.46 | 4.78  | 1.35 | 4.85 | 5.04  | 0.96 | 4.85    | 5.04  | 0.96 |
|              | 35         | 13.87   | 2.92  | 4.74 | 13.05        | 3.10  | 4.21 | 9.83  | 3.83  | 2.57 | 6.59 | 4.11  | 1.61 | 4.60 | 4.29  | 1.07 | 4.60    | 4.29  | 1.07 |
|              | 40         | 11.55   | 2.80  | 4.12 | 11.55        | 2.80  | 4.12 | 9.74  | 3.22  | 3.02 | 6.59 | 3.51  | 1.88 | 4.38 | 3.69  | 1.19 | 4.38    | 3.69  | 1.19 |
|              | 43         | 10.04   | 2.74  | 3.66 | 10.04        | 2.74  | 3.66 | 9.88  | 2.80  | 3.53 | 6.58 | 3.16  | 2.08 | 4.22 | 3.35  | 1.26 | 4.22    | 3.35  | 1.26 |
|              | 46         | 7.80    | 2.61  | 2.99 | 7.80         | 2.61  | 2.99 | 7.80  | 2.61  | 2.99 | 6.53 | 2.84  | 2.30 | 4.12 | 3.10  | 1.33 | 4.12    | 3.10  | 1.33 |
|              | -5         | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 15           | 0          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10         | 9.45    | 9.04  | 1.04 | 9.45         | 9.04  | 1.04 | 9.45  | 9.04  | 1.04 | 7.23 | 9.99  | 0.72 | 7.23 | 9.99  | 0.72 | 7.23    | 9.99  | 0.72 |
|              | 15         | 10.84   | 7.18  | 1.51 | 10.84        | 7.18  | 1.51 | 9.73  | 7.69  | 1.26 | 6.70 | 8.54  | 0.78 | 5.60 | 8.76  | 0.64 | 5.60    | 8.76  | 0.64 |
|              | 19         | 12.36   | 5.95  | 2.08 | 12.36        | 5.95  | 2.08 | 9.74  | 6.86  | 1.42 | 6.63 | 7.54  | 0.88 | 5.48 | 7.63  | 0.72 | 5.48    | 7.63  | 0.72 |
|              | 20         | 12.45   | 5.69  | 2.19 | 12.45        | 5.69  | 2.19 | 9.86  | 6.57  | 1.50 | 6.57 | 7.25  | 0.91 | 5.47 | 7.41  | 0.74 | 5.47    | 7.41  | 0.74 |
|              | 25         | 13.46   | 4.41  | 3.05 | 13.29        | 4.46  | 2.98 | 9.81  | 5.49  | 1.79 | 6.53 | 6.00  | 1.09 | 5.25 | 6.17  | 0.85 | 5.25    | 6.17  | 0.85 |
|              | 30         | 14.27   | 3.56  | 4.01 | 13.20        | 3.82  | 3.45 | 9.87  | 4.72  | 2.09 | 6.67 | 5.09  | 1.31 | 5.00 | 5.21  | 0.96 | 5.00    | 5.21  | 0.96 |
|              | 35         | 14.38   | 3.00  | 4.80 | 13.15        | 3.23  | 4.08 | 9.88  | 3.95  | 2.50 | 6.61 | 4.30  | 1.54 | 4.79 | 4.47  | 1.07 | 4.79    | 4.47  | 1.07 |
|              | 40         | 11.89   | 2.93  | 4.05 | 11.90        | 2.94  | 4.05 | 9.88  | 3.34  | 2.96 | 6.62 | 3.71  | 1.79 | 4.61 | 3.86  | 1.20 | 4.61    | 3.86  | 1.20 |
| 18           | 43         | 10.34   | 2.80  | 3.69 | 10.34        | 2.80  | 3.69 | 9.91  | 2.88  | 3.45 | 6.62 | 3.31  | 2.00 | 4.45 | 3.51  | 1.27 | 4.45    | 3.51  | 1.27 |
|              | 46         | 8.05    | 2.68  | 3.01 | 8.05         | 2.68  | 3.01 | 8.05  | 2.68  | 3.01 | 6.59 | 3.03  | 2.18 | 4.26 | 3.19  | 1.33 | 4.26    | 3.19  | 1.33 |
|              | -5         | 10.12   | 12.81 | 0.79 | 10.12        | 12.81 | 0.79 | 10.12 | 12.81 | 0.79 | 8.90 | 13.65 | 0.65 | 8.90 | 13.65 | 0.65 | 8.90    | 13.65 | 0.65 |
|              | 0          | 10.25   | 12.02 | 0.85 | 10.25        | 12.02 | 0.85 | 10.25 | 12.02 | 0.85 | 8.91 | 12.86 | 0.69 | 8.91 | 12.86 | 0.69 | 8.91    | 12.86 | 0.69 |
|              | 5          | 10.54   | 10.90 | 0.97 | 10.54        | 10.90 | 0.97 | 10.27 | 10.97 | 0.94 | 8.89 | 11.83 | 0.75 | 8.89 | 11.83 | 0.75 | 8.89    | 11.83 | 0.75 |
|              | 10         | 10.63   | 9.39  | 1.13 | 10.63        | 9.39  | 1.13 | 10.36 | 9.70  | 1.07 | 8.19 | 10.75 | 0.76 | 8.19 | 10.75 | 0.76 | 8.19    | 10.75 | 0.76 |
|              | 15         | 12.14   | 8.09  | 1.50 | 12.14        | 8.09  | 1.50 | 10.18 | 9.22  | 1.10 | 6.87 | 10.37 | 0.66 | 6.39 | 10.59 | 0.60 | 6.39    | 10.59 | 0.60 |
|              | 19         | 13.94   | 6.72  | 2.07 | 13.48        | 6.90  | 1.95 | 10.26 | 8.16  | 1.26 | 6.80 | 9.11  | 0.75 | 6.28 | 9.19  | 0.68 | 6.28    | 9.19  | 0.68 |
|              | 20         | 14.02   | 6.39  | 2.19 | 13.58        | 6.58  | 2.06 | 10.16 | 7.87  | 1.29 | 6.75 | 8.72  | 0.77 | 6.22 | 8.77  | 0.71 | 6.22    | 8.77  | 0.71 |
|              | 25         | 15.09   | 4.84  | 3.12 | 13.59        | 5.43  | 2.50 | 10.16 | 6.47  | 1.57 | 6.93 | 7.10  | 0.98 | 5.95 | 7.18  | 0.83 | 5.95    | 7.18  | 0.83 |
| 20           | 30         | 16.05   | 3.89  | 4.12 | 13.63        | 4.58  | 2.97 | 10.31 | 5.59  | 1.84 | 6.89 | 6.13  | 1.12 | 5.69 | 6.34  | 0.90 | 5.69    | 6.34  | 0.90 |
|              | 35         | 16.03   | 3.24  | 4.95 | 13.61        | 3.79  | 3.59 | 10.19 | 4.69  | 2.17 | 6.80 | 5.03  | 1.35 | 5.39 | 5.18  | 1.04 | 5.39    | 5.18  | 1.04 |
|              | 40         | 13.31   | 3.19  | 4.17 | 13.31        | 3.19  | 4.17 | 10.25 | 3.92  | 2.61 | 6.89 | 4.20  | 1.64 | 5.11 | 4.36  | 1.17 | 5.11    | 4.36  | 1.17 |
|              | 43         | 11.58   | 3.05  | 3.80 | 11.58        |       |      |       |       |      |      |       |      |      |       |      |         |       |      |

## 9.7 Prestazioni raffreddamento Mod. 16

| Tw_out<br>°C | DBWB<br>°C | Maximum |       |      | 100%(Normal) |       |      | 75%   |       |      | 50%  |       |      | 25%  |       |      | Minimum |       |      |
|--------------|------------|---------|-------|------|--------------|-------|------|-------|-------|------|------|-------|------|------|-------|------|---------|-------|------|
|              |            | CC      | EER   | PI   | CC           | EER   | PI   | CC    | EER   | PI   | CC   | EER   | PI   | CC   | EER   | PI   | CC      | EER   | PI   |
| 5            | -5         | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 0          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10         | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 15         | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 19         | 11.00   | 4.80  | 2.29 | 11.00        | 4.80  | 2.29 | 10.48 | 4.95  | 2.12 | 7.01 | 5.48  | 1.28 | 4.46 | 5.93  | 0.75 | 4.46    | 5.93  | 0.75 |
|              | 20         | 11.22   | 4.57  | 2.45 | 11.22        | 4.57  | 2.45 | 10.56 | 4.76  | 2.22 | 6.95 | 5.31  | 1.31 | 4.44 | 5.74  | 0.77 | 4.44    | 5.74  | 0.77 |
|              | 25         | 12.16   | 3.62  | 3.36 | 12.16        | 3.62  | 3.36 | 10.54 | 4.04  | 2.61 | 6.98 | 4.52  | 1.54 | 4.27 | 4.92  | 0.87 | 4.27    | 4.92  | 0.87 |
|              | 30         | 13.12   | 2.95  | 4.44 | 13.12        | 2.95  | 4.44 | 10.54 | 3.47  | 3.04 | 7.07 | 3.88  | 1.82 | 4.12 | 4.26  | 0.97 | 4.12    | 4.26  | 0.97 |
|              | 35         | 13.41   | 2.55  | 5.26 | 13.41        | 2.55  | 5.26 | 10.46 | 2.99  | 3.50 | 6.92 | 3.34  | 2.08 | 3.90 | 3.66  | 1.07 | 3.90    | 3.66  | 1.07 |
| 7            | 40         | 10.48   | 2.51  | 4.17 | 10.48        | 2.51  | 4.17 | 10.48 | 2.51  | 4.17 | 6.98 | 2.85  | 2.45 | 3.71 | 3.17  | 1.17 | 3.71    | 3.17  | 1.17 |
|              | 43         | 9.13    | 2.39  | 3.82 | 9.13         | 2.39  | 3.82 | 9.13  | 2.39  | 3.82 | 7.02 | 2.64  | 2.66 | 3.57 | 2.89  | 1.23 | 3.57    | 2.89  | 1.23 |
|              | 46         | 7.02    | 2.28  | 3.08 | 7.02         | 2.28  | 3.08 | 7.02  | 2.28  | 3.08 | 7.02 | 2.28  | 3.08 | 3.48 | 2.67  | 1.30 | 3.48    | 2.67  | 1.30 |
|              | -5         | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 0          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10         | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 15         | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 19         | 11.72   | 5.10  | 2.30 | 11.72        | 5.10  | 2.30 | 10.61 | 5.43  | 1.95 | 7.08 | 5.97  | 1.19 | 4.79 | 6.39  | 0.75 | 4.79    | 6.39  | 0.75 |
|              | 20         | 11.95   | 4.84  | 2.47 | 11.95        | 4.84  | 2.47 | 10.52 | 5.27  | 1.99 | 7.02 | 5.60  | 1.25 | 4.76 | 6.18  | 0.77 | 4.76    | 6.18  | 0.77 |
| 10           | 25         | 12.93   | 3.80  | 3.40 | 12.93        | 3.80  | 3.40 | 10.55 | 4.44  | 2.38 | 7.08 | 4.88  | 1.45 | 4.57 | 5.28  | 0.87 | 4.57    | 5.28  | 0.87 |
|              | 30         | 13.94   | 3.09  | 4.52 | 13.94        | 3.09  | 4.52 | 10.44 | 3.83  | 2.73 | 7.01 | 4.18  | 1.68 | 4.40 | 4.55  | 0.97 | 4.40    | 4.55  | 0.97 |
|              | 35         | 14.26   | 2.60  | 5.48 | 14.00        | 2.75  | 5.09 | 10.72 | 3.23  | 3.32 | 7.06 | 3.64  | 1.94 | 4.17 | 3.90  | 1.07 | 4.17    | 3.90  | 1.07 |
|              | 40         | 11.13   | 2.61  | 4.26 | 11.13        | 2.61  | 4.26 | 10.43 | 2.78  | 3.74 | 6.99 | 3.13  | 2.23 | 3.97 | 3.37  | 1.18 | 3.97    | 3.37  | 1.18 |
|              | 43         | 9.71    | 2.50  | 3.89 | 9.71         | 2.50  | 3.89 | 9.71  | 2.50  | 3.89 | 7.09 | 2.81  | 2.52 | 3.82 | 3.07  | 1.24 | 3.82    | 3.07  | 1.24 |
|              | 46         | 7.48    | 2.38  | 3.15 | 7.48         | 2.38  | 3.15 | 7.48  | 2.38  | 3.15 | 7.00 | 2.43  | 2.89 | 3.72 | 2.83  | 1.31 | 3.72    | 2.83  | 1.31 |
|              | -5         | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 0          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10         | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 11           | 15         | 11.32   | 6.79  | 1.67 | 11.32        | 6.79  | 1.67 | 10.90 | 7.13  | 1.53 | 7.16 | 7.78  | 0.92 | 5.41 | 8.37  | 0.65 | 5.41    | 8.37  | 0.65 |
|              | 19         | 12.86   | 5.57  | 2.31 | 12.86        | 5.57  | 2.31 | 11.05 | 6.21  | 1.78 | 7.36 | 7.03  | 1.05 | 5.33 | 7.37  | 0.72 | 5.33    | 7.37  | 0.72 |
|              | 20         | 13.12   | 5.28  | 2.49 | 13.12        | 5.28  | 2.49 | 10.95 | 6.01  | 1.82 | 7.29 | 6.65  | 1.10 | 5.29 | 7.10  | 0.74 | 5.29    | 7.10  | 0.74 |
|              | 25         | 14.16   | 4.10  | 3.46 | 14.16        | 4.10  | 3.46 | 11.02 | 4.89  | 2.25 | 7.38 | 5.53  | 1.34 | 5.05 | 5.89  | 0.86 | 5.05    | 5.89  | 0.86 |
|              | 30         | 15.23   | 3.29  | 4.62 | 14.50        | 3.45  | 4.20 | 10.94 | 4.28  | 2.56 | 7.33 | 4.64  | 1.58 | 4.85 | 5.04  | 0.96 | 4.85    | 5.04  | 0.96 |
|              | 35         | 15.55   | 2.76  | 5.63 | 14.61        | 2.89  | 5.06 | 11.03 | 3.59  | 3.07 | 7.40 | 4.02  | 1.84 | 4.60 | 4.29  | 1.07 | 4.60    | 4.29  | 1.07 |
|              | 40         | 12.23   | 2.73  | 4.48 | 12.23        | 2.73  | 4.48 | 11.10 | 3.01  | 3.68 | 7.33 | 3.42  | 2.14 | 4.38 | 3.69  | 1.19 | 4.38    | 3.69  | 1.19 |
|              | 43         | 10.67   | 2.69  | 3.97 | 10.67        | 2.69  | 3.97 | 10.67 | 2.69  | 3.97 | 7.31 | 3.06  | 2.39 | 4.22 | 3.35  | 1.26 | 4.22    | 3.35  | 1.26 |
|              | 46         | 8.24    | 2.55  | 3.23 | 8.24         | 2.55  | 3.23 | 8.24  | 2.55  | 3.23 | 7.27 | 2.79  | 2.60 | 4.12 | 3.10  | 1.33 | 4.12    | 3.10  | 1.33 |
|              | -5         | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 15           | 0          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10         | 10.22   | 8.84  | 1.16 | 10.22        | 8.84  | 1.16 | 9.86  | 8.94  | 1.10 | 7.67 | 9.69  | 0.79 | 7.23 | 9.99  | 0.72 | 7.23    | 9.99  | 0.72 |
|              | 15         | 11.67   | 7.00  | 1.67 | 11.67        | 7.00  | 1.67 | 10.83 | 7.18  | 1.51 | 7.38 | 8.08  | 0.91 | 5.60 | 8.76  | 0.64 | 5.60    | 8.76  | 0.64 |
|              | 19         | 13.26   | 5.74  | 2.31 | 13.26        | 5.74  | 2.31 | 11.19 | 6.51  | 1.72 | 7.32 | 7.20  | 1.02 | 5.48 | 7.63  | 0.72 | 5.48    | 7.63  | 0.72 |
|              | 20         | 13.52   | 5.43  | 2.49 | 13.52        | 5.43  | 2.49 | 11.09 | 6.30  | 1.76 | 7.53 | 6.91  | 1.09 | 5.47 | 7.41  | 0.74 | 5.47    | 7.41  | 0.74 |
|              | 25         | 14.58   | 4.19  | 3.48 | 14.58        | 4.19  | 3.48 | 11.17 | 5.21  | 2.15 | 7.40 | 5.80  | 1.28 | 5.25 | 6.17  | 0.85 | 5.25    | 6.17  | 0.85 |
|              | 30         | 15.68   | 3.37  | 4.66 | 14.78        | 3.47  | 4.26 | 11.09 | 4.44  | 2.50 | 7.36 | 4.87  | 1.51 | 5.00 | 5.21  | 0.96 | 5.00    | 5.21  | 0.96 |
|              | 35         | 15.99   | 2.81  | 5.69 | 14.90        | 2.93  | 5.09 | 11.04 | 3.76  | 2.94 | 7.44 | 4.15  | 1.79 | 4.79 | 4.47  | 1.07 | 4.79    | 4.47  | 1.07 |
|              | 40         | 12.64   | 2.78  | 4.55 | 12.64        | 2.78  | 4.55 | 11.19 | 3.17  | 3.53 | 7.41 | 3.54  | 2.09 | 4.61 | 3.86  | 1.20 | 4.61    | 3.86  | 1.20 |
| 18           | 43         | 10.99   | 2.74  | 4.01 | 10.99        | 2.74  | 4.01 | 10.99 | 2.74  | 4.01 | 7.37 | 3.17  | 2.33 | 4.45 | 3.51  | 1.27 | 4.45    | 3.51  | 1.27 |
|              | 46         | 8.51    | 2.60  | 3.27 | 8.51         | 2.60  | 3.27 | 8.51  | 2.60  | 3.27 | 7.36 | 2.92  | 2.52 | 4.26 | 3.19  | 1.33 | 4.26    | 3.19  | 1.33 |
|              | -5         | 10.45   | 12.55 | 0.83 | 10.45        | 12.55 | 0.83 | 10.45 | 12.55 | 0.83 | 8.90 | 13.65 | 0.65 | 8.90 | 13.65 | 0.65 | 8.90    | 13.65 | 0.65 |
|              | 0          | 10.50   | 11.74 | 0.89 | 10.50        | 11.74 | 0.89 | 10.50 | 11.74 | 0.89 | 8.91 | 12.86 | 0.69 | 8.91 | 12.86 | 0.69 | 8.91    | 12.86 | 0.69 |
|              | 5          | 10.81   | 10.66 | 1.01 | 10.81        | 10.66 | 1.01 | 10.81 | 10.66 | 1.01 | 8.89 | 11.83 | 0.75 | 8.89 | 11.83 | 0.75 | 8.89    | 11.83 | 0.75 |
|              | 10         | 11.46   | 9.04  | 1.27 | 11.46        | 9.04  | 1.27 | 10.76 | 9.34  | 1.15 | 8.52 | 10.46 | 0.81 | 8.19 | 10.75 | 0.76 | 8.19    | 10.75 | 0.76 |
|              | 15         | 13.11   | 7.97  | 1.64 | 13.11        | 7.97  | 1.64 | 11.45 | 8.61  | 1.33 | 7.58 | 10.06 | 0.75 | 6.39 | 10.59 | 0.60 | 6.39    | 10.59 | 0.60 |
|              | 19         | 14.94   | 6.46  | 2.31 | 14.94        | 6.46  | 2.31 | 11.64 | 7.90  | 1.47 | 7.82 | 8.95  | 0.87 | 6.28 | 9.19  | 0.68 | 6.28    | 9.19  | 0.68 |
|              | 20         | 15.19   | 6.09  | 2.49 | 15.19        | 6.09  | 2.49 | 11.59 | 7.68  | 1.51 | 7.76 | 8.58  | 0.90 | 6.22 | 8.77  | 0.71 | 6.22    | 8.77  | 0.71 |
|              | 25         | 16.41   | 4.62  | 3.55 | 15.43        | 4.88  | 3.16 | 11.73 | 6.20  | 1.89 | 7.64 | 6.98  | 1.09 | 5.95 | 7.18  | 0.83 | 5.95    | 7.18  | 0.83 |
| 20           | 30         | 17.63   | 3.68  | 4.80 | 15.43        | 4.21  | 3.66 | 11.74 | 5.24  | 2.24 | 7.90 | 5.91  | 1.34 | 5.69 | 6.34  | 0.90 | 5.69    | 6.34  | 0.90 |
|              | 35         | 17.42   | 3.07  | 5.68 | 15.45        | 3.50  | 4.41 | 11.72 | 4.36  | 2.69 | 7.77 | 4.90  | 1.59 | 5.39 | 5.18  | 1.04 | 5.39    | 5.18  | 1.04 |
|              | 40         | 14.12   | 3.14  | 4.50 | 14.12        | 3.14  | 4.50 | 11.63 | 3.65  | 3.19 | 7.78 | 4.10  | 1.90 | 5.11 | 4.36  | 1.17 | 5.11    | 4.36  | 1.17 |
|              |            |         |       |      |              |       |      |       |       |      |      |       |      |      |       |      |         |       |      |



## SUMMARY

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# 1. GENERAL CHARACTERISTICS

## 1.1 System Description

### Unit description

This series of air-water heat pumps meets the needs of winter and summer air conditioning of residential and commercial installations of small and medium power.

All units are suitable for outdoor installation and being able to produce water up to 75°C may be employed in systems with radiant floor, fancoils, radiators and for the indirect production of domestic hot water (DHW) via an external boiler (not provided).

The units are characterized by the use of a DC inverter compressor that allows you to modulate the capacity from 30 to 120% of the rated capacity and are complete with a hydronic kit including all the essential components for a quick and safe installation.

The units are characterized by high energy efficiency and low noise level and they can be used as the sole generator of the system or integrated with other energy sources such as backup electric heaters or boiler.

All units are supplied with temperature probe (supplied as standard, assembled by the installer) for domestic hot water tank (DHW) and with external air temperature probe (already installed on the unit) to realize the climatic control in heating and cooling modes.

All the units are accurately built and individually tested in the factory. The installation only requires the electrical and hydraulic connections.

**REFRIGERANT CIRCUIT:** contained in a compartment protected from the air flow to simplify the maintenance operations, is equipped with DC inverter motor driven compressor twin rotary type to ensure greater dynamic balancing and reduce vibrations. It is placed on vibration-damping rubber supports and wrapped by a double layer of sound-absorbing material to reduce the noise. Furthermore, the compressor is equipped with crankcase oil heater. The circuit is equipped with stainless steel brazed plates heat exchanger complete with antifreeze heater, bi-flow electronic expansion valve, 4-way valve, axial fans with brushless DC motor complete with safety protection grilles, finned coil with anti-corrosion treatment made of copper tubes and aluminium hydrophilic fins. The circuit is controlled by means of temperature probes and pressure transducers and protected by high pressure switches.

All units are equipped with variable fan speed control which allows operation with low external temperatures in cooling and high external temperatures in heating.

**HYDRAULIC CIRCUIT:** contained in a compartment protected from the air flow to simplify the maintenance operations, is equipped with variable electronic circulator (brushless DC motor), managed by the electronic board, water flow switch, automatic air vent, water manometer, expansion vessel, safety valve, Y water filter (supplied as standard, assembled by the installer). The plate heat exchanger and all the hydraulic pipes are thermally insulated to avoid the formation of condensation and reduce heat loss.

For units in version with system electric heater (IBH, (available only on some models), an electric heater is installed in the hydraulic circuit downstream of the plate exchanger which can function in addition (for example with low external temperatures) or as a replacement (for example in the event of a fault) for the thermal power of the heat pump.

### Accessory

- **GAS SEPARATOR.** This is a device to be installed on the system water delivery pipe and has the purpose of guaranteeing the rapid evacuation of the R290 refrigerant in the event of breakage of the plate exchanger. This allows you to comply with the requirements of Annex GG point 6 of the standard IEC 60335-2-40 standard
- **INERTIAL WATER TANK.** It is constituted of a 60-liter tank in painted sheet metal, thermally insulated and placed inside a painted sheet metal box that can be positioned below the unit.
- **WATER TEMPERATURE PROBE.** The temperature probe can be connected to perform the functions T1 / Tbt1 / Tbt2 / T5 / Tw2 / Tsolar (for more details refer to the installation and user manual of the unit).
- **RUBBER ANTIVIBRATION DAMPERS**

## 2. ERP DATA

| Models       |                                           |              | 4    | 6    | 8    | 10   | 12   | 14   | 16   | 12T  | 14T  | 16T  | UM     |
|--------------|-------------------------------------------|--------------|------|------|------|------|------|------|------|------|------|------|--------|
| Heating mode | Low temperature<br>(water outlet at 35°C) | Efficiency   | 200  | 193  | 204  | 200  | 184  | 182  | 181  | 184  | 182  | 181  | ηs (%) |
|              |                                           | Energy class | A+++ |      |      |      |      |      |      |      |      |      | class  |
|              |                                           | SCOP         | 5,07 | 4,89 | 5,19 | 5,07 | 4,67 | 4,63 | 4,59 | 4,67 | 4,63 | 4,59 | W/W    |
|              | Medium temperature<br>(water at 55°C)     | Efficiency   | 149  | 150  | 150  | 150  | 142  | 141  | 140  | 142  | 141  | 140  | ηs (%) |
|              |                                           | Energy class | A++  |      |      |      |      |      |      |      |      |      | class  |
|              |                                           | SCOP         | 3,79 | 3,82 | 3,82 | 3,82 | 3,62 | 3,61 | 3,57 | 3,62 | 3,61 | 3,57 | W/W    |
| Cooling mode | Water at 7°C                              | Efficiency   | 206  | 210  | 231  | 219  | 204  | 204  | 202  | 204  | 204  | 202  | ηs (%) |
|              |                                           | SEER         | 5,23 | 5,32 | 5,86 | 5,55 | 5,19 | 5,18 | 5,12 | 5,19 | 5,18 | 5,12 | W/W    |
|              | Water at 18°C                             | Efficiency   | 251  | 263  | 323  | 323  | 254  | 267  | 263  | 254  | 267  | 263  | ηs (%) |
|              |                                           | SEER         | 6,36 | 6,65 | 8,14 | 8,16 | 6,42 | 6,75 | 6,65 | 6,42 | 6,75 | 6,65 | W/W    |

### NOTA:

The efficiency, SCOP, SEER values indicated in the table refer to the average climate for units without any options or accessories.

The values are referred to units without options and accessories.

### 3. TECHNICAL DATA

| Performances data |                                         |     | 4    | 6    | 8    | 10   | 12   | 14   | 16   | 12T  | 14T  | 16T  | UM  |
|-------------------|-----------------------------------------|-----|------|------|------|------|------|------|------|------|------|------|-----|
| A7W35             | Heating capacity                        | nom | 4,5  | 6,2  | 8,4  | 10   | 12   | 14   | 15   | 12   | 14   | 15   | kW  |
|                   | Power input                             | nom | 0,87 | 1,27 | 1,68 | 2,13 | 2,50 | 3,11 | 3,41 | 2,50 | 3,11 | 3,41 | kW  |
|                   | COP                                     |     | 5,15 | 4,90 | 5,00 | 4,70 | 4,80 | 4,50 | 4,40 | 4,80 | 4,50 | 4,40 | W/W |
|                   | Water flow rate                         |     | 774  | 1066 | 1445 | 1720 | 2064 | 2408 | 2580 | 2064 | 2408 | 2580 | l/h |
|                   | Available static pressure without IBH * |     | 89   | 88   | 79   | 70   | 64   | 54   | 49   | 64   | 54   | 49   | kPa |
| A7W45             | Available static pressure with IBH *    |     | 84   | 82   | 76   | 67   | -    | -    | -    | 53   | 38   | 32   | kPa |
|                   | Heating capacity                        | nom | 4,5  | 6,4  | 8,2  | 10   | 12   | 14   | 15   | 12   | 14   | 15   | kW  |
|                   | Power input                             | nom | 1,11 | 1,68 | 2,13 | 2,74 | 3,24 | 4,00 | 4,48 | 3,24 | 4,00 | 4,48 | kW  |
|                   | COP                                     |     | 4,05 | 3,80 | 3,85 | 3,65 | 3,70 | 3,50 | 3,35 | 3,70 | 3,50 | 3,35 | W/W |
|                   | Water flow rate                         |     | 774  | 1101 | 1410 | 1720 | 2064 | 2408 | 2580 | 2064 | 2408 | 2580 | l/h |
| A7W55             | Available static pressure without IBH * |     | 89   | 87   | 80   | 70   | 64   | 54   | 49   | 64   | 54   | 49   | kPa |
|                   | Available static pressure with IBH *    |     | 84   | 82   | 77   | 67   | -    | -    | -    | 53   | 38   | 32   | kPa |
|                   | Heating capacity                        | nom | 4,6  | 6,2  | 7,8  | 9,5  | 12   | 14   | 15   | 12   | 14   | 15   | kW  |
|                   | Power input                             | nom | 1,44 | 2,00 | 2,44 | 3,12 | 3,87 | 4,67 | 5,26 | 3,87 | 4,67 | 5,26 | kW  |
|                   | COP                                     |     | 3,20 | 3,10 | 3,20 | 3,05 | 3,10 | 3,00 | 2,85 | 3,10 | 3,00 | 2,85 | W/W |
| A35W18            | Water flow rate                         |     | 495  | 667  | 839  | 1021 | 1290 | 1505 | 1613 | 1290 | 1505 | 1613 | l/h |
|                   | Available static pressure without IBH * |     | 89   | 89   | 86   | 86   | 86   | 81   | 78   | 86   | 81   | 78   | kPa |
|                   | Available static pressure with IBH *    |     | 86   | 85   | 83   | 83   | -    | -    | -    | 82   | 76   | 72   | kPa |
|                   | Cooling capacity                        | nom | 4,5  | 6,5  | 8,3  | 10   | 12   | 14   | 16   | 12   | 14   | 16   | kW  |
|                   | Power input                             | nom | 0,82 | 1,28 | 1,61 | 2,11 | 2,67 | 3,33 | 4,10 | 2,67 | 3,33 | 4,10 | kW  |
| A35W7             | EER                                     |     | 5,50 | 5,10 | 5,15 | 4,75 | 4,50 | 4,20 | 3,90 | 4,50 | 4,20 | 3,90 | W/W |
|                   | Water flow rate                         |     | 774  | 1118 | 1428 | 1720 | 2064 | 2408 | 2752 | 2064 | 2408 | 2752 | l/h |
|                   | Available static pressure without IBH * |     | 89   | 87   | 80   | 70   | 64   | 54   | 44   | 64   | 54   | 44   | kPa |
|                   | Available static pressure with IBH *    |     | 84   | 82   | 77   | 67   | -    | -    | -    | 53   | 38   | 24   | kPa |
|                   | Cooling capacity                        | nom | 4,7  | 6,8  | 7,5  | 8,9  | 11,5 | 12,7 | 14   | 11,5 | 12,7 | 14   | kW  |
| A35W7             | Power input                             | nom | 1,29 | 2,19 | 2,17 | 2,74 | 3,77 | 4,38 | 5,09 | 3,77 | 4,38 | 5,09 | kW  |
|                   | EER                                     |     | 3,65 | 3,10 | 3,45 | 3,25 | 3,05 | 2,90 | 2,75 | 3,05 | 2,90 | 2,75 | W/W |
|                   | Water flow rate                         |     | 808  | 1170 | 1290 | 1531 | 1978 | 2184 | 2408 | 1978 | 2184 | 2408 | l/h |
|                   | Available static pressure without IBH * |     | 89   | 87   | 83   | 77   | 66   | 60   | 54   | 66   | 60   | 54   | kPa |
|                   | Available static pressure with IBH *    |     | 84   | 81   | 80   | 74   | -    | -    | -    | 56   | 47   | 38   | kPa |

The values are referred to units without options and accessories.

Data declared according to EN 14511:

**EER** (Energy Efficiency Ratio) = ratio of the total cooling capacity to the effective power input of the unit

**COP** (Coefficient Of Performance) = ratio of the total heating capacity to the effective power input of the unit

**A7W35** = source : air in 7°C d.b. 6°C w.b. / plant : water in 30°C out 35°C

**A7W45** = source : air in 7°C d.b. 6°C w.b. / plant : water in 40°C out 45°C

**A7W55** = source : air in 7°C d.b. 6°C w.b. / plant : water in 47°C out 55°C

**A35W18** = source : air in 35°C d.b. / plant : water in 23°C out 18°C

**A35W7** = source : air in 35°C d.b. / plant : water in 12°C out 7°C

**IBH \*** = Electric heater system

| General data                                                        |        |           | 4                             | 6    | 8             | 10   | 12             | 14   | 16 | 12T                      | 14T | 16T             | UM |            |
|---------------------------------------------------------------------|--------|-----------|-------------------------------|------|---------------|------|----------------|------|----|--------------------------|-----|-----------------|----|------------|
| Power supply                                                        |        |           | 220-240 V ~ / 50 Hz           |      |               |      |                |      |    | 380-415 V~ / 3+N / 50 Hz |     |                 | -  |            |
| Compressor type                                                     |        |           | Twin Rotary DC                |      |               |      |                |      |    |                          |     |                 |    | -          |
| N° compressors / N° refrigerant circuits                            |        |           | 1 / 1                         |      |               |      |                |      |    |                          |     |                 |    | n°         |
| Plant side heat exchanger type                                      |        |           | stainless steel brazed plates |      |               |      |                |      |    |                          |     |                 |    | -          |
| Source side heat exchanger type                                     |        |           | finned coil                   |      |               |      |                |      |    |                          |     |                 |    | -          |
| Fans type                                                           |        |           | DC axial                      |      |               |      |                |      |    |                          |     |                 |    | -          |
| N° fans                                                             |        |           | 1                             |      |               |      |                |      |    |                          |     |                 |    | n°         |
| Expansion tank volume                                               |        |           | 8                             |      |               |      |                |      |    |                          |     |                 |    | l          |
| Water safety valve set                                              |        |           | 3                             |      |               |      |                |      |    |                          |     |                 |    | bar        |
| Electric heater system<br>(available only for version with IBH*)    |        |           | 3,25                          |      |               |      | -              |      |    |                          | 9   |                 |    | kW         |
| Hydraulic fittings                                                  |        |           | 1                             |      | 1 1/4         |      |                |      |    |                          |     |                 | “  |            |
| Minimum water content onf the system                                |        |           | 40                            |      |               |      |                |      |    |                          |     |                 |    | l          |
| DHW boiler - minimum surface of the coil<br>(minimum / recommended) | steel  | 1,4 / 2,5 | 1,75 / 4,0                    |      |               |      |                |      |    |                          | m²  |                 |    |            |
|                                                                     | enamel | 1,7 / 3,0 | 2,5 / 5,6                     |      |               |      |                |      |    |                          | m²  |                 |    |            |
| Refrigerant type                                                    |        |           | R290                          |      |               |      |                |      |    |                          |     |                 |    | tipo       |
| GWP                                                                 |        |           | 3                             |      |               |      |                |      |    |                          |     |                 |    | kg-CO2 eq. |
| Refrigerant charge                                                  |        |           | 0,7<br>0,0021                 |      | 1,1<br>0,0033 |      | 1,25<br>0,0375 |      |    |                          |     | kg<br>t-CO2 eq. |    |            |
| Control type                                                        |        |           | Remote wired                  |      |               |      |                |      |    |                          |     |                 |    | -          |
| SWL - Sound power level Heating *                                   | A7W35  | 56        | 58                            | 60   | 61            | 65   | 65             | 69   | 65 | 65                       | 69  | dB(A)           |    |            |
|                                                                     | Max    | 58        | 60                            | 62   | 63            | 67   | 68             | 70   | 67 | 68                       | 70  | dB(A)           |    |            |
|                                                                     | Sil. 1 | 54        | 56                            | 58   | 59            | 62   | 63             | 64   | 62 | 63                       | 64  | dB(A)           |    |            |
|                                                                     | Sil. 2 | 51        | 53                            | 55   | 56            | 58   | 59             | 60   | 58 | 59                       | 60  | dB(A)           |    |            |
| SWL - Sound power level Cooling *                                   | A35W18 | 56        | 58                            | 60   | 61            | 65   | 66             | 69   | 65 | 66                       | 69  | dB(A)           |    |            |
|                                                                     | Max    | 58        | 60                            | 62   | 63            | 66   | 67             | 70   | 66 | 67                       | 70  | dB(A)           |    |            |
|                                                                     | Sil. 1 | 54        | 56                            | 57   | 58            | 62   | 62             | 64   | 62 | 62                       | 64  | dB(A)           |    |            |
|                                                                     | Sil. 2 | 51        | 53                            | 56   | 55            | 58   | 59             | 60   | 58 | 59                       | 60  | dB(A)           |    |            |
| Max. current input                                                  |        |           | 12                            | 13,5 | 16            | 17,5 | 25             | 26,5 | 28 | 8,5                      | 9   | 9,5             | A  |            |

\* : **SWL** = Sound power levels, with reference to 1x10-12 W with unit operating in conditions:

**A7W35** = source : air in 7°C d.b. 6°C w.b. / plant : water in 30°C out 35°C

**A35W18** = source : air in 35°C d.b. / plant : water in 23°C out 18°C

**Max** = at maximum conditions in heating / cooling mode

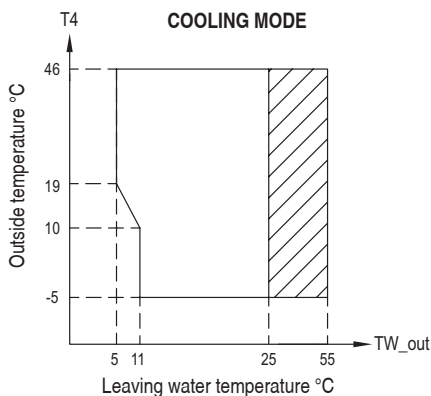
**Sil. 1** = if silent level 1 active in heating / cooling mode

**Sil. 2** = if silent level 2 active in heating / cooling mode

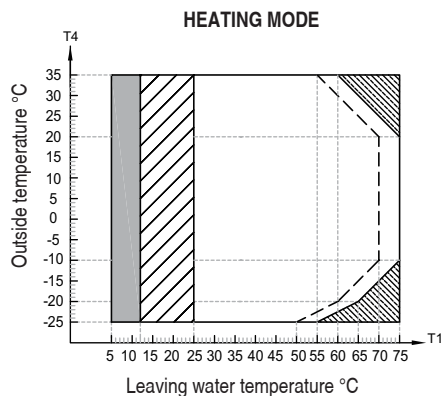
The Total sound power level in dB(A) measured in compliance with ISO 9614 standards.

**IBH \*** = Electric heater system

## 4. OPERATING LIMITS



Operation range by heat pump with possible limitation and protection.

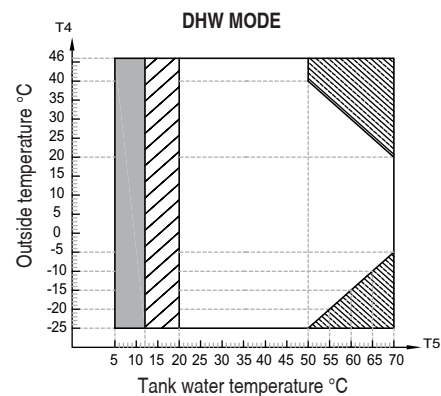


If the IBH (backup heater) / AHS (boiler) setting is enabled, only IBH / AHS is activated; If the IBH / AHS setting is not enabled, only the heat pump is activated with possible limitations and protections.

Operation range by heat pump with possible limitation and protection.

The heat pump remains off and only IBH (backup heater) / AHS (boiler) turns on

Maximum inlet water temperature line for heat pump operation.



If the TBH (tank backup heater) / IBH (backup heater) / AHS (boiler) setting is enabled, only TBH / IBH / AHS is activated.

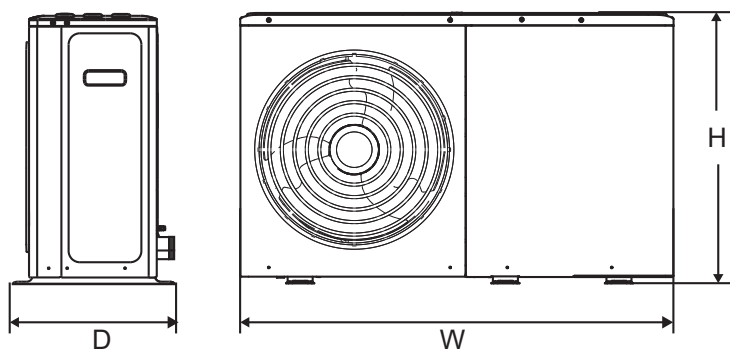
If the TBH / IBH / AHS setting is not enabled, only the heat pump is activated with possible limitations and protections.

Operation range by heat pump with possible limitation and protection.

The heat pump remains off and only TBH (backup heater) / IBH (electric heater) / AHS (boiler) turns on

**NOTE DHW MODE:** the maximum water temperature indicated in the graph is based on correct sizing of the coil surface of any DHW boiler.

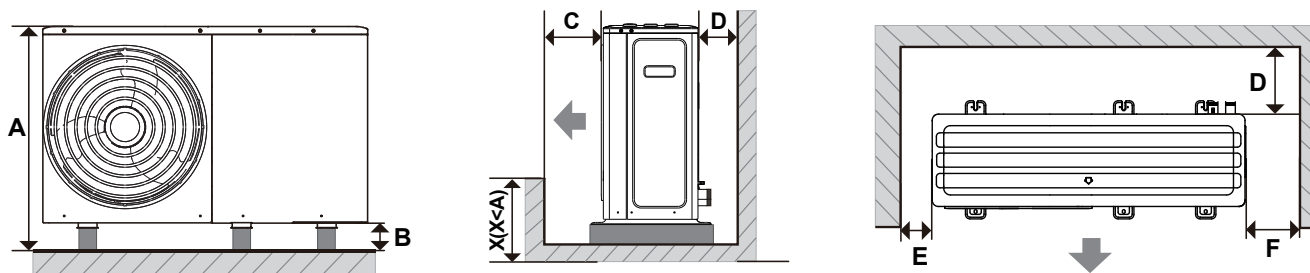
## 5. DIMENSIONAL DATA



| Modelli                                 | 4                | 6 | 8         | 10 | 12        | 14 | 16                | 12T | 14T | 16T | UM |
|-----------------------------------------|------------------|---|-----------|----|-----------|----|-------------------|-----|-----|-----|----|
| Unit (WxHxD)                            | 1299 x 717 x 426 |   |           |    |           |    | 1385 x 865 x 523  |     |     |     | mm |
| Packaging (WxHxD)                       | 1375 x 885 x 475 |   |           |    |           |    | 1465 x 1035 x 560 |     |     |     | mm |
| Weight Net \ Gross version without IBH* | 90 / 110         |   | 117 / 139 |    | 135 / 157 |    | 137 / 159         |     |     |     | kg |
| Weight Net \ Gross version with IBH*    | 95 / 115         |   | 122 / 144 |    | -         |    | 142 / 164         |     |     |     | kg |

IBH \* = Electric heater system

### 5.1 Minimum installation space



| Models | 4               | 6 | 8 | 10 | 12              | 14 | 16 | 12T | 14T | 16T | UM |
|--------|-----------------|---|---|----|-----------------|----|----|-----|-----|-----|----|
| A      | Unit height + B |   |   |    | Unit height + B |    |    |     |     |     | mm |
| B      | ≥ 100*          |   |   |    | ≥ 100*          |    |    |     |     |     | mm |
| C      | ≥ 1000          |   |   |    | ≥ 1500          |    |    |     |     |     | mm |
| D      | ≥ 300           |   |   |    | ≥ 300           |    |    |     |     |     | mm |
| E      | ≥ 500           |   |   |    | ≥ 500           |    |    |     |     |     | mm |
| F      | ≥ 500           |   |   |    | ≥ 500           |    |    |     |     |     | mm |

\* In case of cold, take into account the presence of snow on the ground.

## 6. THE CONTROL SYSTEM

The user interface consists of a wired remote controller (2 wires, max length 50 m from the unit) with menu in 11 languages (IT Italian, EN English, FR French, ES Spanish, DE German, NL Dutch, PL Polish, RO Romanian, PT Portuguese, RU Russian, TR Turkish, GR Greek, SE Swedish, SI Slovenian, CS Czech, SK Slovak, HU Hungarian, HR Croatian) which allows the management of:

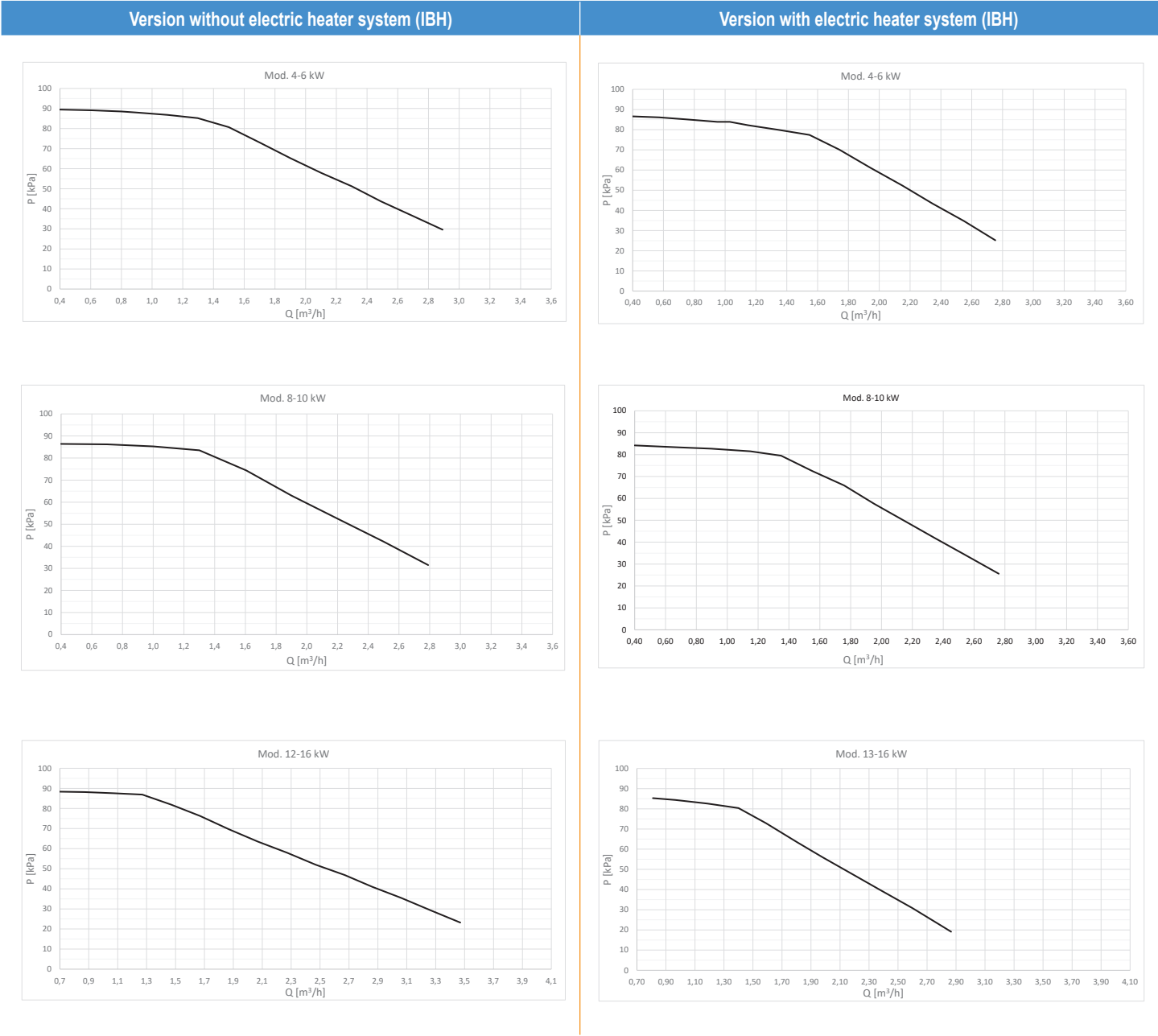
- **HEATING AND COOLING SYSTEM**, where the heat pump is the sole energy source. The unit, if activated in heat or cool mode, works by modulating the frequency of the compressor to maintain the temperature of the produced water to the setpoint value set by the controller. Through parameter you can use the remote controller (eg. For single-zone systems) as a room thermostat.
- **DOMESTIC HOT WATER PRODUCTION (DHW)**. The unit is activated in a heat mode to keep the temperature of a DHW tank (not supplied) to the setpoint value. It requires a 3-way diverter valve (not supplied) and a temperature sensor (T5 probe, L = 10m, provided) to be inserted into one well of the DHW tank.
- **ADDITIONAL SOURCES OF ENERGY** (boiler or electrical heater). Depending on the parameters set, these sources can be activated in integration or replacement of the heat pump when the system is used for space heating or for DHW production. The controller also activate additional energy sources in case the heat pump is not working. A temperature probe (available as an accessory) is required to be inserted into a well of the common pipe or storage tank to measure the temperature of the system water.
- **ELECTRIC HEATER OF THE DHW TANK**. The controller can manage the activation of an electric heater inserted in the DHW tank as a heat integration to the heat pump, for disinfecting function, or as a source of energy reserve for DHW production in case the heat pump is not working.
- **FAST DHW**. This function can be activated manually and it allows you to give priority to DHW production by activating all energy sources (heat pumps, electric heaters, boiler) available for DHW heating to bring in the shortest time possible the DHW tank to the setpoint required.
- **DISINFECT FUNCTION**. You can set from the controller weekly cycles for disinfecting the water in the DhW tank. In order to successfully execute these cycles, the heat pump must be integrated with DHW electric heater or boiler.
- **SILENT MODE**. If active it allows a reduction of the maximum frequency of the compressor and of the fan speed in order to reduce the noise emitted and the power absorbed by the unit. There are 2 levels of silencing. Through time programming, it is possible to define for 2 daily time bands in which the silent mode will be active (e.g. at night). There are 2 mute levels available. The selected level will then be applied to both set time bands
- **ON / OFF** using an external contact. The unit can be turned on and off (eg. thermostat / remote switch) via an external contact: in this case the unit will operate in the mode set by the controller keyboard.
- **HEAT / COOL** via external contacts. The unit can be activated in heat or cool mode via two external contacts (eg. thermostat that manages the heat and cool demand / remote switch).
- **ECO MODE**. For heating mode it is possible to define daily time band and corresponding set point for ECO mode.
- **WEEKLY SCHEDULING**. It allows a scheduling of 6 time bands for each day of the week: for each time band it is possible to define the mode (COOL / HEAT / DHW) and the required setpoint.
- **ANTIFREEZE PROTECTION**. Guaranteed for outdoor air temperature down to -25°C, thanks to the management of the electronic board of the unit which allows you to heat water using antifreeze heater (standard on the plate heat exchanger), the heat pump itself working in heating mode and the electric booster (if installed).
- **CASCADE FUNCTION** – up to 6 units can be controlled together in cascade mode (1 master unit, 5 slave units), with only one remote controller connected to the master unit. The master unit can be dedicated to the production of DHW (domestic hot water). In case of failure of a slave unit, all other units can operate normally.
- **MANAGEMENT OF UP TO 2 SYSTEM ZONES (1 MIXED AND 1 DIRECT)**. The unit is able to manage the pumps (not supplied) for both zones and, for the mixed zone only, also the mixing valve (not supplied) and the water delivery temperature probe (available as an accessory).
- **SOLAR THERMAL**. Forced circulation solar thermal management (circulator not supplied, probe available as an accessory) guaranteed by the operating logic of the heat pump.
- **PHOTOVOLTAIC INPUT AND SMART GRID INPUT**. The unit is equipped with 2 digital inputs for the management of an input from a photovoltaic system and from the electricity grid. Working logic:
  - SG=OFF; EVU=OFF > Normal operation
  - SG=ON; EVU=OFF > Electricity has a high cost, PDC and any electric heaters are turned off.
  - SG=OFF; EVU=ON > Electricity available at low price, priority DHW and auxiliary electric heaters ON.
  - SG=ON; EVU=ON > the unit activates DHW mode with DHW setpoint = 70°C and (if available) the electric heater of the DHW tank will turn on. The unit operates in cooling/heating mode like normal logic.
- **ENERGY MONITORING**. It is possible to monitor consumption, the power output and the share of renewable energy produced by the use of the heat pump, visible both from the user interface and from the APP.
- **CURRENT INPUT LIMITATION BY PARAMETER.**
- **REMOTE CONTROL OF THE UNIT VIA APP** (available for IOS and Android).
- **DETAILED ALARMS DIAGNOSTICS WITH ALARMS HISTORY.**
- **DISPLAY OF ALL OPERATING PARAMETERS.**



7. WORKING HEAD

The useful head is understood to be that at the unit outlet minus all the pressure drops inside the unit.  
The operating range is limited by the minimum value shown in the following table.

| Models                    | 4   | 6 | 8 | 10 | 12  | 14 | 16 | 12T | 14T | 16T | UM |      |
|---------------------------|-----|---|---|----|-----|----|----|-----|-----|-----|----|------|
| Minimum required capacity | 0,4 |   |   |    | 0,7 |    |    |     |     |     |    | m³/h |



NOTA  
P: Working head  
Q: Water flow rate

## 8. PERFORMANCE TABLES - HEATING MODE

### 8.1 Heating performance Mod. 04

| Tw_out<br>°C |     | Maximum |      |      | 100%(Normal) |             |             | 75%  |       |      | 50%  |       |      | 25%  |       |      | Minimum |       |      |  |
|--------------|-----|---------|------|------|--------------|-------------|-------------|------|-------|------|------|-------|------|------|-------|------|---------|-------|------|--|
| DB/WB<br>°C  | °C  | HC      | COP  | PI   | HC           | COP         | PI          | HC   | COP   | PI   | HC   | COP   | PI   | HC   | COP   | PI   | HC      | COP   | PI   |  |
| 25           | -25 | 3.24    | 2.17 | 1.50 | 3.24         | 2.17        | 1.50        | 2.42 | 2.43  | 1.00 | 1.60 | 2.48  | 0.65 | 1.34 | 2.50  | 0.54 | 1.34    | 2.50  | 0.54 |  |
|              | -20 | 3.92    | 2.50 | 1.57 | 3.92         | 2.50        | 1.57        | 2.95 | 2.77  | 1.06 | 1.97 | 2.80  | 0.70 | 1.63 | 2.82  | 0.58 | 1.63    | 2.82  | 0.58 |  |
|              | -15 | 4.63    | 2.85 | 1.63 | 4.45         | 2.89        | 1.54        | 3.28 | 2.97  | 1.11 | 2.20 | 3.12  | 0.70 | 1.92 | 3.15  | 0.61 | 1.92    | 3.15  | 0.61 |  |
|              | -10 | 5.54    | 3.26 | 1.70 | 4.43         | 3.43        | 1.29        | 3.32 | 3.60  | 0.92 | 2.35 | 3.62  | 0.65 | 2.35 | 3.62  | 0.65 | 2.35    | 3.62  | 0.65 |  |
|              | -7  | 5.78    | 3.56 | 1.62 | 4.41         | 3.75        | 1.18        | 3.28 | 3.90  | 0.84 | 2.55 | 3.94  | 0.65 | 2.55 | 3.94  | 0.65 | 2.55    | 3.94  | 0.65 |  |
|              | -5  | 5.95    | 3.74 | 1.59 | 4.49         | 4.10        | 1.09        | 3.39 | 4.21  | 0.81 | 2.73 | 4.28  | 0.64 | 2.73 | 4.28  | 0.64 | 2.73    | 4.28  | 0.64 |  |
|              | -2  | 6.07    | 4.06 | 1.49 | 4.38         | 4.50        | 0.97        | 3.31 | 4.61  | 0.72 | 2.43 | 4.64  | 0.52 | 2.43 | 4.64  | 0.52 | 2.43    | 4.64  | 0.52 |  |
|              | 0   | 6.16    | 4.32 | 1.43 | 4.42         | 4.78        | 0.92        | 3.37 | 4.90  | 0.69 | 2.59 | 4.97  | 0.52 | 2.59 | 4.97  | 0.52 | 2.59    | 4.97  | 0.52 |  |
|              | 2   | 6.29    | 4.58 | 1.37 | 4.40         | 5.06        | 0.87        | 3.33 | 5.18  | 0.64 | 2.74 | 5.27  | 0.52 | 2.74 | 5.27  | 0.52 | 2.74    | 5.27  | 0.52 |  |
|              | 5   | 6.80    | 5.31 | 1.28 | 4.38         | 5.86        | 0.75        | 3.28 | 6.02  | 0.54 | 2.79 | 6.08  | 0.46 | 2.79 | 6.08  | 0.46 | 2.79    | 6.08  | 0.46 |  |
|              | 7   | 7.18    | 5.74 | 1.25 | 4.55         | 6.08        | 0.75        | 3.32 | 6.32  | 0.53 | 3.01 | 6.38  | 0.47 | 3.01 | 6.38  | 0.47 | 3.01    | 6.38  | 0.47 |  |
|              | 10  | 7.04    | 6.25 | 1.13 | 4.46         | 6.89        | 0.65        | 3.34 | 7.17  | 0.47 | 3.34 | 7.17  | 0.47 | 3.34 | 7.17  | 0.47 | 3.34    | 7.17  | 0.47 |  |
|              | 12  | 7.26    | 6.79 | 1.07 | 4.43         | 7.57        | 0.59        | 3.53 | 7.77  | 0.45 | 3.53 | 7.77  | 0.45 | 3.53 | 7.77  | 0.45 | 3.53    | 7.77  | 0.45 |  |
|              | 15  | 6.90    | 7.25 | 0.95 | 4.42         | 7.88        | 0.56        | 3.62 | 8.14  | 0.44 | 3.62 | 8.14  | 0.44 | 3.62 | 8.14  | 0.44 | 3.62    | 8.14  | 0.44 |  |
|              | 20  | 6.48    | 8.11 | 0.80 | 4.52         | 8.79        | 0.51        | 3.40 | 9.23  | 0.37 | 3.40 | 9.23  | 0.37 | 3.40 | 9.23  | 0.37 | 3.40    | 9.23  | 0.37 |  |
|              | 25  | 6.42    | 9.45 | 0.68 | 4.44         | 10.23       | 0.43        | 3.75 | 10.57 | 0.35 | 3.75 | 10.57 | 0.35 | 3.75 | 10.57 | 0.35 | 3.75    | 10.57 | 0.35 |  |
|              | 30  |         |      |      |              |             |             |      |       |      |      |       |      |      |       |      |         |       |      |  |
|              | 35  |         |      |      |              |             |             |      |       |      |      |       |      |      |       |      |         |       |      |  |
| 30           | -25 | 3.22    | 1.99 | 1.62 | 3.22         | 1.99        | 1.62        | 2.39 | 2.04  | 1.17 | 1.60 | 2.10  | 0.76 | 1.29 | 2.12  | 0.61 | 1.29    | 2.12  | 0.61 |  |
|              | -20 | 3.88    | 2.29 | 1.70 | 3.88         | 2.29        | 1.70        | 2.91 | 2.37  | 1.23 | 1.94 | 2.45  | 0.79 | 1.58 | 2.48  | 0.64 | 1.58    | 2.48  | 0.64 |  |
|              | -15 | 4.56    | 2.65 | 1.72 | 4.56         | 2.65        | 1.72        | 3.30 | 2.69  | 1.23 | 2.18 | 2.78  | 0.78 | 1.85 | 2.85  | 0.65 | 1.85    | 2.85  | 0.65 |  |
|              | -10 | 5.45    | 3.02 | 1.80 | 4.42         | 3.17        | 1.39        | 3.32 | 3.23  | 1.03 | 2.20 | 3.28  | 0.67 | 2.20 | 3.28  | 0.67 | 2.20    | 3.28  | 0.67 |  |
|              | -7  | 5.67    | 3.29 | 1.72 | 4.42         | 3.44        | 1.29        | 3.30 | 3.52  | 0.94 | 2.52 | 3.63  | 0.69 | 2.52 | 3.63  | 0.69 | 2.52    | 3.63  | 0.69 |  |
|              | -5  | 5.83    | 3.45 | 1.69 | 4.36         | 3.69        | 1.18        | 3.26 | 3.75  | 0.87 | 2.63 | 3.79  | 0.69 | 2.63 | 3.79  | 0.69 | 2.63    | 3.79  | 0.69 |  |
|              | -2  | 5.95    | 3.74 | 1.59 | 4.40         | 4.02        | 1.09        | 3.33 | 4.08  | 0.82 | 2.39 | 4.18  | 0.57 | 2.39 | 4.18  | 0.57 | 2.39    | 4.18  | 0.57 |  |
|              | 0   | 6.04    | 3.97 | 1.52 | 4.42         | 4.19        | 1.05        | 3.31 | 4.30  | 0.77 | 2.49 | 4.38  | 0.57 | 2.49 | 4.38  | 0.57 | 2.49    | 4.38  | 0.57 |  |
|              | 2   | 6.17    | 4.19 | 1.47 | 4.42         | 4.51        | 0.98        | 3.36 | 4.58  | 0.73 | 2.64 | 4.65  | 0.57 | 2.64 | 4.65  | 0.57 | 2.64    | 4.65  | 0.57 |  |
|              | 5   | 6.66    | 4.84 | 1.38 | 4.36         | 5.15        | 0.85        | 3.26 | 5.25  | 0.62 | 2.79 | 5.37  | 0.52 | 2.79 | 5.37  | 0.52 | 2.79    | 5.37  | 0.52 |  |
|              | 7   | 7.03    | 5.20 | 1.35 | 4.54         | 5.63        | 0.81        | 3.36 | 5.79  | 0.58 | 3.04 | 5.85  | 0.52 | 3.04 | 5.85  | 0.52 | 3.04    | 5.85  | 0.52 |  |
|              | 10  | 7.18    | 5.81 | 1.24 | 4.44         | 6.34        | 0.70        | 3.33 | 6.44  | 0.52 | 3.20 | 6.46  | 0.50 | 3.20 | 6.46  | 0.50 | 3.20    | 6.46  | 0.50 |  |
|              | 12  | 7.42    | 6.10 | 1.21 | 4.42         | 6.58        | 0.67        | 3.44 | 6.78  | 0.51 | 3.44 | 6.78  | 0.51 | 3.44 | 6.78  | 0.51 | 3.44    | 6.78  | 0.51 |  |
|              | 15  | 6.74    | 6.37 | 1.06 | 4.53         | 6.80        | 0.67        | 3.50 | 7.02  | 0.50 | 3.50 | 7.02  | 0.50 | 3.50 | 7.02  | 0.50 | 3.50    | 7.02  | 0.50 |  |
|              | 20  | 6.35    | 7.10 | 0.89 | 4.53         | 7.55        | 0.60        | 3.29 | 7.90  | 0.42 | 3.29 | 7.90  | 0.42 | 3.29 | 7.90  | 0.42 | 3.29    | 7.90  | 0.42 |  |
|              | 25  | 6.31    | 8.18 | 0.77 | 4.44         | 8.78        | 0.51        | 3.62 | 9.07  | 0.40 | 3.62 | 9.07  | 0.40 | 3.62 | 9.07  | 0.40 | 3.62    | 9.07  | 0.40 |  |
|              | 30  | 6.63    | 9.33 | 0.71 | 4.44         | 9.61        | 0.46        | 3.98 | 9.77  | 0.41 | 3.98 | 9.77  | 0.41 | 3.98 | 9.77  | 0.41 | 3.98    | 9.77  | 0.41 |  |
|              | 35  |         |      |      |              |             |             |      |       |      |      |       |      |      |       |      |         |       |      |  |
| 35           | -25 | 3.19    | 1.86 | 1.71 | 3.19         | 1.86        | 1.71        | 2.40 | 1.92  | 1.26 | 1.58 | 2.01  | 0.79 | 1.24 | 2.05  | 0.61 | 1.24    | 2.05  | 0.61 |  |
|              | -20 | 3.82    | 2.14 | 1.79 | 3.82         | 2.14        | 1.79        | 2.87 | 2.18  | 1.31 | 1.92 | 2.28  | 0.84 | 1.52 | 2.32  | 0.65 | 1.52    | 2.32  | 0.65 |  |
|              | -15 | 4.47    | 2.46 | 1.82 | 4.42         | 2.40        | 1.84        | 3.31 | 2.49  | 1.33 | 2.20 | 2.64  | 0.83 | 1.79 | 2.66  | 0.67 | 1.79    | 2.66  | 0.67 |  |
|              | -10 | 5.35    | 2.81 | 1.90 | 4.42         | 2.86        | 1.55        | 3.27 | 2.97  | 1.10 | 2.18 | 3.08  | 0.71 | 2.18 | 3.08  | 0.71 | 2.18    | 3.08  | 0.71 |  |
|              | -7  | 5.56    | 3.04 | 1.83 | 4.50         | 3.10        | 1.45        | 3.32 | 3.12  | 1.06 | 2.36 | 3.21  | 0.73 | 2.36 | 3.21  | 0.73 | 2.36    | 3.21  | 0.73 |  |
|              | -5  | 5.70    | 3.18 | 1.80 | 4.37         | 3.32        | 1.32        | 3.30 | 3.37  | 0.98 | 2.53 | 3.43  | 0.74 | 2.53 | 3.43  | 0.74 | 2.53    | 3.43  | 0.74 |  |
|              | -2  | 5.82    | 3.44 | 1.69 | 4.27         | 3.58        | 1.19        | 3.18 | 3.67  | 0.87 | 2.31 | 3.74  | 0.62 | 2.31 | 3.74  | 0.62 | 2.31    | 3.74  | 0.62 |  |
|              | 0   | 5.90    | 3.64 | 1.62 | 4.38         | 3.78        | 1.16        | 3.28 | 3.88  | 0.85 | 2.39 | 3.95  | 0.61 | 2.39 | 3.95  | 0.61 | 2.39    | 3.95  | 0.61 |  |
|              | 2   | 6.05    | 3.84 | 1.57 | 4.40         | 4.10        | 1.07        | 3.34 | 4.14  | 0.81 | 2.53 | 4.19  | 0.60 | 2.53 | 4.19  | 0.60 | 2.53    | 4.19  | 0.60 |  |
|              | 5   | 6.50    | 4.40 | 1.48 | 4.33         | 4.61        | 0.94        | 3.24 | 4.70  | 0.69 | 2.68 | 4.75  | 0.56 | 2.68 | 4.75  | 0.56 | 2.68    | 4.75  | 0.56 |  |
|              | 7*  | 6.86    | 4.72 | 1.46 | <b>4.50</b>  | <b>5.15</b> | <b>0.87</b> | 3.34 | 5.25  | 0.64 | 2.93 | 5.32  | 0.55 | 2.93 | 5.32  | 0.55 | 2.93    | 5.32  | 0.55 |  |
|              | 10  | 7.00    | 5.21 | 1.34 | 4.43         | 5.58        | 0.79        | 3.37 | 5.67  | 0.59 | 3.24 | 5.78  | 0.56 | 3.24 | 5.78  | 0.56 | 3.24    | 5.78  | 0.56 |  |
|              | 12  | 7.23    | 5.59 | 1.29 | 4.49         | 6.01        | 0.75        | 3.48 | 6.14  | 0.57 | 3.45 | 6.22  | 0.56 | 3.45 | 6.22  | 0.56 | 3.45    | 6.22  | 0.56 |  |
|              | 15  | 6.86    | 5.87 | 1.17 | 4.49         | 6.20        | 0.72        | 3.55 | 6.36  | 0.56 | 3.55 | 6.36  | 0.56 | 3.55 | 6.36  | 0.56 | 3.55    | 6.36  | 0.56 |  |
|              | 20  | 6.20    | 6.23 | 0.99 | 4.52         | 6.52        | 0.69        | 3.17 | 6.66  | 0.48 | 3.17 | 6.66  | 0.48 | 3.17 | 6.66  | 0.48 | 3.17    | 6.66  | 0.48 |  |
|              | 25  | 6.31    | 7.10 | 0.89 | 4.46         | 7.48        | 0.60        | 3.50 | 7.69  | 0.46 | 3.50 | 7.69  | 0.46 | 3.50 | 7.69  | 0.46 | 3.50    | 7.69  | 0.46 |  |
|              | 30  | 6.51    | 8.05 | 0.81 | 4.45         | 8.58        | 0.52        | 3.84 | 8.75  | 0.44 | 3.84 | 8.75  | 0.44 | 3.84 | 8.75  | 0.44 | 3.84    | 8.75  | 0.44 |  |
|              | 35  | 6.76    | 9.10 | 0.74 | 4.40         | 10.04       | 0.44        | 4.20 | 10.05 | 0.42 | 4.20 | 10.05 | 0.42 | 4.20 | 10.05 | 0.42 | 4.20    | 10.05 | 0.42 |  |
| 40           | -25 | 3.15    | 1.79 | 1.76 | 3.15         | 1.79        | 1.76        | 2.36 | 1.84  | 1.29 | 1.56 | 1.94  | 0.81 | 1.26 | 1.95  | 0.65 | 1.26    | 1.95  | 0.65 |  |
|              | -20 | 3.77    | 2.04 | 1.85 | 3.77         | 2.04        | 1.85        | 2.83 | 2.08  | 1.36 | 1.89 | 2.19  | 0.87 | 1.54 | 2.22  | 0.69 | 1.54    | 2.22  | 0.69 |  |
|              | -15 | 4.38    | 2.29 | 1.92 | 4.38         | 2.29        | 1.92        | 3.27 | 2.32  | 1.41 | 2.17 | 2.43  | 0.89 | 1.82 | 2.46  | 0.74 | 1.82    | 2.46  | 0.74 |  |
|              | -10 | 5.22    | 2.63 | 1.99 | 4.50         | 2.65        | 1.70        | 3.40 | 2.69  | 1.26 | 2.26 | 2.73  | 0.83 | 2.16 | 2.75  | 0.79 | 2.16    | 2.75  | 0.79 |  |
|              | -7  | 5.44    | 2.81 | 1.94 | 4.50         | 2.89        | 1.56        | 3.34 | 2.97  | 1.13 | 2.39 | 3.07  | 0.78 | 2.39 | 3.07  | 0.78 | 2.39    | 3.07  | 0.78 |  |
|              | -5  | 5.59    | 2.93 | 1.91 | 4.45         | 3.02        | 1.47        | 3.31 | 3.06  | 1.08 | 2.56 | 3.20  | 0.80 | 2.56 | 3.20  | 0.80 | 2.56    | 3.20  | 0.80 |  |
|              | -2  | 5.70    | 3.17 | 1.80 | 4.29         | 3.28        | 1.31        | 3.22 | 3.34  | 0.97 | 2.57 | 3.36  | 0.76 | 2.57 | 3.36  | 0.76 |         |       |      |  |

| Tw_out<br>°C | DBWB<br>°C | Maximum |      |      | 100%(Normal) |      |      | 75%  |      |      | 50%  |      |      | 25%  |      |      | Minimum |      |      |
|--------------|------------|---------|------|------|--------------|------|------|------|------|------|------|------|------|------|------|------|---------|------|------|
|              |            | HC      | COP  | PI   | HC           | COP  | PI   | HC   | COP  | PI   | HC   | COP  | PI   | HC   | COP  | PI   | HC      | COP  | PI   |
| 50           | -25        | 3.03    | 1.58 | 1.92 | 3.03         | 1.58 | 1.92 | 2.27 | 1.61 | 1.41 | 1.52 | 1.66 | 0.92 | 1.52 | 1.66 | 0.92 | 1.52    | 1.66 | 0.92 |
|              | -20        | 3.61    | 1.78 | 2.03 | 3.61         | 1.78 | 2.03 | 2.69 | 1.81 | 1.48 | 1.81 | 1.88 | 0.96 | 1.81 | 1.88 | 0.96 | 1.81    | 1.88 | 0.96 |
|              | -15        | 4.16    | 1.96 | 2.12 | 4.16         | 1.96 | 2.12 | 3.12 | 1.99 | 1.57 | 2.12 | 2.06 | 1.03 | 2.12 | 2.06 | 1.03 | 2.12    | 2.06 | 1.03 |
|              | -10        | 4.96    | 2.22 | 2.24 | 4.61         | 2.28 | 2.02 | 3.45 | 2.32 | 1.49 | 2.52 | 2.37 | 1.06 | 2.52 | 2.37 | 1.06 | 2.52    | 2.37 | 1.06 |
|              | -7         | 5.17    | 2.38 | 2.17 | 4.64         | 2.42 | 1.92 | 3.41 | 2.50 | 1.37 | 2.77 | 2.55 | 1.09 | 2.77 | 2.55 | 1.09 | 2.77    | 2.55 | 1.09 |
|              | -5         | 5.31    | 2.47 | 2.15 | 4.51         | 2.57 | 1.75 | 3.41 | 2.60 | 1.31 | 2.99 | 2.63 | 1.13 | 2.99 | 2.63 | 1.13 | 2.99    | 2.63 | 1.13 |
|              | -2         | 5.41    | 2.66 | 2.04 | 4.43         | 2.71 | 1.64 | 3.33 | 2.74 | 1.22 | 3.04 | 2.79 | 1.09 | 3.04 | 2.79 | 1.09 | 3.04    | 2.79 | 1.09 |
|              | 0          | 5.50    | 2.80 | 1.97 | 4.54         | 2.90 | 1.56 | 3.43 | 2.87 | 1.19 | 3.24 | 2.94 | 1.10 | 3.24 | 2.94 | 1.10 | 3.24    | 2.94 | 1.10 |
|              | 2          | 5.63    | 2.96 | 1.90 | 4.50         | 3.00 | 1.50 | 3.40 | 3.03 | 1.12 | 3.40 | 3.03 | 1.12 | 3.40 | 3.03 | 1.12 | 3.40    | 3.03 | 1.12 |
|              | 5          | 6.02    | 3.20 | 1.88 | 4.44         | 3.34 | 1.33 | 3.90 | 3.34 | 1.17 | 3.90 | 3.34 | 1.17 | 3.90 | 3.34 | 1.17 | 3.90    | 3.34 | 1.17 |
|              | 7          | 6.35    | 3.51 | 1.81 | 4.57         | 3.54 | 1.29 | 4.26 | 3.58 | 1.19 | 4.26 | 3.58 | 1.19 | 4.26 | 3.58 | 1.19 | 4.26    | 3.58 | 1.19 |
|              | 10         | 6.46    | 3.79 | 1.70 | 4.54         | 3.90 | 1.16 | 4.30 | 3.92 | 1.10 | 4.30 | 3.92 | 1.10 | 4.30 | 3.92 | 1.10 | 4.30    | 3.92 | 1.10 |
|              | 12         | 6.69    | 4.03 | 1.66 | 4.63         | 4.12 | 1.13 | 4.60 | 4.16 | 1.11 | 4.60 | 4.16 | 1.11 | 4.60 | 4.16 | 1.11 | 4.60    | 4.16 | 1.11 |
|              | 15         | 6.31    | 4.17 | 1.51 | 4.77         | 4.25 | 1.12 | 4.73 | 4.29 | 1.10 | 4.73 | 4.29 | 1.10 | 4.73 | 4.29 | 1.10 | 4.73    | 4.29 | 1.10 |
|              | 20         | 5.95    | 4.51 | 1.32 | 4.51         | 4.55 | 0.99 | 4.13 | 4.59 | 0.90 | 4.13 | 4.59 | 0.90 | 4.13 | 4.59 | 0.90 | 4.13    | 4.59 | 0.90 |
|              | 25         | 5.95    | 4.98 | 1.19 | 4.61         | 5.06 | 0.91 | 4.56 | 5.08 | 0.90 | 4.56 | 5.08 | 0.90 | 4.56 | 5.08 | 0.90 | 4.56    | 5.08 | 0.90 |
|              | 30         | 6.26    | 5.46 | 1.15 | 5.08         | 5.70 | 0.89 | 5.03 | 5.71 | 0.88 | 5.03 | 5.71 | 0.88 | 5.03 | 5.71 | 0.88 | 5.03    | 5.71 | 0.88 |
|              | 35         | 6.53    | 6.05 | 1.08 | 5.58         | 6.16 | 0.91 | 5.54 | 6.22 | 0.89 | 5.54 | 6.22 | 0.89 | 5.54 | 6.22 | 0.89 | 5.54    | 6.22 | 0.89 |
|              | -25        | 2.87    | 1.44 | 2.00 | 2.87         | 1.44 | 2.00 | 2.15 | 1.46 | 1.47 | 1.55 | 1.50 | 1.03 | 1.55 | 1.50 | 1.03 | 1.55    | 1.50 | 1.03 |
|              | -20        | 3.33    | 1.59 | 2.10 | 3.33         | 1.59 | 2.10 | 2.56 | 1.65 | 1.55 | 1.87 | 1.69 | 1.11 | 1.87 | 1.69 | 1.11 | 1.87    | 1.69 | 1.11 |
| -15          | 3.93       | 1.77    | 2.22 | 3.93 | 1.77         | 2.22 | 2.93 | 1.81 | 1.62 | 2.19 | 1.85 | 1.18 | 2.19 | 1.85 | 1.18 | 2.19 | 1.85    | 1.18 |      |
| -10          | 4.66       | 1.98    | 2.36 | 4.60 | 1.99         | 2.32 | 3.44 | 2.00 | 1.72 | 2.59 | 2.04 | 1.27 | 2.59 | 2.04 | 1.27 | 2.59 | 2.04    | 1.27 |      |
| -7           | 4.85       | 2.14    | 2.27 | 4.70 | 2.14         | 2.20 | 3.41 | 2.17 | 1.57 | 2.86 | 2.20 | 1.30 | 2.86 | 2.20 | 1.30 | 2.86 | 2.20    | 1.30 |      |
| -5           | 5.00       | 2.22    | 2.25 | 4.55 | 2.25         | 2.02 | 3.42 | 2.28 | 1.50 | 3.07 | 2.30 | 1.33 | 3.07 | 2.30 | 1.33 | 3.07 | 2.30    | 1.33 |      |
| -2           | 5.11       | 2.37    | 2.16 | 4.46 | 2.50         | 1.78 | 3.35 | 2.51 | 1.33 | 3.25 | 2.57 | 1.27 | 3.25 | 2.57 | 1.27 | 3.25 | 2.57    | 1.27 |      |
| 0            | 5.37       | 2.58    | 2.08 | 4.61 | 2.61         | 1.76 | 3.49 | 2.65 | 1.32 | 3.49 | 2.65 | 1.32 | 3.49 | 2.65 | 1.32 | 3.49 | 2.65    | 1.32 |      |
| 2            | 5.50       | 2.70    | 2.04 | 4.60 | 2.70         | 1.70 | 3.67 | 2.73 | 1.34 | 3.67 | 2.73 | 1.34 | 3.67 | 2.73 | 1.34 | 3.67 | 2.73    | 1.34 |      |
| 5            | 5.88       | 2.98    | 1.97 | 4.48 | 3.03         | 1.48 | 4.17 | 3.05 | 1.37 | 4.17 | 3.05 | 1.37 | 4.17 | 3.05 | 1.37 | 4.17 | 3.05    | 1.37 |      |
| 7            | 6.20       | 3.10    | 2.00 | 4.60 | 3.20         | 1.44 | 4.55 | 3.23 | 1.41 | 4.55 | 3.23 | 1.41 | 4.55 | 3.23 | 1.41 | 4.55 | 3.23    | 1.41 |      |
| 10           | 6.32       | 3.38    | 1.87 | 4.59 | 3.45         | 1.33 | 4.26 | 3.53 | 1.21 | 4.26 | 3.53 | 1.21 | 4.26 | 3.53 | 1.21 | 4.26 | 3.53    | 1.21 |      |
| 12           | 6.50       | 3.55    | 1.83 | 4.69 | 3.69         | 1.27 | 4.55 | 3.74 | 1.22 | 4.55 | 3.74 | 1.22 | 4.55 | 3.74 | 1.22 | 4.55 | 3.74    | 1.22 |      |
| 15           | 6.15       | 3.77    | 1.63 | 4.83 | 3.80         | 1.27 | 4.68 | 3.85 | 1.22 | 4.68 | 3.85 | 1.22 | 4.68 | 3.85 | 1.22 | 4.68 | 3.85    | 1.22 |      |
| 20           | 5.78       | 4.04    | 1.43 | 4.60 | 4.10         | 1.12 | 4.11 | 4.07 | 1.01 | 4.11 | 4.07 | 1.01 | 4.11 | 4.07 | 1.01 | 4.11 | 4.07    | 1.01 |      |
| 25           | 5.76       | 4.42    | 1.30 | 4.71 | 4.51         | 1.04 | 4.56 | 4.55 | 1.00 | 4.56 | 4.55 | 1.00 | 4.56 | 4.55 | 1.00 | 4.56 | 4.55    | 1.00 |      |
| 30           | 6.07       | 4.85    | 1.25 | 5.19 | 4.90         | 1.06 | 5.03 | 4.91 | 1.02 | 5.03 | 4.91 | 1.02 | 5.03 | 4.91 | 1.02 | 5.03 | 4.91    | 1.02 |      |
| 35           | 6.32       | 5.00    | 1.26 | 5.73 | 5.06         | 1.13 | 5.54 | 5.08 | 1.09 | 5.54 | 5.08 | 1.09 | 5.54 | 5.08 | 1.09 | 5.54 | 5.08    | 1.09 |      |
| -25          | /          | /       | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
| -20          | 3.29       | 1.49    | 2.21 | 3.29 | 1.49         | 2.21 | 2.47 | 1.52 | 1.63 | 1.94 | 1.53 | 1.27 | 1.94 | 1.53 | 1.27 | 1.94 | 1.53    | 1.27 |      |
| -15          | 3.78       | 1.62    | 2.34 | 3.78 | 1.62         | 2.34 | 2.83 | 1.66 | 1.70 | 2.27 | 1.69 | 1.35 | 2.27 | 1.69 | 1.35 | 2.27 | 1.69    | 1.35 |      |
| -10          | 4.47       | 1.80    | 2.49 | 4.47 | 1.80         | 2.49 | 3.43 | 1.84 | 1.86 | 2.67 | 1.86 | 1.43 | 2.67 | 1.86 | 1.43 | 2.67 | 1.86    | 1.43 |      |
| -7           | 4.68       | 1.93    | 2.43 | 4.61 | 1.93         | 2.39 | 3.45 | 1.96 | 1.76 | 2.94 | 1.99 | 1.48 | 2.94 | 1.99 | 1.48 | 2.94 | 1.99    | 1.48 |      |
| -5           | 4.82       | 2.00    | 2.40 | 4.58 | 2.03         | 2.26 | 3.47 | 2.10 | 1.65 | 3.16 | 2.13 | 1.48 | 3.16 | 2.13 | 1.48 | 3.16 | 2.13    | 1.48 |      |
| -2           | 4.93       | 2.16    | 2.29 | 4.44 | 2.28         | 1.95 | 3.34 | 2.32 | 1.44 | 3.10 | 2.36 | 1.31 | 3.10 | 2.36 | 1.31 | 3.10 | 2.36    | 1.31 |      |
| 0            | 5.20       | 2.34    | 2.22 | 4.41 | 2.35         | 1.88 | 3.31 | 2.42 | 1.37 | 3.31 | 2.42 | 1.37 | 3.31 | 2.42 | 1.37 | 3.31 | 2.42    | 1.37 |      |
| 2            | 5.32       | 2.44    | 2.18 | 4.37 | 2.51         | 1.74 | 3.50 | 2.55 | 1.37 | 3.50 | 2.55 | 1.37 | 3.50 | 2.55 | 1.37 | 3.50 | 2.55    | 1.37 |      |
| 5            | 5.67       | 2.73    | 2.08 | 4.56 | 2.80         | 1.63 | 3.97 | 2.84 | 1.40 | 3.97 | 2.84 | 1.40 | 3.97 | 2.84 | 1.40 | 3.97 | 2.84    | 1.40 |      |
| 7            | 5.97       | 2.89    | 2.07 | 4.58 | 2.94         | 1.56 | 4.34 | 2.98 | 1.46 | 4.34 | 2.98 | 1.46 | 4.34 | 2.98 | 1.46 | 4.34 | 2.98    | 1.46 |      |
| 10           | 6.06       | 3.02    | 2.01 | 4.57 | 3.10         | 1.47 | 4.22 | 3.15 | 1.34 | 4.22 | 3.15 | 1.34 | 4.22 | 3.15 | 1.34 | 4.22 | 3.15    | 1.34 |      |
| 12           | 6.26       | 3.09    | 2.03 | 4.66 | 3.24         | 1.44 | 4.51 | 3.27 | 1.38 | 4.51 | 3.27 | 1.38 | 4.51 | 3.27 | 1.38 | 4.51 | 3.27    | 1.38 |      |
| 15           | 5.91       | 3.35    | 1.77 | 4.79 | 3.39         | 1.41 | 4.65 | 3.43 | 1.35 | 4.65 | 3.43 | 1.35 | 4.65 | 3.43 | 1.35 | 4.65 | 3.43    | 1.35 |      |
| 20           | 5.52       | 3.56    | 1.55 | 4.62 | 3.66         | 1.26 | 3.91 | 3.70 | 1.06 | 3.91 | 3.70 | 1.06 | 3.91 | 3.70 | 1.06 | 3.91 | 3.70    | 1.06 |      |
| 25           | 5.48       | 3.88    | 1.41 | 4.60 | 3.92         | 1.17 | 4.32 | 3.97 | 1.09 | 4.32 | 3.97 | 1.09 | 4.32 | 3.97 | 1.09 | 4.32 | 3.97    | 1.09 |      |
| 30           | 5.78       | 4.17    | 1.39 | 4.95 | 4.25         | 1.17 | 4.78 | 4.28 | 1.11 | 4.78 | 4.28 | 1.11 | 4.78 | 4.28 | 1.11 | 4.78 | 4.28    | 1.11 |      |
| 35           | /          | /       | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
| -25          | /          | /       | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
| -20          | 3.14       | 1.36    | 2.31 | 3.14 | 1.36         | 2.31 | 2.35 | 1.37 | 1.71 | 1.86 | 1.39 | 1.34 | 1.86 | 1.39 | 1.34 | 1.86 | 1.39    | 1.34 |      |
| -15          | 3.61       | 1.47    | 2.45 | 3.61 | 1.47         | 2.45 | 2.72 | 1.51 | 1.81 | 2.16 | 1.51 | 1.43 | 2.16 | 1.51 | 1.43 | 2.16 | 1.51    | 1.43 |      |
| -10          | 4.26       | 1.62    | 2.62 | 4.26 | 1.62         | 2.62 | 3.19 | 1.67 | 1.91 | 2.55 | 1.69 | 1.51 | 2.55 | 1    |      |      |         |      |      |

## 8.2 Heating performance Mod. 06

| Tw_out<br>°C | DB/WB<br>°C | Maximum |      |      | 100%(Normal) |             |             | 75%  |       |      | 50%  |       |      | 25%  |       |      | Minimum |       |      |
|--------------|-------------|---------|------|------|--------------|-------------|-------------|------|-------|------|------|-------|------|------|-------|------|---------|-------|------|
|              |             | HC      | COP  | PI   | HC           | COP         | PI          | HC   | COP   | PI   | HC   | COP   | PI   | HC   | COP   | PI   | HC      | COP   | PI   |
| 25           | -25         | 3.50    | 2.14 | 1.64 | 3.50         | 2.14        | 1.64        | 2.64 | 2.40  | 1.10 | 1.75 | 2.47  | 0.71 | 1.34 | 2.50  | 0.54 | 1.34    | 2.50  | 0.54 |
|              | -20         | 4.35    | 2.41 | 1.80 | 4.35         | 2.41        | 1.80        | 3.26 | 2.72  | 1.20 | 2.18 | 2.80  | 0.78 | 1.63 | 2.82  | 0.58 | 1.63    | 2.82  | 0.58 |
|              | -15         | 5.18    | 2.77 | 1.87 | 5.18         | 2.77        | 1.87        | 3.89 | 2.91  | 1.34 | 2.63 | 2.98  | 0.88 | 1.92 | 3.15  | 0.61 | 1.92    | 3.15  | 0.61 |
|              | -10         | 6.08    | 3.16 | 1.92 | 5.83         | 3.21        | 1.81        | 4.34 | 3.48  | 1.25 | 2.90 | 3.61  | 0.80 | 2.35 | 3.62  | 0.65 | 2.35    | 3.62  | 0.65 |
|              | -7          | 6.36    | 3.41 | 1.86 | 5.79         | 3.56        | 1.63        | 4.32 | 3.78  | 1.14 | 2.90 | 3.92  | 0.74 | 2.55 | 3.94  | 0.65 | 2.55    | 3.94  | 0.65 |
|              | -5          | 6.55    | 3.59 | 1.83 | 5.66         | 3.85        | 1.47        | 4.25 | 4.14  | 1.03 | 2.83 | 4.27  | 0.66 | 2.73 | 4.28  | 0.64 | 2.73    | 4.28  | 0.64 |
|              | -2          | 6.84    | 3.87 | 1.77 | 5.47         | 4.28        | 1.28        | 4.18 | 4.52  | 0.93 | 2.83 | 4.62  | 0.61 | 2.43 | 4.64  | 0.52 | 2.43    | 4.64  | 0.52 |
|              | 0           | 6.99    | 4.10 | 1.70 | 5.66         | 4.49        | 1.26        | 4.29 | 4.79  | 0.90 | 2.78 | 4.91  | 0.57 | 2.59 | 4.97  | 0.52 | 2.59    | 4.97  | 0.52 |
|              | 2           | 7.09    | 4.34 | 1.64 | 5.65         | 4.78        | 1.18        | 4.22 | 5.09  | 0.83 | 2.85 | 5.20  | 0.55 | 2.74 | 5.27  | 0.52 | 2.74    | 5.27  | 0.52 |
|              | 5           | 7.72    | 5.06 | 1.53 | 5.85         | 5.65        | 1.03        | 4.41 | 5.85  | 0.75 | 2.90 | 6.07  | 0.48 | 2.79 | 6.08  | 0.46 | 2.79    | 6.08  | 0.46 |
|              | 7           | 8.00    | 5.50 | 1.45 | 6.24         | 5.92        | 1.05        | 4.68 | 6.07  | 0.77 | 3.14 | 6.35  | 0.49 | 3.01 | 6.38  | 0.47 | 3.01    | 6.38  | 0.47 |
|              | 10          | 7.95    | 5.89 | 1.35 | 6.24         | 6.55        | 0.95        | 4.62 | 6.87  | 0.67 | 3.34 | 7.17  | 0.47 | 3.34 | 7.17  | 0.47 | 3.34    | 7.17  | 0.47 |
|              | 12          | 8.16    | 6.40 | 1.27 | 6.16         | 7.14        | 0.86        | 4.57 | 7.53  | 0.61 | 3.53 | 7.77  | 0.45 | 3.53 | 7.77  | 0.45 | 3.53    | 7.77  | 0.45 |
|              | 15          | 7.92    | 6.79 | 1.17 | 6.27         | 7.39        | 0.85        | 4.65 | 7.81  | 0.60 | 3.62 | 8.14  | 0.44 | 3.62 | 8.14  | 0.44 | 3.62    | 8.14  | 0.44 |
|              | 20          | 7.56    | 7.89 | 0.96 | 6.25         | 8.21        | 0.76        | 4.64 | 8.76  | 0.53 | 3.40 | 9.23  | 0.37 | 3.40 | 9.23  | 0.37 | 3.40    | 9.23  | 0.37 |
|              | 25          | 7.12    | 9.21 | 0.77 | 6.30         | 9.54        | 0.66        | 4.71 | 10.02 | 0.47 | 3.75 | 10.57 | 0.35 | 3.75 | 10.57 | 0.35 | 3.75    | 10.57 | 0.35 |
|              | 30          | /       | /    | /    | /            | /           | /           | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 35          | /       | /    | /    | /            | /           | /           | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 30           | -25         | 3.46    | 1.89 | 1.83 | 3.46         | 1.89        | 1.83        | 2.59 | 2.02  | 1.28 | 1.73 | 2.06  | 0.84 | 1.29 | 2.12  | 0.61 | 1.29    | 2.12  | 0.61 |
|              | -20         | 4.30    | 2.25 | 1.91 | 4.30         | 2.25        | 1.91        | 3.23 | 2.36  | 1.37 | 2.15 | 2.40  | 0.90 | 1.58 | 2.48  | 0.64 | 1.58    | 2.48  | 0.64 |
|              | -15         | 5.11    | 2.58 | 1.98 | 5.11         | 2.58        | 1.98        | 3.83 | 2.66  | 1.44 | 2.56 | 2.71  | 0.94 | 1.85 | 2.85  | 0.65 | 1.85    | 2.85  | 0.65 |
|              | -10         | 6.00    | 2.96 | 2.03 | 5.77         | 2.98        | 1.94        | 4.34 | 3.22  | 1.35 | 2.91 | 3.27  | 0.89 | 2.20 | 3.28  | 0.67 | 2.20    | 3.28  | 0.67 |
|              | -7          | 6.27    | 3.19 | 1.97 | 5.76         | 3.29        | 1.75        | 4.33 | 3.49  | 1.24 | 2.92 | 3.56  | 0.82 | 2.52 | 3.63  | 0.69 | 2.52    | 3.63  | 0.69 |
|              | -5          | 6.45    | 3.34 | 1.93 | 5.68         | 3.51        | 1.62        | 4.26 | 3.72  | 1.15 | 2.87 | 3.76  | 0.76 | 2.63 | 3.79  | 0.69 | 2.63    | 3.79  | 0.69 |
|              | -2          | 6.73    | 3.59 | 1.87 | 5.48         | 3.90        | 1.41        | 4.13 | 4.03  | 1.02 | 2.80 | 4.13  | 0.68 | 2.39 | 4.18  | 0.57 | 2.39    | 4.18  | 0.57 |
|              | 0           | 6.87    | 3.79 | 1.81 | 5.66         | 4.08        | 1.39        | 4.29 | 4.21  | 1.02 | 2.85 | 4.34  | 0.66 | 2.49 | 4.38  | 0.57 | 2.49    | 4.38  | 0.57 |
|              | 2           | 6.96    | 4.00 | 1.74 | 5.65         | 4.31        | 1.31        | 4.28 | 4.53  | 0.95 | 2.89 | 4.62  | 0.63 | 2.64 | 4.65  | 0.57 | 2.64    | 4.65  | 0.57 |
|              | 5           | 7.58    | 4.64 | 1.63 | 5.87         | 4.99        | 1.18        | 4.36 | 5.15  | 0.85 | 2.91 | 5.28  | 0.55 | 2.79 | 5.37  | 0.52 | 2.79    | 5.37  | 0.52 |
|              | 7           | 7.85    | 5.04 | 1.56 | 6.18         | 5.42        | 1.14        | 4.66 | 5.61  | 0.83 | 3.04 | 5.85  | 0.52 | 3.04 | 5.85  | 0.52 | 3.04    | 5.85  | 0.52 |
|              | 10          | 8.16    | 5.59 | 1.46 | 6.22         | 6.04        | 1.03        | 4.59 | 6.21  | 0.74 | 3.20 | 6.46  | 0.50 | 3.20 | 6.46  | 0.50 | 3.20    | 6.46  | 0.50 |
|              | 12          | 8.35    | 6.00 | 1.39 | 6.23         | 6.23        | 1.00        | 4.68 | 6.54  | 0.71 | 3.44 | 6.78  | 0.51 | 3.44 | 6.78  | 0.51 | 3.44    | 6.78  | 0.51 |
|              | 15          | 7.74    | 6.05 | 1.28 | 6.25         | 6.45        | 0.97        | 4.64 | 6.78  | 0.68 | 3.50 | 7.02  | 0.50 | 3.50 | 7.02  | 0.50 | 3.50    | 7.02  | 0.50 |
|              | 20          | 7.38    | 6.90 | 1.07 | 6.23         | 7.14        | 0.87        | 4.64 | 7.54  | 0.62 | 3.29 | 7.90  | 0.42 | 3.29 | 7.90  | 0.42 | 3.29    | 7.90  | 0.42 |
|              | 25          | 7.04    | 7.95 | 0.89 | 6.31         | 8.18        | 0.77        | 4.65 | 8.71  | 0.53 | 3.62 | 9.07  | 0.40 | 3.62 | 9.07  | 0.40 | 3.62    | 9.07  | 0.40 |
|              | 30          | 7.07    | 9.19 | 0.77 | 6.17         | 9.24        | 0.67        | 4.65 | 9.56  | 0.49 | 3.98 | 9.77  | 0.41 | 3.98 | 9.77  | 0.41 | 3.98    | 9.77  | 0.41 |
|              | 35          | /       | /    | /    | /            | /           | /           | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 35           | -25         | 3.40    | 1.76 | 1.94 | 3.40         | 1.76        | 1.94        | 2.56 | 1.91  | 1.34 | 1.74 | 1.94  | 0.89 | 1.24 | 2.05  | 0.61 | 1.24    | 2.05  | 0.61 |
|              | -20         | 4.24    | 2.10 | 2.01 | 4.24         | 2.10        | 2.01        | 3.18 | 2.15  | 1.48 | 2.12 | 2.21  | 0.96 | 1.52 | 2.32  | 0.65 | 1.52    | 2.32  | 0.65 |
|              | -15         | 5.05    | 2.43 | 2.08 | 5.05         | 2.43        | 2.08        | 3.79 | 2.46  | 1.54 | 2.53 | 2.51  | 1.01 | 1.79 | 2.66  | 0.67 | 1.79    | 2.66  | 0.67 |
|              | -10         | 5.91    | 2.77 | 2.13 | 5.81         | 2.78        | 2.09        | 4.34 | 2.94  | 1.48 | 2.93 | 2.98  | 0.98 | 2.18 | 3.08  | 0.71 | 2.18    | 3.08  | 0.71 |
|              | -7          | 6.18    | 2.93 | 2.11 | 5.90         | 2.95        | 2.00        | 4.34 | 3.11  | 1.39 | 2.89 | 3.14  | 0.92 | 2.36 | 3.21  | 0.73 | 2.36    | 3.21  | 0.73 |
|              | -5          | 6.36    | 3.12 | 2.04 | 5.70         | 3.20        | 1.78        | 4.28 | 3.36  | 1.27 | 2.89 | 3.41  | 0.85 | 2.53 | 3.43  | 0.74 | 2.53    | 3.43  | 0.74 |
|              | -2          | 6.61    | 3.32 | 1.99 | 5.51         | 3.55        | 1.55        | 4.16 | 3.62  | 1.15 | 2.76 | 3.69  | 0.75 | 2.31 | 3.74  | 0.62 | 2.31    | 3.74  | 0.62 |
|              | 0           | 6.75    | 3.49 | 1.93 | 5.64         | 3.73        | 1.51        | 4.28 | 3.84  | 1.11 | 2.84 | 3.92  | 0.73 | 2.39 | 3.95  | 0.61 | 2.39    | 3.95  | 0.61 |
|              | 2           | 6.83    | 3.67 | 1.86 | 5.60         | 3.90        | 1.44        | 4.17 | 4.12  | 1.01 | 2.76 | 4.18  | 0.66 | 2.53 | 4.19  | 0.60 | 2.53    | 4.19  | 0.60 |
|              | 5           | 7.42    | 4.26 | 1.74 | 5.92         | 4.48        | 1.32        | 4.37 | 4.60  | 0.95 | 2.89 | 4.72  | 0.61 | 2.68 | 4.75  | 0.56 | 2.68    | 4.75  | 0.56 |
|              | 7*          | 7.70    | 4.61 | 1.67 | <b>6.20</b>  | <b>4.90</b> | <b>1.27</b> | 4.64 | 5.12  | 0.91 | 3.15 | 5.26  | 0.60 | 2.93 | 5.32  | 0.55 | 2.93    | 5.32  | 0.55 |
|              | 10          | 8.00    | 5.08 | 1.57 | 6.19         | 5.36        | 1.15        | 4.69 | 5.56  | 0.84 | 3.24 | 5.78  | 0.56 | 3.24 | 5.78  | 0.56 | 3.24    | 5.78  | 0.56 |
|              | 12          | 8.17    | 5.43 | 1.50 | 6.24         | 5.77        | 1.08        | 4.63 | 5.97  | 0.78 | 3.45 | 6.22  | 0.56 | 3.45 | 6.22  | 0.56 | 3.45    | 6.22  | 0.56 |
|              | 15          | 7.92    | 5.66 | 1.40 | 6.25         | 5.96        | 1.05        | 4.62 | 6.18  | 0.75 | 3.55 | 6.36  | 0.56 | 3.55 | 6.36  | 0.56 | 3.55    | 6.36  | 0.56 |
|              | 20          | 7.53    | 6.21 | 1.21 | 5.97         | 6.30        | 0.95        | 4.62 | 6.51  | 0.71 | 3.17 | 6.66  | 0.48 | 3.17 | 6.66  | 0.48 | 3.17    | 6.66  | 0.48 |
|              | 25          | 6.96    | 6.92 | 1.01 | 6.31         | 7.10        | 0.89        | 4.59 | 7.43  | 0.62 | 3.50 | 7.69  | 0.46 | 3.50 | 7.69  | 0.46 | 3.50    | 7.69  | 0.46 |
|              | 30          | 6.94    | 7.95 | 0.87 | 6.17         | 8.10        | 0.76        | 4.59 | 8.55  | 0.54 | 3.98 | 8.74  | 0.46 | 3.84 | 8.75  | 0.44 | 3.84    | 8.75  | 0.44 |
|              | 35          | 7.12    | 9.07 | 0.79 | 6.14         | 9.51        | 0.65        | 4.73 | 9.88  | 0.48 | 4.20 | 10.05 | 0.42 | 4.20 | 10.05 | 0.42 | 4.20    | 10.05 | 0.42 |
| 40           | -25         | 3.34    | 1.65 | 2.03 | 3.34         | 1.65        | 2.03        | 2.51 | 1.82  | 1.38 | 1.73 | 1.86  | 0.93 | 1.26 | 1.95  | 0.65 | 1.26    | 1.95  | 0.65 |
|              | -20         | 4.17    | 1.98 | 2.11 | 4.17         | 1.98        | 2.11        | 3.20 | 2.06  | 1.55 | 2.08 | 2.09  | 1.00 | 1.54 | 2.22  | 0.69 | 1.54    | 2.22  | 0.69 |
|              | -15         | 4.96    | 2.21 | 2.24 | 4.96         | 2.21        | 2.24        | 3.72 | 2.31  | 1.61 | 2.48 | 2.33  | 1.06 | 1.82 | 2.46  | 0.74 | 1.82    | 2.46  | 0.74 |
|              | -10         | 5.83    | 2.53 | 2.30 | 5.60         | 2.61        | 2.15        | 4.21 | 2.67  | 1.58 | 2.82 | 2.72  | 1.04 | 2.16 |       |      |         |       |      |

| Tw_out<br>°C | DBWB<br>°C | Maximum |      |      | 100%(Normal) |      |      | 75%  |      |      | 50%  |      |      | 25%  |      |      | Minimum |      |      |
|--------------|------------|---------|------|------|--------------|------|------|------|------|------|------|------|------|------|------|------|---------|------|------|
|              |            | HC      | COP  | PI   | HC           | COP  | PI   | HC   | COP  | PI   | HC   | COP  | PI   | HC   | COP  | PI   | HC      | COP  | PI   |
| 55           | -25        | 3.00    | 1.40 | 2.14 | 3.00         | 1.40 | 2.14 | 2.25 | 1.44 | 1.56 | 1.55 | 1.50 | 1.03 | 1.55 | 1.50 | 1.03 | 1.55    | 1.50 | 1.03 |
|              | -20        | 3.66    | 1.57 | 2.34 | 3.66         | 1.57 | 2.34 | 2.77 | 1.63 | 1.70 | 1.87 | 1.69 | 1.11 | 1.87 | 1.69 | 1.11 | 1.87    | 1.69 | 1.11 |
|              | -15        | 4.46    | 1.71 | 2.61 | 4.46         | 1.71 | 2.61 | 3.34 | 1.79 | 1.87 | 2.19 | 1.85 | 1.18 | 2.19 | 1.85 | 1.18 | 2.19    | 1.85 | 1.18 |
|              | -10        | 5.23    | 1.91 | 2.74 | 5.12         | 1.92 | 2.67 | 3.81 | 1.99 | 1.92 | 2.59 | 2.04 | 1.27 | 2.59 | 2.04 | 1.27 | 2.59    | 2.04 | 1.27 |
|              | -7         | 5.47    | 2.13 | 2.57 | 5.20         | 2.15 | 2.42 | 3.74 | 2.16 | 1.73 | 2.86 | 2.20 | 1.30 | 2.86 | 2.20 | 1.30 | 2.86    | 2.20 | 1.30 |
|              | -5         | 5.59    | 2.15 | 2.60 | 5.17         | 2.20 | 2.35 | 3.88 | 2.27 | 1.71 | 3.07 | 2.30 | 1.33 | 3.07 | 2.30 | 1.33 | 3.07    | 2.30 | 1.33 |
|              | -2         | 5.81    | 2.30 | 2.53 | 5.28         | 2.35 | 2.24 | 4.01 | 2.51 | 1.60 | 3.25 | 2.57 | 1.27 | 3.25 | 2.57 | 1.27 | 3.25    | 2.57 | 1.27 |
|              | 0          | 6.16    | 2.50 | 2.46 | 5.34         | 2.60 | 2.06 | 4.04 | 2.63 | 1.53 | 3.49 | 2.65 | 1.32 | 3.49 | 2.65 | 1.32 | 3.49    | 2.65 | 1.32 |
|              | 2          | 6.22    | 2.60 | 2.39 | 5.80         | 2.65 | 2.19 | 4.41 | 2.72 | 1.62 | 3.67 | 2.73 | 1.34 | 3.67 | 2.73 | 1.34 | 3.67    | 2.73 | 1.34 |
|              | 5          | 6.72    | 2.88 | 2.33 | 5.99         | 2.95 | 2.03 | 4.42 | 3.04 | 1.46 | 4.17 | 3.05 | 1.37 | 4.17 | 3.05 | 1.37 | 4.17    | 3.05 | 1.37 |
|              | 7          | 6.97    | 3.00 | 2.32 | 6.20         | 3.10 | 2.00 | 4.72 | 3.19 | 1.48 | 4.55 | 3.23 | 1.41 | 4.55 | 3.23 | 1.41 | 4.55    | 3.23 | 1.41 |
|              | 10         | 7.24    | 3.32 | 2.18 | 6.22         | 3.40 | 1.83 | 4.59 | 3.48 | 1.32 | 4.26 | 3.53 | 1.21 | 4.26 | 3.53 | 1.21 | 4.26    | 3.53 | 1.21 |
|              | 12         | 7.40    | 3.52 | 2.10 | 6.20         | 3.58 | 1.73 | 4.69 | 3.69 | 1.27 | 4.55 | 3.74 | 1.22 | 4.55 | 3.74 | 1.22 | 4.55    | 3.74 | 1.22 |
|              | 15         | 7.14    | 3.67 | 1.95 | 6.25         | 3.72 | 1.68 | 4.83 | 3.80 | 1.27 | 4.68 | 3.85 | 1.22 | 4.68 | 3.85 | 1.22 | 4.68    | 3.85 | 1.22 |
|              | 20         | 6.77    | 3.95 | 1.71 | 6.24         | 3.98 | 1.57 | 4.60 | 4.06 | 1.13 | 4.11 | 4.07 | 1.01 | 4.11 | 4.07 | 1.01 | 4.11    | 4.07 | 1.01 |
|              | 25         | 6.39    | 4.32 | 1.48 | 6.26         | 4.35 | 1.44 | 4.71 | 4.51 | 1.04 | 4.56 | 4.55 | 1.00 | 4.56 | 4.55 | 1.00 | 4.56    | 4.55 | 1.00 |
|              | 30         | 6.50    | 4.72 | 1.38 | 6.22         | 4.75 | 1.31 | 5.03 | 4.91 | 1.02 | 5.03 | 4.91 | 1.02 | 5.03 | 4.91 | 1.02 | 5.03    | 4.91 | 1.02 |
|              | 35         | 6.69    | 4.95 | 1.35 | 6.18         | 4.98 | 1.24 | 5.54 | 5.08 | 1.09 | 5.54 | 5.08 | 1.09 | 5.54 | 5.08 | 1.09 | 5.54    | 5.08 | 1.09 |
|              | 60         | -25     | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
| -20          |            | 3.60    | 1.41 | 2.55 | 3.60         | 1.41 | 2.55 | 2.70 | 1.50 | 1.81 | 1.94 | 1.53 | 1.27 | 1.94 | 1.53 | 1.27 | 1.94    | 1.53 | 1.27 |
| -15          |            | 4.31    | 1.60 | 2.70 | 4.31         | 1.60 | 2.70 | 3.23 | 1.65 | 1.97 | 2.27 | 1.69 | 1.35 | 2.27 | 1.69 | 1.35 | 2.27    | 1.69 | 1.35 |
| -10          |            | 5.05    | 1.78 | 2.83 | 5.05         | 1.78 | 2.83 | 3.83 | 1.83 | 2.10 | 2.67 | 1.86 | 1.43 | 2.67 | 1.86 | 1.43 | 2.67    | 1.86 | 1.43 |
| -7           |            | 5.27    | 1.84 | 2.87 | 5.10         | 1.86 | 2.74 | 3.79 | 1.94 | 1.95 | 2.94 | 1.99 | 1.48 | 2.94 | 1.99 | 1.48 | 2.94    | 1.99 | 1.48 |
| -5           |            | 5.39    | 1.90 | 2.84 | 5.15         | 1.91 | 2.70 | 3.93 | 2.05 | 1.92 | 3.16 | 2.13 | 1.48 | 3.16 | 2.13 | 1.48 | 3.16    | 2.13 | 1.48 |
| -2           |            | 5.61    | 2.01 | 2.79 | 5.29         | 2.04 | 2.60 | 3.99 | 2.29 | 1.74 | 3.10 | 2.36 | 1.31 | 3.10 | 2.36 | 1.31 | 3.10    | 2.36 | 1.31 |
| 0            |            | 5.92    | 2.17 | 2.73 | 5.66         | 2.22 | 2.55 | 4.29 | 2.36 | 1.82 | 3.31 | 2.42 | 1.37 | 3.31 | 2.42 | 1.37 | 3.31    | 2.42 | 1.37 |
| 2            |            | 6.01    | 2.26 | 2.66 | 5.44         | 2.34 | 2.32 | 4.10 | 2.52 | 1.63 | 3.50 | 2.55 | 1.37 | 3.50 | 2.55 | 1.37 | 3.50    | 2.55 | 1.37 |
| 5            |            | 6.49    | 2.59 | 2.51 | 5.96         | 2.65 | 2.25 | 4.41 | 2.81 | 1.57 | 3.97 | 2.84 | 1.40 | 3.97 | 2.84 | 1.40 | 3.97    | 2.84 | 1.40 |
| 7            |            | 6.75    | 2.80 | 2.41 | 6.20         | 2.82 | 2.20 | 4.58 | 2.94 | 1.56 | 4.34 | 2.98 | 1.46 | 4.34 | 2.98 | 1.46 | 4.34    | 2.98 | 1.46 |
| 10           |            | 7.00    | 2.95 | 2.37 | 6.21         | 2.98 | 2.08 | 4.69 | 3.09 | 1.52 | 4.22 | 3.15 | 1.34 | 4.22 | 3.15 | 1.34 | 4.22    | 3.15 | 1.34 |
| 12           |            | 7.13    | 3.03 | 2.35 | 6.18         | 3.10 | 1.99 | 4.66 | 3.24 | 1.44 | 4.51 | 3.27 | 1.38 | 4.51 | 3.27 | 1.38 | 4.51    | 3.27 | 1.38 |
| 15           |            | 6.88    | 3.10 | 2.22 | 6.34         | 3.25 | 1.95 | 4.79 | 3.39 | 1.41 | 4.65 | 3.43 | 1.35 | 4.65 | 3.43 | 1.35 | 4.65    | 3.43 | 1.35 |
| 20           |            | 6.49    | 3.26 | 1.99 | 6.27         | 3.46 | 1.81 | 4.62 | 3.66 | 1.26 | 3.91 | 3.70 | 1.06 | 3.91 | 3.70 | 1.06 | 3.91    | 3.70 | 1.06 |
| 25           |            | 6.09    | 3.76 | 1.62 | 6.09         | 3.76 | 1.62 | 4.60 | 3.92 | 1.17 | 4.32 | 3.97 | 1.09 | 4.32 | 3.97 | 1.09 | 4.32    | 3.97 | 1.09 |
| 30           |            | 6.18    | 4.13 | 1.50 | 6.18         | 4.13 | 1.50 | 4.78 | 4.28 | 1.11 | 4.78 | 4.28 | 1.11 | 4.78 | 4.28 | 1.11 | 4.78    | 4.28 | 1.11 |
| 35           |            | /       | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
| 65           |            | -25     | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
|              | -20        | 3.44    | 1.29 | 2.66 | 3.44         | 1.29 | 2.66 | 2.58 | 1.37 | 1.89 | 1.86 | 1.39 | 1.34 | 1.86 | 1.39 | 1.34 | 1.86    | 1.39 | 1.34 |
|              | -15        | 4.12    | 1.38 | 2.99 | 4.12         | 1.38 | 2.99 | 3.09 | 1.50 | 2.06 | 2.16 | 1.51 | 1.43 | 2.16 | 1.51 | 1.43 | 2.16    | 1.51 | 1.43 |
|              | -10        | 4.83    | 1.56 | 3.10 | 4.83         | 1.56 | 3.10 | 3.61 | 1.65 | 2.18 | 2.55 | 1.69 | 1.51 | 2.55 | 1.69 | 1.51 | 2.55    | 1.69 | 1.51 |
|              | -7         | 5.05    | 1.69 | 2.99 | 5.05         | 1.69 | 2.99 | 3.81 | 1.73 | 2.20 | 2.79 | 1.79 | 1.56 | 2.79 | 1.79 | 1.56 | 2.79    | 1.79 | 1.56 |
|              | -5         | 4.93    | 1.76 | 2.80 | 4.93         | 1.76 | 2.80 | 3.77 | 1.84 | 2.05 | 3.00 | 1.88 | 1.59 | 3.00 | 1.88 | 1.59 | 3.00    | 1.88 | 1.59 |
|              | -2         | 5.31    | 1.89 | 2.81 | 5.31         | 1.89 | 2.81 | 4.02 | 2.12 | 1.89 | 2.91 | 2.15 | 1.36 | 2.91 | 2.15 | 1.36 | 2.91    | 2.15 | 1.36 |
|              | 0          | 5.68    | 1.96 | 2.90 | 5.35         | 1.99 | 2.68 | 4.05 | 2.17 | 1.87 | 3.11 | 2.20 | 1.41 | 3.11 | 2.20 | 1.41 | 3.11    | 2.20 | 1.41 |
|              | 2          | 5.78    | 2.04 | 2.84 | 5.41         | 2.07 | 2.61 | 4.13 | 2.20 | 1.88 | 3.29 | 2.22 | 1.48 | 3.29 | 2.22 | 1.48 | 3.29    | 2.22 | 1.48 |
|              | 5          | 6.23    | 2.30 | 2.71 | 5.96         | 2.32 | 2.57 | 4.44 | 2.43 | 1.83 | 3.74 | 2.46 | 1.52 | 3.74 | 2.46 | 1.52 | 3.74    | 2.46 | 1.52 |
|              | 7          | 6.46    | 2.45 | 2.64 | 6.19         | 2.47 | 2.51 | 4.63 | 2.57 | 1.80 | 4.10 | 2.59 | 1.58 | 4.10 | 2.59 | 1.58 | 4.10    | 2.59 | 1.58 |
|              | 10         | 6.68    | 2.60 | 2.57 | 6.22         | 2.63 | 2.37 | 4.58 | 2.75 | 1.67 | 3.94 | 2.80 | 1.41 | 3.94 | 2.80 | 1.41 | 3.94    | 2.80 | 1.41 |
|              | 12         | 6.82    | 2.72 | 2.51 | 6.20         | 2.75 | 2.25 | 4.58 | 2.85 | 1.60 | 4.21 | 2.89 | 1.46 | 4.21 | 2.89 | 1.46 | 4.21    | 2.89 | 1.46 |
|              | 15         | 6.57    | 2.80 | 2.35 | 6.25         | 2.82 | 2.22 | 4.72 | 2.90 | 1.63 | 4.33 | 2.97 | 1.46 | 4.33 | 2.97 | 1.46 | 4.33    | 2.97 | 1.46 |
|              | 20         | 6.18    | 3.05 | 2.03 | 6.18         | 3.05 | 2.03 | 4.64 | 3.15 | 1.47 | 3.66 | 3.17 | 1.16 | 3.66 | 3.17 | 1.16 | 3.66    | 3.17 | 1.16 |
|              | 25         | 5.74    | 3.27 | 1.76 | 5.74         | 3.27 | 1.76 | 4.55 | 3.38 | 1.34 | 4.05 | 3.41 | 1.19 | 4.05 | 3.41 | 1.19 | 4.05    | 3.41 | 1.19 |
|              | 30         | 5.78    | 3.55 | 1.63 | 5.78         | 3.55 | 1.63 | 4.66 | 3.70 | 1.26 | 4.47 | 3.75 | 1.19 | 4.47 | 3.75 | 1.19 | 4.47    | 3.75 | 1.19 |
|              | 35         | /       | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
|              | 70         | -25     | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
| -20          |            | /       | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
| -15          |            | 3.96    | 1.30 | 3.04 | 3.96         | 1.30 | 3.04 | 2.97 | 1.38 | 2.16 | 2.05 | 1.48 | 1.39 | 2.05 | 1.48 | 1.39 | 2.05    | 1.48 | 1.39 |
| -10          |            | 4.18    | 1.39 | 3.00 | 4.18         | 1.39 | 3.00 | 3.13 | 1.48 | 2.12 | 2.54 | 1.57 | 1.62 | 2.54 | 1.57 | 1.62 | 2.54    | 1.57 | 1.62 |
| -7           |            | 4.48    | 1.48 | 3.03 | 4.48         | 1.48 | 3.03 | 3.44 | 1.59 | 2.16 | 2.83 | 1.63 | 1.74 | 2.83 | 1.63 | 1.74 | 2.83    | 1.63 | 1.74 |
| -5           |            | 4.50    | 1.54 | 2.91 | 4.50         | 1.54 | 2.91 | 3.62 | 1.65 | 2.19 | 2.77 | 1.79 | 1.55 | 2.77 | 1.79 | 1.55 | 2.77    | 1.79 | 1.55 |
| -2           |            | 5.13    | 1.70 | 3.01 | 5.13         | 1.70 | 3.01 | 4.01 | 1.93 | 2.08 | 2.69 | 2.13 | 1.26 | 2.69 | 2.13 | 1.26 | 2.69    | 2.13 | 1.26 |
| 0            |            | 5.35    | 1.76 | 3.04 | 5.35         | 1.76 | 3.04 | 4.05 | 1.98 | 2.04 | 2.88 | 2.16 |      |      |      |      |         |      |      |

### 8.3 Heating performance Mod. 08

|    | Tw_out<br>°C | DBWB<br>°C | Maximum |      |      | 100%(Normal) |      |      | 75%   |      |      | 50%   |      |      | 25%   |      |      | Minimum |      |    |
|----|--------------|------------|---------|------|------|--------------|------|------|-------|------|------|-------|------|------|-------|------|------|---------|------|----|
|    |              |            | HC      | COP  | PI   | HC           | COP  | PI   | HC    | COP  | PI   | HC    | COP  | PI   | HC    | COP  | PI   | HC      | COP  | PI |
| 25 | -25          | 5.15       | 2.16    | 2.39 | 5.15 | 2.16         | 2.39 | 3.86 | 2.49  | 1.55 | 2.57 | 2.79  | 0.92 | 2.19 | 2.85  | 0.77 | 2.19 | 2.85    | 0.77 |    |
|    | -20          | 6.00       | 2.46    | 2.44 | 6.00 | 2.46         | 2.44 | 4.63 | 2.70  | 1.71 | 3.09 | 3.01  | 1.02 | 2.61 | 3.10  | 0.84 | 2.61 | 3.10    | 0.84 |    |
|    | -15          | 7.43       | 2.86    | 2.60 | 7.13 | 2.88         | 2.47 | 5.15 | 3.10  | 1.66 | 3.43 | 3.50  | 0.98 | 3.05 | 3.56  | 0.85 | 3.05 | 3.56    | 0.85 |    |
|    | -10          | 8.61       | 3.07    | 2.80 | 7.01 | 3.65         | 1.92 | 5.38 | 3.83  | 1.41 | 3.65 | 4.23  | 0.86 | 3.36 | 4.25  | 0.79 | 3.36 | 4.25    | 0.79 |    |
|    | -7           | 8.93       | 3.41    | 2.62 | 6.95 | 3.81         | 1.83 | 5.29 | 4.03  | 1.31 | 3.80 | 4.42  | 0.86 | 3.80 | 4.42  | 0.86 | 3.80 | 4.42    | 0.86 |    |
|    | -5           | 9.07       | 3.61    | 2.51 | 7.03 | 4.02         | 1.75 | 5.27 | 4.35  | 1.21 | 4.07 | 4.62  | 0.88 | 4.07 | 4.62  | 0.88 | 4.07 | 4.62    | 0.88 |    |
|    | -2           | 9.34       | 3.97    | 2.35 | 7.10 | 4.42         | 1.60 | 5.40 | 4.82  | 1.12 | 3.64 | 5.25  | 0.69 | 3.64 | 5.25  | 0.69 | 3.64 | 5.25    | 0.69 |    |
|    | 0            | 9.29       | 4.16    | 2.24 | 7.08 | 4.69         | 1.51 | 5.39 | 4.98  | 1.08 | 3.82 | 5.31  | 0.72 | 3.82 | 5.31  | 0.72 | 3.82 | 5.31    | 0.72 |    |
|    | 2            | 9.42       | 4.46    | 2.11 | 7.11 | 5.02         | 1.42 | 5.41 | 5.32  | 1.02 | 4.03 | 5.71  | 0.71 | 4.03 | 5.71  | 0.71 | 4.03 | 5.71    | 0.71 |    |
|    | 5            | 10.18      | 5.39    | 1.89 | 7.66 | 5.89         | 1.30 | 5.77 | 6.27  | 0.92 | 4.23 | 6.52  | 0.65 | 4.23 | 6.52  | 0.65 | 4.23 | 6.52    | 0.65 |    |
|    | 7            | 10.76      | 5.84    | 1.84 | 8.28 | 6.53         | 1.27 | 6.25 | 6.95  | 0.90 | 4.62 | 7.15  | 0.65 | 4.62 | 7.15  | 0.65 | 4.62 | 7.15    | 0.65 |    |
|    | 10           | 11.16      | 6.79    | 1.64 | 8.36 | 7.50         | 1.12 | 6.25 | 7.84  | 0.80 | 5.08 | 8.11  | 0.63 | 5.08 | 8.11  | 0.63 | 5.08 | 8.11    | 0.63 |    |
|    | 12           | 11.40      | 7.62    | 1.50 | 8.28 | 8.41         | 0.99 | 6.31 | 8.85  | 0.71 | 5.42 | 9.01  | 0.60 | 5.42 | 9.01  | 0.60 | 5.42 | 9.01    | 0.60 |    |
|    | 15           | 11.17      | 8.07    | 1.38 | 8.23 | 8.83         | 0.93 | 6.12 | 9.38  | 0.65 | 5.50 | 9.46  | 0.58 | 5.50 | 9.46  | 0.58 | 5.50 | 9.46    | 0.58 |    |
|    | 20           | 10.82      | 8.80    | 1.23 | 8.10 | 9.64         | 0.84 | 6.18 | 10.02 | 0.62 | 5.17 | 10.24 | 0.50 | 5.17 | 10.24 | 0.50 | 5.17 | 10.24   | 0.50 |    |
|    | 25           | 10.65      | 9.75    | 1.09 | 8.14 | 10.30        | 0.79 | 6.16 | 10.68 | 0.58 | 5.70 | 10.83 | 0.53 | 5.70 | 10.83 | 0.53 | 5.70 | 10.83   | 0.53 |    |
|    | 30           | /          | /       | /    | /    | /            | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /    | /       | /    |    |
|    | 35           | /          | /       | /    | /    | /            | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /    | /       | /    |    |
| 30 | -25          | 5.00       | 2.04    | 2.45 | 5.00 | 2.04         | 2.45 | 3.75 | 2.33  | 1.61 | 2.52 | 2.58  | 0.98 | 2.14 | 2.64  | 0.81 | 2.14 | 2.64    | 0.81 |    |
|    | -20          | 5.95       | 2.34    | 2.54 | 5.95 | 2.34         | 2.54 | 4.55 | 2.63  | 1.73 | 3.03 | 2.89  | 1.05 | 2.56 | 3.01  | 0.85 | 2.56 | 3.01    | 0.85 |    |
|    | -15          | 7.36       | 2.56    | 2.87 | 7.18 | 2.58         | 2.78 | 5.15 | 2.92  | 1.76 | 3.43 | 3.20  | 1.07 | 3.00 | 3.30  | 0.91 | 3.00 | 3.30    | 0.91 |    |
|    | -10          | 8.53       | 2.88    | 2.96 | 7.08 | 3.11         | 2.28 | 5.39 | 3.43  | 1.57 | 3.26 | 3.80  | 0.86 | 3.26 | 3.80  | 0.86 | 3.26 | 3.80    | 0.86 |    |
|    | -7           | 8.78       | 3.10    | 2.84 | 6.93 | 3.43         | 2.02 | 5.31 | 3.70  | 1.44 | 3.70 | 4.04  | 0.92 | 3.70 | 4.04  | 0.92 | 3.70 | 4.04    | 0.92 |    |
|    | -5           | 8.99       | 3.30    | 2.72 | 6.97 | 3.70         | 1.89 | 5.32 | 3.99  | 1.33 | 3.98 | 4.35  | 0.92 | 3.98 | 4.35  | 0.92 | 3.98 | 4.35    | 0.92 |    |
|    | -2           | 9.26       | 3.54    | 2.62 | 7.10 | 4.10         | 1.73 | 5.35 | 4.42  | 1.21 | 3.56 | 4.70  | 0.76 | 3.56 | 4.70  | 0.76 | 3.56 | 4.70    | 0.76 |    |
|    | 0            | 9.18       | 3.74    | 2.46 | 7.10 | 4.26         | 1.67 | 5.31 | 4.59  | 1.16 | 3.73 | 4.92  | 0.76 | 3.73 | 4.92  | 0.76 | 3.73 | 4.92    | 0.76 |    |
|    | 2            | 9.30       | 4.00    | 2.32 | 7.10 | 4.49         | 1.58 | 5.35 | 4.86  | 1.10 | 3.94 | 5.18  | 0.76 | 3.94 | 5.18  | 0.76 | 3.94 | 5.18    | 0.76 |    |
|    | 5            | 10.03      | 4.58    | 2.19 | 7.70 | 5.02         | 1.53 | 5.82 | 5.41  | 1.08 | 4.13 | 5.87  | 0.70 | 4.13 | 5.87  | 0.70 | 4.13 | 5.87    | 0.70 |    |
|    | 7            | 10.60      | 5.01    | 2.11 | 8.34 | 5.42         | 1.54 | 6.31 | 5.88  | 1.07 | 4.52 | 6.12  | 0.74 | 4.52 | 6.12  | 0.74 | 4.52 | 6.12    | 0.74 |    |
|    | 10           | 11.02      | 5.64    | 1.95 | 8.41 | 6.21         | 1.35 | 6.30 | 6.60  | 0.96 | 4.98 | 6.79  | 0.73 | 4.98 | 6.79  | 0.73 | 4.98 | 6.79    | 0.73 |    |
|    | 12           | 11.27      | 5.97    | 1.89 | 8.32 | 6.50         | 1.28 | 6.24 | 6.82  | 0.92 | 5.32 | 7.12  | 0.75 | 5.32 | 7.12  | 0.75 | 5.32 | 7.12    | 0.75 |    |
|    | 15           | 11.03      | 6.47    | 1.70 | 8.28 | 7.06         | 1.17 | 6.22 | 7.41  | 0.84 | 5.41 | 7.59  | 0.71 | 5.41 | 7.59  | 0.71 | 5.41 | 7.59    | 0.71 |    |
|    | 20           | 10.71      | 7.48    | 1.43 | 8.24 | 7.89         | 1.04 | 6.13 | 8.25  | 0.74 | 5.06 | 8.30  | 0.61 | 5.06 | 8.30  | 0.61 | 5.06 | 8.30    | 0.61 |    |
|    | 25           | 10.51      | 8.70    | 1.21 | 8.05 | 9.06         | 0.89 | 6.10 | 9.77  | 0.62 | 5.61 | 9.75  | 0.58 | 5.61 | 9.75  | 0.58 | 5.61 | 9.75    | 0.58 |    |
|    | 30           | 10.50      | 9.80    | 1.07 | 8.21 | 10.35        | 0.79 | 6.21 | 10.78 | 0.58 | 6.16 | 11.02 | 0.56 | 6.16 | 11.02 | 0.56 | 6.16 | 11.02   | 0.56 |    |
|    | 35           | /          | /       | /    | /    | /            | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /    | /       | /    |    |
| 35 | -25          | 4.71       | 1.98    | 2.38 | 4.71 | 1.98         | 2.38 | 3.67 | 2.05  | 1.79 | 2.41 | 2.25  | 1.07 | 2.09 | 2.35  | 0.89 | 2.09 | 2.35    | 0.89 |    |
|    | -20          | 5.91       | 2.22    | 2.67 | 5.91 | 2.22         | 2.67 | 4.46 | 2.34  | 1.90 | 2.93 | 2.62  | 1.12 | 2.50 | 2.64  | 0.95 | 2.50 | 2.64    | 0.95 |    |
|    | -15          | 7.25       | 2.40    | 3.02 | 7.25 | 2.40         | 3.02 | 5.40 | 2.56  | 2.11 | 3.60 | 2.87  | 1.25 | 2.91 | 2.96  | 0.98 | 2.91 | 2.96    | 0.98 |    |
|    | -10          | 8.02       | 2.75    | 2.91 | 7.22 | 2.86         | 2.53 | 5.44 | 3.04  | 1.79 | 3.16 | 3.28  | 0.96 | 3.16 | 3.28  | 0.96 | 3.16 | 3.28    | 0.96 |    |
|    | -7           | 8.32       | 2.85    | 2.92 | 7.00 | 3.00         | 2.33 | 5.35 | 3.43  | 1.56 | 3.60 | 3.52  | 1.02 | 3.60 | 3.52  | 1.02 | 3.60 | 3.52    | 1.02 |    |
|    | -5           | 8.57       | 3.11    | 2.76 | 7.03 | 3.36         | 2.09 | 5.25 | 3.62  | 1.45 | 3.88 | 3.72  | 1.04 | 3.88 | 3.72  | 1.04 | 3.88 | 3.72    | 1.04 |    |
|    | -2           | 8.88       | 3.37    | 2.64 | 7.10 | 3.70         | 1.92 | 5.40 | 3.97  | 1.36 | 3.47 | 4.16  | 0.84 | 3.47 | 4.16  | 0.84 | 3.47 | 4.16    | 0.84 |    |
|    | 0            | 8.77       | 3.45    | 2.54 | 7.12 | 3.80         | 1.87 | 5.36 | 4.08  | 1.31 | 3.66 | 4.37  | 0.84 | 3.66 | 4.37  | 0.84 | 3.66 | 4.37    | 0.84 |    |
|    | 2            | 8.90       | 3.60    | 2.47 | 7.10 | 3.85         | 1.84 | 5.38 | 4.32  | 1.25 | 3.84 | 4.48  | 0.86 | 3.84 | 4.48  | 0.86 | 3.84 | 4.48    | 0.86 |    |
|    | 5            | 9.63       | 4.17    | 2.31 | 7.72 | 4.55         | 1.70 | 5.90 | 4.85  | 1.22 | 4.01 | 5.16  | 0.78 | 4.01 | 5.16  | 0.78 | 4.01 | 5.16    | 0.78 |    |
|    | 7*           | 10.17      | 4.59    | 2.22 | 8.40 | 5.00         | 1.68 | 6.37 | 5.39  | 1.18 | 4.40 | 5.64  | 0.78 | 4.40 | 5.64  | 0.78 | 4.40 | 5.64    | 0.78 |    |
|    | 10           | 10.63      | 5.10    | 2.09 | 8.45 | 5.60         | 1.51 | 6.36 | 5.94  | 1.07 | 4.86 | 6.43  | 0.76 | 4.86 | 6.43  | 0.76 | 4.86 | 6.43    | 0.76 |    |
|    | 12           | 10.92      | 5.50    | 1.99 | 8.34 | 6.14         | 1.36 | 6.31 | 6.51  | 0.97 | 5.19 | 6.61  | 0.79 | 5.19 | 6.61  | 0.79 | 5.19 | 6.61    | 0.79 |    |
|    | 15           | 10.69      | 5.83    | 1.83 | 8.31 | 6.35         | 1.31 | 6.30 | 6.70  | 0.94 | 5.31 | 6.90  | 0.77 | 5.31 | 6.90  | 0.77 | 5.31 | 6.90    | 0.77 |    |
|    | 20           | 10.41      | 6.58    | 1.58 | 8.26 | 7.01         | 1.18 | 6.23 | 7.36  | 0.85 | 4.92 | 7.57  | 0.65 | 4.92 | 7.57  | 0.65 | 4.92 | 7.57    | 0.65 |    |
|    | 25           | 10.25      | 7.65    | 1.34 | 8.12 | 8.17         | 0.99 | 6.22 | 8.48  | 0.73 | 5.46 | 8.64  | 0.63 | 5.46 | 8.64  | 0.63 | 5.46 | 8.64    | 0.63 |    |
|    | 30           | 10.43      | 8.94    | 1.17 | 8.10 | 9.55         | 0.85 | 6.10 | 9.94  | 0.61 | 6.00 | 10.06 | 0.60 | 6.00 | 10.06 | 0.60 | 6.00 | 10.06   | 0.60 |    |
|    | 35           | 10.30      | 10.10   | 1.02 | 8.19 | 10.58        | 0.77 | 6.60 | 11.13 | 0.59 | 6.60 | 11.13 | 0.59 | 6.60 | 11.13 | 0.59 | 6.60 | 11.13   | 0.59 |    |
| 40 | -25          | 4.72       | 1.67    | 2.82 | 4.31 | 1.70         | 2.54 | 3.23 | 1.83  | 1.77 | 2.14 | 2.19  | 0.98 | 2.14 | 2.19  | 0.98 | 2.14 | 2.19    | 0.98 |    |
|    | -20          | 6.33       | 2.12    | 2.98 | 5.82 | 2.14         | 2.71 | 4.37 | 2.26  | 1.93 | 2.52 | 2.47  | 1.02 | 2.52 | 2.47  | 1.02 | 2.52 | 2.47    | 1.02 |    |
|    | -15          | 7.22       | 2.25    | 3.21 | 7.22 | 2.25         | 3.21 | 5.44 | 2.35  | 2.31 | 2.98 | 2.86  | 1.04 | 2.98 | 2.86  | 1.04 | 2.98 | 2.86    | 1.04 |    |
|    | -10          | 7.92       | 2.59    | 3.06 | 7.18 | 2.67         | 2.69 | 5.54 | 2.79  | 1.98 | 3.21 | 3.    |      |      |       |      |      |         |      |    |

| Tw_out<br>°C | DBWB<br>°C | Maximum |      |      | 100%(Normal) |      |      | 75%  |      |      | 50%  |      |      | 25%  |      |      | Minimum |      |      |
|--------------|------------|---------|------|------|--------------|------|------|------|------|------|------|------|------|------|------|------|---------|------|------|
|              |            | HC      | COP  | PI   | HC           | COP  | PI   | HC   | COP  | PI   | HC   | COP  | PI   | HC   | COP  | PI   | HC      | COP  | PI   |
| 55           | -25        | 4.34    | 1.37 | 3.16 | 4.34         | 1.37 | 3.16 | 3.25 | 1.56 | 2.09 | 2.59 | 1.67 | 1.55 | 2.59 | 1.67 | 1.55 | 2.59    | 1.67 | 1.55 |
|              | -20        | 6.08    | 1.63 | 3.72 | 5.50         | 1.68 | 3.29 | 4.13 | 1.78 | 2.32 | 2.77 | 2.07 | 1.34 | 2.77 | 2.07 | 1.34 | 2.77    | 2.07 | 1.34 |
|              | -15        | 6.84    | 1.78 | 3.85 | 6.27         | 1.80 | 3.49 | 4.77 | 1.92 | 2.49 | 3.29 | 2.12 | 1.55 | 3.29 | 2.12 | 1.55 | 3.29    | 2.12 | 1.55 |
|              | -10        | 7.56    | 1.98 | 3.82 | 7.23         | 2.05 | 3.53 | 5.46 | 2.22 | 2.46 | 4.08 | 2.38 | 1.71 | 4.08 | 2.38 | 1.71 | 4.08    | 2.38 | 1.71 |
|              | -7         | 7.89    | 2.06 | 3.83 | 6.90         | 2.15 | 3.21 | 5.14 | 2.39 | 2.15 | 4.61 | 2.43 | 1.90 | 4.61 | 2.43 | 1.90 | 4.61    | 2.43 | 1.90 |
|              | -5         | 8.18    | 2.16 | 3.78 | 7.10         | 2.27 | 3.12 | 5.56 | 2.52 | 2.21 | 4.92 | 2.60 | 1.89 | 4.92 | 2.60 | 1.89 | 4.92    | 2.60 | 1.89 |
|              | -2         | 8.56    | 2.20 | 3.89 | 7.32         | 2.36 | 3.11 | 5.70 | 2.60 | 2.19 | 5.00 | 2.73 | 1.83 | 5.00 | 2.73 | 1.83 | 5.00    | 2.73 | 1.83 |
|              | 0          | 8.17    | 2.44 | 3.35 | 7.32         | 2.53 | 2.90 | 5.73 | 2.68 | 2.14 | 5.03 | 2.78 | 1.81 | 5.03 | 2.78 | 1.81 | 5.03    | 2.78 | 1.81 |
|              | 2          | 8.25    | 2.51 | 3.28 | 7.80         | 2.55 | 3.06 | 5.74 | 2.86 | 2.01 | 5.33 | 2.93 | 1.82 | 5.33 | 2.93 | 1.82 | 5.33    | 2.93 | 1.82 |
|              | 5          | 8.87    | 2.91 | 3.05 | 7.87         | 2.97 | 2.65 | 5.96 | 3.12 | 1.91 | 5.96 | 3.12 | 1.91 | 5.96 | 3.12 | 1.91 | 5.96    | 3.12 | 1.91 |
|              | 7          | 9.27    | 3.13 | 2.97 | 7.80         | 3.20 | 2.44 | 6.52 | 3.39 | 1.92 | 6.52 | 3.42 | 1.91 | 6.52 | 3.42 | 1.91 | 6.52    | 3.42 | 1.91 |
|              | 10         | 9.69    | 3.39 | 2.86 | 8.00         | 3.53 | 2.26 | 6.11 | 3.74 | 1.63 | 6.11 | 3.70 | 1.65 | 6.11 | 3.70 | 1.65 | 6.11    | 3.70 | 1.65 |
|              | 12         | 9.88    | 3.55 | 2.78 | 8.05         | 3.75 | 2.15 | 6.55 | 4.01 | 1.63 | 6.55 | 3.97 | 1.65 | 6.55 | 3.97 | 1.65 | 6.55    | 3.97 | 1.65 |
|              | 15         | 9.67    | 3.68 | 2.63 | 7.85         | 3.91 | 2.01 | 6.71 | 4.10 | 1.64 | 6.71 | 4.06 | 1.65 | 6.71 | 4.06 | 1.65 | 6.71    | 4.06 | 1.65 |
|              | 20         | 9.31    | 4.13 | 2.25 | 7.78         | 4.25 | 1.83 | 6.04 | 4.44 | 1.36 | 6.04 | 4.44 | 1.36 | 6.04 | 4.44 | 1.36 | 6.04    | 4.44 | 1.36 |
|              | 25         | 9.14    | 4.62 | 1.98 | 7.83         | 4.77 | 1.64 | 6.66 | 4.94 | 1.35 | 6.66 | 4.94 | 1.35 | 6.66 | 4.94 | 1.35 | 6.66    | 4.94 | 1.35 |
|              | 30         | 9.30    | 5.19 | 1.79 | 7.86         | 5.36 | 1.47 | 7.35 | 5.47 | 1.34 | 7.35 | 5.47 | 1.34 | 7.35 | 5.47 | 1.34 | 7.35    | 5.47 | 1.34 |
|              | 35         | 9.13    | 5.55 | 1.65 | 8.10         | 5.78 | 1.40 | 8.10 | 5.84 | 1.39 | 8.10 | 5.84 | 1.39 | 8.10 | 5.84 | 1.39 | 8.10    | 5.84 | 1.39 |
| 60           | -25        | /       | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
|              | -20        | 6.00    | 1.52 | 3.96 | 5.48         | 1.58 | 3.47 | 4.11 | 1.81 | 2.27 | 2.92 | 1.99 | 1.47 | 2.92 | 1.99 | 1.47 | 2.92    | 1.99 | 1.47 |
|              | -15        | 6.21    | 1.71 | 3.63 | 6.21         | 1.71 | 3.63 | 4.65 | 1.93 | 2.42 | 3.45 | 2.04 | 1.69 | 3.45 | 2.04 | 1.69 | 3.45    | 2.04 | 1.69 |
|              | -10        | 7.52    | 1.96 | 3.84 | 6.96         | 1.99 | 3.49 | 5.22 | 2.25 | 2.32 | 4.27 | 2.46 | 1.73 | 4.27 | 2.46 | 1.73 | 4.27    | 2.46 | 1.73 |
|              | -7         | 7.86    | 2.01 | 3.91 | 6.98         | 2.08 | 3.36 | 5.30 | 2.35 | 2.26 | 4.80 | 2.50 | 1.92 | 4.80 | 2.50 | 1.92 | 4.80    | 2.50 | 1.92 |
|              | -5         | 8.13    | 2.13 | 3.81 | 7.17         | 2.22 | 3.24 | 5.45 | 2.41 | 2.26 | 5.14 | 2.57 | 2.00 | 5.14 | 2.57 | 2.00 | 5.14    | 2.57 | 2.00 |
|              | -2         | 8.38    | 2.19 | 3.83 | 7.29         | 2.30 | 3.17 | 5.58 | 2.45 | 2.28 | 4.83 | 2.60 | 1.86 | 4.83 | 2.60 | 1.86 | 4.83    | 2.60 | 1.86 |
|              | 0          | 7.97    | 2.32 | 3.44 | 7.44         | 2.36 | 3.15 | 5.75 | 2.61 | 2.20 | 4.82 | 2.73 | 1.77 | 4.82 | 2.73 | 1.77 | 4.82    | 2.73 | 1.77 |
|              | 2          | 8.05    | 2.40 | 3.36 | 7.66         | 2.43 | 3.15 | 5.82 | 2.71 | 2.15 | 5.10 | 2.87 | 1.78 | 5.10 | 2.87 | 1.78 | 5.10    | 2.87 | 1.78 |
|              | 5          | 8.63    | 2.73 | 3.16 | 7.94         | 2.79 | 2.85 | 5.99 | 3.05 | 1.96 | 5.99 | 3.05 | 1.96 | 5.99 | 3.05 | 1.96 | 5.99    | 3.05 | 1.96 |
|              | 7          | 8.99    | 2.92 | 3.08 | 8.31         | 2.98 | 2.79 | 6.24 | 3.29 | 1.89 | 6.24 | 3.29 | 1.89 | 6.24 | 3.29 | 1.89 | 6.24    | 3.29 | 1.89 |
|              | 10         | 9.37    | 3.13 | 3.00 | 8.04         | 3.30 | 2.43 | 6.10 | 3.61 | 1.69 | 6.10 | 3.61 | 1.69 | 6.10 | 3.61 | 1.69 | 6.10    | 3.61 | 1.69 |
|              | 12         | 9.59    | 3.37 | 2.85 | 8.07         | 3.56 | 2.27 | 6.51 | 3.82 | 1.70 | 6.51 | 3.82 | 1.70 | 6.51 | 3.82 | 1.70 | 6.51    | 3.82 | 1.70 |
|              | 15         | 9.36    | 3.51 | 2.67 | 7.89         | 3.65 | 2.16 | 6.68 | 3.98 | 1.68 | 6.68 | 3.98 | 1.68 | 6.68 | 3.98 | 1.68 | 6.68    | 3.98 | 1.68 |
|              | 20         | 9.01    | 3.79 | 2.37 | 8.01         | 3.97 | 2.02 | 5.99 | 4.36 | 1.37 | 5.99 | 4.36 | 1.37 | 5.99 | 4.36 | 1.37 | 5.99    | 4.36 | 1.37 |
|              | 25         | 8.79    | 4.21 | 2.09 | 7.90         | 4.38 | 1.80 | 6.35 | 4.63 | 1.37 | 6.35 | 4.63 | 1.37 | 6.35 | 4.63 | 1.37 | 6.35    | 4.63 | 1.37 |
|              | 30         | 8.91    | 4.63 | 1.92 | 7.88         | 4.83 | 1.63 | 7.01 | 4.99 | 1.41 | 7.01 | 4.99 | 1.41 | 7.01 | 4.99 | 1.41 | 7.01    | 4.99 | 1.41 |
|              | 35         | /       | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
| 65           | -25        | /       | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
|              | -20        | 5.43    | 1.48 | 3.67 | 5.43         | 1.48 | 3.67 | 4.07 | 1.53 | 2.66 | 2.82 | 1.62 | 1.74 | 2.82 | 1.62 | 1.74 | 2.82    | 1.62 | 1.74 |
|              | -15        | 6.15    | 1.60 | 3.84 | 6.15         | 1.60 | 3.84 | 4.62 | 1.66 | 2.78 | 3.45 | 1.87 | 1.84 | 3.45 | 1.87 | 1.84 | 3.45    | 1.87 | 1.84 |
|              | -10        | 7.49    | 1.83 | 4.08 | 6.96         | 1.87 | 3.73 | 5.22 | 1.89 | 2.76 | 4.10 | 2.12 | 1.93 | 4.10 | 2.12 | 1.93 | 4.10    | 2.12 | 1.93 |
|              | -7         | 7.80    | 1.96 | 3.97 | 7.21         | 2.02 | 3.57 | 5.47 | 2.11 | 2.59 | 4.63 | 2.17 | 2.13 | 4.63 | 2.17 | 2.13 | 4.63    | 2.17 | 2.13 |
|              | -5         | 8.09    | 2.02 | 4.00 | 7.21         | 2.06 | 3.50 | 5.40 | 2.16 | 2.50 | 4.98 | 2.29 | 2.18 | 4.98 | 2.29 | 2.18 | 4.98    | 2.29 | 2.18 |
|              | -2         | 8.27    | 2.08 | 3.97 | 7.33         | 2.12 | 3.46 | 5.63 | 2.20 | 2.56 | 4.64 | 2.41 | 1.93 | 4.64 | 2.41 | 1.93 | 4.64    | 2.41 | 1.93 |
|              | 0          | 7.80    | 2.19 | 3.56 | 7.45         | 2.19 | 3.40 | 5.68 | 2.23 | 2.55 | 4.58 | 2.51 | 1.82 | 4.58 | 2.51 | 1.82 | 4.58    | 2.51 | 1.82 |
|              | 2          | 7.84    | 2.26 | 3.47 | 7.48         | 2.27 | 3.30 | 5.75 | 2.35 | 2.45 | 4.86 | 2.48 | 1.96 | 4.86 | 2.48 | 1.96 | 4.86    | 2.48 | 1.96 |
|              | 5          | 8.37    | 2.41 | 3.47 | 8.02         | 2.44 | 3.28 | 6.05 | 2.56 | 2.36 | 5.44 | 2.71 | 2.01 | 5.44 | 2.71 | 2.01 | 5.44    | 2.71 | 2.01 |
|              | 7          | 8.71    | 2.57 | 3.39 | 8.11         | 2.67 | 3.03 | 6.17 | 2.79 | 2.21 | 5.95 | 2.89 | 2.06 | 5.95 | 2.89 | 2.06 | 5.95    | 2.89 | 2.06 |
|              | 10         | 9.05    | 2.78 | 3.26 | 8.06         | 2.90 | 2.78 | 6.11 | 3.02 | 2.02 | 5.75 | 3.14 | 1.83 | 5.75 | 3.14 | 1.83 | 5.75    | 3.14 | 1.83 |
|              | 12         | 9.28    | 2.94 | 3.16 | 8.07         | 3.08 | 2.62 | 6.16 | 3.23 | 1.91 | 6.16 | 3.20 | 1.92 | 6.16 | 3.20 | 1.92 | 6.16    | 3.20 | 1.92 |
|              | 15         | 9.02    | 3.04 | 2.97 | 7.87         | 3.17 | 2.48 | 6.31 | 3.30 | 1.91 | 6.31 | 3.30 | 1.91 | 6.31 | 3.30 | 1.91 | 6.31    | 3.30 | 1.91 |
|              | 20         | 8.61    | 3.29 | 2.62 | 7.96         | 3.35 | 2.38 | 5.99 | 3.55 | 1.69 | 5.39 | 3.67 | 1.47 | 5.39 | 3.67 | 1.47 | 5.39    | 3.67 | 1.47 |
|              | 25         | 8.38    | 3.66 | 2.29 | 8.03         | 3.72 | 2.16 | 5.99 | 3.96 | 1.51 | 5.99 | 3.96 | 1.51 | 5.99 | 3.96 | 1.51 | 5.99    | 3.96 | 1.51 |
|              | 30         | 8.40    | 3.95 | 2.13 | 7.86         | 3.99 | 1.97 | 6.61 | 4.01 | 1.65 | 6.61 | 4.01 | 1.65 | 6.61 | 4.01 | 1.65 | 6.61    | 4.01 | 1.65 |
|              | 35         | /       | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
| 70           | -25        | /       | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
|              | -20        | /       | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
|              | -15        | 6.16    | 1.53 | 4.03 | 6.16         | 1.53 | 4.03 | 4.62 | 1.66 | 2.78 | 3.19 | 1.93 | 1.66 | 3.19 | 1.93 | 1.66 | 3.19    | 1.93 | 1.66 |
|              | -10        | 7.46    | 1.76 | 4.24 | 6.96         | 1.80 | 3.87 | 5.28 | 1.92 | 2.75 | 3.99 | 2.12 | 1.88 | 3.99 | 2.12 | 1.88 | 3.99    | 2.12 | 1.88 |
|              | -7         | 7.79    | 1.86 | 4.20 | 6.98         | 1.87 | 3.73 | 5.36 | 1.99 | 2.69 | 4.55 | 2.13 | 2.13 | 4.55 | 2.13 | 2.13 | 4.55    | 2.13 | 2.13 |
|              | -5         | 8.04    | 1.94 | 4.14 | 7.10         | 1.98 | 3.58 | 5.46 | 2.10 | 2.59 | 4.87 | 2.25 | 2.17 | 4.87 | 2.25 | 2.17 | 4.87    | 2.25 | 2.17 |
|              | -2         | 8.25    | 1.98 | 4.16 | 7.39         | 1.99 | 3.71 | 5.61 | 2.13 | 2.63 | 4.46 | 2.32 | 1.92 | 4.46 | 2.32 | 1.92 | 4.46    | 2.32 | 1.92 |
|              | 0          | 7.71    | 2.04 | 3.78 | 7.46         | 2.05 | 3.63 | 5.69 | 2.19 | 2.60 | 4.50 |      |      |      |      |      |         |      |      |

## 8.4 Heating performance Mod. 10

| Tw_out<br>°C | DBWB<br>°C | Maximum |      |      | 100%(Normal) |       |      | 75%  |       |      | 50%  |       |      | 25%  |       |      | Minimum |       |      |
|--------------|------------|---------|------|------|--------------|-------|------|------|-------|------|------|-------|------|------|-------|------|---------|-------|------|
|              |            | HC      | COP  | PI   | HC           | COP   | PI   | HC   | COP   | PI   | HC   | COP   | PI   | HC   | COP   | PI   | HC      | COP   | PI   |
| 25           | -25        | 5.55    | 2.14 | 2.59 | 5.55         | 2.14  | 2.59 | 4.16 | 2.48  | 1.83 | 2.75 | 2.77  | 1.17 | 2.19 | 2.85  | 0.77 | 2.19    | 2.85  | 0.77 |
|              | -20        | 6.71    | 2.34 | 2.87 | 6.71         | 2.34  | 2.87 | 5.03 | 2.67  | 1.88 | 3.34 | 3.00  | 1.23 | 2.61 | 3.10  | 0.84 | 2.61    | 3.10  | 0.84 |
|              | -15        | 7.66    | 2.70 | 2.83 | 7.66         | 2.70  | 2.83 | 5.74 | 3.01  | 1.93 | 3.83 | 3.45  | 1.24 | 3.05 | 3.56  | 0.85 | 3.05    | 3.56  | 0.85 |
|              | -10        | 8.76    | 3.05 | 2.87 | 8.05         | 3.34  | 2.41 | 6.06 | 3.74  | 1.60 | 4.04 | 4.17  | 1.03 | 3.36 | 4.25  | 0.79 | 3.36    | 4.25  | 0.79 |
|              | -7         | 9.12    | 3.26 | 2.80 | 8.03         | 3.60  | 2.23 | 6.11 | 3.95  | 1.56 | 4.12 | 4.40  | 0.97 | 3.80 | 4.42  | 0.86 | 3.80    | 4.42  | 0.86 |
|              | -5         | 9.34    | 3.45 | 2.71 | 8.03         | 3.81  | 2.11 | 6.09 | 4.22  | 1.44 | 4.12 | 4.61  | 0.89 | 4.07 | 4.62  | 0.88 | 4.07    | 4.62  | 0.88 |
|              | -2         | 9.70    | 3.90 | 2.49 | 8.09         | 4.28  | 1.89 | 6.11 | 4.71  | 1.30 | 4.10 | 5.20  | 0.79 | 3.64 | 5.25  | 0.69 | 3.64    | 5.25  | 0.69 |
|              | 0          | 9.68    | 4.08 | 2.37 | 8.20         | 4.44  | 1.84 | 6.15 | 4.85  | 1.27 | 4.13 | 5.28  | 0.78 | 3.82 | 5.31  | 0.72 | 3.82    | 5.31  | 0.72 |
|              | 2          | 9.84    | 4.37 | 2.25 | 8.21         | 4.78  | 1.72 | 6.21 | 5.22  | 1.19 | 4.17 | 5.70  | 0.73 | 4.03 | 5.71  | 0.71 | 4.03    | 5.71  | 0.71 |
|              | 5          | 10.73   | 5.28 | 2.03 | 8.80         | 5.69  | 1.55 | 6.65 | 6.13  | 1.08 | 4.50 | 6.51  | 0.69 | 4.23 | 6.52  | 0.65 | 4.23    | 6.52  | 0.65 |
|              | 7          | 11.46   | 5.70 | 2.01 | 9.97         | 6.26  | 1.59 | 7.48 | 6.70  | 1.12 | 5.07 | 7.06  | 0.72 | 4.62 | 7.15  | 0.65 | 4.62    | 7.15  | 0.65 |
|              | 10         | 11.86   | 6.65 | 1.78 | 9.67         | 7.22  | 1.34 | 7.41 | 7.66  | 0.97 | 5.08 | 8.11  | 0.63 | 5.08 | 8.11  | 0.63 | 5.08    | 8.11  | 0.63 |
|              | 12         | 12.16   | 7.55 | 1.61 | 9.70         | 8.21  | 1.18 | 7.44 | 8.59  | 0.87 | 5.42 | 9.01  | 0.60 | 5.42 | 9.01  | 0.60 | 5.42    | 9.01  | 0.60 |
|              | 15         | 11.93   | 7.92 | 1.51 | 9.54         | 8.55  | 1.12 | 7.15 | 9.11  | 0.78 | 5.50 | 9.46  | 0.58 | 5.50 | 9.46  | 0.58 | 5.50    | 9.46  | 0.58 |
|              | 20         | 11.73   | 8.77 | 1.34 | 9.62         | 9.33  | 1.03 | 7.21 | 9.75  | 0.74 | 5.17 | 10.24 | 0.50 | 5.17 | 10.24 | 0.50 | 5.17    | 10.24 | 0.50 |
|              | 25         | 11.87   | 9.45 | 1.26 | 9.67         | 9.98  | 0.97 | 7.19 | 10.51 | 0.68 | 5.70 | 10.83 | 0.48 | 5.70 | 10.83 | 0.53 | 5.70    | 10.83 | 0.53 |
|              | 30         | /       | /    | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 35         | /       | /    | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 30           | -25        | 5.46    | 2.01 | 2.71 | 5.46         | 2.01  | 2.71 | 4.09 | 2.30  | 1.78 | 2.69 | 2.58  | 1.04 | 2.14 | 2.64  | 0.81 | 2.14    | 2.64  | 0.81 |
|              | -20        | 6.62    | 2.31 | 2.86 | 6.62         | 2.31  | 2.86 | 4.97 | 2.59  | 1.92 | 3.26 | 2.87  | 1.13 | 2.56 | 3.01  | 0.85 | 2.56    | 3.01  | 0.85 |
|              | -15        | 7.57    | 2.54 | 2.98 | 7.57         | 2.54  | 2.98 | 5.67 | 2.85  | 1.99 | 3.78 | 3.18  | 1.19 | 3.00 | 3.30  | 0.91 | 3.00    | 3.30  | 0.91 |
|              | -10        | 8.65    | 2.86 | 3.03 | 8.05         | 2.94  | 2.73 | 6.06 | 3.28  | 1.85 | 4.04 | 3.65  | 1.11 | 3.26 | 3.80  | 0.86 | 3.26    | 3.80  | 0.86 |
|              | -7         | 9.00    | 3.06 | 2.94 | 8.03         | 3.20  | 2.51 | 6.11 | 3.60  | 1.70 | 4.12 | 3.93  | 1.05 | 3.70 | 4.04  | 0.92 | 3.70    | 4.04  | 0.92 |
|              | -5         | 9.24    | 3.23 | 2.86 | 8.03         | 3.45  | 2.33 | 6.09 | 3.84  | 1.59 | 4.12 | 4.20  | 0.98 | 3.98 | 4.35  | 0.92 | 3.98    | 4.35  | 0.92 |
|              | -2         | 9.60    | 3.48 | 2.76 | 8.04         | 3.86  | 2.08 | 6.14 | 4.22  | 1.45 | 4.16 | 4.57  | 0.91 | 3.56 | 4.70  | 0.76 | 3.56    | 4.70  | 0.76 |
|              | 0          | 9.56    | 3.68 | 2.60 | 8.20         | 4.00  | 2.05 | 6.20 | 4.34  | 1.43 | 4.20 | 4.73  | 0.89 | 3.73 | 4.92  | 0.76 | 3.73    | 4.92  | 0.76 |
|              | 2          | 9.70    | 3.91 | 2.48 | 8.21         | 4.27  | 1.92 | 6.26 | 4.63  | 1.35 | 4.25 | 5.02  | 0.85 | 3.94 | 5.18  | 0.76 | 3.94    | 5.18  | 0.76 |
|              | 5          | 10.56   | 4.49 | 2.35 | 8.77         | 4.82  | 1.82 | 6.62 | 5.27  | 1.25 | 4.53 | 5.69  | 0.80 | 4.13 | 5.87  | 0.70 | 4.13    | 5.87  | 0.70 |
|              | 7          | 11.24   | 4.82 | 2.33 | 9.92         | 5.16  | 1.92 | 7.46 | 5.68  | 1.31 | 4.96 | 5.97  | 0.83 | 4.52 | 6.12  | 0.74 | 4.52    | 6.12  | 0.74 |
|              | 10         | 11.65   | 5.50 | 2.12 | 9.66         | 5.94  | 1.63 | 7.38 | 6.36  | 1.16 | 4.98 | 6.79  | 0.73 | 4.98 | 6.79  | 0.73 | 4.98    | 6.79  | 0.73 |
|              | 12         | 11.95   | 5.85 | 2.04 | 9.71         | 6.28  | 1.55 | 7.45 | 6.55  | 1.14 | 5.32 | 7.12  | 0.75 | 5.32 | 7.12  | 0.75 | 5.32    | 7.12  | 0.75 |
|              | 15         | 11.73   | 6.35 | 1.85 | 9.53         | 6.78  | 1.41 | 7.17 | 7.22  | 0.99 | 5.41 | 7.59  | 0.71 | 5.41 | 7.59  | 0.71 | 5.41    | 7.59  | 0.71 |
|              | 20         | 11.56   | 7.33 | 1.58 | 9.62         | 7.77  | 1.24 | 7.24 | 8.01  | 0.90 | 5.06 | 8.30  | 0.61 | 5.06 | 8.30  | 0.61 | 5.06    | 8.30  | 0.61 |
|              | 25         | 11.71   | 8.45 | 1.39 | 9.68         | 8.80  | 1.10 | 7.25 | 9.50  | 0.76 | 5.61 | 9.75  | 0.58 | 5.61 | 9.75  | 0.58 | 5.61    | 9.75  | 0.58 |
|              | 30         | 11.47   | 9.49 | 1.21 | 9.57         | 10.06 | 0.95 | 7.35 | 10.48 | 0.70 | 6.16 | 11.02 | 0.56 | 6.16 | 11.02 | 0.56 | 6.16    | 11.02 | 0.56 |
|              | 35         | /       | /    | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 35           | -25        | 5.37    | 1.92 | 2.80 | 5.37         | 1.92  | 2.80 | 3.96 | 2.01  | 1.97 | 2.62 | 2.20  | 1.19 | 2.09 | 2.35  | 0.89 | 2.09    | 2.35  | 0.89 |
|              | -20        | 6.54    | 2.19 | 2.99 | 6.54         | 2.19  | 2.99 | 4.83 | 2.27  | 2.13 | 3.22 | 2.57  | 1.25 | 2.50 | 2.64  | 0.95 | 2.50    | 2.64  | 0.95 |
|              | -15        | 7.45    | 2.39 | 3.12 | 7.45         | 2.39  | 3.12 | 5.78 | 2.50  | 2.31 | 3.88 | 2.72  | 1.43 | 2.91 | 2.96  | 0.98 | 2.91    | 2.96  | 0.98 |
|              | -10        | 8.54    | 2.69 | 3.18 | 7.60         | 2.82  | 2.70 | 5.74 | 3.01  | 1.91 | 3.92 | 3.12  | 1.26 | 3.16 | 3.28  | 0.96 | 3.16    | 3.28  | 0.96 |
|              | -7         | 8.89    | 2.78 | 3.20 | 8.00         | 2.85  | 2.81 | 6.00 | 3.22  | 1.86 | 4.00 | 3.45  | 1.16 | 3.60 | 3.52  | 1.02 | 3.60    | 3.52  | 1.02 |
|              | -5         | 9.12    | 3.04 | 3.00 | 8.00         | 3.22  | 2.49 | 6.32 | 3.49  | 1.81 | 4.24 | 3.71  | 1.14 | 3.88 | 3.72  | 1.04 | 3.88    | 3.72  | 1.04 |
|              | -2         | 9.49    | 3.29 | 2.88 | 8.02         | 3.55  | 2.26 | 6.40 | 3.83  | 1.67 | 4.26 | 4.05  | 1.05 | 3.47 | 4.16  | 0.84 | 3.47    | 4.16  | 0.84 |
|              | 0          | 9.42    | 3.37 | 2.80 | 8.06         | 3.65  | 2.21 | 6.55 | 3.90  | 1.68 | 4.36 | 4.18  | 1.04 | 3.66 | 4.37  | 0.84 | 3.66    | 4.37  | 0.84 |
|              | 2          | 9.57    | 3.55 | 2.69 | 8.20         | 3.65  | 2.25 | 6.10 | 4.03  | 1.52 | 4.18 | 4.44  | 0.94 | 3.84 | 4.48  | 0.86 | 3.84    | 4.48  | 0.86 |
|              | 5          | 10.40   | 4.00 | 2.60 | 8.81         | 4.35  | 2.02 | 6.59 | 4.70  | 1.40 | 4.53 | 5.12  | 0.88 | 4.01 | 5.16  | 0.78 | 4.01    | 5.16  | 0.78 |
|              | 7*         | 11.05   | 4.40 | 2.51 | 10.00        | 4.70  | 2.13 | 7.50 | 5.12  | 1.47 | 4.97 | 5.52  | 0.90 | 4.40 | 5.64  | 0.78 | 4.40    | 5.64  | 0.78 |
|              | 10         | 11.44   | 4.93 | 2.32 | 9.63         | 5.32  | 1.81 | 7.37 | 5.71  | 1.29 | 5.02 | 6.40  | 0.78 | 4.86 | 6.43  | 0.76 | 4.86    | 6.43  | 0.76 |
|              | 12         | 11.73   | 5.43 | 2.16 | 9.70         | 5.81  | 1.67 | 7.43 | 6.26  | 1.19 | 5.19 | 6.61  | 0.79 | 5.19 | 6.61  | 0.79 | 5.19    | 6.61  | 0.79 |
|              | 15         | 11.53   | 5.66 | 2.04 | 9.82         | 6.03  | 1.63 | 7.16 | 6.44  | 1.11 | 5.31 | 6.90  | 0.77 | 5.31 | 6.90  | 0.77 | 5.31    | 6.90  | 0.77 |
|              | 20         | 11.35   | 6.40 | 1.78 | 9.60         | 6.67  | 1.44 | 7.25 | 7.11  | 1.02 | 4.92 | 7.57  | 0.65 | 4.92 | 7.57  | 0.65 | 4.92    | 7.57  | 0.65 |
|              | 25         | 11.48   | 7.46 | 1.54 | 9.66         | 7.84  | 1.23 | 7.28 | 8.19  | 0.89 | 5.46 | 8.64  | 0.63 | 5.46 | 8.64  | 0.63 | 5.46    | 8.64  | 0.63 |
|              | 30         | 11.25   | 8.83 | 1.27 | 9.66         | 9.25  | 1.04 | 7.21 | 9.69  | 0.74 | 6.00 | 10.06 | 0.60 | 6.00 | 10.06 | 0.60 | 6.00    | 10.06 | 0.60 |
|              | 35         | 10.81   | 9.95 | 1.09 | 9.67         | 10.92 | 0.89 | 7.41 | 10.95 | 0.68 | 6.60 | 11.13 | 0.59 | 6.60 | 11.13 | 0.59 | 6.60    | 11.13 | 0.59 |
| 40           | -25        | 5.29    | 1.84 | 3.23 | 5.29         | 1.84  | 3.23 | 3.88 | 1.72  | 2.25 | 2.56 | 2.08  | 1.23 | 2.14 | 2.19  | 0.98 | 2.14    | 2.19  | 0.98 |
|              | -20        | 6.46    | 2.10 | 3.08 | 6.46         | 2.10  | 3.08 | 4.89 | 2.21  | 2.21 | 3.19 | 2.40  | 1.33 | 2.52 | 2.47  | 1.02 | 2.52    | 2.47  | 1.02 |
|              | -15        | 7.35    | 2.25 | 3.28 | 7.35         | 2.25  | 3.28 | 5.65 | 2.33  | 2.42 | 3.77 | 2.72  | 1.39 | 2.98 | 2.86  | 1.04 | 2.98    | 2.86  | 1.04 |
|              | -10        | 8.43    | 2.51 | 3.35 | 7.86         | 2     |      |      |       |      |      |       |      |      |       |      |         |       |      |

| Tw_out<br>°C | DBWB<br>°C | Maximum |      |      | 100%(Normal) |      |      | 75%  |      |      | 50%  |      |      | 25%  |      |      | Minimum |      |      |
|--------------|------------|---------|------|------|--------------|------|------|------|------|------|------|------|------|------|------|------|---------|------|------|
|              |            | HC      | COP  | PI   | HC           | COP  | PI   | HC   | COP  | PI   | HC   | COP  | PI   | HC   | COP  | PI   | HC      | COP  | PI   |
| 55           | -25        | 5.00    | 1.36 | 3.67 | 5.00         | 1.36 | 3.67 | 3.78 | 1.53 | 2.47 | 2.59 | 1.67 | 1.55 | 2.59 | 1.67 | 1.55 | 2.59    | 1.67 | 1.55 |
|              | -20        | 6.21    | 1.55 | 3.99 | 6.21         | 1.55 | 3.99 | 4.66 | 1.74 | 2.68 | 3.10 | 1.87 | 1.66 | 2.77 | 2.07 | 1.34 | 2.77    | 2.07 | 1.34 |
|              | -15        | 6.98    | 1.69 | 4.13 | 6.98         | 1.69 | 4.13 | 5.33 | 1.89 | 2.82 | 3.58 | 1.98 | 1.81 | 3.29 | 2.12 | 1.55 | 3.29    | 2.12 | 1.55 |
|              | -10        | 8.14    | 1.89 | 4.31 | 7.34         | 2.02 | 3.64 | 5.48 | 2.20 | 2.49 | 4.08 | 2.38 | 1.71 | 4.08 | 2.38 | 1.71 | 4.08    | 2.38 | 1.71 |
|              | -7         | 8.61    | 2.01 | 4.28 | 7.40         | 2.10 | 3.52 | 5.54 | 2.34 | 2.37 | 4.61 | 2.43 | 1.90 | 4.61 | 2.43 | 1.90 | 4.61    | 2.43 | 1.90 |
|              | -5         | 8.96    | 2.14 | 4.20 | 7.67         | 2.18 | 3.52 | 5.84 | 2.50 | 2.33 | 4.92 | 2.60 | 1.89 | 4.92 | 2.60 | 1.89 | 4.92    | 2.60 | 1.89 |
|              | -2         | 9.22    | 2.19 | 4.22 | 8.01         | 2.23 | 3.59 | 6.07 | 2.58 | 2.35 | 5.00 | 2.73 | 1.83 | 5.00 | 2.73 | 1.83 | 5.00    | 2.73 | 1.83 |
|              | 0          | 8.85    | 2.40 | 3.68 | 8.01         | 2.46 | 3.26 | 6.10 | 2.67 | 2.28 | 5.03 | 2.78 | 1.81 | 5.03 | 2.78 | 1.81 | 5.03    | 2.78 | 1.81 |
|              | 2          | 8.96    | 2.46 | 3.65 | 8.40         | 2.50 | 3.36 | 6.32 | 2.81 | 2.25 | 5.33 | 2.93 | 1.82 | 5.33 | 2.93 | 1.82 | 5.33    | 2.93 | 1.82 |
|              | 5          | 9.68    | 2.86 | 3.39 | 8.62         | 2.92 | 2.95 | 6.53 | 3.12 | 2.09 | 5.96 | 3.12 | 1.91 | 5.96 | 3.12 | 1.91 | 5.96    | 3.12 | 1.91 |
|              | 7          | 10.17   | 3.00 | 3.39 | 9.50         | 3.05 | 3.11 | 7.13 | 3.33 | 2.14 | 6.52 | 3.42 | 1.91 | 6.52 | 3.42 | 1.91 | 6.52    | 3.42 | 1.91 |
|              | 10         | 10.37   | 3.15 | 3.29 | 9.17         | 3.35 | 2.74 | 7.02 | 3.60 | 1.95 | 6.11 | 3.70 | 1.65 | 6.11 | 3.70 | 1.65 | 6.11    | 3.70 | 1.65 |
|              | 12         | 10.72   | 3.30 | 3.25 | 9.24         | 3.50 | 2.64 | 6.93 | 3.95 | 1.75 | 6.55 | 3.97 | 1.65 | 6.55 | 3.97 | 1.65 | 6.55    | 3.97 | 1.65 |
|              | 15         | 10.48   | 3.53 | 2.97 | 9.24         | 3.79 | 2.44 | 6.92 | 4.04 | 1.71 | 6.71 | 4.06 | 1.65 | 6.71 | 4.06 | 1.65 | 6.71    | 4.06 | 1.65 |
|              | 20         | 10.25   | 3.85 | 2.66 | 9.07         | 4.10 | 2.21 | 6.81 | 4.37 | 1.56 | 6.04 | 4.44 | 1.36 | 6.04 | 4.44 | 1.36 | 6.04    | 4.44 | 1.36 |
|              | 25         | 10.35   | 4.20 | 2.46 | 9.08         | 4.60 | 1.97 | 6.81 | 4.92 | 1.38 | 6.66 | 4.94 | 1.35 | 6.66 | 4.94 | 1.35 | 6.66    | 4.94 | 1.35 |
|              | 30         | 10.04   | 4.87 | 2.06 | 9.09         | 5.16 | 1.76 | 7.35 | 5.47 | 1.34 | 7.35 | 5.47 | 1.34 | 7.35 | 5.47 | 1.34 | 7.35    | 5.47 | 1.34 |
|              | 35         | 9.59    | 5.45 | 1.76 | 9.13         | 5.49 | 1.66 | 8.10 | 5.84 | 1.39 | 8.10 | 5.84 | 1.39 | 8.10 | 5.84 | 1.39 | 8.10    | 5.84 | 1.39 |
|              | 60         | -25     | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
| -20          |            | 6.12    | 1.50 | 4.08 | 5.97         | 1.57 | 3.80 | 4.57 | 1.77 | 2.58 | 3.04 | 1.97 | 1.54 | 2.92 | 1.99 | 1.47 | 2.92    | 1.99 | 1.47 |
| -15          |            | 6.94    | 1.61 | 4.31 | 6.94         | 1.61 | 4.31 | 5.33 | 1.81 | 2.95 | 3.57 | 2.02 | 1.77 | 3.45 | 2.04 | 1.69 | 3.45    | 2.04 | 1.69 |
| -10          |            | 8.12    | 1.89 | 4.30 | 7.10         | 1.97 | 3.61 | 5.42 | 2.20 | 2.46 | 4.27 | 2.46 | 1.73 | 4.27 | 2.46 | 1.73 | 4.27    | 2.46 | 1.73 |
| -7           |            | 8.58    | 1.95 | 4.40 | 7.12         | 2.03 | 3.51 | 5.31 | 2.35 | 2.26 | 4.80 | 2.50 | 1.92 | 4.80 | 2.50 | 1.92 | 4.80    | 2.50 | 1.92 |
| -5           |            | 8.90    | 2.04 | 4.35 | 7.46         | 2.16 | 3.46 | 5.68 | 2.38 | 2.39 | 5.14 | 2.57 | 2.00 | 5.14 | 2.57 | 2.00 | 5.14    | 2.57 | 2.00 |
| -2           |            | 9.18    | 2.06 | 4.44 | 7.46         | 2.22 | 3.36 | 5.68 | 2.42 | 2.35 | 4.83 | 2.60 | 1.86 | 4.83 | 2.60 | 1.86 | 4.83    | 2.60 | 1.86 |
| 0            |            | 8.73    | 2.27 | 3.85 | 8.01         | 2.32 | 3.45 | 6.10 | 2.53 | 2.42 | 4.82 | 2.73 | 1.77 | 4.82 | 2.73 | 1.77 | 4.82    | 2.73 | 1.77 |
| 2            |            | 8.82    | 2.33 | 3.78 | 8.35         | 2.36 | 3.54 | 6.37 | 2.60 | 2.45 | 5.10 | 2.87 | 1.78 | 5.10 | 2.87 | 1.78 | 5.10    | 2.87 | 1.78 |
| 5            |            | 9.46    | 2.56 | 3.69 | 8.65         | 2.65 | 3.27 | 6.52 | 2.90 | 2.24 | 5.99 | 3.05 | 1.96 | 5.99 | 3.05 | 1.96 | 5.99    | 3.05 | 1.96 |
| 7            |            | 9.90    | 2.75 | 3.60 | 9.27         | 2.82 | 3.28 | 6.99 | 3.13 | 2.23 | 6.24 | 3.29 | 1.89 | 6.24 | 3.29 | 1.89 | 6.24    | 3.29 | 1.89 |
| 10           |            | 10.18   | 2.94 | 3.46 | 9.28         | 3.01 | 3.08 | 7.01 | 3.45 | 2.03 | 6.10 | 3.61 | 1.69 | 6.10 | 3.61 | 1.69 | 6.10    | 3.61 | 1.69 |
| 12           |            | 10.41   | 3.10 | 3.36 | 9.23         | 3.29 | 2.81 | 6.90 | 3.78 | 1.83 | 6.51 | 3.82 | 1.70 | 6.51 | 3.82 | 1.70 | 6.51    | 3.82 | 1.70 |
| 15           |            | 10.17   | 3.21 | 3.16 | 9.22         | 3.40 | 2.71 | 6.68 | 3.98 | 1.68 | 6.68 | 3.98 | 1.68 | 6.68 | 3.98 | 1.68 | 6.68    | 3.98 | 1.68 |
| 20           |            | 9.93    | 3.45 | 2.87 | 9.31         | 3.67 | 2.54 | 6.85 | 4.18 | 1.64 | 5.99 | 4.36 | 1.37 | 5.99 | 4.36 | 1.37 | 5.99    | 4.36 | 1.37 |
| 25           |            | 9.99    | 3.76 | 2.65 | 9.10         | 4.04 | 2.25 | 6.89 | 4.58 | 1.50 | 6.35 | 4.63 | 1.37 | 6.35 | 4.63 | 1.37 | 6.35    | 4.63 | 1.37 |
| 30           |            | 9.64    | 4.36 | 2.21 | 9.26         | 4.49 | 2.06 | 7.01 | 4.99 | 1.41 | 7.01 | 4.99 | 1.41 | 7.01 | 4.99 | 1.41 | 7.01    | 4.99 | 1.41 |
| 35           |            | /       | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
| 65           |            | -25     | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
|              | -20        | 6.10    | 1.47 | 4.16 | 6.10         | 1.47 | 4.16 | 4.62 | 1.50 | 3.08 | 3.18 | 1.60 | 1.99 | 2.82 | 1.62 | 1.74 | 2.82    | 1.62 | 1.74 |
|              | -15        | 6.81    | 1.57 | 4.34 | 6.81         | 1.57 | 4.34 | 5.10 | 1.63 | 3.13 | 3.45 | 1.87 | 1.84 | 3.45 | 1.87 | 1.84 | 3.45    | 1.87 | 1.84 |
|              | -10        | 8.01    | 1.78 | 4.49 | 7.40         | 1.84 | 4.03 | 5.60 | 1.87 | 2.99 | 4.10 | 2.12 | 1.93 | 4.10 | 2.12 | 1.93 | 4.10    | 2.12 | 1.93 |
|              | -7         | 8.56    | 1.87 | 4.58 | 7.39         | 1.99 | 3.71 | 5.62 | 2.09 | 2.69 | 4.63 | 2.17 | 2.13 | 4.63 | 2.17 | 2.13 | 4.63    | 2.17 | 2.13 |
|              | -5         | 8.82    | 1.94 | 4.55 | 7.60         | 2.05 | 3.72 | 5.78 | 2.12 | 2.73 | 4.98 | 2.29 | 2.18 | 4.98 | 2.29 | 2.18 | 4.98    | 2.29 | 2.18 |
|              | -2         | 9.14    | 2.01 | 4.55 | 7.87         | 2.10 | 3.75 | 6.00 | 2.16 | 2.77 | 4.64 | 2.41 | 1.93 | 4.64 | 2.41 | 1.93 | 4.64    | 2.41 | 1.93 |
|              | 0          | 8.58    | 2.09 | 4.10 | 8.15         | 2.18 | 3.74 | 6.21 | 2.20 | 2.82 | 4.58 | 2.51 | 1.82 | 4.58 | 2.51 | 1.82 | 4.58    | 2.51 | 1.82 |
|              | 2          | 8.64    | 2.19 | 3.94 | 8.28         | 2.24 | 3.69 | 6.32 | 2.30 | 2.74 | 4.86 | 2.48 | 1.96 | 4.86 | 2.48 | 1.96 | 4.86    | 2.48 | 1.96 |
|              | 5          | 9.21    | 2.37 | 3.89 | 8.66         | 2.41 | 3.60 | 6.53 | 2.53 | 2.58 | 5.44 | 2.71 | 2.01 | 5.44 | 2.71 | 2.01 | 5.44    | 2.71 | 2.01 |
|              | 7          | 9.61    | 2.50 | 3.85 | 9.26         | 2.55 | 3.64 | 6.98 | 2.69 | 2.59 | 5.95 | 2.89 | 2.06 | 5.95 | 2.89 | 2.06 | 5.95    | 2.89 | 2.06 |
|              | 10         | 9.84    | 2.72 | 3.62 | 9.20         | 2.77 | 3.32 | 6.98 | 2.91 | 2.40 | 5.75 | 3.14 | 1.83 | 5.75 | 3.14 | 1.83 | 5.75    | 3.14 | 1.83 |
|              | 12         | 10.05   | 2.88 | 3.49 | 9.28         | 2.94 | 3.16 | 7.00 | 3.08 | 2.27 | 6.16 | 3.20 | 1.92 | 6.16 | 3.20 | 1.92 | 6.16    | 3.20 | 1.92 |
|              | 15         | 9.81    | 2.97 | 3.31 | 9.30         | 3.03 | 3.07 | 6.84 | 3.27 | 2.09 | 6.31 | 3.30 | 1.91 | 6.31 | 3.30 | 1.91 | 6.31    | 3.30 | 1.91 |
|              | 20         | 9.56    | 3.21 | 2.98 | 9.30         | 3.26 | 2.86 | 6.91 | 3.45 | 2.00 | 5.39 | 3.67 | 1.47 | 5.39 | 3.67 | 1.47 | 5.39    | 3.67 | 1.47 |
|              | 25         | 9.60    | 3.50 | 2.74 | 9.22         | 3.54 | 2.61 | 7.00 | 3.74 | 1.87 | 5.99 | 3.96 | 1.51 | 5.99 | 3.96 | 1.51 | 5.99    | 3.96 | 1.51 |
|              | 30         | 9.18    | 3.90 | 2.36 | 9.18         | 3.90 | 2.36 | 6.61 | 4.01 | 1.65 | 6.61 | 4.01 | 1.65 | 6.61 | 4.01 | 1.65 | 6.61    | 4.01 | 1.65 |
|              | 35         | /       | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
|              | 70         | -25     | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
| -20          |            | /       | /    | /    | /            | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /    | /       | /    | /    |
| -15          |            | 6.86    | 1.45 | 4.72 | 6.86         | 1.45 | 4.72 | 5.32 | 1.61 | 3.30 | 3.19 | 1.93 | 1.66 | 3.19 | 1.93 | 1.66 | 3.19    | 1.93 | 1.66 |
| -10          |            | 7.86    | 1.59 | 4.94 | 7.11         | 1.75 | 4.06 | 5.43 | 1.91 | 2.84 | 3.99 | 2.12 | 1.88 | 3.99 | 2.12 | 1.88 | 3.99    | 2.12 | 1.88 |
| -7           |            | 8.21    | 1.69 | 4.85 | 7.01         | 1.87 | 3.76 | 5.36 | 1.99 | 2.69 | 4.55 | 2.13 | 2.13 | 4.55 | 2.13 | 2.13 | 4.55    | 2.13 | 2.13 |
| -5           |            | 8.43    | 1.78 | 4.73 | 7.20         | 1.96 | 3.67 | 5.46 | 2.10 | 2.59 | 4.87 | 2.25 | 2.17 | 4.87 | 2.25 | 2.17 | 4.87    | 2.25 | 2.17 |
| -2           |            | 8.78    | 1.82 | 4.83 | 7.39         | 1.99 | 3.71 | 5.61 | 2.13 | 2.63 | 4.46 | 2.32 | 1.92 | 4.46 | 2.32 | 1.92 | 4.46    | 2.32 | 1.92 |
| 0            |            | 8.37    | 1.91 | 4.38 | 7.58         |      |      |      |      |      |      |      |      |      |      |      |         |      |      |

## 8.5 Heating performance Mod. 12

| Tw_out<br>°C | DBWB<br>°C | Maximum |      |      | 100%(Normal) |       |      | 75%  |       |      | 50%  |       |      | 25%  |       |      | Minimum |       |      |
|--------------|------------|---------|------|------|--------------|-------|------|------|-------|------|------|-------|------|------|-------|------|---------|-------|------|
|              |            | HC      | COP  | PI   | HC           | COP   | PI   | HC   | COP   | PI   | HC   | COP   | PI   | HC   | COP   | PI   | HC      | COP   | PI   |
| 25           | -25        | 7.62    | 2.27 | 3.36 | 7.62         | 2.27  | 3.36 | 7.62 | 2.27  | 3.36 | 5.06 | 2.42  | 2.09 | 2.98 | 2.56  | 1.17 | 2.98    | 2.56  | 1.17 |
|              | -20        | 9.15    | 2.60 | 3.52 | 9.15         | 2.60  | 3.52 | 7.54 | 2.74  | 2.75 | 5.00 | 2.95  | 1.70 | 3.77 | 3.11  | 1.21 | 3.77    | 3.11  | 1.21 |
|              | -15        | 10.59   | 2.90 | 3.66 | 10.06        | 2.99  | 3.37 | 7.52 | 3.28  | 2.29 | 5.02 | 3.44  | 1.46 | 4.37 | 3.77  | 1.16 | 4.37    | 3.77  | 1.16 |
|              | -10        | 11.28   | 3.03 | 3.72 | 9.93         | 3.19  | 3.11 | 7.57 | 3.54  | 2.14 | 5.18 | 3.96  | 1.31 | 4.69 | 3.99  | 1.18 | 4.69    | 3.99  | 1.18 |
|              | -7         | 11.50   | 3.26 | 3.53 | 9.92         | 3.61  | 2.75 | 7.57 | 4.11  | 1.84 | 5.41 | 4.44  | 1.22 | 5.41 | 4.44  | 1.22 | 5.41    | 4.44  | 1.22 |
|              | -5         | 11.55   | 3.43 | 3.36 | 9.62         | 3.75  | 2.57 | 7.28 | 4.11  | 1.77 | 5.60 | 4.55  | 1.23 | 5.60 | 4.55  | 1.23 | 5.60    | 4.55  | 1.23 |
|              | -2         | 12.14   | 3.73 | 3.25 | 9.52         | 4.22  | 2.26 | 7.16 | 4.59  | 1.56 | 4.86 | 5.22  | 0.93 | 4.86 | 5.22  | 0.93 | 4.86    | 5.22  | 0.93 |
|              | 0          | 12.40   | 3.96 | 3.13 | 9.36         | 4.63  | 2.02 | 7.00 | 5.01  | 1.40 | 5.02 | 5.42  | 0.93 | 5.02 | 5.42  | 0.93 | 5.02    | 5.42  | 0.93 |
|              | 2          | 12.73   | 4.20 | 3.03 | 9.12         | 5.00  | 1.83 | 6.90 | 5.35  | 1.29 | 5.38 | 5.80  | 0.93 | 5.38 | 5.80  | 0.93 | 5.38    | 5.80  | 0.93 |
|              | 5          | 14.25   | 4.89 | 2.91 | 10.79        | 5.64  | 1.91 | 8.19 | 6.44  | 1.27 | 5.85 | 7.00  | 0.84 | 5.85 | 7.00  | 0.84 | 5.85    | 7.00  | 0.84 |
|              | 7          | 15.69   | 5.47 | 2.87 | 12.09        | 6.29  | 1.92 | 8.94 | 6.86  | 1.30 | 6.24 | 7.44  | 0.84 | 6.24 | 7.44  | 0.84 | 6.24    | 7.44  | 0.84 |
|              | 10         | 17.24   | 6.38 | 2.70 | 12.08        | 7.65  | 1.58 | 9.13 | 8.47  | 1.08 | 7.25 | 8.95  | 0.81 | 7.25 | 8.95  | 0.81 | 7.25    | 8.95  | 0.81 |
|              | 12         | 17.67   | 7.04 | 2.51 | 11.95        | 8.48  | 1.41 | 9.16 | 9.37  | 0.98 | 7.71 | 9.79  | 0.79 | 7.71 | 9.79  | 0.79 | 7.71    | 9.79  | 0.79 |
|              | 15         | 16.62   | 7.45 | 2.23 | 12.08        | 8.76  | 1.38 | 9.03 | 9.77  | 0.92 | 7.78 | 10.26 | 0.76 | 7.78 | 10.26 | 0.76 | 7.78    | 10.26 | 0.76 |
|              | 20         | 14.83   | 8.21 | 1.81 | 12.03        | 9.39  | 1.28 | 9.09 | 10.71 | 0.85 | 6.62 | 11.24 | 0.59 | 6.62 | 11.24 | 0.59 | 6.62    | 11.24 | 0.59 |
|              | 25         | 14.66   | 9.04 | 1.62 | 11.92        | 10.23 | 1.17 | 9.14 | 11.50 | 0.79 | 7.40 | 11.79 | 0.63 | 7.40 | 11.79 | 0.63 | 7.40    | 11.79 | 0.63 |
|              | 30         | /       | /    | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 35         | /       | /    | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 30           | -25        | 7.20    | 2.03 | 3.55 | 7.20         | 2.03  | 3.55 | 7.20 | 2.03  | 3.55 | 4.88 | 2.12  | 2.31 | 2.78 | 2.25  | 1.24 | 2.78    | 2.25  | 1.24 |
|              | -20        | 8.71    | 2.34 | 3.72 | 8.71         | 2.34  | 3.72 | 7.53 | 2.44  | 3.09 | 4.99 | 2.55  | 1.96 | 3.56 | 2.76  | 1.29 | 3.56    | 2.76  | 1.29 |
|              | -15        | 10.38   | 2.68 | 3.87 | 9.87         | 2.74  | 3.60 | 7.39 | 2.92  | 2.53 | 5.05 | 3.15  | 1.61 | 4.19 | 3.36  | 1.25 | 4.19    | 3.36  | 1.25 |
|              | -10        | 10.65   | 2.72 | 3.91 | 9.86         | 2.82  | 3.50 | 7.41 | 3.10  | 2.39 | 4.98 | 3.44  | 1.45 | 4.58 | 3.58  | 1.28 | 4.58    | 3.58  | 1.28 |
|              | -7         | 11.30   | 3.01 | 3.75 | 9.92         | 3.17  | 3.13 | 7.62 | 3.59  | 2.13 | 5.39 | 4.03  | 1.34 | 5.39 | 4.03  | 1.34 | 5.39    | 4.03  | 1.34 |
|              | -5         | 11.11   | 3.10 | 3.59 | 9.54         | 3.29  | 2.90 | 7.17 | 3.67  | 1.95 | 5.38 | 4.10  | 1.31 | 5.38 | 4.10  | 1.31 | 5.38    | 4.10  | 1.31 |
|              | -2         | 11.67   | 3.35 | 3.48 | 9.37         | 3.70  | 2.53 | 7.16 | 3.98  | 1.80 | 4.84 | 4.30  | 1.13 | 4.66 | 4.56  | 1.02 | 4.66    | 4.56  | 1.02 |
|              | 0          | 11.93   | 3.56 | 3.35 | 9.23         | 4.05  | 2.28 | 6.94 | 4.31  | 1.61 | 4.74 | 4.63  | 1.02 | 4.74 | 4.63  | 1.02 | 4.74    | 4.63  | 1.02 |
|              | 2          | 12.25   | 3.76 | 3.26 | 9.00         | 4.35  | 2.07 | 6.93 | 4.55  | 1.52 | 5.16 | 5.02  | 1.03 | 5.16 | 5.02  | 1.03 | 5.16    | 5.02  | 1.03 |
|              | 5          | 13.72   | 4.36 | 3.15 | 10.84        | 5.13  | 2.11 | 8.16 | 5.45  | 1.50 | 5.54 | 5.92  | 0.93 | 5.54 | 5.92  | 0.93 | 5.54    | 5.92  | 0.93 |
|              | 7          | 15.02   | 4.86 | 3.09 | 12.02        | 5.43  | 2.21 | 9.10 | 5.80  | 1.57 | 5.98 | 6.33  | 0.94 | 5.98 | 6.33  | 0.94 | 5.98    | 6.33  | 0.94 |
|              | 10         | 16.43   | 5.61 | 2.93 | 11.91        | 6.48  | 1.84 | 9.12 | 7.03  | 1.30 | 6.91 | 7.52  | 0.92 | 6.91 | 7.52  | 0.92 | 6.91    | 7.52  | 0.92 |
|              | 12         | 16.81   | 6.12 | 2.75 | 11.84        | 7.12  | 1.66 | 8.96 | 7.79  | 1.15 | 7.32 | 8.10  | 0.90 | 7.32 | 8.10  | 0.90 | 7.32    | 8.10  | 0.90 |
|              | 15         | 15.87   | 6.43 | 2.47 | 11.82        | 7.45  | 1.59 | 9.07 | 8.44  | 1.07 | 7.59 | 8.73  | 0.87 | 7.59 | 8.73  | 0.87 | 7.59    | 8.73  | 0.87 |
|              | 20         | 14.25   | 7.18 | 1.98 | 12.00        | 8.01  | 1.50 | 8.96 | 9.19  | 0.97 | 6.37 | 9.36  | 0.68 | 6.37 | 9.36  | 0.68 | 6.37    | 9.36  | 0.68 |
|              | 25         | 14.13   | 7.83 | 1.80 | 11.96        | 8.50  | 1.41 | 9.04 | 9.58  | 0.94 | 7.11 | 9.62  | 0.74 | 7.11 | 9.62  | 0.74 | 7.11    | 9.62  | 0.74 |
|              | 30         | 14.12   | 9.33 | 1.51 | 11.93        | 9.98  | 1.20 | 8.97 | 11.20 | 0.80 | 7.78 | 11.27 | 0.69 | 7.78 | 11.27 | 0.69 | 7.78    | 11.27 | 0.69 |
|              | 35         | /       | /    | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 35           | -25        | 7.00    | 1.88 | 3.73 | 7.00         | 1.88  | 3.73 | 7.00 | 1.88  | 3.73 | 4.85 | 1.92  | 2.52 | 2.68 | 2.02  | 1.33 | 2.68    | 2.02  | 1.33 |
|              | -20        | 8.53    | 2.18 | 3.92 | 8.53         | 2.18  | 3.92 | 7.53 | 2.25  | 3.35 | 5.07 | 2.32  | 2.18 | 3.47 | 2.50  | 1.39 | 3.47    | 2.50  | 1.39 |
|              | -15        | 10.21   | 2.50 | 4.08 | 10.07        | 2.51  | 4.01 | 7.55 | 2.66  | 2.83 | 5.13 | 2.79  | 1.84 | 4.16 | 2.97  | 1.40 | 4.16    | 2.97  | 1.40 |
|              | -10        | 10.44   | 2.52 | 4.14 | 9.84         | 2.59  | 3.80 | 7.43 | 2.81  | 2.64 | 5.08 | 3.20  | 1.58 | 4.50 | 3.28  | 1.37 | 4.50    | 3.28  | 1.37 |
|              | -7         | 11.10   | 2.79 | 3.98 | 10.00        | 2.80  | 3.57 | 7.56 | 3.19  | 2.37 | 5.25 | 3.58  | 1.47 | 5.25 | 3.58  | 1.47 | 5.25    | 3.58  | 1.47 |
|              | -5         | 10.91   | 2.86 | 3.81 | 9.66         | 2.98  | 3.24 | 7.25 | 3.25  | 2.23 | 5.08 | 3.62  | 1.40 | 5.08 | 3.62  | 1.40 | 5.08    | 3.62  | 1.40 |
|              | -2         | 11.46   | 3.09 | 3.71 | 9.42         | 3.34  | 2.82 | 7.24 | 3.58  | 2.03 | 4.85 | 4.14  | 1.17 | 4.57 | 4.12  | 1.11 | 4.57    | 4.12  | 1.11 |
|              | 0          | 11.72   | 3.28 | 3.58 | 9.28         | 3.61  | 2.57 | 7.05 | 3.85  | 1.83 | 4.73 | 4.33  | 1.09 | 4.73 | 4.33  | 1.09 | 4.73    | 4.33  | 1.09 |
|              | 2          | 12.04   | 3.45 | 3.49 | 9.10         | 3.80  | 2.39 | 7.05 | 4.04  | 1.75 | 5.07 | 4.50  | 1.13 | 5.07 | 4.50  | 1.13 | 5.07    | 4.50  | 1.13 |
|              | 5          | 13.47   | 3.99 | 3.38 | 10.77        | 4.59  | 2.34 | 8.13 | 4.83  | 1.68 | 5.36 | 5.18  | 1.03 | 5.36 | 5.18  | 1.03 | 5.36    | 5.18  | 1.03 |
|              | 7*         | 14.72   | 4.42 | 3.33 | 12.00        | 4.80  | 2.50 | 9.06 | 5.11  | 1.77 | 5.87 | 5.61  | 1.05 | 5.87 | 5.61  | 1.05 | 5.87    | 5.61  | 1.05 |
|              | 10         | 15.57   | 4.92 | 3.17 | 12.00        | 5.45  | 2.20 | 9.13 | 5.80  | 1.57 | 6.42 | 6.22  | 1.03 | 6.42 | 6.22  | 1.03 | 6.42    | 6.22  | 1.03 |
|              | 12         | 15.92   | 5.33 | 2.99 | 12.12        | 5.91  | 2.05 | 9.04 | 6.35  | 1.42 | 6.82 | 6.71  | 1.02 | 6.82 | 6.71  | 1.02 | 6.82    | 6.71  | 1.02 |
|              | 15         | 15.03   | 5.61 | 2.68 | 12.06        | 6.34  | 1.90 | 8.99 | 6.90  | 1.30 | 7.09 | 7.20  | 0.99 | 7.09 | 7.20  | 0.99 | 7.09    | 7.20  | 0.99 |
|              | 20         | 13.54   | 6.47 | 2.09 | 11.96        | 6.95  | 1.72 | 9.12 | 7.72  | 1.18 | 5.97 | 7.99  | 0.75 | 5.97 | 7.99  | 0.75 | 5.97    | 7.99  | 0.75 |
|              | 25         | 13.43   | 7.08 | 1.90 | 11.84        | 7.50  | 1.58 | 9.06 | 8.23  | 1.10 | 6.68 | 8.47  | 0.79 | 6.68 | 8.47  | 0.79 | 6.68    | 8.47  | 0.79 |
|              | 30         | 13.45   | 8.04 | 1.67 | 12.13        | 8.44  | 1.44 | 9.02 | 8.93  | 1.01 | 7.30 | 9.00  | 0.81 | 7.30 | 9.00  | 0.81 | 7.30    | 9.00  | 0.81 |
|              | 35         | 13.22   | 9.58 | 1.38 | 12.02        | 9.89  | 1.22 | 9.23 | 10.37 | 0.89 | 7.97 | 10.45 | 0.76 | 7.97 | 10.45 | 0.76 | 7.97    | 10.45 | 0.76 |
| 40           | -25        | 6.82    | 1.75 | 3.90 | 6.82         | 1.75  | 3.90 | 6.82 | 1.75  | 3.90 | 5.13 | 1.80  | 2.84 | 2.74 | 1.89  | 1.45 | 2.74    | 1.89  | 1.45 |
|              | -20        | 8.34    | 2.03 | 4.11 | 8.34         | 2.03  | 4.11 | 7.54 | 2.08  | 3.62 | 5.08 | 2.18  | 2.33 | 3.57 | 2.31  | 1.55 | 3.57    | 2.31  | 1.55 |
|              | -15        | 10.03   | 2.34 | 4.29 | 10.03        | 2.34  | 4.29 | 7.59 | 2.45  | 3.10 | 5.20 | 2.50  | 2.08 | 3.99 | 2.59  | 1.54 | 3.99    | 2.59  |      |

| Tw_out<br>°C | DBWB<br>°C | Maximum |      |      | 100%(Normal) |      |      | 75%   |      |      | 50%   |      |      | 25%   |      |      | Minimum |      |      |
|--------------|------------|---------|------|------|--------------|------|------|-------|------|------|-------|------|------|-------|------|------|---------|------|------|
|              |            | HC      | COP  | PI   | HC           | COP  | PI   | HC    | COP  | PI   | HC    | COP  | PI   | HC    | COP  | PI   | HC      | COP  | PI   |
| 55           | -25        | 6.35    | 1.43 | 4.44 | 6.35         | 1.43 | 4.44 | 6.35  | 1.43 | 4.44 | 5.14  | 1.49 | 3.46 | 3.71  | 1.53 | 2.43 | 3.71    | 1.53 | 2.43 |
|              | -20        | 7.93    | 1.69 | 4.70 | 7.93         | 1.69 | 4.70 | 7.81  | 1.69 | 4.63 | 5.11  | 1.74 | 2.93 | 4.73  | 1.85 | 2.55 | 4.73    | 1.85 | 2.55 |
|              | -15        | 9.62    | 1.94 | 4.96 | 9.62         | 1.94 | 4.96 | 7.92  | 1.99 | 3.99 | 5.39  | 2.06 | 2.62 | 5.39  | 2.06 | 2.62 | 5.39    | 2.06 | 2.62 |
|              | -10        | 10.06   | 1.98 | 5.09 | 10.06        | 1.98 | 5.09 | 7.95  | 2.05 | 3.88 | 6.01  | 2.28 | 2.64 | 6.01  | 2.28 | 2.64 | 6.01    | 2.28 | 2.64 |
|              | -7         | 10.40   | 2.15 | 4.84 | 10.40        | 2.15 | 4.84 | 8.01  | 2.22 | 3.61 | 6.53  | 2.32 | 2.82 | 6.53  | 2.32 | 2.82 | 6.53    | 2.32 | 2.82 |
|              | -5         | 10.24   | 2.16 | 4.75 | 10.24        | 2.16 | 4.75 | 7.99  | 2.25 | 3.55 | 6.80  | 2.40 | 2.83 | 6.80  | 2.40 | 2.83 | 6.80    | 2.40 | 2.83 |
|              | -2         | 10.69   | 2.29 | 4.67 | 10.69        | 2.29 | 4.67 | 8.36  | 2.39 | 3.50 | 6.91  | 2.58 | 2.68 | 6.91  | 2.58 | 2.68 | 6.91    | 2.58 | 2.68 |
|              | 0          | 10.91   | 2.40 | 4.54 | 10.91        | 2.40 | 4.54 | 8.59  | 2.51 | 3.42 | 7.28  | 2.69 | 2.71 | 7.28  | 2.69 | 2.71 | 7.28    | 2.69 | 2.71 |
|              | 2          | 11.30   | 2.55 | 4.43 | 11.30        | 2.55 | 4.43 | 8.51  | 2.67 | 3.19 | 7.64  | 2.80 | 2.73 | 7.64  | 2.80 | 2.73 | 7.64    | 2.80 | 2.73 |
|              | 5          | 12.49   | 2.84 | 4.40 | 11.86        | 2.99 | 3.96 | 9.12  | 3.21 | 2.84 | 9.12  | 3.21 | 2.84 | 9.12  | 3.21 | 2.84 | 9.12    | 3.21 | 2.84 |
|              | 7          | 13.61   | 3.11 | 4.37 | 12.00        | 3.10 | 3.87 | 9.75  | 3.39 | 2.88 | 9.75  | 3.39 | 2.88 | 9.75  | 3.39 | 2.88 | 9.75    | 3.39 | 2.88 |
|              | 10         | 14.25   | 3.37 | 4.23 | 12.08        | 3.52 | 3.43 | 9.05  | 3.70 | 2.44 | 9.05  | 3.70 | 2.44 | 9.05  | 3.70 | 2.44 | 9.05    | 3.70 | 2.44 |
|              | 12         | 14.54   | 3.59 | 4.05 | 11.96        | 3.75 | 3.19 | 9.59  | 3.90 | 2.46 | 9.59  | 3.90 | 2.46 | 9.59  | 3.90 | 2.46 | 9.59    | 3.90 | 2.46 |
|              | 15         | 13.74   | 3.71 | 3.71 | 11.92        | 3.90 | 3.06 | 10.36 | 4.13 | 2.51 | 10.06 | 4.14 | 2.43 | 10.06 | 4.14 | 2.43 | 10.06   | 4.14 | 2.43 |
|              | 20         | 12.80   | 4.05 | 3.16 | 11.87        | 4.20 | 2.83 | 9.13  | 4.43 | 2.06 | 7.98  | 4.46 | 1.79 | 7.98  | 4.46 | 1.79 | 7.98    | 4.46 | 1.79 |
|              | 25         | 12.67   | 4.39 | 2.88 | 11.99        | 4.50 | 2.66 | 9.15  | 4.74 | 1.93 | 9.12  | 4.77 | 1.91 | 9.12  | 4.77 | 1.91 | 9.12    | 4.77 | 1.91 |
|              | 30         | 12.64   | 5.02 | 2.52 | 11.88        | 5.13 | 2.32 | 9.92  | 5.19 | 1.91 | 9.92  | 5.19 | 1.91 | 9.92  | 5.19 | 1.91 | 9.92    | 5.19 | 1.91 |
|              | 35         | 12.42   | 5.57 | 2.23 | 11.85        | 5.67 | 2.09 | 10.78 | 5.69 | 1.89 | 10.78 | 5.69 | 1.89 | 10.78 | 5.69 | 1.89 | 10.78   | 5.69 | 1.89 |
|              | 60         | -25     | /    | /    | /            | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /    | /       | /    | /    |
| -20          |            | 7.69    | 1.56 | 4.93 | 7.69         | 1.56 | 4.93 | 7.69  | 1.56 | 4.93 | 5.14  | 1.66 | 3.10 | 4.92  | 1.68 | 2.92 | 4.92    | 1.68 | 2.92 |
| -15          |            | 9.38    | 1.75 | 5.35 | 9.38         | 1.75 | 5.35 | 7.90  | 1.83 | 4.32 | 5.70  | 1.88 | 3.03 | 5.70  | 1.88 | 3.03 | 5.70    | 1.88 | 3.03 |
| -10          |            | 10.01   | 1.87 | 5.36 | 10.01        | 1.87 | 5.36 | 8.04  | 1.94 | 4.14 | 6.39  | 2.12 | 3.01 | 6.39  | 2.12 | 3.01 | 6.39    | 2.12 | 3.01 |
| -7           |            | 10.35   | 1.99 | 5.20 | 10.35        | 1.99 | 5.20 | 7.78  | 2.09 | 3.72 | 7.54  | 2.17 | 3.47 | 7.54  | 2.17 | 3.47 | 7.54    | 2.17 | 3.47 |
| -5           |            | 10.15   | 2.02 | 5.03 | 10.15        | 2.02 | 5.03 | 7.96  | 2.12 | 3.75 | 7.08  | 2.25 | 3.15 | 7.08  | 2.25 | 3.15 | 7.08    | 2.25 | 3.15 |
| -2           |            | 10.44   | 2.11 | 4.96 | 10.44        | 2.11 | 4.96 | 8.18  | 2.15 | 3.80 | 7.12  | 2.31 | 3.08 | 7.12  | 2.31 | 3.08 | 7.12    | 2.31 | 3.08 |
| 0            |            | 10.59   | 2.19 | 4.84 | 10.59        | 2.19 | 4.84 | 8.41  | 2.27 | 3.71 | 7.52  | 2.41 | 3.12 | 7.52  | 2.41 | 3.12 | 7.52    | 2.41 | 3.12 |
| 2            |            | 10.87   | 2.28 | 4.77 | 10.87        | 2.28 | 4.77 | 8.46  | 2.39 | 3.54 | 7.89  | 2.51 | 3.15 | 7.89  | 2.51 | 3.15 | 7.89    | 2.51 | 3.15 |
| 5            |            | 12.75   | 2.65 | 4.81 | 11.76        | 2.68 | 4.38 | 8.91  | 2.87 | 3.10 | 8.91  | 2.87 | 3.10 | 8.91  | 2.87 | 3.10 | 8.91    | 2.87 | 3.10 |
| 7            |            | 13.13   | 2.84 | 4.62 | 12.06        | 2.88 | 4.18 | 9.55  | 3.04 | 3.14 | 9.55  | 3.04 | 3.14 | 9.55  | 3.04 | 3.14 | 9.55    | 3.04 | 3.14 |
| 10           |            | 13.75   | 3.03 | 4.54 | 12.01        | 3.12 | 3.85 | 9.12  | 3.29 | 2.77 | 9.12  | 3.29 | 2.77 | 9.12  | 3.29 | 2.77 | 9.12    | 3.29 | 2.77 |
| 12           |            | 14.03   | 3.22 | 4.36 | 12.18        | 3.32 | 3.67 | 9.66  | 3.45 | 2.80 | 9.66  | 3.45 | 2.80 | 9.66  | 3.45 | 2.80 | 9.66    | 3.45 | 2.80 |
| 15           |            | 13.28   | 3.33 | 3.99 | 12.03        | 3.46 | 3.48 | 10.19 | 3.68 | 2.77 | 10.19 | 3.68 | 2.77 | 10.19 | 3.68 | 2.77 | 10.19   | 3.68 | 2.77 |
| 20           |            | 12.49   | 3.64 | 3.43 | 11.98        | 3.71 | 3.23 | 9.09  | 3.88 | 2.34 | 8.16  | 3.96 | 2.06 | 8.16  | 3.96 | 2.06 | 8.16    | 3.96 | 2.06 |
| 25           |            | 12.36   | 3.95 | 3.13 | 11.91        | 4.00 | 2.98 | 9.40  | 4.25 | 2.21 | 9.40  | 4.25 | 2.21 | 9.40  | 4.25 | 2.21 | 9.40    | 4.25 | 2.21 |
| 30           |            | 12.30   | 4.45 | 2.77 | 11.82        | 4.54 | 2.60 | 10.20 | 4.59 | 2.22 | 10.20 | 4.59 | 2.22 | 10.20 | 4.59 | 2.22 | 10.20   | 4.59 | 2.22 |
| 35           |            | /       | /    | /    | /            | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /    | /       | /    | /    |
| 65           |            | -25     | /    | /    | /            | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /    | /       | /    | /    |
|              | -20        | 7.42    | 1.43 | 5.19 | 7.42         | 1.43 | 5.19 | 7.42  | 1.43 | 5.19 | 5.15  | 1.45 | 3.54 | 4.66  | 1.51 | 3.08 | 4.66    | 1.51 | 3.08 |
|              | -15        | 8.99    | 1.60 | 5.61 | 8.99         | 1.60 | 5.61 | 7.85  | 1.68 | 4.68 | 5.64  | 1.79 | 3.15 | 5.64  | 1.79 | 3.15 | 5.64    | 1.79 | 3.15 |
|              | -10        | 9.76    | 1.78 | 5.48 | 9.76         | 1.78 | 5.48 | 8.07  | 1.82 | 4.43 | 6.46  | 2.01 | 3.22 | 6.46  | 2.01 | 3.22 | 6.46    | 2.01 | 3.22 |
|              | -7         | 10.27   | 1.83 | 5.61 | 10.27        | 1.83 | 5.61 | 7.88  | 1.93 | 4.08 | 6.68  | 2.02 | 3.31 | 6.68  | 2.02 | 3.31 | 6.68    | 2.02 | 3.31 |
|              | -5         | 9.83    | 1.86 | 5.28 | 9.83         | 1.86 | 5.28 | 7.93  | 1.99 | 3.98 | 6.94  | 2.06 | 3.37 | 6.94  | 2.06 | 3.37 | 6.94    | 2.06 | 3.37 |
|              | -2         | 10.06   | 1.96 | 5.14 | 10.06        | 1.96 | 5.14 | 8.09  | 2.04 | 3.97 | 6.90  | 2.10 | 3.29 | 6.90  | 2.10 | 3.29 | 6.90    | 2.10 | 3.29 |
|              | 0          | 9.97    | 2.01 | 4.96 | 9.97         | 2.01 | 4.96 | 8.19  | 2.07 | 3.95 | 7.22  | 2.16 | 3.34 | 7.22  | 2.16 | 3.34 | 7.22    | 2.16 | 3.34 |
|              | 2          | 10.37   | 2.06 | 5.03 | 10.37        | 2.06 | 5.03 | 8.36  | 2.12 | 3.94 | 7.57  | 2.24 | 3.38 | 7.57  | 2.24 | 3.38 | 7.57    | 2.24 | 3.38 |
|              | 5          | 12.26   | 2.39 | 5.14 | 11.77        | 2.39 | 4.91 | 8.74  | 2.47 | 3.55 | 8.52  | 2.56 | 3.32 | 8.52  | 2.56 | 3.32 | 8.52    | 2.56 | 3.32 |
|              | 7          | 12.53   | 2.59 | 4.84 | 12.02        | 2.63 | 4.58 | 9.06  | 2.79 | 3.25 | 9.06  | 2.79 | 3.25 | 9.06  | 2.79 | 3.25 | 9.06    | 2.79 | 3.25 |
|              | 10         | 13.19   | 2.71 | 4.87 | 12.07        | 2.78 | 4.34 | 9.01  | 2.90 | 3.11 | 8.74  | 2.92 | 2.99 | 8.74  | 2.92 | 2.99 | 8.74    | 2.92 | 2.99 |
|              | 12         | 13.45   | 2.87 | 4.69 | 12.04        | 2.93 | 4.11 | 9.28  | 3.07 | 3.03 | 9.28  | 3.07 | 3.03 | 9.28  | 3.07 | 3.03 | 9.28    | 3.07 | 3.03 |
|              | 15         | 12.73   | 2.96 | 4.31 | 11.88        | 3.06 | 3.88 | 10.16 | 3.28 | 3.10 | 9.84  | 3.28 | 3.00 | 9.84  | 3.28 | 3.00 | 9.84    | 3.28 | 3.00 |
|              | 20         | 12.14   | 3.24 | 3.75 | 11.84        | 3.30 | 3.59 | 9.02  | 3.53 | 2.55 | 7.87  | 3.53 | 2.23 | 7.87  | 3.53 | 2.23 | 7.87    | 3.53 | 2.23 |
|              | 25         | 11.99   | 3.52 | 3.41 | 11.99        | 3.52 | 3.41 | 9.14  | 3.78 | 2.42 | 9.14  | 3.78 | 2.42 | 9.14  | 3.78 | 2.42 | 9.14    | 3.78 | 2.42 |
|              | 30         | 11.95   | 3.96 | 3.02 | 11.95        | 3.96 | 3.02 | 9.92  | 4.07 | 2.44 | 9.92  | 4.07 | 2.44 | 9.92  | 4.07 | 2.44 | 9.92    | 4.07 | 2.44 |
|              | 35         | /       | /    | /    | /            | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /    | /       | /    | /    |
|              | 70         | -25     | /    | /    | /            | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /    | /       | /    | /    |
| -20          |            | /       | /    | /    | /            | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /    | /       | /    | /    |
| -15          |            | 8.45    | 1.57 | 5.40 | 8.45         | 1.57 | 5.40 | 7.90  | 1.61 | 4.92 | 5.47  | 1.65 | 3.32 | 5.47  | 1.65 | 3.32 | 5.47    | 1.65 | 3.32 |
| -10          |            | 9.40    | 1.72 | 5.47 | 9.40         | 1.72 | 5.47 | 8.38  | 1.75 | 4.80 | 6.50  | 1.92 | 3.40 | 6.50  | 1.92 | 3.40 | 6.50    | 1.92 | 3.40 |
| -7           |            | 9.86    | 1.80 | 5.47 | 9.86         | 1.80 | 5.47 | 8.00  | 1.85 | 4.33 | 7.04  | 1.96 | 3.59 | 6.90  | 1.96 | 3.52 | 6.90    | 1.96 | 3.52 |
| -5           |            | 9.52    | 1.82 | 5.24 | 9.52         | 1.82 | 5.24 | 8.04  | 1.87 | 4.30 | 7.13  | 1.97 | 3.61 | 7.13  | 1.97 | 3.61 | 7.13    | 1.97 | 3.61 |
| -2           |            | 9.71    | 1.85 | 5.24 | 9.71         | 1.85 | 5.24 | 8.31  | 1.89 | 4.40 | 6.98  | 2.00 | 3.49 |       |      |      |         |      |      |

## 8.6 Heating performance Mod. 14

| Tw_out<br>°C | DBWB<br>°C | Maximum |      |      | 100%(Normal) |      |      | 75%   |       |      | 50%  |       |      | 25%  |       |      | Minimum |       |      |
|--------------|------------|---------|------|------|--------------|------|------|-------|-------|------|------|-------|------|------|-------|------|---------|-------|------|
|              |            | HC      | COP  | PI   | HC           | COP  | PI   | HC    | COP   | PI   | HC   | COP   | PI   | HC   | COP   | PI   | HC      | COP   | PI   |
| 25           | -25        | 8.42    | 2.18 | 3.87 | 8.42         | 2.18 | 3.87 | 8.42  | 2.18  | 3.87 | 5.74 | 2.40  | 2.40 | 2.98 | 2.56  | 1.17 | 2.98    | 2.56  | 1.17 |
|              | -20        | 10.04   | 2.48 | 4.04 | 10.04        | 2.48 | 4.04 | 8.57  | 2.69  | 3.19 | 5.61 | 2.89  | 1.94 | 3.77 | 3.11  | 1.21 | 3.77    | 3.11  | 1.21 |
|              | -15        | 11.52   | 2.77 | 4.16 | 11.52        | 2.77 | 4.16 | 8.55  | 3.19  | 2.68 | 5.60 | 3.41  | 1.64 | 4.37 | 3.77  | 1.16 | 4.37    | 3.77  | 1.16 |
|              | -10        | 12.24   | 2.90 | 4.22 | 11.59        | 2.95 | 3.93 | 8.58  | 3.38  | 2.54 | 5.83 | 3.69  | 1.58 | 4.69 | 3.99  | 1.18 | 4.69    | 3.99  | 1.18 |
|              | -7         | 12.49   | 3.06 | 4.09 | 11.40        | 3.29 | 3.47 | 8.55  | 3.93  | 2.17 | 5.73 | 4.38  | 1.31 | 5.41 | 4.44  | 1.22 | 5.41    | 4.44  | 1.22 |
|              | -5         | 12.71   | 3.27 | 3.89 | 11.35        | 3.46 | 3.28 | 8.54  | 3.94  | 2.17 | 5.60 | 4.55  | 1.23 | 5.60 | 4.55  | 1.23 | 5.60    | 4.55  | 1.23 |
|              | -2         | 13.13   | 3.58 | 3.67 | 11.12        | 3.90 | 2.85 | 8.49  | 4.42  | 1.92 | 5.53 | 4.92  | 1.13 | 4.86 | 5.22  | 0.93 | 4.86    | 5.22  | 0.93 |
|              | 0          | 13.32   | 3.80 | 3.50 | 10.82        | 4.28 | 2.53 | 8.37  | 4.76  | 1.76 | 5.40 | 5.36  | 1.01 | 5.02 | 5.42  | 0.93 | 5.02    | 5.42  | 0.93 |
|              | 2          | 13.53   | 4.03 | 3.36 | 10.81        | 4.64 | 2.33 | 8.06  | 5.11  | 1.58 | 5.54 | 5.72  | 0.97 | 5.38 | 5.80  | 0.93 | 5.38    | 5.80  | 0.93 |
|              | 5          | 15.22   | 4.68 | 3.25 | 12.76        | 5.26 | 2.42 | 9.53  | 6.13  | 1.55 | 6.35 | 6.94  | 0.91 | 5.85 | 7.00  | 0.84 | 5.85    | 7.00  | 0.84 |
|              | 7          | 16.96   | 5.23 | 3.24 | 13.97        | 5.88 | 2.38 | 10.49 | 6.55  | 1.60 | 6.99 | 7.42  | 0.94 | 6.24 | 7.44  | 0.84 | 6.24    | 7.44  | 0.84 |
|              | 10         | 18.26   | 6.13 | 2.98 | 14.12        | 7.28 | 1.94 | 10.55 | 8.00  | 1.32 | 7.25 | 8.95  | 0.81 | 7.25 | 8.95  | 0.81 | 7.25    | 8.95  | 0.81 |
|              | 12         | 18.97   | 6.64 | 2.86 | 14.04        | 8.00 | 1.75 | 10.45 | 8.90  | 1.17 | 7.71 | 9.79  | 0.79 | 7.71 | 9.79  | 0.79 | 7.71    | 9.79  | 0.79 |
|              | 15         | 17.72   | 7.06 | 2.51 | 13.93        | 8.15 | 1.71 | 10.58 | 9.26  | 1.14 | 7.78 | 10.26 | 0.76 | 7.78 | 10.26 | 0.76 | 7.78    | 10.26 | 0.76 |
|              | 20         | 15.66   | 8.00 | 1.96 | 13.97        | 8.66 | 1.61 | 10.45 | 10.28 | 1.02 | 6.95 | 11.20 | 0.62 | 6.62 | 11.24 | 0.59 | 6.62    | 11.24 | 0.59 |
|              | 25         | 15.59   | 8.79 | 1.77 | 13.92        | 9.59 | 1.45 | 10.55 | 11.10 | 0.95 | 7.40 | 11.79 | 0.63 | 7.40 | 11.79 | 0.63 | 7.40    | 11.79 | 0.63 |
|              | 30         | /       | /    | /    | /            | /    | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 35         | /       | /    | /    | /            | /    | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 30           | -25        | 7.97    | 1.97 | 4.05 | 7.97         | 1.97 | 4.05 | 7.97  | 1.97  | 4.05 | 5.53 | 2.10  | 2.63 | 2.78 | 2.25  | 1.24 | 2.78    | 2.25  | 1.24 |
|              | -20        | 9.57    | 2.25 | 4.25 | 9.57         | 2.25 | 4.25 | 8.37  | 2.38  | 3.52 | 5.59 | 2.54  | 2.20 | 3.56 | 2.76  | 1.29 | 3.56    | 2.76  | 1.29 |
|              | -15        | 10.92   | 2.51 | 4.34 | 10.92        | 2.51 | 4.34 | 8.48  | 2.84  | 2.99 | 5.63 | 3.00  | 1.88 | 4.19 | 3.36  | 1.25 | 4.19    | 3.36  | 1.25 |
|              | -10        | 11.57   | 2.62 | 4.42 | 11.57        | 2.62 | 4.42 | 8.59  | 2.96  | 2.90 | 5.84 | 3.24  | 1.80 | 4.58 | 3.58  | 1.28 | 4.58    | 3.58  | 1.28 |
|              | -7         | 12.30   | 2.84 | 4.33 | 11.60        | 3.05 | 3.80 | 8.47  | 3.45  | 2.46 | 5.80 | 3.84  | 1.51 | 5.39 | 4.03  | 1.34 | 5.39    | 4.03  | 1.34 |
|              | -5         | 12.24   | 2.96 | 4.14 | 11.11        | 3.16 | 3.52 | 8.39  | 3.52  | 2.38 | 5.60 | 3.85  | 1.45 | 5.38 | 4.10  | 1.31 | 5.38    | 4.10  | 1.31 |
|              | -2         | 12.65   | 3.23 | 3.92 | 10.90        | 3.46 | 3.15 | 8.33  | 3.87  | 2.15 | 5.57 | 4.18  | 1.33 | 4.66 | 4.56  | 1.02 | 4.66    | 4.56  | 1.02 |
|              | 0          | 12.84   | 3.43 | 3.74 | 10.87        | 3.72 | 2.93 | 8.26  | 4.15  | 1.99 | 5.47 | 4.52  | 1.21 | 4.74 | 4.63  | 1.02 | 4.74    | 4.63  | 1.02 |
|              | 2          | 13.04   | 3.62 | 3.60 | 10.62        | 4.04 | 2.63 | 8.05  | 4.47  | 1.80 | 5.46 | 4.82  | 1.13 | 5.16 | 5.02  | 1.03 | 5.16    | 5.02  | 1.03 |
|              | 5          | 14.66   | 4.20 | 3.50 | 12.70        | 4.55 | 2.79 | 9.56  | 5.25  | 1.82 | 6.42 | 5.75  | 1.12 | 5.54 | 5.92  | 0.93 | 5.54    | 5.92  | 0.93 |
|              | 7          | 16.25   | 4.66 | 3.48 | 14.10        | 5.01 | 2.82 | 10.58 | 5.58  | 1.89 | 7.07 | 6.16  | 1.15 | 5.98 | 6.33  | 0.94 | 5.98    | 6.33  | 0.94 |
|              | 10         | 17.24   | 5.39 | 3.20 | 13.93        | 6.19 | 2.25 | 10.46 | 6.73  | 1.55 | 6.91 | 7.52  | 0.92 | 6.91 | 7.52  | 0.92 | 6.91    | 7.52  | 0.92 |
|              | 12         | 18.09   | 5.84 | 3.10 | 14.05        | 6.76 | 2.08 | 10.43 | 7.41  | 1.41 | 7.32 | 8.10  | 0.90 | 7.32 | 8.10  | 0.90 | 7.32    | 8.10  | 0.90 |
|              | 15         | 16.91   | 6.15 | 2.75 | 13.94        | 6.98 | 2.00 | 10.44 | 7.77  | 1.34 | 7.59 | 8.73  | 0.87 | 7.59 | 8.73  | 0.87 | 7.59    | 8.73  | 0.87 |
|              | 20         | 15.03   | 6.94 | 2.17 | 13.82        | 7.53 | 1.83 | 10.51 | 8.56  | 1.23 | 6.92 | 9.26  | 0.75 | 6.37 | 9.36  | 0.68 | 6.37    | 9.36  | 0.68 |
|              | 25         | 15.04   | 7.58 | 1.98 | 14.13        | 7.93 | 1.78 | 10.39 | 9.18  | 1.13 | 7.11 | 9.62  | 0.74 | 7.11 | 9.62  | 0.74 | 7.11    | 9.62  | 0.74 |
|              | 30         | 14.61   | 9.22 | 1.58 | 14.12        | 9.53 | 1.48 | 10.47 | 10.81 | 0.97 | 7.78 | 11.27 | 0.69 | 7.78 | 11.27 | 0.69 | 7.78    | 11.27 | 0.69 |
|              | 35         | /       | /    | /    | /            | /    | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 35           | -25        | 7.78    | 1.85 | 4.22 | 7.78         | 1.85 | 4.22 | 7.78  | 1.85  | 4.22 | 5.58 | 1.92  | 2.91 | 2.68 | 2.02  | 1.33 | 2.68    | 2.02  | 1.33 |
|              | -20        | 9.38    | 2.11 | 4.45 | 9.38         | 2.11 | 4.45 | 8.42  | 2.19  | 3.84 | 5.61 | 2.31  | 2.43 | 3.47 | 2.50  | 1.39 | 3.47    | 2.50  | 1.39 |
|              | -15        | 10.75   | 2.35 | 4.57 | 10.75        | 2.35 | 4.57 | 8.63  | 2.61  | 3.31 | 5.71 | 2.75  | 2.07 | 4.16 | 2.97  | 1.40 | 4.16    | 2.97  | 1.40 |
|              | -10        | 11.41   | 2.54 | 4.49 | 11.1         | 2.56 | 4.34 | 8.69  | 2.69  | 3.23 | 6.03 | 3.14  | 1.92 | 4.50 | 3.28  | 1.37 | 4.50    | 3.28  | 1.37 |
|              | -7         | 11.75   | 2.73 | 4.30 | 11.5         | 2.76 | 4.17 | 8.60  | 3.03  | 2.84 | 5.76 | 3.50  | 1.64 | 5.25 | 3.58  | 1.47 | 5.25    | 3.58  | 1.47 |
|              | -5         | 12.12   | 2.87 | 4.23 | 11.33        | 2.99 | 3.79 | 8.43  | 3.12  | 2.70 | 5.68 | 3.58  | 1.59 | 5.08 | 3.62  | 1.40 | 5.08    | 3.62  | 1.40 |
|              | -2         | 12.45   | 2.99 | 4.16 | 10.92        | 3.15 | 3.47 | 8.39  | 3.50  | 2.40 | 5.69 | 3.73  | 1.53 | 4.57 | 4.12  | 1.11 | 4.57    | 4.12  | 1.11 |
|              | 0          | 12.63   | 3.17 | 3.99 | 11.14        | 3.35 | 3.33 | 8.34  | 3.72  | 2.24 | 5.46 | 4.01  | 1.36 | 4.73 | 4.33  | 1.09 | 4.73    | 4.33  | 1.09 |
|              | 2          | 12.84   | 3.35 | 3.83 | 10.80        | 3.50 | 3.09 | 8.23  | 3.98  | 2.07 | 5.45 | 4.24  | 1.28 | 5.07 | 4.50  | 1.13 | 5.07    | 4.50  | 1.13 |
|              | 5          | 14.43   | 3.85 | 3.75 | 12.60        | 4.11 | 3.07 | 9.51  | 4.69  | 2.03 | 6.42 | 5.04  | 1.28 | 5.36 | 5.18  | 1.03 | 5.36    | 5.18  | 1.03 |
|              | 7*         | 15.95   | 4.26 | 3.74 | 14.00        | 4.50 | 3.11 | 10.53 | 4.97  | 2.12 | 7.07 | 5.38  | 1.32 | 5.87 | 5.61  | 1.05 | 5.87    | 5.61  | 1.05 |
|              | 10         | 16.52   | 4.76 | 3.47 | 13.53        | 5.27 | 2.57 | 10.31 | 5.63  | 1.83 | 6.96 | 6.15  | 1.13 | 6.42 | 6.22  | 1.03 | 6.42    | 6.22  | 1.03 |
|              | 12         | 17.19   | 5.13 | 3.35 | 13.98        | 5.72 | 2.44 | 10.41 | 6.12  | 1.70 | 7.10 | 6.64  | 1.07 | 6.82 | 6.71  | 1.02 | 6.82    | 6.71  | 1.02 |
|              | 15         | 16.04   | 5.36 | 2.99 | 13.98        | 5.85 | 2.39 | 10.46 | 6.39  | 1.64 | 7.09 | 7.20  | 0.99 | 7.09 | 7.20  | 0.99 | 7.09    | 7.20  | 0.99 |
|              | 20         | 14.32   | 6.15 | 2.33 | 14.13        | 6.47 | 2.18 | 10.36 | 7.26  | 1.43 | 6.93 | 7.84  | 0.88 | 5.97 | 7.99  | 0.75 | 5.97    | 7.99  | 0.75 |
|              | 25         | 14.32   | 6.82 | 2.10 | 13.88        | 6.96 | 1.99 | 10.35 | 7.98  | 1.30 | 7.01 | 8.32  | 0.84 | 6.68 | 8.47  | 0.79 | 6.68    | 8.47  | 0.79 |
|              | 30         | 13.95   | 7.83 | 1.78 | 13.95        | 7.83 | 1.78 | 10.45 | 8.73  | 1.20 | 7.30 | 9.00  | 0.81 | 7.30 | 9.00  | 0.81 | 7.30    | 9.00  | 0.81 |
|              | 35         | 13.22   | 9.58 | 1.38 | 13.22        | 9.58 | 1.38 | 10.47 | 10.32 | 1.01 | 7.97 | 10.45 | 0.76 | 7.97 | 10.45 | 0.76 | 7.97    | 10.45 | 0.76 |
| 40           | -25        | 7.59    | 1.70 | 4.47 | 7.59         | 1.70 | 4.47 | 7.59  | 1.70  | 4.47 | 5.82 | 1.78  | 3.28 | 2.74 | 1.89  | 1.45 | 2.74    | 1.89  | 1.45 |
|              | -20        | 9.22    | 1.98 | 4.65 | 9.22         | 1.98 | 4.65 | 8.57  | 2.02  | 4.24 | 5.67 | 2.11  | 2.68 | 3.57 | 2.31  | 1.55 | 3.57    | 2.31  | 1.55 |
|              | -15        | 10.70   | 2.20 | 4.86 | 10.60        | 2.23 | 4.76 | 8.77  | 2.35  | 3.74 | 5.66 | 2.47  | 2.29 | 3    |       |      |         |       |      |

| Tw_out<br>°C | DBWB<br>°C | Maximum |      |      | 100%(Normal) |      |      | 75%   |      |      | 50%   |      |      | 25%   |      |      | Minimum |      |      |
|--------------|------------|---------|------|------|--------------|------|------|-------|------|------|-------|------|------|-------|------|------|---------|------|------|
|              |            | HC      | COP  | PI   | HC           | COP  | PI   | HC    | COP  | PI   | HC    | COP  | PI   | HC    | COP  | PI   | HC      | COP  | PI   |
| 55           | -25        | 7.14    | 1.41 | 5.08 | 7.14         | 1.41 | 5.08 | 7.14  | 1.41 | 5.08 | 5.87  | 1.47 | 4.00 | 3.71  | 1.53 | 2.43 | 3.71    | 1.53 | 2.43 |
|              | -20        | 8.78    | 1.66 | 5.28 | 8.78         | 1.66 | 5.28 | 8.56  | 1.66 | 5.14 | 5.89  | 1.80 | 3.26 | 4.73  | 1.85 | 2.55 | 4.73    | 1.85 | 2.55 |
|              | -15        | 10.58   | 1.88 | 5.62 | 10.58        | 1.88 | 5.62 | 8.51  | 1.94 | 4.38 | 5.89  | 2.03 | 2.91 | 5.39  | 2.06 | 2.62 | 5.39    | 2.06 | 2.62 |
|              | -10        | 10.79   | 1.93 | 5.58 | 10.79        | 1.93 | 5.58 | 8.72  | 2.01 | 4.34 | 6.01  | 2.28 | 2.64 | 6.01  | 2.28 | 2.64 | 6.01    | 2.28 | 2.64 |
|              | -7         | 11.30   | 2.10 | 5.38 | 11.30        | 2.10 | 5.38 | 8.66  | 2.18 | 3.97 | 6.53  | 2.32 | 2.82 | 6.53  | 2.32 | 2.82 | 6.53    | 2.32 | 2.82 |
|              | -5         | 11.12   | 2.14 | 5.21 | 11.12        | 2.14 | 5.21 | 8.57  | 2.22 | 3.87 | 6.80  | 2.40 | 2.83 | 6.80  | 2.40 | 2.83 | 6.80    | 2.40 | 2.83 |
|              | -2         | 11.64   | 2.22 | 5.24 | 11.64        | 2.22 | 5.24 | 8.86  | 2.37 | 3.74 | 6.91  | 2.58 | 2.68 | 6.91  | 2.58 | 2.68 | 6.91    | 2.58 | 2.68 |
|              | 0          | 11.83   | 2.35 | 5.04 | 11.83        | 2.35 | 5.04 | 8.98  | 2.50 | 3.60 | 7.28  | 2.69 | 2.71 | 7.28  | 2.69 | 2.71 | 7.28    | 2.69 | 2.71 |
|              | 2          | 12.00   | 2.50 | 4.80 | 12.00        | 2.50 | 4.80 | 9.06  | 2.62 | 3.46 | 7.64  | 2.80 | 2.73 | 7.64  | 2.80 | 2.73 | 7.64    | 2.80 | 2.73 |
|              | 5          | 13.46   | 2.79 | 4.83 | 13.33        | 2.80 | 4.77 | 9.72  | 3.08 | 3.15 | 9.12  | 3.21 | 2.84 | 9.12  | 3.21 | 2.84 | 9.12    | 3.21 | 2.84 |
|              | 7          | 14.81   | 3.04 | 4.87 | 14.00        | 3.00 | 4.67 | 10.46 | 3.23 | 3.24 | 9.75  | 3.39 | 2.88 | 9.75  | 3.39 | 2.88 | 9.75    | 3.39 | 2.88 |
|              | 10         | 15.19   | 3.31 | 4.59 | 14.05        | 3.38 | 4.16 | 10.63 | 3.56 | 2.98 | 9.05  | 3.70 | 2.44 | 9.05  | 3.70 | 2.44 | 9.05    | 3.70 | 2.44 |
|              | 12         | 15.75   | 3.50 | 4.50 | 13.93        | 3.64 | 3.83 | 10.50 | 3.81 | 2.75 | 9.59  | 3.90 | 2.46 | 9.59  | 3.90 | 2.46 | 9.59    | 3.90 | 2.46 |
|              | 15         | 14.75   | 3.63 | 4.07 | 14.16        | 3.68 | 3.85 | 10.55 | 4.05 | 2.61 | 10.06 | 4.14 | 2.43 | 10.06 | 4.14 | 2.43 | 10.06   | 4.14 | 2.43 |
|              | 20         | 13.59   | 3.97 | 3.43 | 13.59        | 4.00 | 3.40 | 10.52 | 4.31 | 2.44 | 7.98  | 4.46 | 1.79 | 7.98  | 4.46 | 1.79 | 7.98    | 4.46 | 1.79 |
|              | 25         | 13.54   | 4.27 | 3.17 | 13.54        | 4.27 | 3.17 | 10.61 | 4.69 | 2.26 | 9.12  | 4.77 | 1.91 | 9.12  | 4.77 | 1.91 | 9.12    | 4.77 | 1.91 |
|              | 30         | 13.13   | 4.91 | 2.67 | 13.13        | 4.91 | 2.67 | 10.76 | 5.13 | 2.10 | 9.92  | 5.19 | 1.91 | 9.92  | 5.19 | 1.91 | 9.92    | 5.19 | 1.91 |
|              | 35         | 12.42   | 5.57 | 2.23 | 12.42        | 5.57 | 2.23 | 10.78 | 5.69 | 1.89 | 10.78 | 5.69 | 1.89 | 10.78 | 5.69 | 1.89 | 10.78   | 5.69 | 1.89 |
| 60           | -25        | /       | /    | /    | /            | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /    | /       | /    | /    |
|              | -20        | 8.54    | 1.54 | 5.55 | 8.54         | 1.54 | 5.55 | 8.33  | 1.54 | 5.39 | 5.86  | 1.64 | 3.57 | 4.92  | 1.68 | 2.92 | 4.92    | 1.68 | 2.92 |
|              | -15        | 10.11   | 1.70 | 5.94 | 10.11        | 1.70 | 5.94 | 8.45  | 1.76 | 4.80 | 5.70  | 1.88 | 3.03 | 5.70  | 1.88 | 3.03 | 5.70    | 1.88 | 3.03 |
|              | -10        | 10.52   | 1.84 | 5.71 | 10.52        | 1.84 | 5.71 | 8.65  | 1.92 | 4.51 | 6.39  | 2.12 | 3.01 | 6.39  | 2.12 | 3.01 | 6.39    | 2.12 | 3.01 |
|              | -7         | 11.24   | 1.95 | 5.75 | 11.24        | 1.95 | 5.75 | 8.68  | 2.04 | 4.26 | 7.54  | 2.17 | 3.47 | 7.54  | 2.17 | 3.47 | 7.54    | 2.17 | 3.47 |
|              | -5         | 10.65   | 1.99 | 5.35 | 10.65        | 1.99 | 5.35 | 8.67  | 2.08 | 4.18 | 7.08  | 2.25 | 3.15 | 7.08  | 2.25 | 3.15 | 7.08    | 2.25 | 3.15 |
|              | -2         | 11.17   | 2.07 | 5.39 | 11.17        | 2.07 | 5.39 | 8.79  | 2.14 | 4.10 | 7.12  | 2.31 | 3.08 | 7.12  | 2.31 | 3.08 | 7.12    | 2.31 | 3.08 |
|              | 0          | 11.52   | 2.15 | 5.36 | 11.52        | 2.15 | 5.36 | 8.92  | 2.25 | 3.96 | 7.52  | 2.41 | 3.12 | 7.52  | 2.41 | 3.12 | 7.52    | 2.41 | 3.12 |
|              | 2          | 11.69   | 2.25 | 5.20 | 11.69        | 2.25 | 5.20 | 8.99  | 2.36 | 3.81 | 7.89  | 2.51 | 3.15 | 7.89  | 2.51 | 3.15 | 7.89    | 2.51 | 3.15 |
|              | 5          | 13.37   | 2.58 | 5.18 | 13.37        | 2.58 | 5.18 | 9.90  | 2.76 | 3.58 | 8.91  | 2.87 | 3.10 | 8.91  | 2.87 | 3.10 | 8.91    | 2.87 | 3.10 |
|              | 7          | 14.36   | 2.77 | 5.19 | 14.00        | 2.77 | 5.05 | 10.46 | 2.93 | 3.57 | 9.55  | 3.04 | 3.14 | 9.55  | 3.04 | 3.14 | 9.55    | 3.04 | 3.14 |
|              | 10         | 14.70   | 2.99 | 4.92 | 13.94        | 3.02 | 4.62 | 10.41 | 3.17 | 3.28 | 9.12  | 3.29 | 2.77 | 9.12  | 3.29 | 2.77 | 9.12    | 3.29 | 2.77 |
|              | 12         | 15.23   | 3.15 | 4.83 | 14.02        | 3.22 | 4.36 | 10.49 | 3.37 | 3.11 | 9.66  | 3.45 | 2.80 | 9.66  | 3.45 | 2.80 | 9.66    | 3.45 | 2.80 |
|              | 15         | 14.26   | 3.25 | 4.38 | 14.07        | 3.27 | 4.30 | 10.46 | 3.60 | 2.90 | 10.19 | 3.68 | 2.77 | 10.19 | 3.68 | 2.77 | 10.19   | 3.68 | 2.77 |
|              | 20         | 13.27   | 3.55 | 3.74 | 13.27        | 3.56 | 3.73 | 10.45 | 3.82 | 2.74 | 8.16  | 3.96 | 2.06 | 8.16  | 3.96 | 2.06 | 8.16    | 3.96 | 2.06 |
|              | 25         | 13.23   | 3.84 | 3.44 | 13.23        | 3.84 | 3.44 | 10.56 | 4.17 | 2.53 | 9.40  | 4.25 | 2.21 | 9.40  | 4.25 | 2.21 | 9.40    | 4.25 | 2.21 |
|              | 30         | 12.80   | 4.38 | 2.92 | 12.80        | 4.38 | 2.92 | 10.53 | 4.58 | 2.30 | 10.20 | 4.59 | 2.22 | 10.20 | 4.59 | 2.22 | 10.20   | 4.59 | 2.22 |
|              | 65         | -25     | /    | /    | /            | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /    | /       | /    | /    |
| -20          |            | 8.14    | 1.42 | 5.75 | 8.14         | 1.42 | 5.75 | 8.14  | 1.42 | 5.75 | 5.80  | 1.44 | 4.01 | 4.66  | 1.51 | 3.08 | 4.66    | 1.51 | 3.08 |
| -15          |            | 9.46    | 1.54 | 6.15 | 9.46         | 1.54 | 6.15 | 8.29  | 1.65 | 5.03 | 5.64  | 1.79 | 3.15 | 5.64  | 1.79 | 3.15 | 5.64    | 1.79 | 3.15 |
| -10          |            | 10.25   | 1.75 | 5.85 | 10.25        | 1.75 | 5.85 | 8.71  | 1.80 | 4.85 | 6.46  | 2.01 | 3.22 | 6.46  | 2.01 | 3.22 | 6.46    | 2.01 | 3.22 |
| -7           |            | 10.74   | 1.80 | 5.96 | 10.74        | 1.80 | 5.96 | 8.58  | 1.90 | 4.51 | 6.68  | 2.02 | 3.31 | 6.68  | 2.02 | 3.31 | 6.68    | 2.02 | 3.31 |
| -5           |            | 10.18   | 1.84 | 5.53 | 10.18        | 1.84 | 5.53 | 8.67  | 1.95 | 4.44 | 6.94  | 2.06 | 3.37 | 6.94  | 2.06 | 3.37 | 6.94    | 2.06 | 3.37 |
| -2           |            | 10.77   | 1.93 | 5.59 | 10.77        | 1.93 | 5.59 | 8.86  | 2.01 | 4.40 | 6.90  | 2.10 | 3.29 | 6.90  | 2.10 | 3.29 | 6.90    | 2.10 | 3.29 |
| 0            |            | 10.70   | 1.98 | 5.40 | 10.70        | 1.98 | 5.40 | 8.85  | 2.05 | 4.32 | 7.22  | 2.16 | 3.34 | 7.22  | 2.16 | 3.34 | 7.22    | 2.16 | 3.34 |
| 2            |            | 11.08   | 2.04 | 5.42 | 11.08        | 2.04 | 5.42 | 8.87  | 2.10 | 4.22 | 7.57  | 2.24 | 3.38 | 7.57  | 2.24 | 3.38 | 7.57    | 2.24 | 3.38 |
| 5            |            | 13.21   | 2.36 | 5.60 | 13.21        | 2.36 | 5.60 | 9.95  | 2.45 | 4.07 | 8.52  | 2.56 | 3.32 | 8.52  | 2.56 | 3.32 | 8.52    | 2.56 | 3.32 |
| 7            |            | 13.71   | 2.51 | 5.47 | 13.71        | 2.51 | 5.47 | 10.45 | 2.69 | 3.88 | 9.06  | 2.79 | 3.25 | 9.06  | 2.79 | 3.25 | 9.06    | 2.79 | 3.25 |
| 10           |            | 13.94   | 2.69 | 5.19 | 13.94        | 2.69 | 5.19 | 10.50 | 2.85 | 3.69 | 8.74  | 2.92 | 2.99 | 8.74  | 2.92 | 2.99 | 8.74    | 2.92 | 2.99 |
| 12           |            | 14.63   | 2.79 | 5.24 | 14.04        | 2.82 | 4.98 | 10.53 | 2.96 | 3.55 | 9.28  | 3.07 | 3.03 | 9.28  | 3.07 | 3.03 | 9.28    | 3.07 | 3.03 |
| 15           |            | 13.72   | 2.91 | 4.71 | 13.72        | 2.91 | 4.71 | 10.38 | 3.23 | 3.22 | 9.84  | 3.28 | 3.00 | 9.84  | 3.28 | 3.00 | 9.84    | 3.28 | 3.00 |
| 20           |            | 12.90   | 3.17 | 4.06 | 12.90        | 3.17 | 4.06 | 10.55 | 3.42 | 3.08 | 7.87  | 3.53 | 2.23 | 7.87  | 3.53 | 2.23 | 7.87    | 3.53 | 2.23 |
| 25           |            | 12.85   | 3.44 | 3.74 | 12.85        | 3.44 | 3.74 | 10.47 | 3.71 | 2.82 | 9.14  | 3.78 | 2.42 | 9.14  | 3.78 | 2.42 | 9.14    | 3.78 | 2.42 |
| 30           |            | 12.43   | 3.90 | 3.18 | 12.43        | 3.90 | 3.18 | 10.45 | 4.04 | 2.59 | 9.92  | 4.07 | 2.44 | 9.92  | 4.07 | 2.44 | 9.92    | 4.07 | 2.44 |
| 70           |            | -25     | /    | /    | /            | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /    | /       | /    | /    |
|              | -20        | /       | /    | /    | /            | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /    | /       | /    | /    |
|              | -15        | 8.93    | 1.54 | 5.79 | 8.93         | 1.54 | 5.79 | 8.22  | 1.56 | 5.27 | 5.47  | 1.65 | 3.32 | 5.47  | 1.65 | 3.32 | 5.47    | 1.65 | 3.32 |
|              | -10        | 9.87    | 1.68 | 5.87 | 9.87         | 1.68 | 5.87 | 8.72  | 1.74 | 5.02 | 6.50  | 1.92 | 3.40 | 6.50  | 1.92 | 3.40 | 6.50    | 1.92 | 3.40 |
|              | -7         | 10.07   | 1.75 | 5.74 | 10.07        | 1.75 | 5.74 | 8.60  | 1.82 | 4.73 | 7.04  | 1.96 | 3.59 | 6.90  | 1.96 | 3.52 | 6.90    | 1.96 | 3.52 |
|              | -5         | 9.75    | 1.77 | 5.52 | 9.75         | 1.77 | 5.52 | 8.68  | 1.84 | 4.71 | 7.13  | 1.97 | 3.61 | 7.13  | 1.97 | 3.61 | 7.13    | 1.97 | 3.61 |
|              | -2         | 10.33   | 1.80 | 5.74 | 10.33        | 1.80 | 5.74 | 8.83  | 1.87 | 4.72 | 6.98  | 2.00 | 3.49 | 6.98  | 2.00 | 3.49 | 6.98    | 2.00 | 3.49 |
|              | 0          | 10.33   | 1.84 | 5.63 | 10.33        | 1.84 | 5.63 | 8.77  | 1.93 | 4.54 | 7.26  | 2.05 | 3    |       |      |      |         |      |      |

## 8.7 Heating performance Mod. 16

| Tw_out<br>°C | DBWB<br>°C | Maximum |      |      | 100%(Normal) |      |      | 75%   |       |      | 50%  |       |      | 25%  |       |      | Minimum |       |      |
|--------------|------------|---------|------|------|--------------|------|------|-------|-------|------|------|-------|------|------|-------|------|---------|-------|------|
|              |            | HC      | COP  | PI   | HC           | COP  | PI   | HC    | COP   | PI   | HC   | COP   | PI   | HC   | COP   | PI   | HC      | COP   | PI   |
| 25           | -25        | 9.50    | 1.99 | 4.77 | 9.50         | 1.99 | 4.77 | 9.50  | 1.99  | 4.77 | 6.47 | 2.26  | 2.87 | 2.98 | 2.56  | 1.17 | 2.98    | 2.56  | 1.17 |
|              | -20        | 11.26   | 2.37 | 4.75 | 11.26        | 2.37 | 4.75 | 9.73  | 2.54  | 3.83 | 6.53 | 2.72  | 2.40 | 3.77 | 3.11  | 1.21 | 3.77    | 3.11  | 1.21 |
|              | -15        | 12.59   | 2.63 | 4.78 | 12.59        | 2.63 | 4.78 | 9.45  | 3.07  | 3.08 | 6.18 | 3.26  | 1.90 | 4.37 | 3.77  | 1.16 | 4.37    | 3.77  | 1.16 |
|              | -10        | 13.33   | 2.76 | 4.84 | 12.68        | 2.84 | 4.47 | 9.54  | 3.24  | 2.95 | 6.32 | 3.54  | 1.78 | 4.69 | 3.99  | 1.18 | 4.69    | 3.99  | 1.18 |
|              | -7         | 13.66   | 2.87 | 4.75 | 12.66        | 3.06 | 4.14 | 9.31  | 3.74  | 2.49 | 6.35 | 4.10  | 1.55 | 5.41 | 4.44  | 1.22 | 5.41    | 4.44  | 1.22 |
|              | -5         | 14.12   | 3.11 | 4.53 | 12.76        | 3.30 | 3.87 | 9.48  | 3.75  | 2.53 | 6.34 | 4.23  | 1.50 | 5.60 | 4.55  | 1.23 | 5.60    | 4.55  | 1.23 |
|              | -2         | 14.53   | 3.42 | 4.25 | 12.62        | 3.68 | 3.43 | 9.63  | 4.20  | 2.29 | 6.36 | 4.71  | 1.35 | 4.86 | 5.22  | 0.93 | 4.86    | 5.22  | 0.93 |
|              | 0          | 14.83   | 3.64 | 4.08 | 12.56        | 3.96 | 3.17 | 9.45  | 4.59  | 2.06 | 6.27 | 5.10  | 1.23 | 5.02 | 5.42  | 0.93 | 5.02    | 5.42  | 0.93 |
|              | 2          | 15.03   | 3.84 | 3.91 | 12.65        | 4.24 | 2.98 | 9.46  | 4.93  | 1.92 | 6.45 | 5.41  | 1.19 | 5.38 | 5.80  | 0.93 | 5.38    | 5.80  | 0.93 |
|              | 5          | 16.61   | 4.48 | 3.71 | 14.00        | 4.99 | 2.80 | 10.59 | 5.95  | 1.78 | 7.06 | 6.67  | 1.06 | 5.85 | 7.00  | 0.84 | 5.85    | 7.00  | 0.84 |
|              | 7          | 18.67   | 5.02 | 3.72 | 14.95        | 5.68 | 2.63 | 11.19 | 6.44  | 1.74 | 7.50 | 7.28  | 1.03 | 6.24 | 7.44  | 0.84 | 6.24    | 7.44  | 0.84 |
|              | 10         | 20.15   | 5.81 | 3.47 | 15.07        | 7.12 | 2.12 | 11.31 | 7.86  | 1.44 | 7.58 | 8.85  | 0.86 | 7.25 | 8.95  | 0.81 | 7.25    | 8.95  | 0.81 |
|              | 12         | 20.80   | 6.32 | 3.29 | 15.03        | 7.78 | 1.93 | 11.26 | 8.70  | 1.29 | 7.71 | 9.79  | 0.79 | 7.71 | 9.79  | 0.79 | 7.71    | 9.79  | 0.79 |
|              | 15         | 19.18   | 6.60 | 2.91 | 15.08        | 8.04 | 1.88 | 11.23 | 9.16  | 1.23 | 7.78 | 10.26 | 0.76 | 7.78 | 10.26 | 0.76 | 7.78    | 10.26 | 0.76 |
|              | 20         | 16.84   | 7.51 | 2.24 | 15.04        | 8.19 | 1.84 | 11.32 | 9.88  | 1.15 | 7.46 | 11.00 | 0.68 | 6.62 | 11.24 | 0.59 | 6.62    | 11.24 | 0.59 |
|              | 25         | 17.20   | 8.27 | 2.08 | 14.89        | 9.12 | 1.63 | 11.38 | 10.81 | 1.05 | 7.48 | 11.73 | 0.64 | 7.40 | 11.79 | 0.63 | 7.40    | 11.79 | 0.63 |
|              | 30         | /       | /    | /    | /            | /    | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 35         | /       | /    | /    | /            | /    | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 30           | -25        | 9.03    | 1.82 | 4.96 | 9.03         | 1.82 | 4.96 | 9.03  | 1.82  | 4.96 | 6.06 | 1.99  | 3.05 | 2.78 | 2.25  | 1.24 | 2.78    | 2.25  | 1.24 |
|              | -20        | 10.78   | 2.07 | 5.21 | 10.78        | 2.07 | 5.21 | 9.38  | 2.17  | 4.32 | 6.19 | 2.40  | 2.58 | 3.56 | 2.76  | 1.29 | 3.56    | 2.76  | 1.29 |
|              | -15        | 12.13   | 2.41 | 5.03 | 12.13        | 2.41 | 5.03 | 9.32  | 2.72  | 3.43 | 6.13 | 2.85  | 2.15 | 4.19 | 3.36  | 1.25 | 4.19    | 3.36  | 1.25 |
|              | -10        | 12.81   | 2.50 | 5.12 | 12.71        | 2.52 | 5.05 | 9.40  | 2.86  | 3.29 | 6.24 | 3.10  | 2.01 | 4.58 | 3.58  | 1.28 | 4.58    | 3.58  | 1.28 |
|              | -7         | 13.43   | 2.67 | 5.02 | 12.64        | 2.80 | 4.51 | 9.31  | 3.33  | 2.80 | 6.37 | 3.72  | 1.71 | 5.39 | 4.03  | 1.34 | 5.39    | 4.03  | 1.34 |
|              | -5         | 13.60   | 2.82 | 4.82 | 12.65        | 2.93 | 4.32 | 9.42  | 3.30  | 2.85 | 6.20 | 3.66  | 1.70 | 5.38 | 4.10  | 1.31 | 5.38    | 4.10  | 1.31 |
|              | -2         | 14.02   | 3.10 | 4.53 | 12.40        | 3.29 | 3.77 | 9.47  | 3.68  | 2.57 | 6.47 | 3.99  | 1.62 | 4.66 | 4.56  | 1.02 | 4.66    | 4.56  | 1.02 |
|              | 0          | 14.31   | 3.29 | 4.35 | 12.54        | 3.51 | 3.58 | 9.45  | 3.98  | 2.37 | 6.18 | 4.35  | 1.42 | 4.74 | 4.63  | 1.02 | 4.74    | 4.63  | 1.02 |
|              | 2          | 14.50   | 3.47 | 4.18 | 12.64        | 3.72 | 3.40 | 9.50  | 4.26  | 2.23 | 6.21 | 4.61  | 1.35 | 5.16 | 5.02  | 1.03 | 5.16    | 5.02  | 1.03 |
|              | 5          | 16.01   | 4.03 | 3.97 | 14.18        | 4.33 | 3.28 | 10.37 | 5.16  | 2.01 | 7.04 | 5.64  | 1.25 | 5.54 | 5.92  | 0.93 | 5.54    | 5.92  | 0.93 |
|              | 7          | 17.89   | 4.49 | 3.98 | 15.23        | 4.86 | 3.13 | 11.25 | 5.50  | 2.04 | 7.52 | 6.08  | 1.24 | 5.98 | 6.33  | 0.94 | 5.98    | 6.33  | 0.94 |
|              | 10         | 19.22   | 5.15 | 3.73 | 15.19        | 5.93 | 2.56 | 11.17 | 6.62  | 1.69 | 7.43 | 7.46  | 1.00 | 6.91 | 7.52  | 0.92 | 6.91    | 7.52  | 0.92 |
|              | 12         | 19.84   | 5.59 | 3.55 | 15.03        | 6.65 | 2.26 | 11.20 | 7.29  | 1.54 | 7.72 | 8.01  | 0.96 | 7.32 | 8.10  | 0.90 | 7.32    | 8.10  | 0.90 |
|              | 15         | 18.32   | 5.80 | 3.16 | 15.03        | 6.80 | 2.21 | 11.43 | 7.56  | 1.51 | 7.59 | 8.73  | 0.87 | 7.59 | 8.73  | 0.87 | 7.59    | 8.73  | 0.87 |
|              | 20         | 16.17   | 6.55 | 2.47 | 15.00        | 7.26 | 2.07 | 11.25 | 8.36  | 1.35 | 7.50 | 9.22  | 0.81 | 6.37 | 9.36  | 0.68 | 6.37    | 9.36  | 0.68 |
|              | 25         | 16.57   | 7.17 | 2.31 | 15.04        | 7.58 | 1.98 | 11.19 | 8.99  | 1.24 | 7.46 | 9.53  | 0.78 | 7.11 | 9.62  | 0.74 | 7.11    | 9.62  | 0.74 |
|              | 30         | 15.63   | 8.79 | 1.78 | 14.89        | 9.34 | 1.59 | 11.34 | 10.62 | 1.07 | 7.86 | 11.26 | 0.70 | 7.78 | 11.27 | 0.69 | 7.78    | 11.27 | 0.69 |
|              | 35         | /       | /    | /    | /            | /    | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 35           | -25        | 8.83    | 1.74 | 5.08 | 8.83         | 1.74 | 5.08 | 8.83  | 1.76  | 5.02 | 6.21 | 1.85  | 3.36 | 2.68 | 2.02  | 1.33 | 2.68    | 2.02  | 1.33 |
|              | -20        | 10.56   | 1.96 | 5.39 | 10.56        | 1.96 | 5.39 | 9.30  | 2.07  | 4.50 | 6.25 | 2.20  | 2.85 | 3.47 | 2.50  | 1.39 | 3.47    | 2.50  | 1.39 |
|              | -15        | 11.96   | 2.27 | 5.28 | 11.96        | 2.23 | 5.37 | 9.25  | 2.50  | 3.70 | 6.35 | 2.62  | 2.42 | 4.16 | 2.97  | 1.40 | 4.16    | 2.97  | 1.40 |
|              | -10        | 12.47   | 2.43 | 5.13 | 12.33        | 2.46 | 5.02 | 9.39  | 2.56  | 3.66 | 6.32 | 2.88  | 2.19 | 4.50 | 3.28  | 1.37 | 4.50    | 3.28  | 1.37 |
|              | -7         | 13.06   | 2.71 | 4.82 | 12.43        | 2.74 | 4.54 | 9.41  | 2.90  | 3.24 | 6.44 | 3.35  | 1.92 | 5.25 | 3.58  | 1.47 | 5.25    | 3.58  | 1.47 |
|              | -5         | 13.25   | 2.79 | 4.75 | 12.55        | 2.91 | 4.31 | 9.44  | 3.01  | 3.14 | 6.27 | 3.28  | 1.91 | 5.08 | 3.62  | 1.40 | 5.08    | 3.62  | 1.40 |
|              | -2         | 13.61   | 2.90 | 4.70 | 12.67        | 3.06 | 4.14 | 9.51  | 3.33  | 2.86 | 6.33 | 3.60  | 1.76 | 4.57 | 4.12  | 1.11 | 4.57    | 4.12  | 1.11 |
|              | 0          | 14.05   | 3.03 | 4.64 | 12.80        | 3.17 | 4.03 | 9.64  | 3.56  | 2.71 | 6.30 | 3.87  | 1.63 | 4.73 | 4.33  | 1.09 | 4.73    | 4.33  | 1.09 |
|              | 2          | 14.27   | 3.20 | 4.46 | 12.80        | 3.20 | 4.00 | 9.72  | 3.80  | 2.55 | 6.35 | 4.09  | 1.55 | 5.07 | 4.50  | 1.13 | 5.07    | 4.50  | 1.13 |
|              | 5          | 15.77   | 3.72 | 4.24 | 14.20        | 3.92 | 3.63 | 10.49 | 4.57  | 2.30 | 7.02 | 4.96  | 1.42 | 5.36 | 5.18  | 1.03 | 5.36    | 5.18  | 1.03 |
|              | 7*         | 17.57   | 4.13 | 4.25 | 15.00        | 4.40 | 3.41 | 11.17 | 4.89  | 2.28 | 7.50 | 5.31  | 1.41 | 5.87 | 5.61  | 1.05 | 5.87    | 5.61  | 1.05 |
|              | 10         | 18.31   | 4.58 | 4.00 | 14.94        | 5.04 | 2.97 | 11.24 | 5.51  | 2.04 | 7.49 | 6.09  | 1.23 | 6.42 | 6.22  | 1.03 | 6.42    | 6.22  | 1.03 |
|              | 12         | 18.87   | 4.93 | 3.82 | 15.01        | 5.52 | 2.72 | 11.29 | 5.99  | 1.88 | 8.35 | 6.45  | 1.29 | 6.82 | 6.71  | 1.02 | 6.82    | 6.71  | 1.02 |
|              | 15         | 17.44   | 5.10 | 3.42 | 15.03        | 5.61 | 2.68 | 11.37 | 6.23  | 1.83 | 7.44 | 6.98  | 1.07 | 7.09 | 7.20  | 0.99 | 7.09    | 7.20  | 0.99 |
|              | 20         | 15.44   | 5.70 | 2.71 | 15.00        | 6.09 | 2.46 | 11.25 | 6.92  | 1.62 | 7.50 | 7.48  | 1.00 | 5.97 | 7.99  | 0.75 | 5.97    | 7.99  | 0.75 |
|              | 25         | 15.82   | 6.18 | 2.56 | 14.97        | 6.45 | 2.32 | 11.11 | 7.48  | 1.49 | 7.51 | 8.30  | 0.90 | 6.68 | 8.47  | 0.79 | 6.68    | 8.47  | 0.79 |
|              | 30         | 14.95   | 7.47 | 2.00 | 14.95        | 7.47 | 2.00 | 11.30 | 8.60  | 1.31 | 7.66 | 8.91  | 0.86 | 7.30 | 9.00  | 0.81 | 7.30    | 9.00  | 0.81 |
|              | 35         | 13.82   | 9.45 | 1.46 | 13.82        | 9.45 | 1.46 | 11.09 | 10.10 | 1.10 | 8.07 | 10.43 | 0.77 | 7.97 | 10.45 | 0.76 | 7.97    | 10.45 | 0.76 |
| 40           | -25        | 8.64    | 1.56 | 5.53 | 8.64         | 1.56 | 5.53 | 8.64  | 1.56  | 5.53 | 6.26 | 1.74  | 3.60 | 2.74 | 1.89  | 1.45 | 2.74    | 1.89  | 1.45 |
|              | -20        | 10.39   | 1.83 | 5.68 | 10.39        | 1.83 | 5.68 | 9.32  | 1.90  | 4.91 | 6.33 | 2.09  | 3.03 | 3.57 | 2.31  | 1.55 | 3.57    | 2.31  | 1.55 |
|              | -15        | 11.78   | 2.05 | 5.74 | 11.78        | 2.05 | 5.74 | 9.30  | 2.24  | 4.14 | 6.47 | 2.41  |      |      |       |      |         |       |      |

| Tw_out<br>°C | DBWB<br>°C | Maximum |      |      | 100%(Normal) |      |      | 75%   |      |      | 50%   |      |      | 25%   |      |      | Minimum |      |      |
|--------------|------------|---------|------|------|--------------|------|------|-------|------|------|-------|------|------|-------|------|------|---------|------|------|
|              |            | HC      | COP  | PI   | HC           | COP  | PI   | HC    | COP  | PI   | HC    | COP  | PI   | HC    | COP  | PI   | HC      | COP  | PI   |
| 55           | -25        | 8.18    | 1.30 | 6.30 | 8.18         | 1.30 | 6.30 | 8.18  | 1.30 | 6.30 | 6.36  | 1.44 | 4.42 | 3.71  | 1.53 | 2.43 | 3.71    | 1.53 | 2.43 |
|              | -20        | 9.82    | 1.56 | 6.28 | 9.82         | 1.56 | 6.28 | 9.50  | 1.58 | 6.00 | 6.32  | 1.76 | 3.58 | 4.73  | 1.85 | 2.55 | 4.73    | 1.85 | 2.55 |
|              | -15        | 11.19   | 1.79 | 6.26 | 11.19        | 1.79 | 6.26 | 9.60  | 1.89 | 5.07 | 6.29  | 2.02 | 3.11 | 5.39  | 2.06 | 2.62 | 5.39    | 2.06 | 2.62 |
|              | -10        | 11.30   | 1.90 | 5.95 | 11.30        | 1.90 | 5.95 | 9.43  | 2.02 | 4.67 | 6.20  | 2.24 | 2.77 | 6.01  | 2.28 | 2.64 | 6.01    | 2.28 | 2.64 |
|              | -7         | 12.40   | 2.05 | 6.05 | 12.40        | 2.05 | 6.05 | 9.46  | 2.16 | 4.38 | 7.20  | 2.26 | 3.18 | 6.53  | 2.32 | 2.82 | 6.53    | 2.32 | 2.82 |
|              | -5         | 11.85   | 2.09 | 5.68 | 11.85        | 2.09 | 5.68 | 9.33  | 2.18 | 4.28 | 6.80  | 2.40 | 2.83 | 6.80  | 2.40 | 2.83 | 6.80    | 2.40 | 2.83 |
|              | -2         | 12.28   | 2.18 | 5.64 | 12.28        | 2.18 | 5.64 | 9.53  | 2.33 | 4.08 | 6.91  | 2.58 | 2.68 | 6.91  | 2.58 | 2.68 | 6.91    | 2.58 | 2.68 |
|              | 0          | 12.91   | 2.26 | 5.70 | 12.91        | 2.26 | 5.70 | 9.81  | 2.45 | 4.01 | 7.28  | 2.69 | 2.71 | 7.28  | 2.69 | 2.71 | 7.28    | 2.69 | 2.71 |
|              | 2          | 13.10   | 2.45 | 5.35 | 13.10        | 2.45 | 5.35 | 9.92  | 2.57 | 3.85 | 7.64  | 2.80 | 2.73 | 7.64  | 2.80 | 2.73 | 7.64    | 2.80 | 2.73 |
|              | 5          | 14.74   | 2.66 | 5.54 | 14.27        | 2.75 | 5.19 | 10.82 | 3.05 | 3.55 | 9.12  | 3.21 | 2.84 | 9.12  | 3.21 | 2.84 | 9.12    | 3.21 | 2.84 |
|              | 7          | 16.00   | 2.81 | 5.70 | 15.00        | 2.85 | 5.26 | 11.22 | 3.17 | 3.54 | 9.75  | 3.39 | 2.88 | 9.75  | 3.39 | 2.88 | 9.75    | 3.39 | 2.88 |
|              | 10         | 16.89   | 3.14 | 5.38 | 14.99        | 3.34 | 4.49 | 11.19 | 3.52 | 3.18 | 9.05  | 3.70 | 2.44 | 9.05  | 3.70 | 2.44 | 9.05    | 3.70 | 2.44 |
|              | 12         | 17.36   | 3.38 | 5.13 | 15.00        | 3.56 | 4.22 | 11.20 | 3.78 | 2.97 | 9.59  | 3.90 | 2.46 | 9.59  | 3.90 | 2.46 | 9.59    | 3.90 | 2.46 |
|              | 15         | 16.14   | 3.53 | 4.58 | 14.95        | 3.61 | 4.14 | 11.29 | 3.85 | 2.93 | 10.06 | 4.14 | 2.43 | 10.06 | 4.14 | 2.43 | 10.06   | 4.14 | 2.43 |
|              | 20         | 14.72   | 3.79 | 3.88 | 14.72        | 3.79 | 3.88 | 11.05 | 4.09 | 2.70 | 7.98  | 4.46 | 1.79 | 7.98  | 4.46 | 1.79 | 7.98    | 4.46 | 1.79 |
|              | 25         | 15.00   | 4.14 | 3.63 | 15.00        | 4.14 | 3.63 | 11.31 | 4.62 | 2.45 | 9.12  | 4.77 | 1.91 | 9.12  | 4.77 | 1.91 | 9.12    | 4.77 | 1.91 |
|              | 30         | 14.09   | 4.75 | 2.97 | 14.09        | 4.75 | 2.97 | 11.10 | 5.06 | 2.19 | 9.92  | 5.19 | 1.91 | 9.92  | 5.19 | 1.91 | 9.92    | 5.19 | 1.91 |
|              | 35         | 12.99   | 5.50 | 2.36 | 12.99        | 5.50 | 2.36 | 11.27 | 5.60 | 2.01 | 10.78 | 5.63 | 1.91 | 10.78 | 5.69 | 1.89 | 10.78   | 5.69 | 1.89 |
| 60           | -25        | /       | /    | /    | /            | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /    | /       | /    | /    |
|              | -20        | 9.37    | 1.45 | 6.45 | 9.37         | 1.45 | 6.45 | 9.37  | 1.45 | 6.45 | 6.26  | 1.62 | 3.87 | 4.92  | 1.68 | 2.92 | 4.92    | 1.68 | 2.92 |
|              | -15        | 10.45   | 1.67 | 6.24 | 10.45        | 1.67 | 6.24 | 9.37  | 1.73 | 5.42 | 6.16  | 1.85 | 3.33 | 5.70  | 1.88 | 3.03 | 5.70    | 1.88 | 3.03 |
|              | -10        | 10.85   | 1.75 | 6.20 | 10.85        | 1.79 | 6.07 | 9.49  | 1.87 | 5.08 | 6.39  | 2.12 | 3.01 | 6.39  | 2.12 | 3.01 | 6.39    | 2.12 | 3.01 |
|              | -7         | 11.65   | 1.84 | 6.32 | 11.65        | 1.88 | 6.19 | 9.49  | 1.96 | 4.85 | 7.54  | 2.17 | 3.47 | 7.54  | 2.17 | 3.47 | 7.54    | 2.17 | 3.47 |
|              | -5         | 11.39   | 1.87 | 6.10 | 11.39        | 1.91 | 5.98 | 9.32  | 2.02 | 4.61 | 7.08  | 2.25 | 3.15 | 7.08  | 2.25 | 3.15 | 7.08    | 2.25 | 3.15 |
|              | -2         | 11.54   | 1.97 | 5.86 | 11.54        | 2.01 | 5.74 | 9.46  | 2.08 | 4.56 | 7.12  | 2.31 | 3.08 | 7.12  | 2.31 | 3.08 | 7.12    | 2.31 | 3.08 |
|              | 0          | 11.91   | 2.04 | 5.83 | 11.91        | 2.08 | 5.72 | 9.74  | 2.17 | 4.48 | 7.52  | 2.41 | 3.12 | 7.52  | 2.41 | 3.12 | 7.52    | 2.41 | 3.12 |
|              | 2          | 12.18   | 2.14 | 5.70 | 12.18        | 2.18 | 5.59 | 9.85  | 2.27 | 4.33 | 7.89  | 2.51 | 3.15 | 7.89  | 2.51 | 3.15 | 7.89    | 2.51 | 3.15 |
|              | 5          | 14.19   | 2.49 | 5.70 | 14.19        | 2.49 | 5.70 | 10.75 | 2.67 | 4.03 | 8.91  | 2.87 | 3.10 | 8.91  | 2.87 | 3.10 | 8.91    | 2.87 | 3.10 |
|              | 7          | 15.59   | 2.63 | 5.93 | 15.10        | 2.68 | 5.64 | 11.10 | 2.88 | 3.85 | 9.55  | 3.04 | 3.14 | 9.55  | 3.04 | 3.14 | 9.55    | 3.04 | 3.14 |
|              | 10         | 16.34   | 2.82 | 5.79 | 15.00        | 2.94 | 5.11 | 11.25 | 3.13 | 3.59 | 9.12  | 3.29 | 2.77 | 9.12  | 3.29 | 2.77 | 9.12    | 3.29 | 2.77 |
|              | 12         | 16.78   | 3.05 | 5.50 | 14.97        | 3.17 | 4.72 | 11.17 | 3.35 | 3.34 | 9.66  | 3.45 | 2.80 | 9.66  | 3.45 | 2.80 | 9.66    | 3.45 | 2.80 |
|              | 15         | 15.61   | 3.14 | 4.97 | 15.04        | 3.20 | 4.69 | 11.26 | 3.43 | 3.29 | 10.19 | 3.68 | 2.77 | 10.19 | 3.68 | 2.77 | 10.19   | 3.68 | 2.77 |
|              | 20         | 14.03   | 3.35 | 4.18 | 14.03        | 3.35 | 4.18 | 11.18 | 3.60 | 3.10 | 8.16  | 3.96 | 2.06 | 8.16  | 3.96 | 2.06 | 8.16    | 3.96 | 2.06 |
|              | 25         | 14.67   | 3.73 | 3.93 | 14.67        | 3.73 | 3.93 | 11.25 | 4.09 | 2.75 | 9.40  | 4.25 | 2.21 | 9.40  | 4.25 | 2.21 | 9.40    | 4.25 | 2.21 |
|              | 30         | 13.75   | 4.24 | 3.24 | 13.75        | 4.24 | 3.24 | 11.31 | 4.50 | 2.51 | 10.20 | 4.59 | 2.22 | 10.20 | 4.59 | 2.22 | 10.20   | 4.59 | 2.22 |
|              | 35         | /       | /    | /    | /            | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /    | /       | /    | /    |
| 65           | -25        | /       | /    | /    | /            | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /    | /       | /    | /    |
|              | -20        | 8.66    | 1.39 | 6.21 | 8.66         | 1.39 | 6.21 | 8.66  | 1.39 | 6.21 | 6.29  | 1.41 | 4.47 | 4.66  | 1.51 | 3.08 | 4.66    | 1.51 | 3.08 |
|              | -15        | 9.67    | 1.52 | 6.36 | 9.67         | 1.52 | 6.36 | 9.20  | 1.55 | 5.93 | 6.22  | 1.73 | 3.60 | 5.64  | 1.79 | 3.15 | 5.64    | 1.79 | 3.15 |
|              | -10        | 10.37   | 1.72 | 6.02 | 10.37        | 1.72 | 6.02 | 9.44  | 1.76 | 5.37 | 6.46  | 2.01 | 3.22 | 6.46  | 2.01 | 3.22 | 6.46    | 2.01 | 3.22 |
|              | -7         | 10.85   | 1.79 | 6.05 | 10.85        | 1.79 | 6.05 | 9.40  | 1.82 | 5.16 | 6.68  | 2.02 | 3.31 | 6.68  | 2.02 | 3.31 | 6.68    | 2.02 | 3.31 |
|              | -5         | 10.83   | 1.81 | 5.98 | 10.83        | 1.81 | 5.98 | 9.43  | 1.90 | 4.96 | 6.94  | 2.06 | 3.37 | 6.94  | 2.06 | 3.37 | 6.94    | 2.06 | 3.37 |
|              | -2         | 10.89   | 1.90 | 5.72 | 10.89        | 1.90 | 5.72 | 9.52  | 1.94 | 4.92 | 6.90  | 2.10 | 3.29 | 6.90  | 2.10 | 3.29 | 6.90    | 2.10 | 3.29 |
|              | 0          | 11.29   | 1.94 | 5.83 | 11.29        | 1.94 | 5.83 | 9.67  | 2.00 | 4.84 | 7.22  | 2.16 | 3.34 | 7.22  | 2.16 | 3.34 | 7.22    | 2.16 | 3.34 |
|              | 2          | 11.32   | 2.01 | 5.63 | 11.32        | 2.01 | 5.63 | 9.85  | 2.08 | 4.74 | 7.57  | 2.24 | 3.38 | 7.57  | 2.24 | 3.38 | 7.57    | 2.24 | 3.38 |
|              | 5          | 13.21   | 2.36 | 5.60 | 13.21        | 2.36 | 5.60 | 10.62 | 2.42 | 4.38 | 8.52  | 2.56 | 3.32 | 8.52  | 2.56 | 3.32 | 8.52    | 2.56 | 3.32 |
|              | 7          | 14.71   | 2.43 | 6.06 | 14.71        | 2.43 | 6.06 | 11.39 | 2.55 | 4.46 | 9.06  | 2.79 | 3.25 | 9.06  | 2.79 | 3.25 | 9.06    | 2.79 | 3.25 |
|              | 10         | 15.00   | 2.61 | 5.75 | 15.00        | 2.61 | 5.75 | 11.20 | 2.70 | 4.14 | 8.74  | 2.92 | 2.99 | 8.74  | 2.92 | 2.99 | 8.74    | 2.92 | 2.99 |
|              | 12         | 15.35   | 2.72 | 5.65 | 15.17        | 2.74 | 5.53 | 11.26 | 2.90 | 3.88 | 9.28  | 3.07 | 3.03 | 9.28  | 3.07 | 3.03 | 9.28    | 3.07 | 3.03 |
|              | 15         | 15.04   | 2.81 | 5.36 | 15.04        | 2.81 | 5.36 | 11.17 | 3.03 | 3.69 | 9.84  | 3.28 | 3.00 | 9.84  | 3.28 | 3.00 | 9.84    | 3.28 | 3.00 |
|              | 20         | 13.09   | 3.06 | 4.27 | 13.09        | 3.06 | 4.27 | 11.25 | 3.20 | 3.52 | 7.87  | 3.53 | 2.23 | 7.87  | 3.53 | 2.23 | 7.87    | 3.53 | 2.23 |
|              | 25         | 14.27   | 3.31 | 4.32 | 14.27        | 3.31 | 4.32 | 11.13 | 3.63 | 3.07 | 9.14  | 3.78 | 2.42 | 9.14  | 3.78 | 2.42 | 9.14    | 3.78 | 2.42 |
|              | 30         | 13.37   | 3.79 | 3.52 | 13.37        | 3.79 | 3.52 | 11.21 | 3.98 | 2.82 | 9.92  | 4.07 | 2.44 | 9.92  | 4.07 | 2.44 | 9.92    | 4.07 | 2.44 |
|              | 35         | /       | /    | /    | /            | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /    | /       | /    | /    |
| 70           | -25        | /       | /    | /    | /            | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /    | /       | /    | /    |
|              | -20        | /       | /    | /    | /            | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /    | /       | /    | /    |
|              | -15        | 9.17    | 1.53 | 5.99 | 9.17         | 1.53 | 5.99 | 9.17  | 1.53 | 5.99 | 6.35  | 1.61 | 3.93 | 5.47  | 1.65 | 3.32 | 5.47    | 1.65 | 3.32 |
|              | -10        | 9.99    | 1.67 | 5.98 | 9.99         | 1.67 | 5.98 | 9.96  | 1.69 | 5.88 | 6.50  | 1.92 | 3.40 | 6.50  | 1.92 | 3.40 | 6.50    | 1.92 | 3.40 |
|              | -7         | 10.61   | 1.73 | 6.14 | 10.61        | 1.73 | 6.14 | 9.57  | 1.78 | 5.38 | 7.04  | 1.96 | 3.59 | 6.90  | 1.96 | 3.52 | 6.90    | 1.96 | 3.52 |
|              | -5         | 10.38   | 1.75 | 5.92 | 10.38        | 1.75 | 5.92 | 9.59  | 1.79 | 5.36 | 7.13  | 1.97 | 3.61 | 7.13  | 1.97 | 3.61 | 7.13    | 1.97 | 3.61 |
|              | -2         | 10.45   | 1.79 | 5.84 | 10.45        | 1.79 | 5.84 | 9.64  | 1.84 |      |       |      |      |       |      |      |         |      |      |

9. PERFORMANCE TABLES - COOLING MODE

9.1 Cooling performance Mod. 04

| Tw_out<br>°C | DBWB<br>°C | Maximum |       |      | 100%(Normal) |       |      | 75%  |       |      | 50%  |       |      | 25%  |       |      | Minimum |       |      |
|--------------|------------|---------|-------|------|--------------|-------|------|------|-------|------|------|-------|------|------|-------|------|---------|-------|------|
|              |            | CC      | EER   | PI   | CC           | EER   | PI   | CC   | EER   | PI   | CC   | EER   | PI   | CC   | EER   | PI   | CC      | EER   | PI   |
| 5            | -5         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 0          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 15         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 19         | 4.48    | 5.70  | 0.79 | 4.48         | 5.70  | 0.79 | 3.49 | 6.09  | 0.57 | 2.53 | 6.25  | 0.40 | 2.53 | 6.25  | 0.40 | 2.53    | 6.25  | 0.40 |
|              | 20         | 4.53    | 5.50  | 0.82 | 4.53         | 5.50  | 0.82 | 3.53 | 5.90  | 0.60 | 2.60 | 6.15  | 0.42 | 2.50 | 6.20  | 0.40 | 2.50    | 6.20  | 0.40 |
|              | 25         | 4.96    | 4.59  | 1.08 | 4.69         | 4.68  | 1.00 | 3.55 | 5.02  | 0.71 | 2.45 | 5.29  | 0.46 | 2.38 | 5.32  | 0.45 | 2.38    | 5.32  | 0.45 |
|              | 30         | 5.23    | 3.92  | 1.33 | 4.68         | 4.09  | 1.15 | 3.49 | 4.38  | 0.80 | 2.31 | 4.52  | 0.51 | 2.24 | 4.55  | 0.49 | 2.24    | 4.55  | 0.49 |
|              | 35         | 5.31    | 3.38  | 1.57 | 4.68         | 3.52  | 1.33 | 3.54 | 3.70  | 0.96 | 2.35 | 3.85  | 0.61 | 2.14 | 3.88  | 0.55 | 2.14    | 3.88  | 0.55 |
|              | 40         | 4.99    | 2.96  | 1.69 | 4.70         | 3.02  | 1.56 | 3.50 | 3.26  | 1.07 | 2.35 | 3.38  | 0.70 | 1.99 | 3.45  | 0.58 | 1.99    | 3.45  | 0.58 |
|              | 43         | 4.80    | 2.71  | 1.77 | 4.66         | 2.75  | 1.69 | 3.52 | 2.98  | 1.18 | 2.34 | 3.23  | 0.72 | 1.92 | 3.31  | 0.58 | 1.92    | 3.31  | 0.58 |
| 7            | 46         | 4.49    | 2.55  | 1.76 | 4.49         | 2.55  | 1.76 | 3.37 | 2.68  | 1.26 | 2.24 | 2.85  | 0.79 | 1.88 | 2.93  | 0.64 | 1.88    | 2.93  | 0.64 |
|              | -5         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 0          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 15         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 19         | 4.76    | 6.06  | 0.78 | 4.68         | 6.12  | 0.77 | 3.53 | 6.60  | 0.53 | 2.70 | 6.72  | 0.40 | 2.70 | 6.72  | 0.40 | 2.70    | 6.72  | 0.40 |
|              | 20         | 4.81    | 5.84  | 0.82 | 4.74         | 5.90  | 0.80 | 3.49 | 6.38  | 0.55 | 2.75 | 6.59  | 0.42 | 2.67 | 6.60  | 0.41 | 2.67    | 6.60  | 0.41 |
|              | 25         | 5.28    | 4.85  | 1.09 | 4.72         | 5.06  | 0.93 | 3.51 | 5.39  | 0.65 | 2.62 | 5.72  | 0.46 | 2.54 | 5.73  | 0.44 | 2.54    | 5.73  | 0.44 |
|              | 30         | 5.57    | 4.13  | 1.35 | 4.73         | 4.40  | 1.07 | 3.55 | 4.68  | 0.76 | 2.48 | 4.86  | 0.51 | 2.41 | 4.87  | 0.49 | 2.41    | 4.87  | 0.49 |
|              | 35         | 5.66    | 3.54  | 1.60 | 4.70         | 3.65  | 1.29 | 3.53 | 3.99  | 0.89 | 2.33 | 4.35  | 0.54 | 2.29 | 4.38  | 0.52 | 2.29    | 4.38  | 0.52 |
|              | 40         | 5.33    | 3.10  | 1.72 | 4.71         | 3.24  | 1.45 | 3.50 | 3.49  | 1.00 | 2.36 | 3.65  | 0.65 | 2.15 | 3.67  | 0.59 | 2.15    | 3.67  | 0.59 |
| 10           | 43         | 5.13    | 2.83  | 1.81 | 4.68         | 2.95  | 1.59 | 3.53 | 3.15  | 1.12 | 2.35 | 3.29  | 0.71 | 2.06 | 3.31  | 0.62 | 2.06    | 3.31  | 0.62 |
|              | 46         | 4.79    | 2.67  | 1.80 | 4.79         | 2.67  | 1.80 | 3.47 | 2.91  | 1.19 | 2.33 | 3.01  | 0.78 | 2.03 | 3.03  | 0.67 | 2.03    | 3.03  | 0.67 |
|              | -5         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 0          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 15         | 4.75    | 7.31  | 0.65 | 4.63         | 7.58  | 0.61 | 3.52 | 7.91  | 0.44 | 3.08 | 8.04  | 0.38 | 3.08 | 8.04  | 0.38 | 3.08    | 8.04  | 0.38 |
|              | 19         | 5.23    | 6.53  | 0.80 | 4.61         | 6.69  | 0.69 | 3.44 | 6.98  | 0.49 | 3.07 | 7.31  | 0.42 | 2.99 | 7.34  | 0.41 | 2.99    | 7.34  | 0.41 |
|              | 20         | 5.30    | 6.33  | 0.84 | 4.59         | 6.40  | 0.72 | 3.52 | 6.72  | 0.52 | 3.04 | 7.08  | 0.43 | 2.95 | 7.10  | 0.42 | 2.95    | 7.10  | 0.42 |
|              | 25         | 5.81    | 5.21  | 1.12 | 4.58         | 5.44  | 0.84 | 3.45 | 5.79  | 0.59 | 2.89 | 6.16  | 0.47 | 2.81 | 6.17  | 0.46 | 2.81    | 6.17  | 0.46 |
|              | 30         | 6.13    | 4.46  | 1.37 | 4.62         | 4.70  | 0.98 | 3.51 | 5.10  | 0.69 | 2.75 | 5.38  | 0.51 | 2.67 | 5.39  | 0.50 | 2.67    | 5.39  | 0.50 |
|              | 35         | 6.22    | 3.80  | 1.64 | 4.59         | 4.13  | 1.11 | 3.51 | 4.33  | 0.81 | 2.59 | 4.53  | 0.57 | 2.51 | 4.55  | 0.55 | 2.51    | 4.55  | 0.55 |
| 11           | 40         | 5.86    | 3.32  | 1.77 | 4.59         | 3.57  | 1.28 | 3.50 | 3.79  | 0.92 | 2.43 | 4.03  | 0.60 | 2.39 | 4.05  | 0.59 | 2.39    | 4.05  | 0.59 |
|              | 43         | 5.63    | 3.05  | 1.85 | 4.67         | 3.28  | 1.42 | 3.46 | 3.51  | 0.99 | 2.34 | 3.67  | 0.64 | 2.30 | 3.68  | 0.62 | 2.30    | 3.68  | 0.62 |
|              | 46         | 5.27    | 2.88  | 1.83 | 4.65         | 3.00  | 1.55 | 3.51 | 3.20  | 1.10 | 2.34 | 3.35  | 0.70 | 2.25 | 3.37  | 0.67 | 2.25    | 3.37  | 0.67 |
|              | -5         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 0          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10         | 4.40    | 8.40  | 0.52 | 4.37         | 8.50  | 0.51 | 4.01 | 8.80  | 0.46 | 4.01 | 8.80  | 0.46 | 4.01 | 8.80  | 0.46 | 4.01    | 8.80  | 0.46 |
|              | 15         | 4.89    | 7.51  | 0.65 | 4.58         | 7.68  | 0.60 | 3.52 | 8.00  | 0.44 | 3.17 | 8.18  | 0.39 | 3.17 | 8.18  | 0.39 | 3.17    | 8.18  | 0.39 |
|              | 19         | 5.40    | 6.78  | 0.80 | 4.56         | 7.22  | 0.63 | 3.44 | 7.54  | 0.46 | 3.17 | 7.74  | 0.41 | 3.09 | 7.76  | 0.40 | 3.09    | 7.76  | 0.40 |
|              | 20         | 5.46    | 6.42  | 0.85 | 4.62         | 6.89  | 0.67 | 3.52 | 7.24  | 0.49 | 3.14 | 7.34  | 0.43 | 3.05 | 7.35  | 0.42 | 3.05    | 7.35  | 0.42 |
|              | 25         | 5.99    | 5.42  | 1.11 | 4.61         | 5.78  | 0.80 | 3.45 | 6.20  | 0.56 | 2.99 | 6.23  | 0.48 | 2.90 | 6.29  | 0.46 | 2.90    | 6.29  | 0.46 |
|              | 30         | 6.32    | 4.57  | 1.38 | 4.66         | 4.92  | 0.95 | 3.52 | 5.25  | 0.67 | 2.84 | 5.56  | 0.51 | 2.76 | 5.59  | 0.49 | 2.76    | 5.59  | 0.49 |
| 15           | 35         | 6.41    | 3.89  | 1.65 | 4.64         | 4.31  | 1.08 | 3.43 | 4.65  | 0.74 | 2.67 | 4.78  | 0.56 | 2.60 | 4.79  | 0.54 | 2.60    | 4.79  | 0.54 |
|              | 40         | 6.04    | 3.39  | 1.78 | 4.56         | 3.80  | 1.20 | 3.43 | 4.03  | 0.85 | 2.52 | 4.17  | 0.60 | 2.47 | 4.20  | 0.59 | 2.47    | 4.20  | 0.59 |
|              | 43         | 5.82    | 3.12  | 1.86 | 4.65         | 3.41  | 1.36 | 3.50 | 3.64  | 0.96 | 2.42 | 3.80  | 0.64 | 2.37 | 3.82  | 0.62 | 2.37    | 3.82  | 0.62 |
|              | 46         | 5.44    | 2.92  | 1.86 | 4.63         | 3.11  | 1.49 | 3.45 | 3.31  | 1.04 | 2.33 | 3.45  | 0.68 | 2.33 | 3.45  | 0.68 | 2.33    | 3.45  | 0.68 |
|              | -5         | 5.34    | 11.68 | 0.46 | 5.34         | 11.68 | 0.46 | 5.11 | 11.85 | 0.43 | 5.11 | 11.85 | 0.43 | 5.11 | 11.85 | 0.43 | 5.11    | 11.85 | 0.43 |
|              | 0          | 5.27    | 11.35 | 0.46 | 5.27         | 11.35 | 0.46 | 4.70 | 11.60 | 0.41 | 4.70 | 11.60 | 0.41 | 4.70 | 11.60 | 0.41 | 4.70    | 11.60 | 0.41 |
|              | 5          | 5.36    | 10.88 | 0.49 | 5.36         | 10.88 | 0.49 | 4.60 | 11.03 | 0.42 | 4.60 | 11.03 | 0.42 | 4.60 | 11.03 | 0.42 | 4.60    | 11.03 | 0.42 |
|              | 10         | 5.16    | 10.44 | 0.49 | 5.16         | 10.44 | 0.49 | 4.46 | 10.66 | 0.42 | 4.46 | 10.66 | 0.42 | 4.46 | 10.66 | 0.42 | 4.46    | 10.66 | 0.42 |
|              | 15         | 5.39    | 9.21  | 0.59 | 4.60         | 9.34  | 0.49 | 3.62 | 9.69  | 0.37 | 3.50 | 9.71  | 0.36 | 3.50 | 9.71  | 0.36 | 3.50    | 9.71  | 0.36 |
|              | 19         | 5.96    | 8.12  | 0.73 | 4.53         | 8.29  | 0.55 | 3.55 | 8.66  | 0.41 | 3.42 | 8.68  | 0.39 | 3.42 | 8.68  | 0.39 | 3.42    | 8.68  | 0.39 |
|              | 20         | 6.03    | 7.42  | 0.81 | 4.49         | 7.61  | 0.59 | 3.51 | 7.94  | 0.44 | 3.37 | 7.96  | 0.42 | 3.37 | 7.96  | 0.42 | 3.37    | 7.96  | 0.42 |
|              | 25         | 6.75    | 6.01  | 1.12 | 4.50         | 6.60  | 0.68 | 3.45 | 6.96  | 0.50 | 3.28 | 7.09  | 0.46 | 3.28 | 7.09  | 0.46 | 3.28    | 7.09  | 0.46 |
| 18           | 30         | 7.13    | 5.04  | 1.41 | 4.56         | 5.78  | 0.79 | 3.41 | 6.20  | 0.55 | 3.11 | 6.29  | 0.49 | 3.05 | 6.31  | 0.48 | 3.05    | 6.31  | 0.48 |
|              | 35         | 7.18    | 4.24  | 1.70 | 4.57         | 4.93  | 0.93 | 3.43 | 5.29  | 0.65 | 3.04 | 5.47  | 0.56 | 2.98 | 5.50  | 0.54 | 2.98    | 5.50  | 0.54 |
|              | 40         | 6.79    | 3.69  | 1.84 | 4.54         | 4.27  | 1.06 | 3.47 | 4.50  | 0.76 | 2.87 | 4.64  | 0.62 | 2.80 | 4.65  | 0.60 | 2.80    | 4.65  | 0.60 |
|              | 43         | 6.51    | 3.33  | 1.96 | 4.58         | 3.90  | 1.17 | 3.47 | 4.18  | 0.83 | 2.77 | 4.21  | 0.66 | 2.71 | 4.22  | 0.64 | 2.71    | 4.22  | 0.64 |
|              | 46         | 6.14    | 3.18  | 1.93 | 4.62         | 3.58  | 1.29 | 3.45 | 3.79  | 0.91 | 2.68 | 3.83  | 0.70 | 2.68 | 3.83  | 0.70 | 2.68    | 3.83  | 0.70 |
|              | -5         | 5.85    | 12.71 | 0.46 | 5.85         | 12.71 | 0.46 | 5.35 | 13.01 | 0.41 | 5.35 | 13.01 | 0.41 | 5.35 | 13.01 | 0.41 | 5.35    | 13.01 | 0.41 |
|              | 0          | 5.75    | 12.26 | 0.47 | 5.75         | 12.26 | 0.47 | 5.45 | 12.56 | 0.43 | 5.45 | 12.56 | 0.43 | 5.45 | 12.56 | 0.43 | 5.45    | 12.56 | 0.43 |
|              | 5          | 5.80    | 10.92 | 0.53 | 5.80         | 10.92 | 0.53 | 5.60 | 11.02 | 0.51 | 5.60 | 11.02 | 0.51 | 5.60 | 11.02 | 0.51 | 5.60    | 11.02 | 0.51 |
|              | 10         | 5.66    | 10.31 | 0.55 | 4.98         | 10.81 | 0.46 | 4.81 | 10.84 | 0.44 | 4.81 | 10.84 | 0.44 | 4.81 | 10.84 | 0.44 | 4.81    | 10.84 | 0.44 |
|              | 15         | 5.91    | 9.85  | 0.60 | 4.50         | 10.21 | 0.44 | 3.90 | 10.63 | 0.37 | 3.82 | 10.66 | 0.36 | 3.82 | 10.66 | 0.36 | 3.82    | 10.66 | 0.36 |
|              | 19         | 6.51    | 8.54  | 0.76 | 4.43         | 9.25  | 0.48 | 3.82 | 9.61  | 0.40 | 3.74 | 9.64  | 0.39 | 3.74 | 9.64  | 0.39 | 3.74    | 9.64  | 0.39 |
|              | 20         | 6.59    | 8.00  | 0.82 | 4.52         | 8.67  | 0.52 | 3.80 | 9.17  | 0.41 | 3.70 | 9.19  | 0.40 | 3.70 | 9.19  | 0.40 | 3.70    | 9.19  | 0.40 |

## 9.2 Cooling performance Mod. 06

| Tw_out<br>°C | DB/WB<br>°C | Maximum |       |      | 100%(Normal) |       |      | 75%  |       |      | 50%  |       |      | 25%  |       |      | Minimum |       |      |
|--------------|-------------|---------|-------|------|--------------|-------|------|------|-------|------|------|-------|------|------|-------|------|---------|-------|------|
|              |             | CC      | EER   | PI   | CC           | EER   | PI   | CC   | EER   | PI   | CC   | EER   | PI   | CC   | EER   | PI   | CC      | EER   | PI   |
| 5            | -5          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 0           | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5           | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 15          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 19          | 5.40    | 5.30  | 1.02 | 5.40         | 5.30  | 1.02 | 4.05 | 5.71  | 0.71 | 2.70 | 6.22  | 0.43 | 2.53 | 6.25  | 0.40 | 2.53    | 6.25  | 0.40 |
|              | 20          | 5.54    | 5.12  | 1.08 | 5.54         | 5.12  | 1.08 | 4.15 | 5.53  | 0.75 | 2.77 | 6.10  | 0.45 | 2.50 | 6.20  | 0.40 | 2.50    | 6.20  | 0.40 |
|              | 25          | 6.10    | 4.22  | 1.45 | 6.10         | 4.22  | 1.45 | 4.57 | 4.69  | 0.98 | 3.05 | 5.15  | 0.59 | 2.38 | 5.32  | 0.45 | 2.38    | 5.32  | 0.45 |
|              | 30          | 6.75    | 3.48  | 1.94 | 6.75         | 3.48  | 1.94 | 5.09 | 3.83  | 1.33 | 3.40 | 4.39  | 0.77 | 2.24 | 4.55  | 0.49 | 2.24    | 4.55  | 0.49 |
|              | 35          | 6.77    | 2.94  | 2.30 | 6.77         | 2.94  | 2.30 | 5.04 | 3.30  | 1.53 | 3.40 | 3.68  | 0.92 | 2.14 | 3.88  | 0.55 | 2.14    | 3.88  | 0.55 |
| 7            | 40          | 5.44    | 2.80  | 1.95 | 5.44         | 2.80  | 1.95 | 4.08 | 3.06  | 1.33 | 2.72 | 3.32  | 0.82 | 1.99 | 3.45  | 0.58 | 1.99    | 3.45  | 0.58 |
|              | 43          | 4.86    | 2.70  | 1.80 | 4.86         | 2.70  | 1.80 | 3.64 | 2.90  | 1.25 | 2.43 | 3.20  | 0.76 | 1.92 | 3.31  | 0.58 | 1.92    | 3.31  | 0.58 |
|              | 46          | 4.49    | 2.55  | 1.76 | 4.49         | 2.55  | 1.76 | 3.37 | 2.68  | 1.26 | 2.24 | 2.85  | 0.79 | 1.88 | 2.93  | 0.64 | 1.88    | 2.93  | 0.64 |
|              | -5          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 0           | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5           | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 15          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 19          | 5.76    | 5.68  | 1.01 | 5.76         | 5.68  | 1.01 | 4.32 | 6.13  | 0.70 | 2.88 | 6.65  | 0.43 | 2.70 | 6.72  | 0.40 | 2.70    | 6.72  | 0.40 |
|              | 20          | 5.90    | 5.47  | 1.08 | 5.90         | 5.47  | 1.08 | 4.43 | 5.91  | 0.75 | 2.95 | 6.41  | 0.46 | 2.67 | 6.60  | 0.41 | 2.67    | 6.60  | 0.41 |
| 10           | 25          | 6.50    | 4.52  | 1.44 | 6.50         | 4.52  | 1.44 | 4.88 | 4.93  | 0.99 | 3.25 | 5.42  | 0.60 | 2.54 | 5.73  | 0.44 | 2.54    | 5.73  | 0.44 |
|              | 30          | 7.11    | 3.69  | 1.93 | 6.75         | 3.77  | 1.79 | 5.09 | 4.14  | 1.23 | 3.37 | 4.69  | 0.72 | 2.41 | 4.87  | 0.49 | 2.41    | 4.87  | 0.49 |
|              | 35          | 7.14    | 3.01  | 2.38 | 6.80         | 3.10  | 2.19 | 5.05 | 3.55  | 1.42 | 3.38 | 4.01  | 0.84 | 2.29 | 4.38  | 0.52 | 2.29    | 4.38  | 0.52 |
|              | 40          | 5.82    | 2.94  | 1.98 | 5.82         | 2.94  | 1.98 | 4.36 | 3.26  | 1.34 | 2.91 | 3.61  | 0.81 | 2.15 | 3.67  | 0.59 | 2.15    | 3.67  | 0.59 |
|              | 43          | 5.19    | 2.82  | 1.84 | 5.19         | 2.82  | 1.84 | 3.89 | 3.07  | 1.27 | 2.59 | 3.27  | 0.79 | 2.06 | 3.31  | 0.62 | 2.06    | 3.31  | 0.62 |
|              | 46          | 4.79    | 2.67  | 1.80 | 4.79         | 2.67  | 1.80 | 3.59 | 2.89  | 1.24 | 2.33 | 3.01  | 0.78 | 2.03 | 3.03  | 0.67 | 2.03    | 3.03  | 0.67 |
|              | -5          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 0           | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5           | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 11           | 15          | 5.72    | 7.05  | 0.81 | 5.72         | 7.05  | 0.81 | 4.33 | 7.60  | 0.57 | 3.08 | 8.04  | 0.38 | 3.08 | 8.04  | 0.38 | 3.08    | 8.04  | 0.38 |
|              | 19          | 6.31    | 6.17  | 1.02 | 6.31         | 6.17  | 1.02 | 4.73 | 6.65  | 0.71 | 3.20 | 7.30  | 0.44 | 2.99 | 7.34  | 0.41 | 2.99    | 7.34  | 0.41 |
|              | 20          | 6.47    | 5.94  | 1.09 | 6.47         | 5.94  | 1.09 | 4.89 | 6.37  | 0.77 | 3.26 | 7.05  | 0.46 | 2.95 | 7.10  | 0.42 | 2.95    | 7.10  | 0.42 |
|              | 25          | 7.16    | 4.80  | 1.49 | 6.64         | 4.90  | 1.36 | 4.96 | 5.42  | 0.92 | 3.40 | 5.99  | 0.57 | 2.81 | 6.17  | 0.46 | 2.81    | 6.17  | 0.46 |
|              | 30          | 7.80    | 3.96  | 1.97 | 6.75         | 4.17  | 1.62 | 5.04 | 4.67  | 1.08 | 3.36 | 5.23  | 0.64 | 2.67 | 5.39  | 0.50 | 2.67    | 5.39  | 0.50 |
|              | 35          | 7.80    | 3.27  | 2.39 | 6.65         | 3.58  | 1.86 | 5.05 | 3.97  | 1.27 | 3.37 | 4.35  | 0.77 | 2.51 | 4.55  | 0.55 | 2.51    | 4.55  | 0.55 |
|              | 40          | 6.40    | 3.17  | 2.02 | 6.40         | 3.17  | 2.02 | 4.80 | 3.48  | 1.38 | 3.20 | 3.81  | 0.84 | 2.39 | 4.05  | 0.59 | 2.39    | 4.05  | 0.59 |
|              | 43          | 5.70    | 3.01  | 1.89 | 5.70         | 3.01  | 1.89 | 4.26 | 3.32  | 1.28 | 2.81 | 3.55  | 0.79 | 2.30 | 3.68  | 0.62 | 2.30    | 3.68  | 0.62 |
|              | 46          | 5.27    | 2.88  | 1.83 | 5.27         | 2.88  | 1.83 | 3.96 | 3.15  | 1.25 | 2.64 | 3.22  | 0.82 | 2.25 | 3.37  | 0.67 | 2.25    | 3.37  | 0.67 |
|              | -5          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 15           | 0           | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5           | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10          | 4.94    | 8.30  | 0.59 | 4.94         | 8.30  | 0.59 | 4.01 | 8.80  | 0.46 | 4.01 | 8.80  | 0.46 | 4.01 | 8.80  | 0.46 | 4.01    | 8.80  | 0.46 |
|              | 15          | 5.89    | 7.19  | 0.82 | 5.89         | 7.19  | 0.82 | 4.42 | 7.73  | 0.57 | 3.17 | 8.18  | 0.39 | 3.17 | 8.18  | 0.39 | 3.17    | 8.18  | 0.39 |
|              | 19          | 6.50    | 6.30  | 1.03 | 6.50         | 6.60  | 0.98 | 4.87 | 7.14  | 0.68 | 3.30 | 7.71  | 0.43 | 3.09 | 7.76  | 0.40 | 3.09    | 7.76  | 0.40 |
|              | 20          | 6.68    | 5.93  | 1.13 | 6.68         | 6.23  | 1.07 | 5.05 | 6.73  | 0.75 | 3.38 | 7.27  | 0.46 | 3.05 | 7.35  | 0.42 | 3.05    | 7.35  | 0.42 |
|              | 25          | 7.39    | 4.87  | 1.52 | 6.65         | 5.26  | 1.26 | 5.02 | 5.65  | 0.89 | 3.31 | 6.23  | 0.53 | 2.90 | 6.29  | 0.46 | 2.90    | 6.29  | 0.46 |
|              | 30          | 7.98    | 4.01  | 1.99 | 6.64         | 4.31  | 1.54 | 5.01 | 4.87  | 1.03 | 3.37 | 5.44  | 0.62 | 2.76 | 5.59  | 0.49 | 2.76    | 5.59  | 0.49 |
|              | 35          | 8.04    | 3.30  | 2.44 | 6.67         | 3.71  | 1.80 | 5.06 | 4.15  | 1.22 | 3.39 | 4.70  | 0.72 | 2.60 | 4.79  | 0.54 | 2.60    | 4.79  | 0.54 |
|              | 40          | 6.61    | 3.21  | 2.06 | 6.61         | 3.21  | 2.06 | 4.95 | 3.60  | 1.37 | 3.30 | 4.05  | 0.82 | 2.47 | 4.20  | 0.59 | 2.47    | 4.20  | 0.59 |
| 18           | 43          | 5.88    | 3.05  | 1.93 | 5.88         | 3.05  | 1.93 | 4.41 | 3.45  | 1.28 | 2.94 | 3.70  | 0.79 | 2.37 | 3.82  | 0.62 | 2.37    | 3.82  | 0.62 |
|              | 46          | 5.44    | 2.92  | 1.86 | 5.44         | 2.92  | 1.86 | 4.08 | 3.27  | 1.25 | 2.72 | 3.41  | 0.80 | 2.33 | 3.45  | 0.68 | 2.33    | 3.45  | 0.68 |
|              | -5          | 5.89    | 11.26 | 0.52 | 5.89         | 11.26 | 0.52 | 5.11 | 11.85 | 0.43 | 5.11 | 11.85 | 0.43 | 5.11 | 11.85 | 0.43 | 5.11    | 11.85 | 0.43 |
|              | 0           | 5.78    | 11.00 | 0.53 | 5.78         | 11.00 | 0.53 | 5.00 | 11.56 | 0.43 | 4.70 | 11.60 | 0.41 | 4.70 | 11.60 | 0.41 | 4.70    | 11.60 | 0.41 |
|              | 5           | 5.54    | 10.52 | 0.53 | 5.54         | 10.52 | 0.53 | 4.80 | 11.00 | 0.44 | 4.60 | 11.03 | 0.42 | 4.60 | 11.03 | 0.42 | 4.60    | 11.03 | 0.42 |
|              | 10          | 5.40    | 10.18 | 0.53 | 5.40         | 10.18 | 0.53 | 4.46 | 10.66 | 0.42 | 4.46 | 10.66 | 0.42 | 4.46 | 10.66 | 0.42 | 4.46    | 10.66 | 0.42 |
|              | 15          | 6.45    | 8.91  | 0.72 | 6.45         | 8.91  | 0.72 | 4.84 | 9.21  | 0.53 | 3.50 | 9.71  | 0.36 | 3.50 | 9.71  | 0.36 | 3.50    | 9.71  | 0.36 |
|              | 19          | 7.13    | 7.66  | 0.93 | 6.55         | 7.73  | 0.85 | 4.89 | 8.17  | 0.60 | 3.42 | 8.68  | 0.39 | 3.42 | 8.68  | 0.39 | 3.42    | 8.68  | 0.39 |
|              | 20          | 7.32    | 7.00  | 1.05 | 6.49         | 7.28  | 0.89 | 4.96 | 7.52  | 0.66 | 3.37 | 7.96  | 0.42 | 3.37 | 7.96  | 0.42 | 3.37    | 7.96  | 0.42 |
|              | 25          | 8.30    | 5.47  | 1.52 | 6.50         | 6.03  | 1.08 | 4.86 | 6.57  | 0.74 | 3.28 | 7.09  | 0.46 | 3.28 | 7.09  | 0.46 | 3.28    | 7.09  | 0.46 |
| 20           | 30          | 8.99    | 4.51  | 1.99 | 6.57         | 5.12  | 1.28 | 4.91 | 5.74  | 0.85 | 3.27 | 6.28  | 0.52 | 3.05 | 6.31  | 0.48 | 3.05    | 6.31  | 0.48 |
|              | 35          | 8.99    | 3.70  | 2.43 | 6.52         | 4.42  | 1.48 | 4.92 | 4.91  | 1.00 | 3.31 | 5.40  | 0.61 | 2.98 | 5.50  | 0.54 | 2.98    | 5.50  | 0.54 |
|              | 40          | 7.42    | 3.44  | 2.16 | 6.58         | 3.70  | 1.78 | 4.95 | 4.16  | 1.19 | 3.33 | 4.61  | 0.72 | 2.80 | 4.65  | 0.60 | 2.80    | 4.65  | 0.60 |
|              | 43          | 6.60    | 3.30  | 2.00 | 6.51         | 3.33  | 1.96 | 4.88 | 3.79  | 1.29 | 3.31 | 4.18  | 0.79 | 2.71 | 4.22  | 0.64 | 2.71    | 4.22  | 0.64 |
|              | 46          |         |       |      |              |       |      |      |       |      |      |       |      |      |       |      |         |       |      |

### 9.3 Cooling performance Mod. 08

| Tw_out<br>°C | DBWB<br>°C | Maximum |       |      | 100%(Normal) |       |      | 75%  |       |      | 50%  |       |      | 25%  |       |      | Minimum |       |      |
|--------------|------------|---------|-------|------|--------------|-------|------|------|-------|------|------|-------|------|------|-------|------|---------|-------|------|
|              |            | CC      | EER   | PI   | CC           | EER   | PI   | CC   | EER   | PI   | CC   | EER   | PI   | CC   | EER   | PI   | CC      | EER   | PI   |
| 5            | -5         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 0          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 15         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 19         | 6.76    | 5.39  | 1.26 | 6.76         | 5.39  | 1.26 | 5.07 | 6.11  | 0.83 | 3.50 | 6.81  | 0.51 | 3.50 | 6.81  | 0.51 | 3.50    | 6.81  | 0.51 |
|              | 20         | 6.82    | 5.18  | 1.32 | 6.82         | 5.18  | 1.32 | 5.11 | 5.80  | 0.88 | 3.47 | 6.56  | 0.53 | 3.47 | 6.56  | 0.53 | 3.47    | 6.56  | 0.53 |
|              | 25         | 7.29    | 4.26  | 1.71 | 7.29         | 4.26  | 1.71 | 5.62 | 4.85  | 1.16 | 3.74 | 5.51  | 0.68 | 3.35 | 5.54  | 0.61 | 3.35    | 5.54  | 0.61 |
|              | 30         | 7.67    | 3.57  | 2.15 | 7.48         | 3.63  | 2.06 | 5.61 | 4.40  | 1.28 | 3.72 | 4.82  | 0.77 | 3.21 | 4.92  | 0.65 | 3.21    | 4.92  | 0.65 |
|              | 35         | 7.69    | 3.02  | 2.55 | 7.44         | 3.09  | 2.41 | 5.63 | 3.55  | 1.59 | 3.75 | 4.02  | 0.93 | 3.03 | 4.35  | 0.70 | 3.03    | 4.35  | 0.70 |
|              | 40         | 6.76    | 2.77  | 2.44 | 6.76         | 2.77  | 2.44 | 4.92 | 3.10  | 1.59 | 3.14 | 3.43  | 0.91 | 2.85 | 3.62  | 0.79 | 2.85    | 3.62  | 0.79 |
| 7            | 43         | 6.18    | 2.63  | 2.35 | 6.18         | 2.63  | 2.35 | 4.57 | 2.95  | 1.55 | 3.08 | 3.05  | 1.01 | 2.73 | 3.13  | 0.87 | 2.73    | 3.13  | 0.87 |
|              | 46         | 5.26    | 2.55  | 2.06 | 5.26         | 2.55  | 2.06 | 3.95 | 2.75  | 1.44 | 2.61 | 2.96  | 0.88 | 2.61 | 2.96  | 0.88 | 2.61    | 2.96  | 0.88 |
|              | -5         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 0          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 15         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 19         | 7.18    | 5.73  | 1.25 | 7.18         | 5.73  | 1.25 | 5.50 | 6.60  | 0.83 | 3.75 | 7.42  | 0.51 | 3.75 | 7.42  | 0.51 | 3.75    | 7.42  | 0.51 |
|              | 20         | 7.24    | 5.51  | 1.31 | 7.24         | 5.51  | 1.31 | 5.53 | 6.38  | 0.87 | 3.72 | 7.14  | 0.52 | 3.72 | 7.14  | 0.52 | 3.72    | 7.14  | 0.52 |
|              | 25         | 7.78    | 4.50  | 1.73 | 7.44         | 4.63  | 1.61 | 5.55 | 5.34  | 1.04 | 3.73 | 5.96  | 0.63 | 3.59 | 5.98  | 0.60 | 3.59    | 5.98  | 0.60 |
|              | 30         | 8.17    | 3.76  | 2.17 | 7.48         | 3.99  | 1.87 | 5.56 | 4.51  | 1.24 | 3.71 | 4.93  | 0.75 | 3.44 | 4.98  | 0.69 | 3.44    | 4.98  | 0.69 |
| 10           | 35         | 8.19    | 3.17  | 2.59 | 7.50         | 3.45  | 2.17 | 5.58 | 3.88  | 1.44 | 3.78 | 4.21  | 0.90 | 3.25 | 4.42  | 0.74 | 3.25    | 4.42  | 0.74 |
|              | 40         | 7.23    | 2.92  | 2.48 | 7.23         | 2.92  | 2.48 | 5.58 | 3.30  | 1.69 | 3.69 | 3.66  | 1.01 | 3.06 | 3.80  | 0.80 | 3.06    | 3.80  | 0.80 |
|              | 43         | 6.60    | 2.76  | 2.39 | 6.60         | 2.76  | 2.39 | 4.95 | 3.15  | 1.57 | 3.30 | 3.33  | 0.99 | 2.94 | 3.37  | 0.87 | 2.94    | 3.37  | 0.87 |
|              | 46         | 5.63    | 2.68  | 2.10 | 5.63         | 2.68  | 2.10 | 4.22 | 2.95  | 1.43 | 2.81 | 3.10  | 0.91 | 2.81 | 3.10  | 0.91 | 2.81    | 3.10  | 0.91 |
|              | -5         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 0          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 15         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 19         | 7.20    | 7.00  | 1.03 | 7.20         | 7.00  | 1.03 | 5.40 | 8.00  | 0.68 | 4.19 | 8.85  | 0.47 | 4.19 | 8.85  | 0.47 | 4.19    | 8.85  | 0.47 |
|              | 20         | 7.91    | 6.26  | 1.26 | 7.63         | 6.30  | 1.21 | 5.73 | 7.47  | 0.77 | 4.15 | 8.10  | 0.51 | 4.15 | 8.10  | 0.51 | 4.15    | 8.10  | 0.51 |
| 11           | 25         | 7.97    | 6.06  | 1.32 | 7.71         | 6.11  | 1.26 | 5.68 | 7.00  | 0.81 | 4.12 | 7.25  | 0.57 | 4.12 | 7.25  | 0.57 | 4.12    | 7.25  | 0.57 |
|              | 30         | 8.54    | 4.88  | 1.75 | 7.67         | 5.24  | 1.46 | 5.75 | 6.21  | 0.93 | 3.98 | 6.75  | 0.59 | 3.98 | 6.75  | 0.59 | 3.98    | 6.75  | 0.59 |
|              | 35         | 9.00    | 3.90  | 2.31 | 7.62         | 4.51  | 1.69 | 5.79 | 5.13  | 1.13 | 3.81 | 5.80  | 0.66 | 3.81 | 5.80  | 0.66 | 3.81    | 5.80  | 0.66 |
|              | 40         | 8.98    | 3.34  | 2.69 | 7.67         | 3.79  | 2.03 | 5.73 | 4.38  | 1.31 | 3.90 | 4.82  | 0.81 | 3.61 | 4.96  | 0.73 | 3.61    | 4.96  | 0.73 |
|              | 43         | 7.95    | 3.13  | 2.54 | 7.64         | 3.19  | 2.39 | 5.72 | 3.70  | 1.55 | 3.81 | 4.34  | 0.88 | 3.40 | 4.38  | 0.78 | 3.40    | 4.38  | 0.78 |
|              | 46         | 7.26    | 2.96  | 2.45 | 7.26         | 2.96  | 2.45 | 5.44 | 3.43  | 1.59 | 3.63 | 4.04  | 0.90 | 3.25 | 4.09  | 0.79 | 3.25    | 4.09  | 0.79 |
|              | 46         | 6.20    | 2.89  | 2.14 | 6.20         | 2.89  | 2.14 | 4.65 | 3.12  | 1.49 | 3.12 | 3.61  | 0.86 | 3.12 | 3.61  | 0.86 | 3.12    | 3.61  | 0.86 |
|              | -5         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 0          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10         | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 15           | 15         | 6.21    | 8.12  | 0.77 | 5.49         | 8.22  | 0.67 | 5.15 | 9.21  | 0.56 | 5.15 | 9.21  | 0.56 | 5.15 | 9.21  | 0.56 | 5.15    | 9.21  | 0.56 |
|              | 19         | 7.33    | 7.34  | 1.00 | 6.20         | 7.57  | 0.82 | 4.65 | 8.40  | 0.55 | 4.09 | 8.96  | 0.46 | 4.09 | 8.96  | 0.46 | 4.09    | 8.96  | 0.46 |
|              | 20         | 8.04    | 6.28  | 1.28 | 7.10         | 6.70  | 1.06 | 5.33 | 7.92  | 0.67 | 4.06 | 8.48  | 0.48 | 4.06 | 8.48  | 0.48 | 4.06    | 8.48  | 0.48 |
|              | 25         | 8.29    | 6.12  | 1.35 | 7.72         | 6.25  | 1.23 | 5.91 | 7.01  | 0.84 | 4.26 | 7.62  | 0.56 | 4.26 | 7.62  | 0.56 | 4.26    | 7.62  | 0.56 |
|              | 30         | 8.83    | 5.02  | 1.76 | 7.80         | 5.27  | 1.48 | 5.81 | 6.24  | 0.93 | 4.12 | 7.02  | 0.59 | 4.12 | 7.02  | 0.59 | 4.12    | 7.02  | 0.59 |
|              | 35         | 9.25    | 4.03  | 2.29 | 7.74         | 4.48  | 1.73 | 5.85 | 5.19  | 1.13 | 3.94 | 6.02  | 0.66 | 3.94 | 6.02  | 0.66 | 3.94    | 6.02  | 0.66 |
|              | 40         | 9.26    | 3.47  | 2.67 | 7.80         | 3.82  | 2.04 | 5.79 | 4.54  | 1.27 | 3.88 | 5.06  | 0.77 | 3.74 | 5.10  | 0.73 | 3.74    | 5.10  | 0.73 |
|              | 43         | 8.17    | 3.19  | 2.57 | 7.78         | 3.30  | 2.36 | 5.80 | 3.83  | 1.51 | 3.94 | 4.20  | 0.94 | 3.52 | 4.25  | 0.83 | 3.52    | 4.25  | 0.83 |
|              | 46         | 7.50    | 3.03  | 2.47 | 7.50         | 3.03  | 2.47 | 5.80 | 3.46  | 1.67 | 3.94 | 3.92  | 1.00 | 3.38 | 3.96  | 0.85 | 3.38    | 3.96  | 0.85 |
|              | 46         | 7.20    | 2.92  | 2.47 | 7.20         | 2.92  | 2.47 | 5.40 | 3.23  | 1.67 | 3.60 | 3.43  | 1.05 | 3.23 | 3.45  | 0.94 | 3.23    | 3.45  | 0.94 |
|              | 46         | 7.35    | 12.19 | 0.60 | 7.35         | 12.19 | 0.60 | 6.54 | 12.60 | 0.52 | 6.54 | 12.60 | 0.52 | 6.54 | 12.60 | 0.52 | 6.54    | 12.60 | 0.52 |
| 18           | 0          | 6.42    | 10.86 | 0.59 | 6.42         | 10.86 | 0.59 | 5.71 | 11.45 | 0.50 | 5.71 | 11.45 | 0.50 | 5.71 | 11.45 | 0.50 | 5.71    | 11.45 | 0.50 |
|              | 5          | 6.23    | 9.90  | 0.63 | 6.23         | 9.90  | 0.63 | 5.58 | 10.52 | 0.53 | 5.58 | 10.52 | 0.53 | 5.58 | 10.52 | 0.53 | 5.58    | 10.52 | 0.53 |
|              | 10         | 6.48    | 9.12  | 0.71 | 6.26         | 9.20  | 0.68 | 5.14 | 9.85  | 0.53 | 5.14 | 9.85  | 0.53 | 5.14 | 9.85  | 0.53 | 5.14    | 9.85  | 0.53 |
|              | 15         | 8.27    | 7.98  | 1.04 | 8.08         | 8.17  | 0.99 | 6.06 | 9.22  | 0.66 | 4.71 | 9.43  | 0.50 | 4.71 | 9.43  | 0.50 | 4.71    | 9.43  | 0.50 |
|              | 19         | 8.48    | 7.05  | 1.20 | 8.08         | 7.20  | 1.12 | 6.05 | 8.24  | 0.73 | 4.87 | 8.91  | 0.55 | 4.87 | 8.91  | 0.55 | 4.87    | 8.91  | 0.55 |
|              | 20         | 8.71    | 6.45  | 1.35 | 8.08         | 6.76  | 1.20 | 6.06 | 7.78  | 0.78 | 4.84 | 8.40  | 0.58 | 4.84 | 8.40  | 0.58 | 4.84    | 8.40  | 0.58 |
|              | 25         | 9.93    | 5.59  | 1.78 | 8.04         | 6.20  | 1.30 | 6.12 | 7.23  | 0.85 | 4.67 | 7.71  | 0.61 | 4.67 | 7.71  | 0.61 | 4.67    | 7.71  | 0.61 |
|              | 30         | 10.38   | 4.55  | 2.28 | 8.01         | 5.19  | 1.54 | 6.04 | 6.20  | 0.97 | 4.48 | 6.52  | 0.69 | 4.48 | 6.52  | 0.69 | 4.48    | 6.52  | 0.69 |
|              | 35         | 10.45   | 3.79  | 2.76 | 8.03         | 4.60  | 1.75 | 6.05 | 5.24  | 1.15 | 4.27 | 5.62  | 0.76 | 4.27 | 5.62  | 0.76 | 4.27    | 5.62  | 0.76 |
|              | 40         | 9.25    | 3.49  | 2.65 | 8.10         | 3.82  | 2.12 | 6.10 | 4.44  | 1.37 | 4.03 | 4.73  | 0.85 | 4.03 |       |      |         |       |      |

9.4 Cooling performance Mod. 10

| Tw_out<br>°C | DB/WB<br>°C | Maximum |       |      | 100%(Normal) |       |      | 75%  |       |      | 50%  |       |      | 25%  |       |      | Minimum |       |      |
|--------------|-------------|---------|-------|------|--------------|-------|------|------|-------|------|------|-------|------|------|-------|------|---------|-------|------|
|              |             | CC      | EER   | PI   | CC           | EER   | PI   | CC   | EER   | PI   | CC   | EER   | PI   | CC   | EER   | PI   | CC      | EER   | PI   |
| 5            | -5          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 0           | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5           | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 15          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 19          | 7.06    | 5.21  | 1.36 | 7.06         | 5.21  | 1.36 | 5.30 | 6.05  | 0.88 | 3.50 | 6.81  | 0.51 | 3.50 | 6.81  | 0.51 | 3.50    | 6.81  | 0.51 |
|              | 20          | 7.23    | 4.97  | 1.45 | 7.23         | 4.97  | 1.45 | 5.41 | 5.75  | 0.94 | 3.61 | 6.53  | 0.55 | 3.47 | 6.56  | 0.53 | 3.47    | 6.56  | 0.53 |
|              | 25          | 7.81    | 4.09  | 1.91 | 7.81         | 4.09  | 1.91 | 5.86 | 4.78  | 1.22 | 3.88 | 5.48  | 0.71 | 3.35 | 5.54  | 0.61 | 3.35    | 5.54  | 0.61 |
|              | 30          | 8.20    | 3.41  | 2.40 | 8.20         | 3.41  | 2.40 | 6.15 | 4.32  | 1.42 | 4.06 | 4.76  | 0.85 | 3.21 | 4.92  | 0.65 | 3.21    | 4.92  | 0.65 |
|              | 35          | 8.23    | 2.88  | 2.86 | 8.23         | 2.88  | 2.86 | 6.14 | 3.42  | 1.79 | 4.06 | 3.98  | 1.02 | 3.03 | 4.35  | 0.70 | 3.03    | 4.35  | 0.70 |
| 7            | 40          | 6.76    | 2.77  | 2.44 | 6.76         | 2.77  | 2.44 | 5.02 | 3.06  | 1.64 | 3.32 | 3.42  | 0.97 | 2.85 | 3.62  | 0.79 | 2.85    | 3.62  | 0.79 |
|              | 43          | 6.18    | 2.63  | 2.35 | 6.18         | 2.63  | 2.35 | 4.57 | 2.95  | 1.55 | 3.08 | 3.05  | 1.01 | 2.73 | 3.13  | 0.87 | 2.73    | 3.13  | 0.87 |
|              | 46          | 5.26    | 2.55  | 2.06 | 5.26         | 2.55  | 2.06 | 3.95 | 2.75  | 1.44 | 2.61 | 2.96  | 0.88 | 2.61 | 2.96  | 0.88 | 2.61    | 2.96  | 0.88 |
|              | -5          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 0           | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5           | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 15          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 19          | 7.54    | 5.55  | 1.36 | 7.54         | 5.55  | 1.36 | 5.70 | 6.52  | 0.87 | 3.75 | 7.42  | 0.51 | 3.75 | 7.42  | 0.51 | 3.75    | 7.42  | 0.51 |
|              | 20          | 7.72    | 5.29  | 1.46 | 7.72         | 5.29  | 1.46 | 5.65 | 6.32  | 0.89 | 3.86 | 7.07  | 0.55 | 3.72 | 7.14  | 0.52 | 3.72    | 7.14  | 0.52 |
| 10           | 25          | 8.33    | 4.32  | 1.93 | 8.33         | 4.32  | 1.93 | 6.25 | 5.20  | 1.20 | 4.15 | 5.90  | 0.70 | 3.59 | 5.98  | 0.60 | 3.59    | 5.98  | 0.60 |
|              | 30          | 8.73    | 3.58  | 2.44 | 8.73         | 3.58  | 2.44 | 6.55 | 4.33  | 1.51 | 4.37 | 4.90  | 0.89 | 3.44 | 4.98  | 0.69 | 3.44    | 4.98  | 0.69 |
|              | 35          | 9.23    | 3.06  | 3.01 | 8.90         | 3.25  | 2.74 | 6.57 | 3.62  | 1.81 | 4.38 | 4.15  | 1.06 | 3.25 | 4.42  | 0.74 | 3.25    | 4.42  | 0.74 |
|              | 40          | 7.23    | 2.92  | 2.48 | 7.23         | 2.92  | 2.48 | 5.58 | 3.30  | 1.69 | 3.69 | 3.66  | 1.01 | 3.06 | 3.80  | 0.80 | 3.06    | 3.80  | 0.80 |
|              | 43          | 6.60    | 2.76  | 2.39 | 6.60         | 2.76  | 2.39 | 4.95 | 3.15  | 1.57 | 3.30 | 3.33  | 0.99 | 2.94 | 3.37  | 0.87 | 2.94    | 3.37  | 0.87 |
|              | 46          | 5.63    | 2.68  | 2.10 | 5.63         | 2.68  | 2.10 | 4.22 | 2.95  | 1.43 | 2.81 | 3.10  | 0.91 | 2.81 | 3.10  | 0.91 | 2.81    | 3.10  | 0.91 |
|              | -5          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 0           | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5           | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 11           | 15          | 7.48    | 6.95  | 1.08 | 7.48         | 6.95  | 1.08 | 5.55 | 7.96  | 0.70 | 4.19 | 8.85  | 0.47 | 4.19 | 8.85  | 0.47 | 4.19    | 8.85  | 0.47 |
|              | 19          | 8.30    | 6.11  | 1.36 | 8.30         | 6.11  | 1.36 | 6.16 | 7.28  | 0.85 | 4.15 | 8.10  | 0.51 | 4.15 | 8.10  | 0.51 | 4.15    | 8.10  | 0.51 |
|              | 20          | 8.49    | 5.83  | 1.46 | 8.49         | 5.83  | 1.46 | 6.37 | 6.82  | 0.93 | 4.28 | 7.21  | 0.59 | 4.12 | 7.25  | 0.57 | 4.12    | 7.25  | 0.57 |
|              | 25          | 9.15    | 4.68  | 1.96 | 9.15         | 4.68  | 1.96 | 6.76 | 5.90  | 1.15 | 4.59 | 6.61  | 0.69 | 3.98 | 6.75  | 0.59 | 3.98    | 6.75  | 0.59 |
|              | 30          | 9.58    | 3.84  | 2.49 | 9.21         | 3.89  | 2.37 | 6.88 | 4.84  | 1.42 | 4.56 | 5.53  | 0.82 | 3.81 | 5.80  | 0.66 | 3.81    | 5.80  | 0.66 |
|              | 35          | 9.60    | 3.22  | 2.98 | 9.20         | 3.30  | 2.79 | 6.87 | 4.06  | 1.69 | 4.57 | 4.63  | 0.99 | 3.61 | 4.96  | 0.73 | 3.61    | 4.96  | 0.73 |
|              | 40          | 7.95    | 3.13  | 2.54 | 7.95         | 3.13  | 2.54 | 5.94 | 3.64  | 1.63 | 3.95 | 4.30  | 0.92 | 3.40 | 4.38  | 0.78 | 3.40    | 4.38  | 0.78 |
|              | 43          | 7.26    | 2.96  | 2.45 | 7.26         | 2.96  | 2.45 | 5.55 | 3.42  | 1.62 | 3.67 | 4.00  | 0.92 | 3.25 | 4.09  | 0.79 | 3.25    | 4.09  | 0.79 |
|              | 46          | 6.20    | 2.89  | 2.14 | 6.20         | 2.89  | 2.14 | 4.65 | 3.12  | 1.49 | 3.12 | 3.61  | 0.86 | 3.12 | 3.61  | 0.86 | 3.12    | 3.61  | 0.86 |
|              | -5          | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 15           | 0           | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5           | /       | /     | /    | /            | /     | /    | /    | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10          | 6.74    | 7.92  | 0.85 | 6.74         | 7.92  | 0.85 | 5.15 | 9.21  | 0.56 | 5.15 | 9.21  | 0.56 | 5.15 | 9.21  | 0.56 | 5.15    | 9.21  | 0.56 |
|              | 15          | 7.71    | 7.21  | 1.07 | 7.71         | 7.21  | 1.07 | 5.10 | 8.37  | 0.61 | 4.33 | 8.92  | 0.49 | 4.09 | 8.96  | 0.46 | 4.09    | 8.96  | 0.46 |
|              | 19          | 8.56    | 6.16  | 1.39 | 8.56         | 6.16  | 1.39 | 6.36 | 7.65  | 0.83 | 4.29 | 8.41  | 0.51 | 4.06 | 8.48  | 0.48 | 4.06    | 8.48  | 0.48 |
|              | 20          | 8.76    | 5.97  | 1.47 | 8.76         | 5.97  | 1.47 | 6.59 | 6.89  | 0.96 | 4.43 | 7.60  | 0.58 | 4.26 | 7.62  | 0.56 | 4.26    | 7.62  | 0.56 |
|              | 25          | 9.43    | 4.88  | 1.93 | 9.43         | 4.88  | 1.93 | 6.93 | 5.96  | 1.16 | 4.59 | 6.87  | 0.67 | 4.12 | 7.02  | 0.59 | 4.12    | 7.02  | 0.59 |
|              | 30          | 9.86    | 3.93  | 2.51 | 9.30         | 4.02  | 2.31 | 6.98 | 4.87  | 1.43 | 4.71 | 5.74  | 0.82 | 3.94 | 6.02  | 0.66 | 3.94    | 6.02  | 0.66 |
|              | 35          | 9.90    | 3.29  | 3.01 | 9.26         | 3.52  | 2.63 | 6.98 | 4.26  | 1.64 | 4.60 | 4.83  | 0.95 | 3.74 | 5.10  | 0.73 | 3.74    | 5.10  | 0.73 |
|              | 40          | 8.17    | 3.19  | 2.57 | 8.17         | 3.19  | 2.57 | 6.13 | 3.68  | 1.67 | 4.09 | 4.13  | 0.99 | 3.52 | 4.25  | 0.83 | 3.52    | 4.25  | 0.83 |
| 18           | 43          | 7.50    | 3.03  | 2.47 | 7.50         | 3.03  | 2.47 | 6.13 | 3.45  | 1.78 | 4.08 | 3.87  | 1.05 | 3.38 | 3.96  | 0.85 | 3.38    | 3.96  | 0.85 |
|              | 46          | 7.20    | 2.92  | 2.47 | 7.20         | 2.92  | 2.47 | 5.40 | 3.23  | 1.67 | 3.60 | 3.43  | 1.05 | 3.23 | 3.45  | 0.94 | 3.23    | 3.45  | 0.94 |
|              | -5          | 8.12    | 12.09 | 0.67 | 8.12         | 12.09 | 0.67 | 6.54 | 12.60 | 0.52 | 6.54 | 12.60 | 0.52 | 6.54 | 12.60 | 0.52 | 6.54    | 12.60 | 0.52 |
|              | 0           | 8.05    | 10.55 | 0.76 | 8.05         | 10.55 | 0.76 | 5.71 | 11.45 | 0.50 | 5.71 | 11.45 | 0.50 | 5.71 | 11.45 | 0.50 | 5.71    | 11.45 | 0.50 |
|              | 5           | 7.82    | 9.60  | 0.81 | 7.82         | 9.60  | 0.81 | 5.58 | 10.52 | 0.53 | 5.58 | 10.52 | 0.53 | 5.58 | 10.52 | 0.53 | 5.58    | 10.52 | 0.53 |
|              | 10          | 7.65    | 8.84  | 0.86 | 7.65         | 8.84  | 0.86 | 5.14 | 9.65  | 0.53 | 5.14 | 9.65  | 0.53 | 5.14 | 9.65  | 0.53 | 5.14    | 9.65  | 0.53 |
|              | 15          | 8.79    | 7.79  | 1.13 | 8.79         | 8.02  | 1.10 | 7.25 | 8.86  | 0.82 | 4.71 | 9.43  | 0.50 | 4.71 | 9.43  | 0.50 | 4.71    | 9.43  | 0.50 |
|              | 19          | 9.69    | 6.82  | 1.42 | 9.69         | 6.82  | 1.42 | 7.21 | 7.87  | 0.92 | 4.87 | 8.91  | 0.55 | 4.87 | 8.91  | 0.55 | 4.87    | 8.91  | 0.55 |
|              | 20          | 9.91    | 6.10  | 1.62 | 9.61         | 6.32  | 1.52 | 7.31 | 7.35  | 0.99 | 4.84 | 8.40  | 0.58 | 4.84 | 8.40  | 0.58 | 4.84    | 8.40  | 0.58 |
|              | 25          | 10.66   | 5.35  | 1.99 | 9.67         | 5.74  | 1.68 | 7.29 | 6.85  | 1.06 | 4.85 | 7.65  | 0.63 | 4.67 | 7.71  | 0.61 | 4.67    | 7.71  | 0.61 |
| 20           | 30          | 11.11   | 4.31  | 2.58 | 9.68         | 4.87  | 1.99 | 7.31 | 5.83  | 1.25 | 4.82 | 6.41  | 0.75 | 4.48 | 6.72  | 0.67 | 4.48    | 6.72  | 0.67 |
|              | 35          | 11.16   | 3.58  | 3.12 | 9.67         | 4.02  | 2.41 | 7.20 | 4.79  | 1.50 | 4.91 | 5.31  | 0.92 | 4.27 | 5.62  | 0.76 | 4.27    | 5.62  | 0.76 |
|              | 40          | 9.25    | 3.49  | 2.65 | 9.25         | 3.49  | 2.65 | 6.82 | 4.22  | 1.62 | 4.66 | 4.59  | 1.02 | 4.03 | 4.73  | 0.85 | 4.03    | 4.73  | 0.85 |
|              | 43          | 8.49    | 3.33  | 2.55 | 8.49         | 3.33  | 2.55 | 6.34 | 3.92  | 1.62 | 4.17 | 4.06  | 1.03 | 3.87 | 4.27  |      |         |       |      |

9.5 Cooling performance Mod. 12

| Tw. out<br>°C | DBWB<br>°C | Maximum |       |      | 100%(Normal) |        |      | 75%   |       |      | 50%  |       |      | 25%  |       |      | Minimum |       |      |
|---------------|------------|---------|-------|------|--------------|--------|------|-------|-------|------|------|-------|------|------|-------|------|---------|-------|------|
|               |            | CC      | EER   | PI   | CC           | EER    | PI   | CC    | EER   | PI   | CC   | EER   | PI   | CC   | EER   | PI   | CC      | EER   | PI   |
| 5             | -5         | /       | /     | /    | /            | /      | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|               | 0          | /       | /     | /    | /            | /      | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|               | 5          | /       | /     | /    | /            | /      | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|               | 10         | /       | /     | /    | /            | /      | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|               | 15         | /       | /     | /    | /            | /      | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|               | 19         | 9.56    | 5.20  | 1.84 | 9.56         | 5.20   | 1.84 | 8.60  | 5.37  | 1.60 | 5.84 | 5.72  | 1.02 | 4.46 | 5.93  | 0.75 | 4.46    | 5.93  | 0.75 |
|               | 20         | 9.82    | 4.94  | 1.99 | 9.82         | 4.94   | 1.99 | 8.59  | 5.21  | 1.65 | 5.79 | 5.53  | 1.05 | 4.44 | 5.74  | 0.77 | 4.44    | 5.74  | 0.77 |
|               | 25         | 10.63   | 3.97  | 2.68 | 10.63        | 3.97   | 2.68 | 8.71  | 4.41  | 1.97 | 5.71 | 4.71  | 1.21 | 4.27 | 4.92  | 0.87 | 4.27    | 4.92  | 0.87 |
|               | 30         | 11.13   | 3.28  | 3.39 | 11.13        | 3.28   | 3.39 | 8.64  | 3.78  | 2.28 | 5.85 | 4.01  | 1.46 | 4.12 | 4.26  | 0.97 | 4.12    | 4.26  | 0.97 |
|               | 35         | 11.25   | 2.79  | 4.03 | 11.25        | 2.79   | 4.03 | 8.62  | 3.21  | 2.69 | 5.76 | 3.51  | 1.64 | 3.90 | 3.66  | 1.07 | 3.90    | 3.66  | 1.07 |
| 7             | 40         | 9.88    | 2.56  | 3.86 | 9.88         | 2.56   | 3.86 | 8.66  | 2.76  | 3.14 | 5.75 | 3.02  | 1.90 | 3.71 | 3.17  | 1.17 | 3.71    | 3.17  | 1.17 |
|               | 43         | 8.60    | 2.45  | 3.51 | 8.60         | 2.45   | 3.51 | 8.55  | 2.48  | 3.45 | 5.73 | 2.79  | 2.05 | 3.57 | 2.89  | 1.23 | 3.57    | 2.89  | 1.23 |
|               | 46         | 6.64    | 2.31  | 2.88 | 6.64         | 2.31   | 2.88 | 6.64  | 2.31  | 2.88 | 5.75 | 2.45  | 2.34 | 3.48 | 2.67  | 1.30 | 3.48    | 2.67  | 1.30 |
|               | -5         | /       | /     | /    | /            | /      | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|               | 0          | /       | /     | /    | /            | /      | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|               | 5          | /       | /     | /    | /            | /      | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|               | 10         | /       | /     | /    | /            | /      | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|               | 15         | /       | /     | /    | /            | /      | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|               | 19         | 10.19   | 5.53  | 1.84 | 10.19        | 5.53   | 1.84 | 8.59  | 6.02  | 1.43 | 5.69 | 6.18  | 0.92 | 4.79 | 6.39  | 0.75 | 4.79    | 6.39  | 0.75 |
|               | 20         | 10.46   | 5.24  | 1.99 | 10.46        | 5.24   | 1.99 | 8.51  | 5.64  | 1.51 | 5.68 | 6.04  | 0.94 | 4.76 | 6.18  | 0.77 | 4.76    | 6.18  | 0.77 |
| 10            | 25         | 11.32   | 4.18  | 2.71 | 11.32        | 4.18   | 2.71 | 8.67  | 4.90  | 1.77 | 5.71 | 5.11  | 1.12 | 4.57 | 5.28  | 0.87 | 4.57    | 5.28  | 0.87 |
|               | 30         | 11.85   | 3.44  | 3.44 | 11.56        | 3.51   | 3.29 | 8.71  | 4.22  | 2.06 | 5.68 | 4.35  | 1.31 | 4.40 | 4.55  | 0.97 | 4.40    | 4.55  | 0.97 |
|               | 35         | 11.96   | 2.99  | 4.00 | 11.50        | 3.05   | 3.77 | 8.57  | 3.60  | 2.38 | 5.81 | 3.76  | 1.54 | 4.17 | 3.90  | 1.07 | 4.17    | 3.90  | 1.07 |
|               | 40         | 10.57   | 2.70  | 3.91 | 10.57        | 2.70   | 3.91 | 8.58  | 3.05  | 2.81 | 5.82 | 3.23  | 1.80 | 3.97 | 3.37  | 1.18 | 3.97    | 3.37  | 1.18 |
|               | 43         | 9.13    | 2.55  | 3.58 | 9.13         | 2.55   | 3.58 | 8.61  | 2.64  | 3.26 | 5.82 | 2.98  | 1.95 | 3.82 | 3.07  | 1.24 | 3.82    | 3.07  | 1.24 |
|               | 46         | 7.08    | 2.42  | 2.93 | 7.08         | 2.42   | 2.93 | 7.08  | 2.42  | 2.93 | 5.75 | 2.60  | 2.21 | 3.72 | 2.83  | 1.31 | 3.72    | 2.83  | 1.31 |
|               | -5         | /       | /     | /    | /            | /      | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|               | 0          | /       | /     | /    | /            | /      | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|               | 5          | /       | /     | /    | /            | /      | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|               | 10         | /       | /     | /    | /            | /      | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 11            | 15         | 10.09   | 7.20  | 1.40 | 10.09        | 7.20   | 1.40 | 8.78  | 7.75  | 1.13 | 5.85 | 8.34  | 0.70 | 5.41 | 8.37  | 0.65 | 5.41    | 8.37  | 0.65 |
|               | 19         | 11.19   | 6.08  | 1.84 | 11.19        | 6.08   | 1.84 | 8.80  | 6.85  | 1.28 | 5.70 | 7.27  | 0.78 | 5.33 | 7.37  | 0.72 | 5.33    | 7.37  | 0.72 |
|               | 20         | 11.49   | 5.74  | 2.00 | 11.49        | 5.74   | 2.00 | 8.72  | 6.62  | 1.32 | 5.85 | 7.01  | 0.84 | 5.29 | 7.10  | 0.74 | 5.29    | 7.10  | 0.74 |
|               | 25         | 12.40   | 4.51  | 2.75 | 11.69        | 4.73   | 2.47 | 8.73  | 5.56  | 1.57 | 5.89 | 5.83  | 1.01 | 5.05 | 5.89  | 0.86 | 5.05    | 5.89  | 0.86 |
|               | 30         | 12.97   | 3.70  | 3.51 | 11.71        | 4.03   | 2.91 | 8.81  | 4.68  | 1.88 | 5.85 | 4.86  | 1.20 | 4.85 | 5.04  | 0.96 | 4.85    | 5.04  | 0.96 |
|               | 35         | 13.05   | 3.10  | 4.21 | 11.63        | 3.43   | 3.39 | 8.71  | 4.08  | 2.13 | 5.81 | 4.20  | 1.38 | 4.60 | 4.29  | 1.07 | 4.60    | 4.29  | 1.07 |
|               | 40         | 11.55   | 2.80  | 4.12 | 10.57        | 3.02   | 3.50 | 8.80  | 3.45  | 2.55 | 5.88 | 3.57  | 1.65 | 4.38 | 3.69  | 1.19 | 4.38    | 3.69  | 1.19 |
|               | 43         | 10.04   | 2.74  | 3.66 | 10.04        | 2.74   | 3.66 | 8.69  | 2.99  | 2.91 | 5.90 | 3.28  | 1.80 | 4.22 | 3.35  | 1.26 | 4.22    | 3.35  | 1.26 |
|               | 46         | 7.80    | 2.61  | 2.99 | 7.80         | 2.61   | 2.99 | 7.80  | 2.61  | 2.99 | 5.82 | 2.99  | 1.95 | 4.12 | 3.10  | 1.33 | 4.12    | 3.10  | 1.33 |
|               | -5         | /       | /     | /    | /            | /      | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 15            | 0          | /       | /     | /    | /            | /      | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|               | 5          | /       | /     | /    | /            | /      | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|               | 10         | 8.97    | 9.13  | 0.98 | 8.97         | 9.13   | 0.98 | 8.76  | 9.40  | 0.93 | 7.23 | 9.99  | 0.72 | 7.23 | 9.99  | 0.72 | 7.23    | 9.99  | 0.72 |
|               | 15         | 10.40   | 7.44  | 1.40 | 10.40        | 7.44   | 1.40 | 8.83  | 8.15  | 1.08 | 6.05 | 8.72  | 0.69 | 5.60 | 8.76  | 0.64 | 5.60    | 8.76  | 0.64 |
|               | 19         | 11.54   | 6.27  | 1.84 | 11.54        | 6.27   | 1.84 | 8.85  | 7.17  | 1.24 | 5.77 | 7.60  | 0.76 | 5.48 | 7.63  | 0.72 | 5.48    | 7.63  | 0.72 |
|               | 20         | 11.84   | 5.91  | 2.00 | 11.64        | 6.00   | 1.94 | 8.77  | 6.92  | 1.27 | 5.81 | 7.37  | 0.79 | 5.47 | 7.41  | 0.74 | 5.47    | 7.41  | 0.74 |
|               | 25         | 12.78   | 4.63  | 2.76 | 11.67        | 4.97   | 2.35 | 8.80  | 5.78  | 1.52 | 5.78 | 6.13  | 0.94 | 5.25 | 6.17  | 0.85 | 5.25    | 6.17  | 0.85 |
|               | 30         | 13.36   | 3.78  | 3.53 | 11.72        | 4.22   | 2.78 | 8.69  | 5.01  | 1.73 | 5.71 | 5.16  | 1.10 | 5.00 | 5.21  | 0.96 | 5.00    | 5.21  | 0.96 |
|               | 35         | 13.44   | 3.17  | 4.24 | 11.68        | 3.59   | 3.25 | 8.81  | 4.25  | 2.07 | 5.64 | 4.41  | 1.28 | 4.79 | 4.47  | 1.07 | 4.79    | 4.47  | 1.07 |
|               | 40         | 11.89   | 2.93  | 4.05 | 11.76        | 2.96   | 3.97 | 8.74  | 3.52  | 2.48 | 5.78 | 3.79  | 1.52 | 4.61 | 3.86  | 1.20 | 4.61    | 3.86  | 1.20 |
| 18            | 43         | 10.34   | 2.80  | 3.69 | 10.34        | 2.80   | 3.69 | 8.81  | 3.10  | 2.84 | 5.74 | 3.42  | 1.68 | 4.45 | 3.51  | 1.27 | 4.45    | 3.51  | 1.27 |
|               | 46         | 8.05    | 2.68  | 3.01 | 8.05         | 2.68   | 3.01 | 8.05  | 2.68  | 3.01 | 5.84 | 3.08  | 1.90 | 4.26 | 3.19  | 1.33 | 4.26    | 3.19  | 1.33 |
|               | -5         | 9.88    | 12.96 | 0.76 | 9.88         | 12.96  | 0.76 | 9.88  | 12.96 | 0.76 | 8.90 | 13.65 | 0.65 | 8.90 | 13.65 | 0.65 | 8.90    | 13.65 | 0.65 |
|               | 0          | 10.00   | 12.18 | 0.82 | 10.00        | 12.18  | 0.82 | 10.00 | 12.34 | 0.81 | 8.91 | 12.86 | 0.69 | 8.91 | 12.86 | 0.69 | 8.91    | 12.86 | 0.69 |
|               | 5          | 10.27   | 10.97 | 0.94 | 10.27        | 10.97  | 0.94 | 8.89  | 11.83 | 0.75 | 8.89 | 11.83 | 0.75 | 8.89 | 11.83 | 0.75 | 8.89    | 11.83 | 0.75 |
|               | 10         | 10.10   | 9.60  | 1.05 | 10.10        | 9.60   | 1.05 | 8.80  | 10.40 | 0.85 | 8.19 | 10.75 | 0.76 | 8.19 | 10.75 | 0.76 | 8.19    | 10.75 | 0.76 |
|               | 15         | 11.68   | 8.26  | 1.41 | 11.68        | 8.26   | 1.41 | 8.94  | 10.20 | 0.88 | 6.39 | 10.59 | 0.60 | 6.39 | 10.59 | 0.60 | 6.39    | 10.59 | 0.60 |
|               | 19         | 13.03   | 7.14  | 1.83 | 11.73        | 7.55   | 1.55 | 8.99  | 8.74  | 1.03 | 6.28 | 9.19  | 0.68 | 6.28 | 9.19  | 0.68 | 6.28    | 9.19  | 0.68 |
|               | 20         | 13.35   | 6.68  | 2.00 | 11.85        | 7.16   | 1.65 | 8.91  | 8.41  | 1.06 | 6.22 | 8.77  | 0.71 | 6.22 | 8.77  | 0.71 | 6.22    | 8.77  | 0.71 |
|               | 25         | 14.35   | 5.11  | 2.81 | 11.86        | 6.04   | 1.97 | 9.00  | 6.85  | 1.31 | 5.95 | 7.18  | 0.83 | 5.95 | 7.18  | 0.83 | 5.95    | 7.18  | 0.83 |
| 20            | 30         | 15.04   | 4.16  | 3.62 | 11.83        | 5.12   | 2.31 | 8.95  | 5.86  | 1.53 | 6.02 | 6.21  | 0.97 | 5.69 | 6.34  | 0.90 | 5.69    | 6.34  | 0.90 |
|               | 35         | 14.86   | 3.50  | 4.24 | 11.84        | 4.17   | 2.84 | 8.93  | 4.95  | 1.81 | 5.95 | 5.12  | 1.16 | 5.39 | 5.18  | 1.04 | 5.39    | 5.18  | 1.04 |
|               | 40         | 13.31   | 3.19  | 4.17 | 12.01        | 3.47   | 3.46 | 8.91  | 4.14  | 2.15 | 6.05 | 4.30  | 1.41 | 5.11 | 4.36  | 1.17 | 5.11    | 4.36  | 1.17 |
|               | 43         | 11.58   | 3.05  | 3.80 | 11.58        | 3.05</ |      |       |       |      |      |       |      |      |       |      |         |       |      |

## 9.6 Cooling performance Mod. 14

| Tw_out<br>°C | DB/WB<br>°C | Maximum |       |      | 100%(Normal) |       |      | 75%   |       |      | 50%  |       |      | 25%  |       |      | Minimum |       |      |
|--------------|-------------|---------|-------|------|--------------|-------|------|-------|-------|------|------|-------|------|------|-------|------|---------|-------|------|
|              |             | CC      | EER   | PI   | CC           | EER   | PI   | CC    | EER   | PI   | CC   | EER   | PI   | CC   | EER   | PI   | CC      | EER   | PI   |
| 5            | -5          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 0           | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5           | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 15          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 19          | 10.25   | 4.96  | 2.06 | 10.25        | 4.96  | 2.06 | 9.55  | 5.19  | 1.84 | 6.43 | 5.59  | 1.15 | 4.46 | 5.93  | 0.75 | 4.46    | 5.93  | 0.75 |
|              | 20          | 10.32   | 4.77  | 2.16 | 10.32        | 4.77  | 2.16 | 9.64  | 4.99  | 1.93 | 6.37 | 5.41  | 1.18 | 4.44 | 5.74  | 0.77 | 4.44    | 5.74  | 0.77 |
|              | 25          | 11.21   | 3.79  | 2.96 | 11.21        | 3.79  | 2.96 | 9.52  | 4.25  | 2.24 | 6.44 | 4.60  | 1.40 | 4.27 | 4.92  | 0.87 | 4.27    | 4.92  | 0.87 |
|              | 30          | 11.92   | 3.11  | 3.83 | 11.92        | 3.11  | 3.83 | 9.56  | 3.63  | 2.63 | 6.36 | 3.97  | 1.60 | 4.12 | 4.26  | 0.97 | 4.12    | 4.26  | 0.97 |
|              | 35          | 11.98   | 2.65  | 4.52 | 11.98        | 2.65  | 4.52 | 9.54  | 3.11  | 3.07 | 6.40 | 3.45  | 1.86 | 3.90 | 3.66  | 1.07 | 3.90    | 3.66  | 1.07 |
| 7            | 40          | 9.88    | 2.56  | 3.86 | 9.88         | 2.56  | 3.86 | 9.54  | 2.63  | 3.62 | 6.34 | 2.98  | 2.13 | 3.71 | 3.17  | 1.17 | 3.71    | 3.17  | 1.17 |
|              | 43          | 8.60    | 2.45  | 3.51 | 8.60         | 2.45  | 3.51 | 8.60  | 2.45  | 3.51 | 6.30 | 2.72  | 2.31 | 3.57 | 2.89  | 1.23 | 3.57    | 2.89  | 1.23 |
|              | 46          | 6.64    | 2.31  | 2.88 | 6.64         | 2.31  | 2.88 | 6.64  | 2.31  | 2.88 | 6.35 | 2.37  | 2.68 | 3.48 | 2.67  | 1.30 | 3.48    | 2.67  | 1.30 |
|              | -5          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 0           | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5           | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 15          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 19          | 10.92   | 5.26  | 2.07 | 10.92        | 5.26  | 2.07 | 9.62  | 5.74  | 1.68 | 6.44 | 6.11  | 1.05 | 4.79 | 6.39  | 0.75 | 4.79    | 6.39  | 0.75 |
|              | 20          | 11.00   | 5.06  | 2.17 | 11.00        | 5.06  | 2.17 | 9.54  | 5.57  | 1.71 | 6.38 | 5.78  | 1.10 | 4.76 | 6.18  | 0.77 | 4.76    | 6.18  | 0.77 |
| 10           | 25          | 11.93   | 3.99  | 2.99 | 11.93        | 3.99  | 2.99 | 9.63  | 4.65  | 2.07 | 6.31 | 5.00  | 1.26 | 4.57 | 5.28  | 0.87 | 4.57    | 5.28  | 0.87 |
|              | 30          | 12.67   | 3.25  | 3.89 | 12.67        | 3.25  | 3.89 | 9.54  | 3.99  | 2.39 | 6.43 | 4.26  | 1.51 | 4.40 | 4.55  | 0.97 | 4.40    | 4.55  | 0.97 |
|              | 35          | 12.70   | 2.90  | 4.38 | 12.70        | 2.90  | 4.38 | 9.56  | 3.39  | 2.82 | 6.32 | 3.71  | 1.71 | 4.17 | 3.90  | 1.07 | 4.17    | 3.90  | 1.07 |
|              | 40          | 10.57   | 2.70  | 3.91 | 10.57        | 2.70  | 3.91 | 9.51  | 2.90  | 3.28 | 6.30 | 3.19  | 1.98 | 3.97 | 3.37  | 1.18 | 3.97    | 3.37  | 1.18 |
|              | 43          | 9.13    | 2.55  | 3.58 | 9.13         | 2.55  | 3.58 | 9.13  | 2.55  | 3.58 | 6.28 | 2.88  | 2.18 | 3.82 | 3.07  | 1.24 | 3.82    | 3.07  | 1.24 |
|              | 46          | 7.08    | 2.42  | 2.93 | 7.08         | 2.42  | 2.93 | 7.08  | 2.42  | 2.93 | 6.35 | 2.50  | 2.54 | 3.72 | 2.83  | 1.31 | 3.72    | 2.83  | 1.31 |
|              | -5          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 0           | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5           | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 11           | 15          | 10.50   | 6.95  | 1.51 | 10.50        | 6.95  | 1.51 | 9.86  | 7.26  | 1.36 | 6.49 | 8.21  | 0.79 | 5.41 | 8.37  | 0.65 | 5.41    | 8.37  | 0.65 |
|              | 19          | 11.99   | 5.77  | 2.08 | 11.99        | 5.77  | 2.08 | 9.86  | 6.51  | 1.52 | 6.64 | 7.15  | 0.93 | 5.33 | 7.37  | 0.72 | 5.33    | 7.37  | 0.72 |
|              | 20          | 12.07   | 5.53  | 2.18 | 12.07        | 5.53  | 2.18 | 9.77  | 6.30  | 1.55 | 6.59 | 6.89  | 0.96 | 5.29 | 7.10  | 0.74 | 5.29    | 7.10  | 0.74 |
|              | 25          | 13.07   | 4.30  | 3.04 | 13.07        | 4.30  | 3.04 | 9.71  | 5.25  | 1.85 | 6.50 | 5.74  | 1.13 | 5.05 | 5.89  | 0.86 | 5.05    | 5.89  | 0.86 |
|              | 30          | 13.86   | 3.48  | 3.98 | 13.11        | 3.65  | 3.59 | 9.76  | 4.54  | 2.15 | 6.46 | 4.78  | 1.35 | 4.85 | 5.04  | 0.96 | 4.85    | 5.04  | 0.96 |
|              | 35          | 13.87   | 2.92  | 4.74 | 13.05        | 3.10  | 4.21 | 9.83  | 3.83  | 2.57 | 6.59 | 4.11  | 1.61 | 4.60 | 4.29  | 1.07 | 4.60    | 4.29  | 1.07 |
|              | 40          | 11.55   | 2.80  | 4.12 | 11.55        | 2.80  | 4.12 | 9.74  | 3.22  | 3.02 | 6.59 | 3.51  | 1.88 | 4.38 | 3.69  | 1.19 | 4.38    | 3.69  | 1.19 |
|              | 43          | 10.04   | 2.74  | 3.66 | 10.04        | 2.74  | 3.66 | 9.88  | 2.80  | 3.53 | 6.58 | 3.16  | 2.08 | 4.22 | 3.35  | 1.26 | 4.22    | 3.35  | 1.26 |
|              | 46          | 7.80    | 2.61  | 2.99 | 7.80         | 2.61  | 2.99 | 7.80  | 2.61  | 2.99 | 6.53 | 2.84  | 2.30 | 4.12 | 3.10  | 1.33 | 4.12    | 3.10  | 1.33 |
|              | -5          | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 15           | 0           | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 5           | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|              | 10          | 9.45    | 9.04  | 1.04 | 9.45         | 9.04  | 1.04 | 9.45  | 9.04  | 1.04 | 7.23 | 9.99  | 0.72 | 7.23 | 9.99  | 0.72 | 7.23    | 9.99  | 0.72 |
|              | 15          | 10.84   | 7.18  | 1.51 | 10.84        | 7.18  | 1.51 | 9.73  | 7.69  | 1.26 | 6.70 | 8.54  | 0.78 | 5.60 | 8.76  | 0.64 | 5.60    | 8.76  | 0.64 |
|              | 19          | 12.36   | 5.95  | 2.08 | 12.36        | 5.95  | 2.08 | 9.74  | 6.86  | 1.42 | 6.63 | 7.54  | 0.88 | 5.48 | 7.63  | 0.72 | 5.48    | 7.63  | 0.72 |
|              | 20          | 12.45   | 5.69  | 2.19 | 12.45        | 5.69  | 2.19 | 9.86  | 6.57  | 1.50 | 6.57 | 7.25  | 0.91 | 5.47 | 7.41  | 0.74 | 5.47    | 7.41  | 0.74 |
|              | 25          | 13.46   | 4.41  | 3.05 | 13.29        | 4.46  | 2.98 | 9.81  | 5.49  | 1.79 | 6.53 | 6.00  | 1.09 | 5.25 | 6.17  | 0.85 | 5.25    | 6.17  | 0.85 |
|              | 30          | 14.27   | 3.56  | 4.01 | 13.20        | 3.82  | 3.45 | 9.87  | 4.72  | 2.09 | 6.67 | 5.09  | 1.31 | 5.00 | 5.21  | 0.96 | 5.00    | 5.21  | 0.96 |
|              | 35          | 14.38   | 3.00  | 4.80 | 13.15        | 3.23  | 4.08 | 9.88  | 3.95  | 2.50 | 6.61 | 4.30  | 1.54 | 4.79 | 4.47  | 1.07 | 4.79    | 4.47  | 1.07 |
|              | 40          | 11.89   | 2.93  | 4.05 | 11.90        | 2.94  | 4.05 | 9.88  | 3.34  | 2.96 | 6.62 | 3.71  | 1.79 | 4.61 | 3.86  | 1.20 | 4.61    | 3.86  | 1.20 |
| 18           | 43          | 10.34   | 2.80  | 3.69 | 10.34        | 2.80  | 3.69 | 9.91  | 2.88  | 3.45 | 6.62 | 3.31  | 2.00 | 4.45 | 3.51  | 1.27 | 4.45    | 3.51  | 1.27 |
|              | 46          | 8.05    | 2.68  | 3.01 | 8.05         | 2.68  | 3.01 | 8.05  | 2.68  | 3.01 | 6.59 | 3.03  | 2.18 | 4.26 | 3.19  | 1.33 | 4.26    | 3.19  | 1.33 |
|              | -5          | 10.12   | 12.81 | 0.79 | 10.12        | 12.81 | 0.79 | 10.12 | 12.81 | 0.79 | 8.90 | 13.65 | 0.65 | 8.90 | 13.65 | 0.65 | 8.90    | 13.65 | 0.65 |
|              | 0           | 10.25   | 12.02 | 0.85 | 10.25        | 12.02 | 0.85 | 10.25 | 12.02 | 0.85 | 8.91 | 12.86 | 0.69 | 8.91 | 12.86 | 0.69 | 8.91    | 12.86 | 0.69 |
|              | 5           | 10.54   | 10.90 | 0.97 | 10.54        | 10.90 | 0.97 | 10.27 | 10.97 | 0.94 | 8.89 | 11.83 | 0.75 | 8.89 | 11.83 | 0.75 | 8.89    | 11.83 | 0.75 |
|              | 10          | 10.63   | 9.39  | 1.13 | 10.63        | 9.39  | 1.13 | 10.36 | 9.70  | 1.07 | 8.19 | 10.75 | 0.76 | 8.19 | 10.75 | 0.76 | 8.19    | 10.75 | 0.76 |
|              | 15          | 12.14   | 8.09  | 1.50 | 12.14        | 8.09  | 1.50 | 10.18 | 9.22  | 1.10 | 6.87 | 10.37 | 0.66 | 6.39 | 10.59 | 0.60 | 6.39    | 10.59 | 0.60 |
|              | 19          | 13.94   | 6.72  | 2.07 | 13.48        | 6.90  | 1.95 | 10.26 | 8.16  | 1.26 | 6.80 | 9.11  | 0.75 | 6.28 | 9.19  | 0.68 | 6.28    | 9.19  | 0.68 |
|              | 20          | 14.02   | 6.39  | 2.19 | 13.58        | 6.58  | 2.06 | 10.16 | 7.87  | 1.29 | 6.75 | 8.72  | 0.77 | 6.22 | 8.77  | 0.71 | 6.22    | 8.77  | 0.71 |
|              | 25          | 15.09   | 4.84  | 3.12 | 13.59        | 5.43  | 2.50 | 10.16 | 6.47  | 1.57 | 6.93 | 7.10  | 0.98 | 5.95 | 7.18  | 0.83 | 5.95    | 7.18  | 0.83 |
| 20           | 30          | 16.05   | 3.89  | 4.12 | 13.63        | 4.58  | 2.97 | 10.31 | 5.59  | 1.84 | 6.89 | 6.13  | 1.12 | 5.69 | 6.34  | 0.90 | 5.69    | 6.34  | 0.90 |
|              | 35          | 16.03   | 3.24  | 4.95 | 13.61        | 3.79  | 3.59 | 10.19 | 4.69  | 2.17 | 6.80 | 5.03  | 1.35 | 5.39 | 5.18  | 1.04 | 5.39    | 5.18  | 1.04 |
|              | 40          | 13.31   | 3.19  | 4.17 | 13.31        | 3.19  | 4.17 | 10.25 | 3.92  | 2.61 | 6.89 | 4.20  | 1.64 | 5.11 | 4.36  | 1.17 | 5.11    | 4.36  | 1.17 |
|              | 43          | 11.58   | 3.05  | 3.80 | 11.58        | 3.05  | 3.80 | 10.26 | 3.51  | 2.92 |      |       |      |      |       |      |         |       |      |

## 9.7 Cooling performance Mod. 16

|                    |       | Maximum |       |      | 100%(Normal) |       |      | 75%   |       |      | 50%  |       |      | 25%  |       |      | Minimum |       |      |
|--------------------|-------|---------|-------|------|--------------|-------|------|-------|-------|------|------|-------|------|------|-------|------|---------|-------|------|
| T <sub>w,out</sub> | DB/WB | CC      | EER   | PI   | CC           | EER   | PI   | CC    | EER   | PI   | CC   | EER   | PI   | CC   | EER   | PI   | CC      | EER   | PI   |
| 5                  | -5    | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|                    | 0     | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|                    | 5     | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|                    | 10    | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|                    | 15    | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|                    | 19    | 11.00   | 4.80  | 2.29 | 11.00        | 4.80  | 2.29 | 10.48 | 4.95  | 2.12 | 7.01 | 5.48  | 1.28 | 4.46 | 5.93  | 0.75 | 4.46    | 5.93  | 0.75 |
|                    | 20    | 11.22   | 4.57  | 2.45 | 11.22        | 4.57  | 2.45 | 10.56 | 4.76  | 2.22 | 6.95 | 5.31  | 1.31 | 4.44 | 5.74  | 0.77 | 4.44    | 5.74  | 0.77 |
|                    | 25    | 12.16   | 3.62  | 3.36 | 12.16        | 3.62  | 3.36 | 10.54 | 4.04  | 2.61 | 6.98 | 4.52  | 1.54 | 4.27 | 4.92  | 0.87 | 4.27    | 4.92  | 0.87 |
|                    | 30    | 13.12   | 2.95  | 4.44 | 13.12        | 2.95  | 4.44 | 10.54 | 3.47  | 3.04 | 7.07 | 3.88  | 1.82 | 4.12 | 4.26  | 0.97 | 4.12    | 4.26  | 0.97 |
|                    | 35    | 13.41   | 2.55  | 5.26 | 13.41        | 2.55  | 5.26 | 10.46 | 2.99  | 3.50 | 6.92 | 3.34  | 2.08 | 3.90 | 3.66  | 1.07 | 3.90    | 3.66  | 1.07 |
| 7                  | 40    | 10.48   | 2.51  | 4.17 | 10.48        | 2.51  | 4.17 | 10.48 | 2.51  | 4.17 | 6.98 | 2.85  | 2.45 | 3.71 | 3.17  | 1.17 | 3.71    | 3.17  | 1.17 |
|                    | 43    | 9.13    | 2.39  | 3.82 | 9.13         | 2.39  | 3.82 | 9.13  | 2.39  | 3.82 | 7.02 | 2.64  | 2.66 | 3.57 | 2.89  | 1.23 | 3.57    | 2.89  | 1.23 |
|                    | 46    | 7.02    | 2.28  | 3.08 | 7.02         | 2.28  | 3.08 | 7.02  | 2.28  | 3.08 | 7.02 | 2.28  | 3.08 | 3.48 | 2.67  | 1.30 | 3.48    | 2.67  | 1.30 |
|                    | -5    | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|                    | 0     | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|                    | 5     | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|                    | 10    | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|                    | 15    | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|                    | 19    | 11.72   | 5.10  | 2.30 | 11.72        | 5.10  | 2.30 | 10.61 | 5.43  | 1.95 | 7.08 | 5.97  | 1.19 | 4.79 | 6.39  | 0.75 | 4.79    | 6.39  | 0.75 |
|                    | 20    | 11.95   | 4.84  | 2.47 | 11.95        | 4.84  | 2.47 | 10.52 | 5.27  | 1.99 | 7.02 | 5.60  | 1.25 | 4.76 | 6.18  | 0.77 | 4.76    | 6.18  | 0.77 |
| 10                 | 25    | 12.93   | 3.80  | 3.40 | 12.93        | 3.80  | 3.40 | 10.55 | 4.44  | 2.38 | 7.08 | 4.88  | 1.45 | 4.57 | 5.28  | 0.87 | 4.57    | 5.28  | 0.87 |
|                    | 30    | 13.94   | 3.09  | 4.52 | 13.94        | 3.09  | 4.52 | 10.44 | 3.83  | 2.73 | 7.01 | 4.18  | 1.68 | 4.40 | 4.55  | 0.97 | 4.40    | 4.55  | 0.97 |
|                    | 35    | 14.26   | 2.60  | 5.48 | 14.00        | 2.75  | 5.09 | 10.72 | 3.23  | 3.32 | 7.06 | 3.64  | 1.94 | 4.17 | 3.90  | 1.07 | 4.17    | 3.90  | 1.07 |
|                    | 40    | 11.13   | 2.61  | 4.26 | 11.13        | 2.61  | 4.26 | 10.43 | 2.78  | 3.74 | 6.99 | 3.13  | 2.23 | 3.97 | 3.37  | 1.18 | 3.97    | 3.37  | 1.18 |
|                    | 43    | 9.71    | 2.50  | 3.89 | 9.71         | 2.50  | 3.89 | 9.71  | 2.50  | 3.89 | 7.09 | 2.81  | 2.52 | 3.82 | 3.07  | 1.24 | 3.82    | 3.07  | 1.24 |
|                    | 46    | 7.48    | 2.38  | 3.15 | 7.48         | 2.38  | 3.15 | 7.48  | 2.38  | 3.15 | 7.00 | 2.43  | 2.89 | 3.72 | 2.83  | 1.31 | 3.72    | 2.83  | 1.31 |
|                    | -5    | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|                    | 0     | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|                    | 5     | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|                    | 10    | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 11                 | 15    | 11.32   | 6.79  | 1.67 | 11.32        | 6.79  | 1.67 | 10.90 | 7.13  | 1.53 | 7.16 | 7.78  | 0.92 | 5.41 | 8.37  | 0.65 | 5.41    | 8.37  | 0.65 |
|                    | 19    | 12.86   | 5.57  | 2.31 | 12.86        | 5.57  | 2.31 | 11.05 | 6.21  | 1.78 | 7.36 | 7.03  | 1.05 | 5.33 | 7.37  | 0.72 | 5.33    | 7.37  | 0.72 |
|                    | 20    | 13.12   | 5.28  | 2.49 | 13.12        | 5.28  | 2.49 | 10.95 | 6.01  | 1.82 | 7.29 | 6.65  | 1.10 | 5.29 | 7.10  | 0.74 | 5.29    | 7.10  | 0.74 |
|                    | 25    | 14.16   | 4.10  | 3.46 | 14.16        | 4.10  | 3.46 | 11.02 | 4.89  | 2.25 | 7.38 | 5.53  | 1.34 | 5.05 | 5.89  | 0.86 | 5.05    | 5.89  | 0.86 |
|                    | 30    | 15.23   | 3.29  | 4.62 | 14.50        | 3.45  | 4.20 | 10.94 | 4.28  | 2.56 | 7.33 | 4.64  | 1.58 | 4.85 | 5.04  | 0.96 | 4.85    | 5.04  | 0.96 |
|                    | 35    | 15.55   | 2.76  | 5.63 | 14.61        | 2.89  | 5.06 | 11.03 | 3.59  | 3.07 | 7.40 | 4.02  | 1.84 | 4.60 | 4.29  | 1.07 | 4.60    | 4.29  | 1.07 |
|                    | 40    | 12.23   | 2.73  | 4.48 | 12.23        | 2.73  | 4.48 | 11.10 | 3.01  | 3.68 | 7.33 | 3.42  | 2.14 | 4.38 | 3.69  | 1.19 | 4.38    | 3.69  | 1.19 |
|                    | 43    | 10.67   | 2.69  | 3.97 | 10.67        | 2.69  | 3.97 | 10.67 | 2.69  | 3.97 | 7.31 | 3.06  | 2.39 | 4.22 | 3.35  | 1.26 | 4.22    | 3.35  | 1.26 |
|                    | 46    | 8.24    | 2.55  | 3.23 | 8.24         | 2.55  | 3.23 | 8.24  | 2.55  | 3.23 | 7.27 | 2.79  | 2.60 | 4.12 | 3.10  | 1.33 | 4.12    | 3.10  | 1.33 |
|                    | -5    | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
| 15                 | 0     | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|                    | 5     | /       | /     | /    | /            | /     | /    | /     | /     | /    | /    | /     | /    | /    | /     | /    | /       | /     | /    |
|                    | 10    | 10.22   | 8.84  | 1.16 | 10.22        | 8.84  | 1.16 | 9.86  | 8.94  | 1.10 | 7.67 | 9.69  | 0.79 | 7.23 | 9.99  | 0.72 | 7.23    | 9.99  | 0.72 |
|                    | 15    | 11.67   | 7.00  | 1.67 | 11.67        | 7.00  | 1.67 | 10.83 | 7.18  | 1.51 | 7.38 | 8.08  | 0.91 | 5.60 | 8.76  | 0.64 | 5.60    | 8.76  | 0.64 |
|                    | 19    | 13.26   | 5.74  | 2.31 | 13.26        | 5.74  | 2.31 | 11.19 | 6.51  | 1.72 | 7.32 | 7.20  | 1.02 | 5.48 | 7.63  | 0.72 | 5.48    | 7.63  | 0.72 |
|                    | 20    | 13.52   | 5.43  | 2.49 | 13.52        | 5.43  | 2.49 | 11.09 | 6.30  | 1.76 | 7.53 | 6.91  | 1.09 | 5.47 | 7.41  | 0.74 | 5.47    | 7.41  | 0.74 |
|                    | 25    | 14.58   | 4.19  | 3.48 | 14.58        | 4.19  | 3.48 | 11.17 | 5.21  | 2.15 | 7.40 | 5.80  | 1.28 | 5.25 | 6.17  | 0.85 | 5.25    | 6.17  | 0.85 |
|                    | 30    | 15.68   | 3.37  | 4.66 | 14.78        | 3.47  | 4.26 | 11.09 | 4.44  | 2.50 | 7.36 | 4.87  | 1.51 | 5.00 | 5.21  | 0.96 | 5.00    | 5.21  | 0.96 |
|                    | 35    | 15.99   | 2.81  | 5.69 | 14.90        | 2.93  | 5.09 | 11.04 | 3.76  | 2.94 | 7.44 | 4.15  | 1.79 | 4.79 | 4.47  | 1.07 | 4.79    | 4.47  | 1.07 |
|                    | 40    | 12.64   | 2.78  | 4.55 | 12.64        | 2.78  | 4.55 | 11.19 | 3.17  | 3.53 | 7.41 | 3.54  | 2.09 | 4.61 | 3.86  | 1.20 | 4.61    | 3.86  | 1.20 |
| 18                 | 43    | 10.99   | 2.74  | 4.01 | 10.99        | 2.74  | 4.01 | 10.99 | 2.74  | 4.01 | 7.37 | 3.17  | 2.33 | 4.45 | 3.51  | 1.27 | 4.45    | 3.51  | 1.27 |
|                    | 46    | 8.51    | 2.60  | 3.27 | 8.51         | 2.60  | 3.27 | 8.51  | 2.60  | 3.27 | 7.36 | 2.92  | 2.52 | 4.26 | 3.19  | 1.33 | 4.26    | 3.19  | 1.33 |
|                    | -5    | 10.45   | 12.55 | 0.83 | 10.45        | 12.55 | 0.83 | 10.45 | 12.55 | 0.83 | 8.90 | 13.65 | 0.65 | 8.90 | 13.65 | 0.65 | 8.90    | 13.65 | 0.65 |
|                    | 0     | 10.50   | 11.74 | 0.89 | 10.50        | 11.74 | 0.89 | 10.50 | 11.74 | 0.89 | 8.91 | 12.86 | 0.69 | 8.91 | 12.86 | 0.69 | 8.91    | 12.86 | 0.69 |
|                    | 5     | 10.81   | 10.66 | 1.01 | 10.81        | 10.66 | 1.01 | 10.81 | 10.66 | 1.01 | 8.89 | 11.83 | 0.75 | 8.89 | 11.83 | 0.75 | 8.89    | 11.83 | 0.75 |
|                    | 10    | 11.46   | 9.04  | 1.27 | 11.46        | 9.04  | 1.27 | 10.76 | 9.34  | 1.15 | 8.52 | 10.46 | 0.81 | 8.19 | 10.75 | 0.76 | 8.19    | 10.75 | 0.76 |
|                    | 15    | 13.11   | 7.97  | 1.64 | 13.11        | 7.97  | 1.64 | 11.45 | 8.61  | 1.33 | 7.58 | 10.06 | 0.75 | 6.39 | 10.59 | 0.60 | 6.39    | 10.59 | 0.60 |
|                    | 19    | 14.94   | 6.46  | 2.31 | 14.94        | 6.46  | 2.31 | 11.64 | 7.90  | 1.47 | 7.82 | 8.95  | 0.87 | 6.28 | 9.19  | 0.68 | 6.28    | 9.19  | 0.68 |
|                    | 20    | 15.19   | 6.09  | 2.49 | 15.19        | 6.09  | 2.49 | 11.59 | 7.68  | 1.51 | 7.76 | 8.58  | 0.90 | 6.22 | 8.77  | 0.71 | 6.22    | 8.77  | 0.71 |
|                    | 25    | 16.41   | 4.62  | 3.55 | 15.43        | 4.88  | 3.16 | 11.73 | 6.20  | 1.89 | 7.64 | 6.98  | 1.09 | 5.95 | 7.18  | 0.83 | 5.95    | 7.18  | 0.83 |
| 20                 | 30    | 17.63   | 3.68  | 4.80 | 15.43        | 4.21  | 3.66 | 11.74 | 5.24  | 2.24 | 7.90 | 5.91  | 1.34 | 5.69 | 6.34  | 0.90 | 5.69    | 6.34  | 0.90 |
|                    | 35    | 17.42   | 3.07  | 5.68 | 15.45        | 3.50  | 4.41 | 11.72 | 4.36  | 2.69 | 7.77 | 4.90  | 1.59 | 5.39 | 5.18  | 1.04 | 5.39    | 5.18  | 1.04 |
|                    | 40    | 14.12   | 3.14  | 4.50 | 14.12        | 3.14  | 4.50 | 11.63 | 3.65  | 3.19 | 7.78 | 4.10  | 1.90 | 5.11 | 4.36  | 1.17 | 5.11    | 4.36  | 1.17 |
|                    | 43    | 12.33   | 3.00  | 4.12 | 12.33        | 3.00  | 4.12 | 11.7  |       |      |      |       |      |      |       |      |         |       |      |







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